Challenges and Advances
in Computational Chemistry and Physics 36
Series Editor: Jerzy Leszczynski

Chen Qu
Hanchao Liu *Editors*

Machine Learning in Molecular Sciences

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Challenges and Advances in Computational Chemistry and Physics

Volume 36

Series Editor

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Machine Learning in Molecular Sciences

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Preface

In July 2022, DeepMind announced that they have predicted and released structures for nearly all catalogued proteins known to science, with over 200 million structures, using their deep learning-powered AlphaFold system. This breakthrough dramatically accelerates research in human health, drug development, and almost every field in biology. This is also a highlight of the numerous research achievements of applying machine learning in the broad field of molecular science. With great pleasure and excitement, we take the opportunity to put forth this book and bring it under the series *Challenges and Advances in Computational Chemistry and Physics*. We aim to present some recent advances of machine learning research in the broad field of molecular sciences and to provide readers an overview of this vibrant research area.

This book covers both machine learning methodologies and machine learning applications in a wide range of topics in molecular sciences, from electronic structure theory, to nuclear dynamics of small molecules, to the design and synthesis of large organic and biological molecules. Chapter 1 is an introduction to the basic concepts of machine learning, followed by a pedagogical example and discussions on strengths and challenges when applying machine learning tools to molecular sciences. The following two chapters are focused on methodologies. Chapter 2 gives a comprehensive review of the graph neural network and its applications in molecular sciences. Chapter 3 talks about a voxelized representation of the material structure that could be used to construct machine-learned interatomic potentials and structure-property relationships. Each of the remaining chapters describes a specific application of machine learning in molecular sciences. Chapters 4 and 5 are both about using machine learning to solve electronic structure problem; Chap. 4 reviews how ML is used to develop exchange-correlation functionals of density functional theory, while Chap. 5 is focused on two surrogate models for electronic structure theory: One learns the local density of states and the other uses a physics-informed neural network as a partial differential equation solver to solve time-dependent Schrödinger equation. Chapter 6 is a tutorial on developing machine-learned potential energy surfaces from three aspects: data quality, data sampling, and data fitting. Chapter 7 reviews the application of machine learning in thermochemistry and chemical kinetics, such

as predicting thermodynamical properties and rate constants, and searching for transition states. Chapter 8 is focused on applications of machine learning in organic synthesis, including retrosynthesis planning, reaction discovery and optimization, and lab automation. Chapter 9 describes the use of machine learning in directed evolution and protein engineering.

We acknowledge that the machine learning research in molecular sciences is a vast, deep, and rapidly evolving topic. It would be impossible to cover all the area within one book. We hope this book can serve as an general pedagogical resource for researchers in molecular sciences and in machine learning; meanwhile, it can offer deep diving into several specific topics as mentioned above. The intended audience are relatively broad, from computational researchers in molecular sciences, to any researchers in molecular sciences who are interested in studying machine learning, to machine learning researchers who are interested in applying machine learning to natural science. Finally, we would like to thank all the authors and reviewers of this book. Without their expertise and tremendous dedication, this book would not be possible.

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Chapter 1
An Introduction to Machine Learning in Molecular Sciences



Hanchao Liu and Chen Qu

Abstract This chapter introduces the concept of machine learning and its three main approaches: supervised learning, unsupervised learning, and reinforcement learning. We then provide a pedagogical example of predicting the boiling points of chemicals using machine learning. From the example, we aim to demonstrate (1) the end-to-end process of a machine learning project, from preparing data, to building models, to evaluating the model performance, (2) the characteristics of some commonly used ML models in molecular sciences, (3) the power and challenges of applying machine learning in this typical molecular science problem. We further discuss the strengths of machine learning in the broad field of molecular sciences, and what makes it so successful in many recent applications. Finally, we discuss some typical challenges in applying machine learning in molecular sciences, including data scarcity, model explainability, and uncertainty quantification.

Keywords Machine learning · Graph convolutional neural network · Boiling point · Scalability · Data scarcity · Explainability · Uncertainty quantification

1.1 What Is Machine Learning?

Machine learning (ML) is the collection of methods that learn from data, and allow computers to describe a problem and make predictions without explicitly programming the computers to do so. The early concept of self-learning computers dates back to 1950s. It is since 2000s that machine learning started to flourish and yielded numerous achievements that transformed not only the research community but also our daily lives. Here are some machine learning use cases in our daily lives:

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- Identify whether an email is a spam or not.
- Assist doctors to make diagnosis based on medical images.
- Translate conversations in real time on a smart phone.

For molecular sciences, many of the branches have rich history of mining data and deriving models from data, such as cheminformatics, which dates back at least half a century. Since early 2010s, the broad field of molecular sciences started to widely adopt machine learning and the data-driven approaches, and referred it as "the fourth paradigm of science", in addition to the other three paradigms, experiment, theory and computation [1, 2]. Examples of machine learning in molecular sciences include:

- Predict protein 3D structures.
- Optimize organic synthesis pathways.
- Identify drug candidates.

We begin with discussing the three approaches of machine learning, supervised learning, unsupervised learning, and reinforcement learning.

1.1.1 Supervised Learning

The goal of supervised learning is to learn a function that maps input data to output target, using examples of input data with known outcomes (labels). There are two types of supervised learning, regression and classification. Regression aims to predict a number from the input data. A simple regression example is to predict the house price using the house information such as the size, built year and location. Classification aims to predict the category (or categories), among a total of two or more categories, from the input data. A famous classification example is to recognize handwritten digits. One trains a multi-class classification model to predict the digit from an image of handwriting, among ten possible outcomes from 0 to 9.

The workflow of building a supervised learning model typically includes the following steps:

1. Define the problem and identify the predicting target. The problem is often framed as one of the several classic supervised learning paradigms, such as regression, binary classification, or multi-class classification.
2. Collect the training examples and labels. The training data set may be readily available from previous studies in some cases, but often one needs to generate the training examples via data annotation by human, experimental measurements, or computations from first principles.
3. Determine the model type and the objective function. There are an increasingly rich choices of models. When choosing models, one needs to consider several factors such as the complexity of the problem, the size of the training data, computational cost and the ability to explain the model prediction. Expanded discussion on this topic will be presented in Sects. 1.2 and 1.3.

4. Determine the representation of input data. This step is also known as feature engineering. The goal of this step is to make the input data comprehensible to computers and to enhance the model predictability.
5. Train the model. In this step, the parameters of the model are learned by mathematically minimizing the objective function on the training examples.
6. Evaluate the model. After the model is trained, the model accuracy is usually evaluated against a data set that is unseen during the training step.

These steps are just a simplified overview of a supervised learning process. Often, additional steps are needed when solving a real-world problem. For example, one may have to tune the hyperparameters of the models and the learning process, to learn the best model for the problem. Another example is that one often needs to spend significant amount of efforts to ensure the model does not overfit or underfit the training data set.

Although supervised learning requires annotating data with labels, it is a powerful methodology and is widely applied in many fields, including many areas in molecular sciences. In Sect. 1.2, we will walk you through a concrete example of predicting the boiling points of chemicals to further illustrate how supervised learning is applied in molecular science and to explain how a machine learning project works from end to end.

1.1.2 Unsupervised Learning

In contrast to supervised learning which learns from labeled data, the goal of unsupervised learning is to learn internal structures or patterns from unlabeled data. Common unsupervised learning tasks include cluster analysis, anomaly detection and dimensionality reduction. Generative modeling is another important form of unsupervised learning.

Cluster analysis is a task of grouping objects such that objects in the same group are more similar to each other than to those in other groups. Marketers perform clustering analysis to group similar customers into segments, so that they can target each segment of customers effectively. Evolutional biologists apply clustering algorithms on animal genetics to establish taxonomy and propose evolutional hypotheses at the molecular level.

Anomaly detection is a task of identifying unusual data points that significantly deviate from the majority of the data points and do not conform to a well defined notion of normal behavior. Online stores run algorithms to detect fraudulent transactions. Cybersecurity companies develop advanced anomaly detection algorithms to analyze network traffic and detect cyberattacks. Anomaly detection is also used in data preprocessing to detect outliers in a data set.

Dimensionality reduction is a task of transforming data from a high-dimensional space into a low-dimensional space such that the low-dimensional representation retains some meaningful information of the original data. Dimensionality reduc-

tion is used in data compression, with the goals of saving data storage space and improving data transfer speed, while retaining all or some of the original information. Dimensionality reduction is also used in data visualization. The dimension of a multi-dimensional data set is usually reduced to two or three, so that the data can be projected and visualized in a 2D or 3D plot.

Another important form of unsupervised learning is generative modeling. The generative modeling approaches learn to generate new data with the same statistics as the training set. For example, the generative large language models are trained with enormous amount of text data and are able to summarize and generate text. Variational autoencoder (VAE) and generative adversarial networks (GAN) are two commonly used approaches in generative modeling. To explain how a VAE works, let us first take a look at the autoencoder. An autoencoder learns an encoding function that transforms the input to an encoded representation (also known as the embedding space or latent space), and a decoding function that reproduces the input from the encoded representation. Unlike an autoencoder, which transforms the input to a fixed vector in the latent space, a VAE transforms the input to a probability distribution, usually a Gaussian distribution. By sampling the learned distribution in the latent space, a VAE is able to generate new data samples that are similar to the training samples. VAE has been successfully employed in many areas, including molecular design [3–5]. GAN works by training two contesting models in a game, a generator and a discriminator, in order to "fool" the discriminator to think the generated candidates from the generator are not synthesized. Since its birth in 2014, it has yielded many successful applications such as generating realistic images and composing music. GAN has also been applied in many fields in molecular sciences, including generating candidates for drug discovery and material designs [6–12].

1.1.3 Reinforcement Learning

Reinforcement learning is a type of machine learning where a computer agent learns to perform a task through repeated trial and error. The computer agent learns how to take actions in an environment, in order to maximize its reward over time. The history of reinforcement learning has two threads [13]. One is from the psychology of animal learning, in particular the learning by trial and error. The other one is from control theory, in solving the problem of optimal control using value functions and dynamic programming.

Reinforcement learning has been applied in many fields, such as gaming [14], autonomous cars and robotics. Encouraging applications of reinforcement learning in molecular sciences have been seen in recent years, mainly in chemical robotics and self-driving laboratory [15–19].

1.2 Prediction of Boiling Points: An End-to-End Machine Learning Example

We start with a very simple example to illustrate the use of ML techniques in molecular sciences and its workflow: prediction of the normal boiling point of a chemical compound. Almost everyone has seen how water boils when it is heated up to a certain temperature (100 °C at sea level). This temperature is the boiling point, and every pure substance has its own unique boiling point. An interesting question is, can we estimate the boiling point of a chemical compound when given the chemical formula or molecular structure? In fact this problem is also very import to scientists. For example, synthetic chemists may want to know the boiling point of a compound so that they know at which temperature can they expect the final product during distillation; pharmaceutical companies use boiling point as a means to measure the purity of a substance because a small amount of impurities can change the boiling point significantly.

Before presenting a machine-learned model for this problem, let us first briefly describe the first-principle approaches that have been used to predict the boiling points.

1.2.1 First-Principle Method

In principle, the physical and chemical behaviors of molecules are governed by quantum mechanics, so one can obtain the boiling point through first-principle simulations. Unfortunately, these calculations are prohibitively expensive, even for small molecules with less than ten atoms.

Next we will give a glimpse of the typical procedures to calculate the boiling points with first principles. Note that this approach is based on the fundamental physical laws, and therefore could achieve unprecedented accuracy if every step is treated rigorously without approximations. However, this is impossible with current computers and algorithms, and approximations must be introduced to greatly reduce the computational cost. As a consequence, the accuracy is often lackluster if affordable algorithms are used.

The boiling point cannot be obtained directly by solving a Schrödinger equation; instead multiple steps are involved. The first step is separating the motion of electrons and nuclei (the Born–Oppenheimer approximation), which is based on the fact that electrons are much lighter than the nuclei and move much faster. In order to solve the electronic Schrödinger equation of the system, many electronic structure theories have been developed, and density functional theory (DFT) is probably the most widely used one. After solving the electronic Schrödinger equation at various molecular configurations, what we obtain is a potential energy surface (PES), which describes the energy of a system as a function of nuclear coordinates. This potential energy surface is needed to solve the nuclear Schrödinger equation. Since most atoms

(except hydrogen) are heavy and nuclear quantum effects are small, the nuclei can be treated as classical particles, and their motion can be computed by solving Newton's equation rather than the nuclear Schrödinger equation. This is the molecular dynamics (MD) simulation, and the potential energy surface is a key component of MD because it provides the force acting on each atom (and therefore the term "force field" are often used interchangeably with potential energy surface). In the MD simulation, researchers can determine the temperature at which a phase transition such as vaporization occurs.

So briefly, the two key steps in a first-principle calculation are the electronic structure calculation and the MD simulation. Unfortunately, these calculations are still extremely expensive; it may take days to months to calculate the boiling point of a single chemical compound. Note that ML has already been applied to these two problems to accelerate the computations, e.g., by developing reliable and efficient density functionals for the electronic structure calculation, and by constructing ML-PESs that are accurate and efficient for MD simulations. Some of these applications are described in other chapters of this book.

It is also worth mentioning that some empirical rules have been used to estimate the boiling points. For example, Trouton's rule states that for many liquids, the ratio between the enthalpy of vaporization and the boiling point is nearly a constant. It is possible to compute the enthalpy of vaporization in first principles and then use this to estimate the boiling point.

1.2.2 Machine Learning Approaches

Clearly, the first-principle approach is not feasible for most researchers because not everyone is familiar with quantum chemistry and molecular simulations, or has access to plenty of computational resources to perform these calculations. Furthermore, it is inefficient to spend months of computations to obtain only one property for one molecule. Finally, the accuracy of affordable first-principle methods is still not adequate for satisfactory predictions. A model capable for accurate and fast prediction (in a few seconds) of this property is in demand.

On the other hand, we may be able to address this problem in a data-centric way: when the boiling points of plenty of molecules have been measured experimentally, we might be able to discover the connection between the boiling point and the chemical structure, and then use this connection to predict the boiling point of an unknown molecule. We humans are considered to be good at such kind of tasks: we find patterns from existing data, generalize the rules, and then apply the rules to make predictions. This is how we obtain new knowledge in general. In fact, for a relatively simple task such as predicting the boiling point, we have already obtained a few rules successfully, described below. However, the patterns are not always easy to find and it requires expertise and domain knowledge. So naturally some researchers turned to computers and try to let the computer find these patterns. Unfortunately, computers were considered not good at this type of tasks; they can do the "weight

lifting" exceptionally well provided that the programmers tell them what to do, but they are bad at inventing rules.

Nevertheless, researchers were able to develop simple mathematical models to predict molecular properties. Specifically, it is widely believed that the structure of a molecule determines its physical and chemical properties, so experts with domain knowledge can extract features from a molecule that may be closely related to the targeted property, and then build simple regression models based on the extracted features. This kind of model is called a Quantitative Structure-Property Relationship (QSPR) model, and it is an important subject in cheminformatics [20–22]. From the perspective of machine learning, many early QSPR models are also machine-learned models. These models often employed "classical" machine learning algorithms such as linear or multivariant regression. The accuracy and robustness of these QSPR approaches are often insufficient due to the small size of the data and relatively simple models. In addition, these approaches still highly rely on the experience and intuition of the researchers because a set of features (or descriptors) that are relevant to the problem has to be selected as the input to the ML model.

For boiling point, physicists have discovered some highly relevant factors and these can be chosen as the features. For example, molecules with larger mass are harder to accelerate and escape from the liquid phase into the gas, so their boiling point is higher. Polar molecules have relatively strong inter-molecular electrostatic attraction because they have one slightly positive and one slightly negative end, so they tend to bind together as a liquid and the boiling point is usually higher than non-polar molecules with a similar mass. Based on these findings, many QSPR and "classical" ML models have been developed to predict the boiling points [23–32].

Thanks to the availability of faster computer hardware, high-quality data, and new ML algorithms, the last decade saw a surge in the successful applications of ML, especially deep learning, to many fields, including physical sciences. These modern machine learning techniques have also been applied to predict molecular properties.

Next we will walk you through the steps to train ML models to predict boiling points. Hopefully with this example, we could show what machine learning is and how it works, how machine learning differs from traditional methods, what the advantages and limitations of ML are, and why ML could be a game changer in many areas including molecular sciences.

1.2.2.1 Data Acquiring and Preprocessing

Naturally, data plays a central role in "data-driven" approaches. Quality of data is a key to the success of a ML model. There is a saying "garbage in, garbage out", meaning that supervised learning models trained with human-labeled data can only be as good as the quality of that database.

The quantity of data is also important, especially when training a deep learning model, because a deep learning model contains a large number of parameters, sometimes millions or billions of them. Therefore, a large collection of data is needed to avoid overfitting.

Therefore, obtaining a high-quality data set with sufficient number of samples is a key step to a successful machine learning model. For molecular science, data usually come from two sources: experiments and computations. Experiments, when carefully conducted, can produce high-quality data; however, experiments are often time-consuming, so experimental data are scarce. On the other hand, computational data are more abundant, but the quality may be limited by the level of theory applied in the calculations.

For boiling point, luckily, high quality experimental data are available; for example, the database captured by the National Institute of Standards and Technology (NIST)/Thermodynamics Research Center's (TRC) SOURCE Data Archival System contains more than 12,000 experimental determinations of the normal boiling point for a wide variety of molecules [33]. In the pedagogical example below, however, we will use computational data instead. The data set used in the example is constructed as follows: 5000 molecules are randomly picked from the QM9 data set [34, 35], together with 7 molecular properties (molecular mass, dipole moment, polarizability, electronic spatial extent, vibrational zero point energy, heat capacity, free energy) computed at DFT/B3LYP/6-31G(2df,p) level of theory. These 7 molecular properties can be readily used as features for the ML models. The normal boiling points of these molecules are estimated using the EPI Suite [36]. This computed data set will be used to train the ML models described below.

Before moving on, the data set needs to be split into a training set and a test set. The training set, as suggested by the name, is used to train the model. That is, parameters in the model will be optimized according to this set. The test set, on the other hand, should never be used to tune the parameters. The importance of the test set cannot be overstated because it is used to evaluate how robust the ML model is on unseen data.

1.2.2.2 Model Selection and Training

The next step is to select an appropriate model for the problem, and train the model using the training data set. Once a model is selected, there are a few (or millions of, depending on the complexity of the model) unknown parameters. The training process is to optimize these parameters so that the difference between the predicted and labeled values is as small as possible. We briefly describe three different models for predicting the boiling points.

Linear Regression: Let's start with the simplest ML model, a linear regression with only one feature (the molecular mass). A linear regression model can be written as

$$\tilde{y} = wx + b, \tag{1.1}$$

where x is the feature, i.e., molecular mass in this case, and $\tilde{y}$ is the prediction. There are two unknown parameters w and b to be optimized during the training.

We can add more features in this linear regression model. The linear regression can be generalized as

$$\tilde{y} = \boldsymbol{w}\boldsymbol{x} + b, \tag{1.2}$$

where $\boldsymbol{x} \in \mathcal{R}^n$ is a vector of length n that contains n features. In this case 7 features: molecular mass, dipole moment, polarizability, electronic spatial extent, vibrational zero point energy, heat capacity, and free energy.

The goal is to find an optimal set of $\boldsymbol{w}$ and b that minimizes the objective function

$$L = \frac{1}{N}\sum_i (y_i - \tilde{y}_i)^2 + \lambda R, \tag{1.3}$$

where y is the true value, R is a regularization term, often used to mitigate overfitting, and λ is a scalar to adjust the magnitude of the regularization term. The sum is over all training samples. In ML, L is also called a loss function. In this example, for simplicity, λ is set to zero. This is a classic problem in linear algebra, so mathematicians have already developed robust algorithms to solve this. We will skip the details here.

Unfortunately a linear regression is, indicated by its name, a linear model, so it is often insufficient when the relationship between the features and the output are nonlinear by nature. (Of course a certain level of nonlinearity can be introduced by nonlinear transformation of features. For example, we can transform feature x to $\exp(x)$ first and then apply the linear regression.) In this case, we could use a more sophisticated model, the artificial neural network (ANN).

Artificial Neural Network: In principle, an ANN can be a universal approximation for any mathematical function, thus a suitable model to be used in this case. This can be written as

$$\tilde{y} = \mathrm{ANN}(\boldsymbol{x}), \tag{1.4}$$

where $\boldsymbol{x}$ are the same 7 features used in linear regression, and ANN can be viewed as a sophisticated, nonlinear function. The goal is still to minimize the loss function, but since ANN is a nonlinear function, it is much more challenging to find the optimal parameters. The widely used algorithm to find the optimal parameters is gradient descent, and in neural networks, this algorithm can be carried out efficiently using back propagation [37]. One challenge is that it is often impossible to find the absolute lowest point of the loss function (called the global minimum); instead, the optimization may end up in a shallower well (called a local minima). In practice, fortunately, local minima are often good enough.

In reality the optimization is much more complicated than what we have summarized above, but the gradient descent and back propagation are the two fundamentals of essentially all optimizers used in ML.

In the linear regression and the simple ANN model mentioned above, the features are selected or designed by human, which requires domain knowledge and experience. If a relevant feature is not included in the model, the performance of the model is usually unsatisfactory; on the other hand, if too many irrelevant features are

included, the accuracy of the model may even deteriorate. In this boiling point problem, the other 5 features in addition to molecular mass and dipole are chosen simply because these are commonly used descriptors and available in the QM9 database.

Because the features are so important to the success of the ML model, feature engineering is often carried out before training the model. Naturally one could ask the following question: is it possible to let the system automatically "discover" the representations/features needed for a particular problem? Indeed much effort has been devoted to representation learning.

Representation Learning: Instead of using human-designed features, the input can be very basic information. The hope is that with sufficient data, the computer can figure out good features and representations. For example, here we have also trained a state-of-the-art, graph convolutional neural network (GCNN) to predict the boiling point, following the same procedures described in literature [38]. In this model, the input is basically a graph showing the connectivity of the atoms. Information involved in the input includes atom types, which atoms are in a ring, which two atoms are connected with a chemical bond and the bond order, total number of atoms and bonds in a molecule, and the molecular mass. These are very straightforward descriptors of atoms and bonds without sophisticate feature engineering. Yet it can be seen below that this GCNN model gives much better predictions compared to the linear regression and ANN model.

1.2.2.3 Model Testing

The next step is to test the models we trained, and now it's time to use the testing set held out before training. Figures 1.1, 1.2 and 1.3 show the correlation plots between the "experimental" boiling points (which are actually computed with EPI Suite, but we treat them as the true labels in this example) and the predictions from the ML models on the testing data. Each blue point represents a data point in the test set; the closer a point to the diagonal line, the better the prediction is. We can also use the coefficient of determination, R^2, to evaluate the performance of the model quantitatively.

Figure 1.1 shows the correlation plots for two linear regression models: one uses only one feature (molecular mass) and the other uses molecular mass and dipole as features. The model is basically ineffective when only using molecular mass as the feature, but when the dipole is added as the second feature, the performance is much better, and R^2 increases from 0.064 to 0.31.

Figure 1.2 compares the performance of an ANN model with a linear regression model, using all the 7 features. It can be seen that the linear regression model improves when more features are added. Furthermore, ANN, as a more expressive and flexible model, indeed outperforms the linear regression model.

However, in the linear regression model, adding 5 additional features just raise the R^2 from 0.31 to 0.47 (see right panel of Fig. 1.1 and left panel of Fig. 1.2), so these

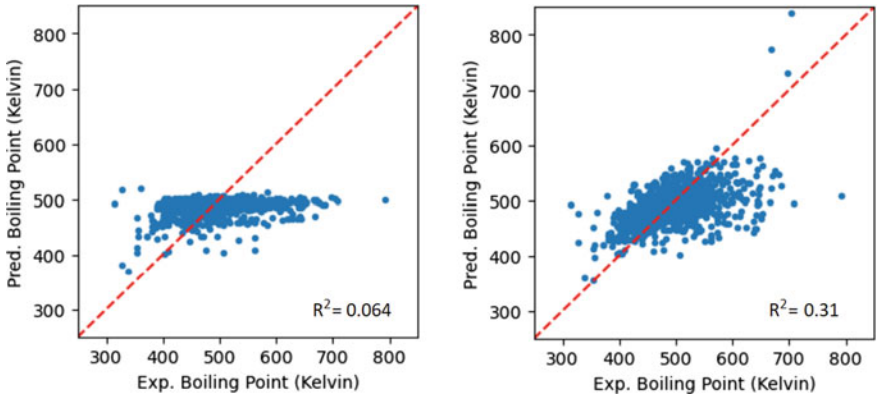


Fig. 1.1 The correlation plots between the prediction of the linear regression model and the "experiments" (left: linear regression using molecular mass as the feature; right: linear regression model using molecular mass and dipole moment as features)

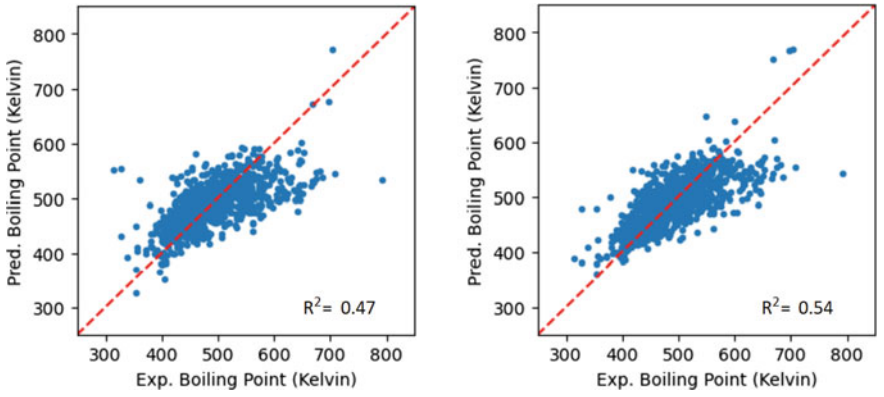


Fig. 1.2 The correlation plots between the "experiments" and the predictions from a linear regression model (left) and a ANN model (right) using 7 features

5 features are likely not quite relevant in this problem. In addition, the accuracy of the ANN model, albeit better than the linear regression, is still not satisfactory. This indicates that the features we choose are probably not good ones, and it's likely that some relevant features are missing in the model. This is not surprising because we simply took the features available in the QM9 data set without sophisticated feature engineering.

Lastly the test accuracy of the GCNN model is shown in Fig. 1.3. Clearly it is much better than the linear regression and ANN, and demonstrates the power of modern machine learning. However, the better performance does not come without a cost. In linear regression and ANN models, the inputs are a set of features/descriptors of the molecule that have physical meanings, and by careful feature engineering, scientists

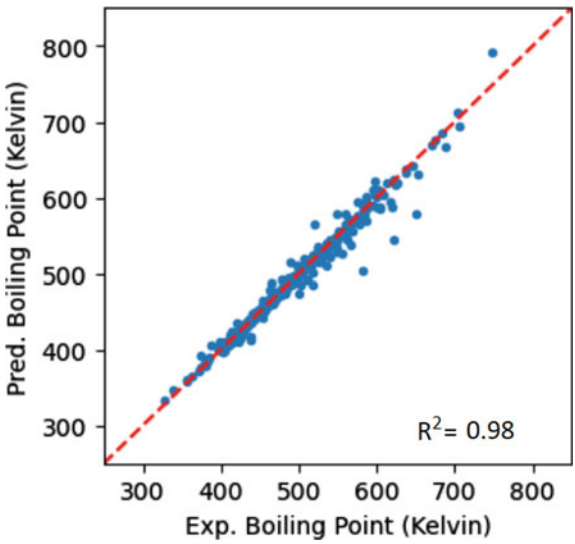


Fig. 1.3 Correlation plots between the "experiments" and the predictions from the GCNN

are able to find the most relevant descriptor and reveal the underlying physics. On the other hand, the GCNN simply takes a molecular connectivity graph as input and outputs a result. It more or less operates like a black box and lacks the explainability that simpler models have.

With this example, we have shown the basic workflow of training a machine learning model, and demonstrated the power of modern deep learning. Meanwhile, this example also illustrates some challenges of the machine learning approach, such as data scarcity and model explainability. We will discuss them in details in the next section.

1.3 Strengths and Challenges of Machine Learning

The above example of boiling point prediction demonstrates some great strengths of machine learning, and alludes to some of its challenges. In this section, we further discuss the strengths and challenges of machine learning, particularly in the context of molecular sciences.

1.3.1 Strengths of Machine Learning

In recent years, we have seen an explosion of interests and successes in applying ML in many disciplines including molecular sciences. This, on one hand, largely thanks to the recent advances in ML algorithms (e.g. transformer, GAN), ML tooling (e.g.

TensorFlow, PyTorch, the rich Python ML libraries), and ML-optimized hardware (e.g. GPUs, TPUs), which all make ML easily available to practitioners. It is also because ML has the strengths to solve many problems in better and more efficient ways. We would like to highlight one key strength of ML, its ability to scale, as shown in the following three aspects.

First, the performance of ML algorithms scales with data. ML algorithms have the ability to learn implicit knowledge from data. The finesse of the learned knowledge usually improves as the size of data increases, provided that the model has sufficiently complex structures (e.g. deep neural network). This quality, in combination with the boost that human beings are accumulating data at an increasingly rapid speed, enables ML to perform exceptionally well in many tasks. ML has improved and automated numerous processes in various fields ranging from research, to manufacturing, to finance, to personal technologies.

Second, ML scales with problems, and the same ML algorithm can be applied to problems in different domains. Let's compare machine learning with computations from first principles. The latter requires deep understanding of a domain in order to develop a computer algorithm that is usually limited to solving a specific set of problems. Machine learning, on the other hand, can learn without being explicitly programmed. This allows ML to be easily applied to many different problems without the need of developing domain-specific algorithms, only requiring that there are enough training examples. For instance, the deep neural network model can be applied in physical, chemical, biological or many engineering problems. What needs to be customized is feature representation, where domain knowledge is heavily applied. This is not to say machine learning can replace algorithms based on first principles or domain knowledge. In fact, they complement each other very well. In many scenarios, computation from first principles or domain knowledge is still the approach to get the most accurate output albeit at a higher computational cost. These computations also generates examples that can be used in machine learning [39].

Third, ML can make predictions at massive scale and enables applications that are not possible before. Once a ML model is trained, it is often computationally efficient to make prediction, making it feasible to generate large amount of predictions. This strength enables high throughput candidate generation and screening in drug discovery [40–43] and material design [44–49]. One can also build a machine learning model that predicts molecular energy at any nuclear configuration (potential energy surface) using highly accurate data from first-principle computations. These potential energy surfaces enabled many accurate simulations of reaction dynamics and molecular spectra for complex molecular systems, where molecular energies are repeatedly needed and efficiently provided by the potential energy surface predictions [2, 50–60]. To further highlight the scalability of ML in prediction, AlphaFold, an artificial intelligence (AI) program developed by DeepMind to predict protein 3D structure, has predicted 200 millions protein 3D structure at an accuracy level that is regularly competitive with experiment [61, 62]. In comparison, experimental methods that are used to determine protein structures, such as X-ray crystallography or cryogenic electron microscopy, would require months to years of work to resolve a single protein structure.

1.3.2 Challenges of Machine Learning

1.3.2.1 Data Scarcity

As shown in the boiling point example, successfully applying ML usually requires sufficient and high-quality training data. As discussed in Sect. 1.2, the two typical ways of acquiring training data are experiments or computations. Experimental data are in general reliable but expensive to acquire. The quality and acquiring cost of computational data vary significantly depending on the underlying theories used in the computation. The limitation of data acquisition often results in the scarcity of high-quality training data in molecular sciences problems. There are several strategies to address this issue. We discuss two commonly used strategies, transfer learning and active learning.

Transfer learning is a strategy to leverage data from other problems. The idea is to learn and store knowledge from one problem and to apply it on other related problems. For example, in training a classifier to predict whether an image contains a letter, one can transfer the learned knowledge and apply the model to other problems such as predicting whether an image contains a traffic sign. The benefit is clear; the data may be limited for one problem, but via transfer learning one can leverage the knowledge learned from related problems with richer data. The popularization of using pre-trained image models and large language models demonstrated the strength of transfer learning. As an example in molecular sciences, Iovanac et al. applied autoencoder and latent space enrichment to several chemical properties of molecules [63, 64]. The recent progress of transfer learning in drug discovery is nicely summarized by Cai et al. [65].

Active learning is a strategy to continuously sample and acquire new training data that are most useful in improving the model. In contrast to the standard workflow of a supervised learning where one would acquire the entire training data upfront, the active learning approach allows one to start with a moderate size of training data. The algorithm samples new input data based on model performance feedback. For example, one can choose to sample more data from the region where the model needs the most improvement. Then, one acquires the labels for the newly sampled input data, usually from experiments or first-principle computations, and feed the new training data to improve the model. After several iterations, the model can usually improve efficiently because the new training samples directly address the weakness of the model in its previous iteration. The process of developing a potential energy surface is a great example of active learning [54, 55, 66–74]. The labeled data are generated from computations from first principles, ab initio electronic structural computations, which can be expensive if using an accurate level of theory. Therefore initially one can start with a moderate size of training data. Then, one can evaluate the model by running molecular dynamic simulations using the model predicted potential energy. Likely the simulations would identify some spurious regions, where the model predicted potential energies are unrealistically high or low. The molecular configurations in these regions can therefore be sampled, labeled from first-principle

computations, and then used as new training data. The process is repeated until the model can accurately predict the energies in the configuration space of interest.

1.3.2.2 Model Explainability

In the previous boiling point example, as we evolved from the basic linear regression model to the complex graph convolutional neural network model, we significantly improved the accuracy but we lost the ability to clearly explain why the model is making a prediction. This is hardly an exception, and in fact over the past decades, the increasing complexity of ML models and the tendencies of using ML models as black boxes have made it increasingly difficult to explain model predictions. Recently, the community has realized the importance of model explainability and more attention has been paid on this topic [75]. Model explainability is valuable and should be desired for the following reasons. First, it provides more transparency to the model, and helps researchers to understand how their models actually work, which in turn makes the development processes much easier. Second, it helps others to trust the model and its outputs when they can see how the output results are being yielded in a glass box, compared to in a black box [76, 77]. Third, while sometimes having the ability to accurately predict is a valuable standalone merit, often the goal of scientific research is to distill scientific knowledge. Black-box models are clearly not enough in this sense. Finally, ML research and applications must comply with the increasingly strict AI regulations, which often require some explainability of the decision making process.

We have seen an increasing emphasis of model explainability and interpretability in recent ML researches in many areas of molecular sciences [78–84]. One main research area focuses on understanding how a particular input lead to the particular output [85–89]. Another area is to break down and analyze the deep neural network itself, and extract insightful information about the hidden layers and its transferability to other applications [90, 91].

1.3.2.3 Uncertainty Quantification

Uncertainty quantification (UQ) is the practice to quantitatively characterize (and therefore reduce) the uncertainties of machine learning models. When working on a ML project, people tend to focus on the model accuracy, and often underappreciate the significance of UQ. We all know that measuring the uncertainty is important in experiments, so does it in model prediction. Only with uncertainty, model results can be compared in a meaningful way. Obtaining and publishing the uncertainty of model predictions also helps other researchers to trace the error sources and eventually build trust in the model. It is encouraging to see more systematic studies on this topic recently [92–97].

There are two main types of uncertainty, the data (also known as aleatoric) and the model (also known as epistemic) uncertainties. The data uncertainty comes from

the underlying uncertainty of the source, such as the measurement uncertainty from experiments or the errors from first-principle computations. The model uncertainty includes the choice of the model and its hyperparameters (whether the model is suitable to generate the statistics of the training data), and how well the model is fit (overfit or underfit). The model uncertainty can also be illustrated with the fact that different models trained on the same data set make somewhat different predictions. We first need to be able to quantify the uncertainty before we can take steps to reduce it. There are several established methods to measure the model uncertainty [98], and here we discuss two common methods, the Bayesian approach and the ensemble method.

The Bayesian approach treats the parameters in a neural network as prior distributions, and predicts their posterior distribution with various methods. Fully utilizing this approach is too computationally intensive, therefore in practice some approximate Bayesian approaches are often applied. One way is to treat the distribution of each parameter as a Gaussian distribution, reducing the prior distribution only to a mean and a variance. With that, the total parameters is only doubled from a regular neural network model. The other approximation is called Monte-Carlo dropout [99], which randomly drops out nodes in the neural network when making prediction. Multiple predictions are made using random dropout, and are used to statistically estimate the uncertainty of the prediction.

Compared to the Bayesian approach, the ensemble method is more straightforward. It works by training several different models on the same training data set. The mean and variance (uncertainty) of the final prediction can be statistically estimated from the predictions of each individual model in the ensemble.

1.4 Summary

To recap, machine learning learns without explicit programming and has three approaches, supervised learning, unsupervised learning and reinforcement learning. ML scales with data, scales with problems, and is able to make predictions at massive scale, which all make it a powerful tool that is transforming many fields including molecular sciences. There are challenges when applying ML in molecular sciences, such as data scarcity, model explainability and uncertainty quantification, and we are seeing increasingly more efforts in addressing them.

We have shown a pedagogical ML example of predicting the boiling points of chemicals. We hope this example explains how ML works from end to end and demonstrates the power of modern ML techniques such as the graph convolutional neural network.

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Chapter 2
Graph Neural Networks for Molecules



Yuyang Wang, Zijie Li, and Amir Barati Farimani

Abstract Graph neural networks (GNNs), which are capable of learning representations from graphical data, are naturally suitable for modeling molecular systems. This review introduces GNNs and their various applications for small organic molecules. GNNs rely on message-passing operations, a generic yet powerful framework, to update node features iteratively. Many researches design GNN architectures to effectively learn topological information of 2D molecule graphs as well as geometric information of 3D molecular systems. GNNs have been implemented in a wide variety of molecular applications, including molecular property prediction, molecular scoring and docking, molecular optimization and *de novo* generation, molecular dynamics simulation, etc. Besides, the review also summarizes the recent development of self-supervised learning for molecules with GNNs.

Keywords Graph neural network · Molecular modeling · Quantitative structure-activity relationship · Molecular generation · Molecular simulation · Self-supervised learning · Graph representation learning

2.1 Message Passing Graph Neural Networks

Graphs are ubiquitous data structure that expresses a set of objects and the relationships between them. Formally, a graph $\mathcal{G}$ is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denotes the set of objects (nodes) and their relationships (edges), respectively [1].

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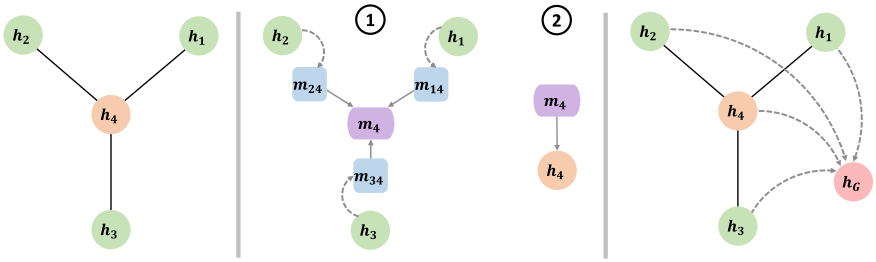
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(a) An example graph (b) Message-passing function (c) Readout function

Fig. 2.1 Illustration of **a** an example graph defined by nodes and edges, **b** messaging-passing function containing ① an aggregation operation from $\boldsymbol{h}_1$, $\boldsymbol{h}_2$, and $\boldsymbol{h}_3$ to obtain message m_4 and ② an update operation to update the node feature $\boldsymbol{h}_4$ from $\boldsymbol{m}_4$, and **c** readout function to obtain the graph feature $\boldsymbol{h}_G$

Figure 2.1a shows an example of a graph containing four nodes and three edges. Graphs are powerful expressions that can model a wide range of systems. For example, the social network can be modeled as a graph where each node represents a person and each edge represents social connections between them, e.g., friendship, spouseship, colleagueship, etc. [2]. Other graphical systems include chemical compounds [3], knowledge graphs [4], physical systems [5], and various other domains [6–8]. Recently, there is growing attention from the machine learning (ML) community to develop ML models, especially deep neural networks (DNNs) [9], to analyze graphical data [10]. Previous deep learning models, including convolutional neural networks (CNNs) [11] and recurrent neural networks (RNNs) [12], fail to directly operate on graphical structures. Graph neural networks (GNNs), a deep learning method, are developed to learn representations from graphs directly [13]. GNNs have been prevalent in various domains and many different tasks. In what follows, we will introduce the basic concepts and operations of GNNs.

Modern GNNs are built upon the message-passing layer that aggregates neighboring information to update each node in an iterative manner. The framework is first formalized by Gilmer et al. [14]. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where the ith node is $u_i \in \mathcal{V}$ and the edge between u_i and u_j is $e_{ij} \in \mathcal{E}$. Each node u_i is initialized as a feature vector $\boldsymbol{h}_i^{(0)} = \text{Emb}_n(u_i)$ via an node embedding function. Similar to node embedding, each edge e_{ij} is mapped to a feature vector via an edge embedding function $\boldsymbol{a}_{ij} = \text{Emb}_e(e_{ij})$. The message-passing function is demonstrated in Fig. 2.1b. Each message-passing layer contains two operations: (1) computation and aggregation of the messages from neighboring nodes and edges, and (2) update of the node feature based on the message and old feature. The two operations at kth layers in a GNN is given in Eqs. 2.1 and 2.2, respectively.

$$\boldsymbol{m}_i^{(k)} = \sum_{u_j \in \mathcal{N}(u_i)} \phi_m^{(k)} \left(\boldsymbol{h}_i^{(k-1)}, \boldsymbol{h}_j^{(k-1)}, \boldsymbol{a}_{ij} \right), \tag{2.1}$$

$$\boldsymbol{h}_i^{(k)} = \phi_f^{(k)}\left(\boldsymbol{h}_i^{(k-1)}, \boldsymbol{m}_i^{(k)}\right), \tag{2.2}$$

where $\boldsymbol{h}_i^{(k)} \in \mathbb{R}^F$ denotes the feature vector of node u_i at kth layer, $\mathcal{N}(u_i)$ models all the neighbors of node u_i. $\phi_m^{(k)}$ and $\phi_u^{(k)}$ are the message and update functions. By conducting the two operations iteratively, node features within the graph are updated and can be utilized for node-level tasks. To conduct graph-level tasks, like molecular property prediction, directly utilizing all the node features might be infeasible as different graphs have different numbers of nodes. To obtain the representation of the whole graph, the readout operation is introduced to down-sample the node features as illustrated in Fig. 2.1c. For a GNN with K layers, after the last message-passing layer, the readout function $R(\cdot)$ is given in Eq. 2.3.

$$\boldsymbol{h}_G = R\left(\{\boldsymbol{h}_i^{(K)} \mid u_i \in \mathcal{V}\}\right). \tag{2.3}$$

The design of the message-passing operation is essential to learning graph representations [15]. Since proposed, different GNNs have been developed with different aggregation and update functions for expressive graph representation learning [16–18]. Early spectral methods are built upon the spectral representation of graphs. Such methods compute the eigenvectors of the graph Laplacian for aggregating neighboring information and apply nonlinear activation functions on the aggregated feature to update each node [19, 20]. Graph convolutional network (GCN) [21] introduces a simple yet generic GNN framework, where the aggregation is implemented as an element-wise mean pooling over the node and its neighbors, and the update is implemented as a linear transformation $\boldsymbol{W}$ followed by nonlinear ReLU function as given in Eq. 2.4.

$$\boldsymbol{h}_i^{(k)} = \text{ReLU}\left(\boldsymbol{W} \cdot \text{MEAN}(\boldsymbol{h}_j^{(k-1)}) \mid u_j \in \mathcal{N}(u_i) \cup \{u_i\}\right). \tag{2.4}$$

GraphSAGE [22] formulates aggregation via multiplication of learnable weight matrix and neighboring features followed by ReLU and max-pooling over the aggregated features as given in Eq. 2.5.

$$\boldsymbol{h}_i^{(k)} = \sigma\left(\boldsymbol{W}' \cdot \left[\text{MAX}\left(\text{ReLU}(\boldsymbol{W} \cdot \boldsymbol{h}_j^{(k-1)}) \mid u_j \in \mathcal{N}(u_i)\right) \parallel h_i^{(k-1)}\right]\right), \tag{2.5}$$

where $[\cdot\|\cdot]$ is the concatenation and σ is a nonlinear activation function like sigmoid. The update in GraphSAGE contains a concatenation of the node and aggregated features succeeded by a linear transformation $\boldsymbol{W}'$. Equation 2.6 shows the message-passing in graph isomorphism network (GIN) [23], another widely used GNN architecture. GIN proposes to sum the node features and all the neighboring features and applies a multi-layer perceptron (MLP) to update the node in the message-passing layer.

$$\boldsymbol{h}_i^{(k)} = \text{MLP}\left((1+\epsilon)\boldsymbol{h}_i^{(k-1)} + \sum_{u_j \in \mathcal{N}(u_i)} \boldsymbol{h}_j^{(k-1)}\right). \tag{2.6}$$

Further, graph attention network (GAT) [24] introduces the attention mechanism to message-passing via computing the attention score as the weight coefficient in aggregation. The standard attention follows $\text{Attn}(Q, K, V) = \frac{QK^\top}{\sqrt{d_k}}V$, where Q, K, and V are query, key, and value matrices of the embedding of each token, respectively, with the square root of the embedding dimension d_k as the scaling factor. The message-passing in GAT adjusts the attention as given in Eqs. 2.7 and 2.8.

$$\boldsymbol{h}_j^{(k)} = \sigma\left(\frac{1}{K}\sum_{k=1}^{K}\sum_{u_j \in \mathcal{N}(u_i)\cup\{u_i\}} \alpha_{ik}^k \boldsymbol{W}^k \boldsymbol{h}_j^{(k-1)}\right), \tag{2.7}$$

$$\alpha_{ij}^k = \frac{\exp\left(\text{ReLU}(\boldsymbol{a}^\top[\boldsymbol{W}^k h_i^{(k-1)} \parallel \boldsymbol{W}^k h_j^{(k-1)}])\right)}{\sum_{u_j \in \mathcal{N}(u_i)} \exp\left(\text{ReLU}(\boldsymbol{a}^\top[\boldsymbol{W}^k h_i^{(k-1)} \parallel \boldsymbol{W}^k h_j^{(k-1)}])\right)}, \tag{2.8}$$

where α_{ij}^k is the scalar that measures attention score, K is the number of attention heads, and $\boldsymbol{a} \in \mathbb{R}^{2F}$ is the weight vector. Multiple works also leverage RNNs to model the message-passing function. Gated graph neural networks (GGNN) by Li et al. [17] employs the gated recurrent unit (GRU) [25] to aggregate neighboring information and update node features as shown in Eq. 2.9.

$$\boldsymbol{h}_i^{(k)} = \text{GRU}\left(\boldsymbol{h}_i^{(k-1)}, \sum_{u_j \in \mathcal{N}(u_i)} \boldsymbol{W}\boldsymbol{h}_j^{(k-1)}\right). \tag{2.9}$$

Long short-term memory (LSTM) [12] is also implemented similarly as GRU for message-passing functions [26].

Recently, there are also efforts on adapting the transformer architecture to graphical data [27]. The standard transformer is designed to learn from sequential data like text [28]. How to encode the structural information of the graph is the major challenge in applying the transformer on graphs. To this end, Graphormer [29] is directly built upon the standard transformer with an effective encoding of the structural information. Specifically, Graphormer includes spatial encoding of the shortest path, edge encoding of edge features on the shortest path, and centrality encoding of the degree centrality of each node. Each encoding is corporated with learnable parameters and added to the attention score. Dwivedi et al. [30] introduce a graph transformer that learns from the local attention of neighboring connections. It also implements the Laplacian eigenvectors of the graph as the positional encoding in place of sinusoidal positional encoding. Recently, tokenized graph transformer (TokenGT) is proposed to demonstrate that pure transformers can be powerful graph representation learners

[31]. Unlike previous works which only tokenize nodes and integrate edge information in attention updates, TokenGT tokenizes both nodes and edges with orthonormal node identifiers encoding the connectivity of the tokens and trainable type identifiers that encode whether a token is a node or an edge. TokenGT allows adaption of linear attention (e.g., Performer [32]) introduced for pure transformer and reaches $\mathcal{O}(N + M)$ cost, where N and M are the number of nodes and edges, respectively.

For the readout operation, the primitive choices are pooling over all the node features, including mean-pooling, max-pooling, and summation-pooling. Xu et al. [23] point out that summation-pooling is more expressive than mean- and max-pooling, which can capture the full multiset while the other two poolings fail. Works have also investigated other readout functions to improve representation learning and computational efficiency. Attention mechanisms have been implemented in place of summation- or mean-pooling as the readout operation [14, 17]. Vinyals et al. [33] propose set2set function which implements an LSTM for unordered and size-variant input sets. Other methods have probed the rearrangement of the nodes to downsample the features. Defferrard et al. [20] coarsen the graph into multi-levels via the Graclus algorithm and then rearrange the nodes into a balanced binary tree. The readout is conducted by aggregating the node features in a bottom-up manner. Following the insight, Zhang et al. [34] propose SortPool which ranks nodes based on their structural roles within the input graph and truncates the size of the graph after ranking. DiffPool [35], on the other hand, develops a differentiable pooling module that generates hierarchical graph representations. It learns a soft clustering assignment at each layer to aggregate node features. SAGPool [36] combines self-attention with end-to-end hierarchical representation learning which includes both node feature and graph topology. In general, the readout operation can be considered as a special message-passing layer that aggregates all the updated node features within the graph. It also plays an important role in learning expressive graph representations.

2.2 Molecular Graph Neural Networks

The previous section has introduced the basic concepts of the message-passing GNN and its prevalent variants. This section focuses on the GNN architectures that are adapted and designed for molecular representation learning. As the focus of this chapter, molecules can be naturally viewed as graphs [37]. Figure 2.2 illustrates examples of how molecule graphs are built. Within the graph, each node models an atom, and each edge models the interatomic interactions. Interactions include two-dimensional (2D) topological information like covalent bonds and 3D geometric information like distances and angles. Graph representations have demonstrated advantages over other molecular featurization techniques. Some methods develop a language that converts a molecule into a one-dimensional (1D) string, e.g., simplified molecular-input line-entry system (SMILES) [38] and recently developed self-referencing embedded strings (SELFIES) [39]. Other works propose rules that can embed each molecule into a feature vector, namely a molecular fingerprint or

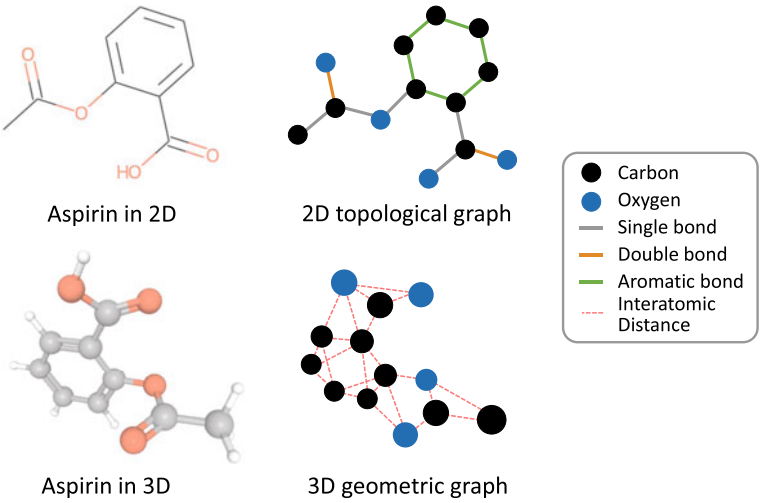


Fig. 2.2 Illustration of building graphs from molecules using 2D topological and 3D geometric information. The aspirin molecule is shown as an example

a descriptor, that usually encodes the presence or absence of certain substructures in the molecule, e.g., extended-connectivity fingerprint (ECFP) [40] and molecular access system (MACCS) keys [41]. Both string-based and fingerprint-based methods have been widely and successfully implemented in many applications including molecular similarity search, clustering, and virtual screening. With the rise of deep learning in recent years, deep learning methods have been applied to learn molecular representations from molecular languages and fingerprints. Though these featurization techniques follow certain rules to encode substructure information, they, however, still struggle to directly model the topological and geometric information of molecules. GNNs, which learn representation from molecule graphs, are capable of encoding the graphical structures. Many works have investigated the GNN architectures for learning from molecule graphs. Also, atomic and bond descriptors, like atomic number, charge, chirality, hybridization, bond type, etc., can be added to the node and edge features. This chapter focuses on the message-passing architecture design for molecule graphs.

Duvenaud et al. [3] introduce one of the pioneering works that learning molecular representations via a GNN. In this work, node features are aggregated through concatenation and are updated through a learnable matrix conditioned on the node degree followed by a sigmoid function. The readout operation computes a weighted summation over node features from all the layers. Another trailblazing model, named Weave, from Kearnes et al. [42] proposes molecular graph convolutions that update edge features in each message-passing layer. Specifically, at each layer, one first updates the edge features from connected nodes and then updates the node features by aggregating the edges. Hu et al. [43] extend the standard GIN by including the edge features in message-passing as given in Eq. 2.10.

$$h_v^{(k)} = \text{MLP}\left(h_v^{l-1} + \sum_{u_j \in \mathcal{N}(u_i)} (h_j^{(k-1)} + a_{ij})\right), \tag{2.10}$$

where a_{ij} is the embedding of the edge between nodes u_i and u_j, containing the information of bond type and direction. Glimer et al. [14] elucidate a simple yet unified GNN framework named message passing neural network (MPNN) which most of the previous models fall into. They then propose enn-s2s, a variant of MPNN that aggregates neighboring information as given in Eq. 2.11.

$$a_i^{(k)} = \sum_{u_j \in \mathcal{N}(u_i)} \text{MLP}(a_{ij}) h_j. \tag{2.11}$$

Besides, it follows set2set [33] as the readout function to obtain molecule representations. Yang et al. [44] further proposed directed MPNN (D-MPNN) that uses information associated with directed edges instead of information associated with vertices in standard MPNN. In addition, D-MPNN implements rich atom and bond features to improve the expressiveness of molecule graphs. Attention mechanisms have been developed for molecular GNNs. Xiong et al. [45] introduce AttentiveFP containing atom embedding layers and molecule embedding layers. The atom embedding layers borrow the attention mechanism introduced in GAT to aggregate local messages and the output attention context is fed into a gated recurrent unit (GRU) together with atom features from the previous layer to obtain updated context. The molecule embedding layers assume a virtual node that connects to all atoms and follows the same pattern as atom embedding layers. The final output of the virtual node is used as the representation of the whole molecule graph. Not only the attention mechanisms but also the transformer architectures are investigated for molecule graphs. Rong et al. [46] introduce GTransformer which combines the message-passing framework with transformer. GTransformer applies a bi-level message-passing strategy that aggregates and updates the information on both nodes and edges. Besides, it employs the residue connection and dynamic message-passing by randomly choosing the number of aggregation hops.

So far, we have discussed GNNs that model molecules as 2D graphs. Namely, these models only consider topological information while ignoring geometric information like distances and angles in the 3D Euclidean space. Such 2D GNNs have demonstrated promising performance in many applications. However, 3D information is crucial as it is closely related to the energy landscape and molecules rely on 3D conformation to function in practice. Nevertheless, simply adding the positional coordinates into GNNs can be problematic as translation or rotation of the molecule will change the output of the models. One would expect to design GNN architectures whose output is immutable to the rotations and translations of 3D molecular structures [47]. To formalize, we borrow the concept from group theory and denote the set of proper rigid transformations (i.e., translations and rotations) in n-dimensional Euclidean space as SE(n) [48]. Many works thus explore SE(3)-invariant GNNs for molecules existing in 3D Euclidean space. Schütt et al. [49] introduce the deep

tensor neural network (DTNN) that model the distances between atoms in message-passing. DTNN extract the graph-level representation by feeding each node feature into an MLP and summing them up. Following DTNN, Schütt et al. further propose SchNet [50] that is composed of well-designed layers to model local correlations between atoms. Equation 2.12 elaborate the atom-wise update while Eqs. 2.13 and 2.14 elaborate continuous-filter convolution at the $(k+1)$th layer in SchNet.

$$h_i^{(k)} = W h_i^{(k)} + b \tag{2.12}$$

$$h_i^{(k)} = \sum_{u_j \in \mathcal{N}(u_i)} h_j^{(k)} \circ W_{\text{cf}}^{(k-1)}, \tag{2.13}$$

$$W_{\text{cf}}^{(k)} = \text{SoftPlus}\left(W_2^{(k)} \cdot \text{SoftPlus}\left(W_1^{(k)} \cdot \Big|\Big|_{u_j \in \mathcal{N}(u_i)} \text{RBF}(d_{ij})\right)\right), \tag{2.14}$$

where $\circ$ is an element-wise multiplication, d_{ij} is the distance in Euclidean space between nodes u_i and u_j, and $\text{RBF}(d_{ij}) = \Big|\Big|_{k=1}^{K} \exp(-\gamma \|d_{ij} - \mu_k\|^2)$ concatenates radial basis functions with $\mu_k = 0.1k$ Å and $\gamma = 10$ Å that expands the interatomic distance from a scalar to a vector of $\mathbb{R}^K$. Besides, it implements SoftPlus, a smooth approximation to ReLU, as nonlinear activation functions. Each message-passing layer starts with an atomise-wise update followed by a continuous-filter convolution operation. It then conducted two atom-wise updates with a SoftPlus activation in between to obtain the combination term $v_i^{(k)}$. The final output node feature at the $(k+1)$th layer is updated by $h_i^{(k+1)} = h_i^{(k)} + v_i^{(k)}$. SchNet effectively encodes 3D distance information to molecular GNN and inspires many follow-up works in this domain. PhysNet by Unke et al. [51] also leverages the interatomic distances to build an SE(3)-invariant GNN, which adapts the interaction blocks to update node features from distances and the residual blocks to learn representations with deeper neural networks. DimeNet by Gasteiger et al. [52] follows the architecture of PhysNet while integrating additional angular information expanded with Fourier–Bessel representations. The message-passing at the kth layer from node u_j to u_i takes in not only the distance d_{ij} but also angles $\angle u_k u_j u_i$ as well as $h_k^{(k-1)}$, where $u_k \in \mathcal{N}(u_j) \backslash \{u_i\}$. The same team further proposes an improved version named DimeNet++ [53] with fast interactions and embedding hierarchy.

Though SE(3)-invariant has merits for GNN in various molecular property predictions, it is still limited for expressing graph representations in some aspects. One limitation is that such invariance requires the message-passing to contain only features of distances or angles but is unable to encode directional information. Besides, for applications like force field prediction where the output is expected as a set of 3D vectors, SE(3)-invariant GNNs give the same output for molecules with different rotations in 3D. However, the force field is expected to rotate together with the rotation of the molecular system. To this end, SE(3)-equivariance is introduced which generalizes the concept of invariance. Formally, equivariance is a kind of symmetry for function. For function $\Phi : \mathcal{X} \to \mathcal{Y}$ (e.g., Φ can be a deep neural network), it is

equivariant with respect to group G such that it commutes with any group actions $g \in G$ on $\mathcal{X}$ and $\mathcal{Y}$ as shown in Eq. 2.15.

$$\rho_g^{\mathcal{Y}}(\Phi(x)) = \Phi\left(\rho_g^{\mathcal{X}}(x)\right), \forall x \in \mathcal{X}, g \in G, \tag{2.15}$$

where $\rho^{\mathcal{X}}(g)$ and $\rho^{\mathcal{Y}}(g)$ are the group representation of group action g on $\mathcal{X}$ and $\mathcal{Y}$ space, respectively. The group representation of group action $g \in G$ on vector space $\mathcal{V}$ is defined as:

$$\rho : G \mapsto GL(V), \tag{2.16}$$

such that $\rho(g_1 g_2) = \rho(g_1)\rho(g_2), \forall g_1, g_2 \in G$ (function satisfies this property is called group homomorphism). The group representation allows operating abstract mathematical object-group action, on the vector space of particular interest. More specifically, for 3D molecules' modeling, we are interested in studying the equivariance with respect to 3D rotations/translations, which corresponds to the 3D special Euclidean group, SE(3). A useful property of the equivariant functions is that composing them will yield another equivariant function, which means the whole network is equivariant if each layer is equivariant. In addition, translation invariance (invariance is also equivariance) is straightforward to achieve in most neural networks as long as they do not use global coordinates as features. Therefore in most of the models, the 3D rotation group SO(3) is the only group that requires special care.

The first major category of equivariant networks is based on irreducible representation, tensor product, and spherical harmonics. For 3D rotations, its representations are orthogonal matrices and can always be decomposed into the irreducible representation of the following form:

$$\rho(g) = Q^T \left[\bigoplus_l D^{(l)}(g) \right] Q, \tag{2.17}$$

where Q is an $N \times N$ orthogonal matrix (for change of basis), $\bigoplus$ is the direct sum, and $D^{(l)}(g)$ is the *Wigner D-matrices* for group action g [54]. Vectors transformed under $D^{(l)}(g)$ are called lth order vectors, which have a length of $2l + 1$. Based on these representations, useful learnable equivariant layers can be developed. A common recipe is first building learnable filters with spherical harmonics and then composing them with input features through the tensor product, which ascribes to the fact that spherical harmonics are the equivariant basis for SO(3) and the tensor product is equivariant. Moreover, the tensor product of the irreducible representations of the vector space can be evaluated by looking up a set of pre-calculated coefficients, which are called Clebsch–Gordon coefficients.

Tensor Field Network [55] proposes a SO(3) equivariant layer under this recipe:

$$h_{\text{out},i}^l = W^{ll} h_{\text{in},i}^l + \sum_{k \geq 0} \sum_{j \in \mathcal{N}(i)}^{n} W^{lk}(\vec{r}_j - \vec{r}_i) h_{\text{in},j}^k, \tag{2.18}$$

where h_i^l denotes the lth order vector of input/output features, $W^{lk}(\vec{r}_j - \vec{r}_i)$ is a learnable filter conditioned on relative position $\vec{r}_j - \vec{r}_i$. More specifically, the learnable filter is derived by multiplying a learnable scalar with Clebsch–Gordon coefficients and spherical harmonic basis function:

$$W^{lk}(\vec{r}_j - \vec{r}_i) = \sum_{J=|k-l|}^{k+l} \psi_J^{lk}\left(||r_{ij}||\right) \sum_{m=-J}^{J} Y_m^{(J)}\left(\frac{\vec{r}_j - \vec{r}_i}{||r_{ij}||}\right) Q_{Jm}^{lk}, \tag{2.19}$$

where $\psi_J^{lk}(\cdot)$ is a learnable function conditioned on the interatomic distance: $||r_{ij}|| = ||\vec{r}_j - \vec{r}_i||_2$, $Y_m^{(J)}$ denotes the mth dimension of Jth spherical harmonics, and Q_{Jm}^{lk} denotes the (J, m)th coefficient in Clebsch–Gordon matrice. The composed learnable filter $W^{lk}(\vec{r}_j - \vec{r}_i)$ with a shape $(2l + 1) \times (2k + 1)$ will map type-k features to type-l features. Tensor Field Network, there is an active line of works [56–58] in using operation on irreducible representation to build SE(3) equivariant network.

Another direction to build an equivariant network is to exploit the properties of vectorial features instead of leveraging tools from group representation theory. The principle is that the operation on the directional features (e.g. velocities) can only be linear. Therefore, when building the message passing layer, the operations on vector features $\vec{h}$ are restricted to: (a) Linear projection $W\vec{h}$; (b) Dot product: $< \vec{h}_1, \vec{h}_2 >$; (c) Tensor product: $\vec{h}_1 \otimes \vec{h}_2$. An example of a such recipe is PaiNN [59], where it proposes an equivariant message passing block by extending the continuous and invariant message passing layer proposed in [50]. Each layer in the PaiNN comprises two sub-blocks, a message passing block, and an update block. Inside the message passing block, the update of scalar features Δs_i are calculated as:

$$\Delta s_i = \sum_{j \in \mathcal{N}(i)} \phi_s(s_j) \odot W_s(||r_{ij}||), \tag{2.20}$$

where $\odot$ denotes Hadamard product, $\phi_s(\cdot)$ is a learnable atom-wise function, $W_s(||r_{ij}||)$ is the learnable continuous filter conditioned on interatomic distance, which is a linear combination of the radial basis function (PaiNN adopts the radial basis function proposed in [60]). Similarly, the update of vector features is defined as:

$$\Delta\vec{h}_i = \sum_{j \in \mathcal{N}(i)} \vec{h}_i \odot \phi_{vv}(s_j) \odot W_{vv}(||r_{ij}||) \tag{2.21}$$

$$+ \sum_{j \in \mathcal{N}(i)} \phi_{vs}(s_j) W_{vs}(||r_{ij}||) \frac{\vec{r}_j - \vec{r}_i}{||r_{ij}||}, \tag{2.22}$$

where the first half of the equation is a convolution with respect to the vector features, and the second half is a convolution with respect to the scalar features using an equivariant filter. After the message passing block, the scalar and vector features are updated with the calculated residuals: $s_i \leftarrow s_i + \Delta s_i$, $\vec{h}_i \leftarrow \vec{h}_i + \Delta\vec{h}_i$ and then fed

into update block. Next, in the update block, the update of scalar features is calculated as:

$$\Delta s_i = a_{ss}(s_i, ||W_v\vec{h}_i||) + a_{sv}(s_i, ||W_v\vec{h}_i||) < W_u\vec{h}_i, W_v\vec{h}_i >, \tag{2.23}$$

where a_{ss} and a_{sv} are learnable function, and W_v, W_u are learnable linear projection matrices. And then the update of vector features is calculated as:

$$\Delta\vec{h}_i = a_{vv}(s_i, ||W_v\vec{h}_i||)W_u\vec{h}_i. \tag{2.24}$$

In practice, a_{ss}, a_{sv}, a_{vv} are derived from the same network a by splitting along the feature dimension of output. The scalar features and vector features are again updated with residuals. Equivariant Transformer [61] extends the above message passing layer to attention. GVP-GNN [62] leverages a similar idea with different design choices of the message passing layer. As demonstrated in Villar et al. [63], such formulation is expressive enough for approximating SE(3) equivariant functions.

2.3 Graph Neural Networks on Molecular Applications

GNNs have been widely implemented to various applications in molecular sciences. This section reviews the molecular applications empowered by GNNs, including molecular property prediction, molecular scoring and docking, molecular dynamics simulation, molecular optimization and generation, and others. Each section introduces the formulation of the problem, prevalent datasets and metrics, as well as works to solve the problem using GNNs (Fig. 2.3).

2.3.1 Molecular Property Prediction

The most straightforward utilization of GNNs is to predict the molecular properties given the molecule graphs. Such a task can be considered as a graph classification or regression task following the empirical risk minimization in supervised learning. For a dataset of N molecule graph and property pairs $\{(\mathcal{G}_1, y_1), \ldots, (\mathcal{G}_N, y_N)\}$, the objective to optimize is given in Eq. 2.25.

$$\min_\theta \sum_i \ell(\text{GNN}_\theta(\mathcal{G}_i), y_i), \tag{2.25}$$

where $\text{GNN}_\theta(\cdot)$ is a GNN model parameterized by θ to predict a certain property from an input molecule graph $\mathcal{G}_i$ and $\ell(\cdot, \cdot)$ measures the difference between the prediction and ground truth label, like cross-entropy loss for classification tasks and mean squared error for regression tasks. Researchers have collected multiple databases containing a wide variety of molecular properties so that different GNN models

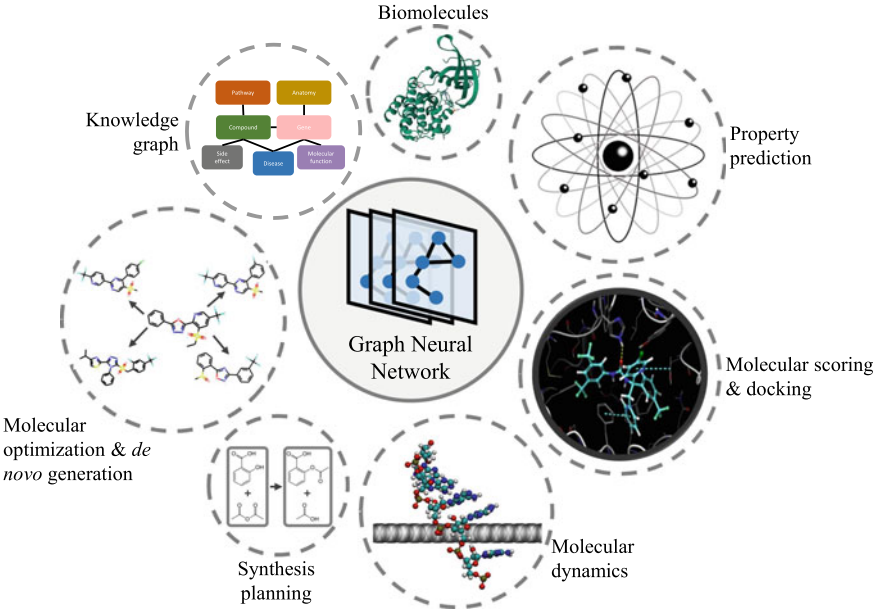


Fig. 2.3 Various applications of GNNs on molecular sciences

can be benchmarked. Table 2.1 summarizes the domain, the number of compounds, the number of tasks, task type, whether containing 3D information and the sources of the popular databases for molecular property predictions. MoleculeNet [64] is a widely used benchmark for molecular property prediction built upon multiple public databases. It contains multi-level properties, including physiology, biophysics, physical chemistry, and quantum mechanics. Besides, benchmarking-GNN [65] creates ZINC and AQSOL datasets for predicting the constrained and aqueous solubility. It should be pointed out that benchmarking-GNN only uses a subset of the original ZINC database (250,000 molecules) [66]. Recent efforts have focused on generating labeled molecular datasets via density functional theory (DFT) calculations. QM9, which is also included in MoleculeNet, contains approximately 134k molecules with their geometric, energetic, electronic, and thermodynamic properties. Alchemy [67] extends the previous dataset to include 12 quantum mechanical properties of 119,487 molecules with up to 14 heavy atoms. PCQM4M and its latest version PCQM4Mv2 in the OGB challenge aim at predicting the DFT-calculated HOMO-LUMO energy gap from 2D or 3D molecule graphs. ANI-1 [68] extends this concept of such datasets by including the energies of off-equilibrium conformations. Specifically, ANI-1 contains 24,687,809 conformations from 57,462 molecules. There are also works that create curated datasets for specific tasks, including quantitative structure-odor relationship (QSOR) [69], Carbon absorption [70], etc. Molecular property prediction is challenging since different datasets in it have numbers of instances of different magnitudes, from less than 200 to about 4,000,000 molecules. Also, most classifi-

Table 2.1 Summary of benchmarks for molecular property predictions

Domain	Dataset	# Molecules	# Tasks	Task type	3D	Source
Physiology	MUTAG	188	1	Classification	No	[71]
	SIDER	1427	27	Classification	No	[64, 72]
	ClinTox	1478	2	Classification	No	[64, 73]
	BBBP	2039	1	Classification	No	[64, 74]
	NCI1	4110	1	Classification	No	[75]
	Tox21	7831	12	Classification	No	[64, 76]
	ToxCast	8575	617	Classification	No	[64, 77]
Biophysics	BACE	1513	1	Classification	No	[64, 78]
	CCRF-CEM	3047	1	Regression	No	[79]
	PC-3	4294	1	Regression	No	[79]
	PDBBind	19,443	1	Regression	Yes	[64, 80, 81]
	HIV	41,127	1	Classification	No	[64, 82]
	MUV	93,087	17	Classification	No	[64, 83]
	PCBA	437,929	128	Classification	No	[64, 84]
Physical chemistry	FreeSolv	642	1	Regression	No	[64, 85]
	ESOL	1128	1	Regression	No	[64, 86]
	Lipophilicity	4200	1	Regression	No	[64, 87]
	AQSOL	9982	1	Regression	No	[65, 88]
	ZINC	12,000	1	Regression	No	[65, 66]
Quantum mechanics	QM7	7165	1	Regression	Yes	[64, 89]
	QM7b	7211	14	Regression	Yes	[64, 90]
	QM8	21,786	12	Regression	Yes	[64, 91]
	QM9	133,885	12	Regression	Yes	[64, 92]
	Alchemy	119,487	12	Regression	Yes	[67]
	PCQM4M	3,803,453	1	Regression	No	[93]
	PCQM4Mv2	3,378,606	1	Regression	Yes	[93]
	ANI-1	24,687,809 (conf.)	1	Regression	Yes	[68]

cation tasks in it have quite imbalanced labels, therefore, ROC-AUC and PRC-AUC are widely used instead of accuracy to measure the performance of different models. Regression benchmarks adapt either root-mean-square error (RMSE) or mean absolute error (MAE) as evaluation metrics. Besides, Wu et al. [64] introduce scaffold splitting to separate structurally different molecules into training, validation, and test subsets. This strategy provides a more challenging yet realistic setting than random splitting and has been leveraged in many works on molecular property prediction [43–45].

Most of the GNN architectures introduced in Sects. 2.1 and 2.2 can be leveraged on molecular property predictions. Apart from those models, a wide range of GNN variants have been developed for the application. Wieder et al. [94] provide a comprehensive survey for the molecular property predictions via GNNs till the year 2020. Early works utilize spectral GNNs for molecular property prediction [95–98]. Since the proposal of GCN [21], a simple yet generic algorithm, many GNNs have been

built upon it for molecular property predictions [99–103]. After the formalization of the message-passing framework by Gilmer et al. [14], researchers have built GNNs systematically following the insight to predict a wide range of molecular properties [44, 104, 105].

Recently, there are emerging researches that leverage DNNs, especially SE(3)-invariant or SE(3)-equivariant GNNs with 3D geometric information, for quantum mechanics (QM) predictions [51, 106]. Many works are evaluated on QM9. Early SE(3)-invariant GNNs, including SchNet [50], HIP-NN [107], DimeNet [52], are built upon interatomic distances and/or angles. Some works have leveraged orbital information to better predict QM properties [108, 109]. GNNs built upon SE(3)-equivariant operations have also been widely applied to this application. Examples include SE(3)-Transformer [56], Cormorant [110], SphereNet [111], GemNet [112], PaiNN [113], and TorchMD-Net [61]. Many SE(3)-equivariant GNNs have also been benchmarked MD17 [114] which contains energy and force field predictions from molecular dynamics simulations of eight molecules. We will discuss GNN applications on MD simulations in Sect. 2.3.3 extensively. Recent works have investigated ab initio solution of the Schrödinger equation to acquire fundamental QM properties equipped with GNNs. Hermann et al. [115] propose PauliNet, a GNN-based wave function ansatz. PauliNet adapts SchNet as the GNN backbone and exploits multireference Hartree–Fock solution as the optimization start point. The whole framework is trained using variational quantum Monte Carlo (VMC). Gao et al. [116] introduce a framework combining a GNN and a neural wave function to solve the Schrödinger equation for multiple geometries via the VMC method simultaneously.

2.3.2 Molecular Scoring and Docking

Molecular scoring mainly refers to the prediction of pharmacological activity of a molecule candidate, which is essential in *in silico* drug discovery [117]. Such a method is widely applied in the virtual screening of drug candidates. It can also be leveraged to narrow down the search space for *de novo* molecular generation. There are two major categories of molecular scoring: ligand-based scoring and structure-based scoring. The ligand-based scoring requires no information about the receptor in pharmacological activities. Such a setting can be considered as a special case of molecular property predictions that model the quantitative structure–activity relationships (QSAR), like drug-likeness, toxicity, solubility, etc. On the contrary, structure-based scoring is based on the structure of the biological target (usually a protein) of a molecule. Such a strategy predicts the drug-target interactions given the drug candidates and the target receptors. Molecular docking predicts the preferred pose of a molecule when bound with a binding site to form a stable protein-ligand complex. It is one of the most popular techniques in structure-based drug discovery. Several works leverage 3D CNN to model the protein-ligand complex [118, 119]. However, 3D CNN requires voxelization of 3D space which adds burdens to the memory. Moreover, such a voxelization loses resolution of the exact pose in 3D

and lacks explicit modeling of interatomic interactions. This chapter reviews the structure-based molecular scoring and molecular docking empowered by GNNs.

PDBBind [80] collects biomolecular complexes with their experimental binding affinity data and is widely used in structure-based scoring with deep learning as a regression task. DUD-E [120] is another dataset that contains 102 targets across different protein families. For each target, both positive (active) and negative (decoy) ligands are provided, which are formalized as classification tasks. Also, BindingDB [121] is a public database of measured binding affinities of proteins and ligands. Drugbank [122], a public online database, that collects FDA-approved drugs with drug target and drug action information. ChEMBL [87] also provides plenty of protein-ligand information including chemical compound and activity assay data.

Structure-based scoring aims at predicting the binding affinity of drug-target interactions (DTIs) [123]. The key to designing GNN models for structure-based molecular scoring is how to integrate the information of the protein and/or the binding pocket to the ligand [124]. Feinberg et al. [125] introduce PotentialNet, a pioneering GNN framework that models the protein-ligand complexes. PotentialNet takes two stages in message-passing: (1) intra-molecular message-passing based only on covalent bonds, and (2) intra- and inter-molecular message-passing based on Euclidean distance and bonds. Gomes et al. [126] introduce a framework contain three parameter-sharing Siamese GNNs that encode ligand, protein, and protein-ligand complex, to features G_{ligand}, G_{protein}, and G_{complex}, respectively. The prediction of binding affinity then utilizes $\Delta G = G_{\text{complex}} - G_{\text{ligand}} - G_{\text{protein}}$. Lim et al. [127] improve the framework by conducting intra-molecular message passing and inter-molecular massage-passing simultaneously. The model utilizes the subtraction between the features of a target protein and the ligand in their complex to predict the binding affinity. Such a strategy that models the interactions within the ligands and interactions between ligands and targets is widely used in later works. InteractionGraphNet [128] by Jiang et al. models the intra- and inter-molecular interactions sequentially with two independent GNN models. Morrone et al. [129] adapt a dual-GNN framework to encode the ligand graph and protein-ligand interactions separately. The output of the two GNNs is concatenated and fed to an MLP to predict the binding affinity of the complex. Son et al. [130] propose GraphBAR, a framework containing multiple graphs whose adjacency matrices cover neighbors within different distance cutoffs. Knutson et al. [131] propose two parallel GNNs with one incorporating domain knowledge of proteins and ligands while the other learning interactions with no domain prior. The works listed above are built upon protein-ligand complexes. However, these models may suffer to make accurate predictions for those complexes without experimental measurements. To this end, Torng and Altman [132] propose to encode protein pocket graphs and 2D molecule graphs independently and combine the output features to predict the DTIs. Recent works have introduced language models to encode protein information in predicting DTIs [133, 134].

Unlike structure-based scoring which predicts binding affinity directly, molecular docking focuses on predicting the posture of a protein-ligand complex. Conventional docking contains a search module to generate massive potential protein-ligand binding poses and a scoring module to evaluate the interaction of binding poses [135].

Such methods rely on heavy candidate sampling and usually involve empirical scoring functions, thus may be time-consuming and inaccurate [136]. Recently, there are emerging works that apply GNNs for molecular docking. Jiang et al. [137] develop a GNN to refine the initial docking pose from conventional docking software. The GNN can be run multiple times to gradually get optimal predictions. Méndez-Lucio et al. [138] report DeepDock, which first learns a statistical potential based on distance likelihood and then samples ligand conformations based on learned potential. Specifically, DeepDock models statistical potentials on torsion angles to generate favorable molecular conformations in the binding site [139]. However, these methods assume prior knowledge of the binding site for the ligands. Some works have investigated predicting protein binding sites using GNNs [6, 140]. To this end, GNN methods have been developed to both locate the binding site and predict the preferred binding conformations. EquiBind [141] by Stärk et al. adapts SE(3) GNN to predict the binding site and applies constraints that only allow change of torsion angles while keeping bond lengths and angles fixed. TANKBind [142] by Lu et al. incorporates trigonometry constraints to GNNs which avoid unrealistic conformations like atom overlapping. It also leverages contrastive learning to build the energy landscape for the inter-molecular interaction of different binding sites.

2.3.3 Molecular Dynamics Simulation

Molecular Dynamics (MD) has a wide range of applications in material science, chemistry, and biophysics [143–145]. It provides a numerical way to study and predict intricate molecular systems. In essence, MD simulation calculates atomic forces and then updates the system states with a discretized equation of motion. The forces can either be modeled by ab initio approaches (AIMD) like density functional theory (DFT) [146] that considers the electronic structure of atoms or through empirical potentials which bypasses electronic structures [147]. AIMD is highly accurate but its computation is prohibitively expensive, which limits its scope of application. On the contrary, empirical force fields are much more efficient, yet with worse accuracy and limited generalizability. The major limitation of MD simulations using empirical force fields stems from the difficulty in how to describe complex interatomic potentials accurately with the appropriate functional form given the diverse types of interactions in the system. In this regard, neural networks have the potential to close the accuracy gap between empirical potentials and AIMD with its function approximation capability [148–150].

Depending on how the environment of the atom is described, neural networks can be categorized into two broad categories. The first kind of approach relies on hand-designed featurization, which builds tailored descriptors that exploit domain knowledge [106, 151, 152]. One of the first examples is Behler–Parrinello Neural Networks (BPNN) [153–155], which proposes atom-centered symmetry functions (ACSFs) for describing the local neighborhood around an atom. Graph neural net-

works, on the other hand, provide a framework to directly learn the atomic representations from the raw atomic coordinates and low-level atomic features, offering an alternative to the manually tailored atomic descriptors [3, 14, 50, 59, 60].

Based on the above frameworks, neural networks can be used to address the efficiency-accuracy tradeoff in MD from the following aspects. The first aspect is to use neural networks to learn the forcefield (i.e. interatomic potential) of a molecular system. In this case, neural networks trained on data with ab initio level accuracy can serve as fast and efficient surrogate models for AIMD. The output of the network can either be the potential energy surface (PES) or atomic forces, in turn forming an energy-based model (also known as neural network potentials) [156] or force-based model [157–159]. These two variants are similar in principle (Chmiela et al. [114] demonstrate that force-based models essentially learn a linearization of the PES), but force-based models are more accurate at predicting forces in practice and can bypass the process of calculating energy gradients. However, force-based models are generally not energy conserving and thus can generate unphysical predictions. Another perspective is learning the pattern of molecular trajectories instead of the dynamics, which treats the simulation as a sampling problem. Here neural networks are used as a probabilistic generative model [160–162] to model the distribution $p(x_{t+1}|x_t)$ with x_t denotes the state of the system at time t. Since this paradigm no longer depends on dynamics to update the system, it does not suffer from the truncation error that arises from the discretization error of the equation of motion and thus can adopt a very large time step size (e.g. at the scale of a nanosecond). The downside is such a method can lead to unphysical prediction and cannot be used to study properties related to energy and dynamics.

2.3.4 Molecular Optimization and Generation

Efficient and effective molecular generation is of crucial importance in practice [163]. Especially, in the pharmaceutical industry, the discovery of new drugs is a long and expensive process that costs more than $2.5 billion and 10–15 years on average [164]. Therefore, it is appealing to develop techniques that automatically and effectively generate plausible molecule candidates. The deep generative models, which learn to approximate the distribution of observed data, have been leveraged for molecule graph optimization and *de novo* generation [165, 166]. The optimization of the molecule usually starts with a hit compound and manipulates the graph to achieve better target properties like toxicity, drug-likeness, binding affinity, etc. The *de novo* molecule generative models generate novel molecules from scratch or conditioned on desired properties or specific fragments. In this case, molecule optimization can be considered as a conditional generation problem conditioned on the hit molecule. Thus, we review these two applications together in this section. There are two major frameworks in molecule graph generation: (1) fragment-based generative models and (2) molecule-based generative models [167]. The fragment-based models generate molecule graph iteratively, where at each step the model choose an action,

including adding, deleting, or editing one or multiple atoms, bonds, or functional groups. Such a fragment-based framework can be incorporated with reinforcement learning (DRL) that defines a Markov decision process (MDP) and learns the best generative policy by maximizing the expected accumulated rewards. It can also be implemented with autoregressive models like RNNs, which determine the action based on statuses at previous steps [168, 169]. The molecule-based model, on the other hand, generates all the attributes (i.e., atoms and bonds) of a molecule at once. Many generative methods equipped with GNNs fall into this category, including variational autoencoder (VAE) [170], generative adversarial network (GAN) [171], flow-based generative model [172], and score-based or diffusion generative model [173, 174]. VAE usually contains an encoder and a decoder that are trained to maximize the evidence lower bound concerning the log-likelihood of training data. GAN introduces two jointly trained components: a generator and a discriminator, where the generator generates samples while the discriminator attempts to distinguish between samples from the training data and those from the generator. Flow-based models are built upon a sequence of invertible operations that directly models the likelihood of the observed data. Score-based or diffusion model defines a Markov chain that adds random noise to data at each step and learns to denoise from the noise to recover the data. The following part in this section introduces molecule graph generative methods in detail.

Metrics and Benchmarks To measure the performance of deep generative models on molecule graphs, one needs to define comprehensive metrics and benchmarks [175, 176]. Simple but effective metrics include validity which assesses whether the generated molecules are valid, uniqueness/diversity which evaluates if a model generates a different molecule at each sampling, and novelty which measures whether generated molecules exist in the training set. Besides, Preuer et al. propose the Fréchet ChemNet distance (FCD) [177] following the widely-used Fréchet Inception distance (FID) [178] in image synthesis, which is built upon the difference of hidden representations from ChemNet, a deep neural network, between generated molecules and those in the training set. Kullback–Leibler (KL) divergence is also used to measure whether the generative model approximates the distribution of the training set. In conditional generation or optimization, certain properties are used to evaluate the generative models. Many works build evaluation metrics on drug-likeness [179]. For example, the synthetic accessibility score (SAS) describes the ease of synthesis of molecules based on fragment contributions and a complexity penalty [180], the octanol-water partition coefficient log P characterizes the drug-likeness of a molecule, the penalized log P is the subtraction of original log P and SAS, and quantitative estimate of drug-likeness (QED) applies desirability functions which provides a multicriteria metric to assess drug-likeness [181]. Maximum mean discrepancy (MMD) [182, 183] is another metric to evaluate the generated graphs. MMD is employed to measure the distribution difference between generated molecule graphs and training set on degrees, clustering coefficients, orbit counts, as well as bond lengths of different types. For candidate drug optimization or generative, the binding affinity of molecules with respect to the target protein pocket computed by molecular docking tools [184] or molecular

dynamic (MD) simulations [185] are also used as the objective [186]. Energy-related properties, like HOMO-LUMO gap and dipole moment, have also been leveraged as the optimization or generation targets [187]. Additionally, Gao et al. [188] point out that sampling efficiency should be another important consideration in real molecular generation applications.

Several works have contributed to creating molecular generation benchmarks and databases. GuacaMol [175] by Brown et al. employ a standardized subset from ChEMBL database [87]. It includes validity, uniqueness, novelty, FCD, and KL divergence as the evaluation measurements. MOSES [176] by Polykovskiy et al. contains 1,936,962 molecular structures selected from ZINC [66] and is split into the training, test, and scaffold test sets containing approximately 1.6M, 176k, and 176k structure, respectively. Aside from the metrics included on GuacaMol, MOSES implements the similarity of fragments and scaffolds and properties distribution to evaluate the generative models. Besides, ZINC [66], ChEMBL [87], GDB databases [189] are also used in training 2D molecule graph generative models. QM9 [92] which contains more than 130K 3D molecular structures, is also widely used for 3D molecular generation. Recently, GEOM [190] by Axelrod et al. introduces GEOM-QM9, an extension to QM9, containing multiple conformations for most molecules, and GEOM-Drugs containing 304,466 drug-like species up to a maximum of 91 heavy atoms.

Fragment-Based Generative Model Fragment-based generative model modifies a molecule graph by adding, removing, or substituting a fragment (i.e., an atom or a motif) sequentially [191]. Such a framework can be directly modeled as an RL problem [192–194]. At each time step t, an agent receives a reward r_t and predicts an action $a_t \in \mathcal{A}$ given the current state $s_t \in \mathcal{S}$, and the next state $s_{t+1} \in \mathcal{S}$ is depended solely on s_t and a_t following the MDP setting. Here, $\mathcal{A}$ and $\mathcal{S}$ denote the action and state space, respectively. In this case, the state is an incomplete molecule graph, the action is what fragment to add, remove, or substitute, and the agent is usually modeled by a deep neural network, including GNNs. Besides, the reward r_t evaluates how well the molecule is generated which could be a chemical metric or an output from property prediction ML models that approximate empirical measurements. The optimization objective of an RL agent (Eq. 2.26) is to maximize the expected accumulated return R_t of a policy defined by the agent.

$$\mathbb{E}_\pi[R_t] = \mathbb{E}_\pi\left[\sum_t^T \gamma^t r_t\right], \tag{2.26}$$

where $\gamma \in (0, 1]$ is a discount rate and T is the maximal length of a trajectory. RL provides a generic framework for molecular design of different targets [195]. You and Liu et al. [196] propose a graph convolutional policy network (GCPN) that successively constructs a molecule by adding an atom, a substructure, or a bond, and is trained via the policy gradient algorithm. It built its agent on GCN [21]. Experiments show that GCPN can generate molecules with optimized or targeted penalized log P or QED properties. Jin et al. [197] present RationaleRL composed of rationale

extraction, graph completion, and rationale distribution. RationaleRL is evaluated on GNK3β and JNK3 that measures the inhibition against glycogen synthase kinase-3 c-Jun N-terminal kinase-3 [198]. It also measures the validity, diversity, and novelty of the generative model. DeepGraphMolGen proposed by Khemchandani et al. [199] adapts GNNs for action prediction as well as property prediction in an RL setting. It designs a multi-objective reward function containing SAS, QED, and predicted binding affinity at the dopamine and norepinephrine transporters. Besides RL, autoregressive models can be incorporated with fragment-based molecule graph generation [168, 200]. GraphINVENT by Mercado et al. [200] leverages a tiered GNN architecture to generate a single bond at each step. Podda et al. [201] and Chen et al. [202] develop autoregressive generative models that operate on the fragments. Lim et al. [203] propose to extend a given molecule scaffold by sequentially adding nodes and edges, which guarantees the generated molecules preserve the certain scaffold. Xie et al. [204] propose a framework named MARS which combines molecule graph editing with MCMC sampling. The method employs a GNN to sample fragment-editing actions at each step adaptively. Shi et al. [205] introduce GraphAF built upon an autoregressive normalized flow-based generative model that adds nodes and edges to a molecule graph sequentially. GraphAF can also be fine-tuned with RL for molecular optimization of certain properties. Following the normalized flow model, GraphDF by Luo et al. [206] samples discrete latent variables which are mapped to additional nodes and edges via invertible modulo shift transforms.

Fragment-based generative models introduced so far have demonstrated the effectiveness of generating valid, unique, and diverse molecules as well as generating or optimizing towards desired SAS, log P, and QED. However, these models only adapt 2D topological information without 3D geometric features. Many recent works have investigated generative models on 3D molecular structures as many properties, like energy and protein-ligand binding, are related to 3D geometries. Gebauer et al. [207] introduce G-SchNet, an autoregressive generative model for 3D molecular structure generation via placing atoms in 3D Euclidean space one by one. Gebauer et al. further introduce cG-SchNet [208], a conditional version of G-SchNet to generate molecules with certain motifs or low-energy (e.g., small HOMO-LUMO gap). Simm et al. [209] propose an RL formulation that adds atoms in 3D euclidean space sequentially and designs a reward function based on the fast quantum-chemical calculated electronic energy. The team also introduce MolGym, an RL environment comprising several molecular design tasks along with baseline models. Flam-Shepherd et al. [210] further introduce a 3D molecule generative RL framework to add fragments instead of single atoms at each step, which is more efficient in creating larger and more complex molecules. Luo et al. [211] present G-SphereNet follows the flow-based generation while determining atoms in 3D space by predicting distances, angles, and torsion to preserve equivariance. Besides, researchers have investigated generating 3D molecules given a designated protein binding site. Docking scores along with other generic metrics are utilized to evaluate the deep generative models. Luo et al. [212] and GraphBP by Liu et al. [213] develop 3D autoregressive generative methods to sample the type and position of a new atom sequentially in the 3D binding pocket. Powers et al. [214] propose a molecular optimization method that expands

a small fragment into a larger molecule within a protein site. At each step, a GNN trained by imitation learning selects the connecting point as well as the type and dihedral angle of a fragment to be connected. To preserve the equivariance in the 3D molecular generation, many works adopt a local spherical coordinate system in the generation process [207, 212, 213]. Besides, Imrie et al. [215] introduce DeLinker which generates linkers given two fragments in 3D Euclidean space.

Molecule-Based Generative model Unlike fragment-based generative models, molecule-based generative models create the node features, edge features, and adjacency matrix of a molecule graph simultaneously. VAE, a popular generative model, has been leveraged for molecule-based generation [216]. A VAE is composed of two sub-models: an encoder that encodes the input into a latent feature domain, and a decoder that maps the feature back to the input [217]. The loss for training a VAE is given in Eq. 2.27.

$$\ell = -\mathbb{E}_{z\sim q_\theta(z|x)}\left[\log p_\phi(x'|z)\right] + \mathrm{KL}\left(q_\theta(z|x)\|p(z)\right), \tag{2.27}$$

where x is a input data, z is the latent vector, q_θ is the encoder, and p_ϕ is the decoder. The first term measures how well the model reconstructs the data and the second term regularizes the latent vector to be similar to a prior Gaussian distribution $p(z)$ through a KL divergence. Such a framework has been leveraged to generate molecule graphs [217–219]. Simonovsky et al. [220] propose GraphVAE, where the encoder is modeled by a GNN and the decoder is modeled by an MLP that outputs the probability of node features, edge features, and adjacency matrix given the predefined number of nodes. The reconstruction loss measures the difference between generated and input molecule graphs in terms of node features, edge features, and adjacency matrix. Kwon et al. [221] improve the framework with an approximate graph matching for efficient reconstruction loss. Besides, the work incorporates RL and an auxiliary property prediction to improve molecule graph generation. Ma et al. [222] investigate regularized VAE to generate semantically valid molecule graphs. Constraints included in the decoder include atomic valence, graph connectivity, and node label compatibility. Bresson et al. [223] design a graph VAE with GCN [21] as the encoder and a novel two-step decoder. In the decoder, an MLP first predicts the molecular formula from the latent vector and then a GCN learns to place bonds between atoms from the same latent vector. Jin et al. [224] propose JT-VAE, a two-phase generative model employing subgraphs. JT-VAE first generates a junction tree (JT) representing the scaffold of chemical substructures and then the substructures are assembled into a complete molecule graph. Jin et al. [225] further improve the framework by leveraging larger and more flexible motifs and a decoder that operates in a hierarchical coarse-to-fine manner. The model demonstrates better performance in larger molecule generation compared to previous works. Li et al. [226] present DeepScaffold that generates molecules based on scaffolds. DeepScaffold contains DeepScaffold a VAE with a GNN encoder to complete atom and bond types for scaffold, a scaffold-based generator, and a filter for pharmacophore constraints. Mahmood et al. [227] also report masked molecular generation which learns the conditional distribution of unobserved

atoms and bonds given observed components. VAE-based model can also be directly leveraged for conditional generation with respect to targeted properties [228, 229]. Besides, given the learned latent representation by the VAE framework, molecular optimization can also be conducted in the latent space [230, 231]. Jin et al. [232] report VJTNN, a molecular optimization framework based on JT-VAE, which includes stochastic latent codes to capture meaningful molecular variations. To avoid infeasible optimized molecules, the model introduces an adversarial training method to align the distribution of graphs on the latent domain. VJTNN achieves competitive performance on QED, penalized log P, and biological activity optimization. LIMO, by Eckmann et al. [233], adapts VAE with an inceptionism-like reverse optimization to optimize generated molecules towards better binding affinity.

Apart from the VAE-based model, GAN has also been leveraged for molecule graph generation [234]. GAN [171] learns to approximate the distribution of actual data $p_x(x)$ through a min-max game between a generator G and a discriminator D. The discriminator is trained to classify real samples from generated fake ones, while the generator learns to generate samples from random noise $z \sim p_z(z)$ that the discriminator fails to recognize. The objective of GAN is given in Eq. 2.28.

$$\min_G \max_D \left(\mathbb{E}_{x\sim p_x(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}\left[\log\left(1 - D(G(z))\right)\right]\right), \tag{2.28}$$

where G attempts to generate plausible samples by minimizing the objective while D attempts to better detect fake samples by maximizing the objective. De Cao et al. [235] implement the architecture of GAN to molecule graph generation and propose MolGAN. The generator in MolGAN generates an adjacency matrix and feature matrix that combine to define a molecule graph. Despite the generator and discriminator as in the standard GAN, MolGAN adapts an RL agent to generate molecules with targeted chemical properties (e.g., QED, SAS, log P). Maziarka et al. [236] introduce Mol-CycleGAN for molecular optimization, following CycleGAN [237]. Mol-CycleGAN generates a structurally similar molecule with preferred properties given a start compound. The model demonstrates effectiveness in optimizing penalized log P. Tsujimoto et al. [238] further present L-MolGAN, which improves MolGAN performance on large molecule generation by penalizing the disconnected generated molecule graphs.

Other generative models have also been investigated for the molecule graph generation. The normalized flow-based model can also be leveraged in molecule-based generation besides fragment-based generation [239].

A flow-based model defines a invertible deterministic transformation f_θ parameterized by θ between the data space X and the latent space Z as $f_\theta : Z \to X$ [172]. Z follows a prior Gaussian distribution p_Z, the log-likelihood of a data x is given in Eq. 2.29.

$$\log p_X(x) = \log p_Z(f_\theta^{-1}(x)) + \log|\det J|, \tag{2.29}$$

where $J = \frac{\partial f_\theta^{-1}(x)}{x}$ is the Jacobian matrix of the function $f_\theta^{-1}(x)$. The objective of training a flow-based model is to maximize the log-likelihood. To efficiently compute

det J, the affine coupling mapping is leveraged in a normalized flow-based model [240, 241]. GraphNVP by Madhawa et al. [242] and MoFlow by Zang et al. [243] employ reversible normalizing flow-based model for the molecule graph generation. Both models decompose the generation into two phases, namely the generation of the adjacency matrix and node attributes, which yields the complete molecule graphs. Score-based [173] or diffusion generative models [174] are also investigated in molecule graph generation [244]. Song et al. [245] propose a unified framework that generalizes score-based or diffusion generative models through stochastic differential equations (SDEs). The forward diffusion process that transforms data x to a simple noise distribution is given in Eq. 2.30.

$$dx = f(x,t)dt + g(t)dw, \tag{2.30}$$

where w is the standard Wiener process, $f(\cdot,t)$ is the drift coefficient of $x(t)$, and $g(\cdot)$ is the diffusion coefficient of $x(t)$. To sample data from random noise, the reversed SDE is given in Eq. 2.31.

$$dx = \left[f(x,t) - g^2(t)\nabla_x \log p_t(x)\right]dt + g(t)d\tilde{w}, \tag{2.31}$$

where $\tilde{w}$ is a standard Wiener process during the reversed flow. The training objective for a score function $s_\theta(x,t)$ parameterized by θ such that $s_\theta(x,t) \approx \nabla_x \log p_t(x)$ is given in Eq. 2.32.

$$\mathbb{E}_{t\sim\mathcal{U}(0,T)}\mathbb{E}_{p_t(x)}\left[\lambda(t)\|\nabla_x \log p_t(x) - s_\theta(x)\|_2^2\right], \tag{2.32}$$

where $\lambda(t)$ is a positive weighting function and $\mathcal{U}(0,T)$ is a uniform distribution over the diffusion time interval. Hoogeboom et al. [187] propose equivariant diffusion model (EDM) which is built upon DDPM [174], a diffusion generative framework, to generate 3D molecules with an equivariant GNN model. EDM treats molecules as atomic types and coordinates without explicitly considering interatomic bonds. The framework has been demonstrated effective for random and conditional generation. Besides, diffusion model has been introduced to biomolecule generation (e.g., proteins) with GNNs [246].

Molecular Conformation Generation Despite molecular optimization and generation via GNN and deep generative models, another important application of generative models is molecular conformation generation, which aims at predicting the ensemble of low-energy 3D conformations of a molecule from its 2D graph solely [247]. Understanding the low-energy conformations, the most stable configurations of molecules in 3D Euclidean space, is of great importance. Since 3D molecular structures determine the functions of chemical and biological processes [248]. Popular metrics for molecular conformation generation include the matching score (MAT) and the coverage score (COV) [249, 250] which are both built up root-mean-square deviation (RMSD) computing the normalized Frobenius norm of the discrepancy of two aligned atomic coordinate systems [251]. Formally, let S_g denote the sets of

generated conformations and S_r denote the one with reference conformations. The expression of MAT and COV are given in Eqs. 2.33 and 2.34, respectively.

$$\mathrm{MAT}(S_g, S_r) = \frac{1}{|S_r|} \sum_{\mathbf{R} \in S_r} \min_{\hat{\mathbf{R}} \in S_g} \mathrm{RMSD}(\mathbf{R}, \hat{\mathbf{R}}), \tag{2.33}$$

$$\mathrm{COV}(S_g, S_r) = \frac{1}{|S_r|} |\{\mathbf{R} \in S_r | \mathrm{RMSD}(\mathbf{R}, \hat{\mathbf{R}}) < \delta, \hat{\mathbf{R}} \in S_g\}|, \tag{2.34}$$

where δ is a pre-defined threshold. Generally, a lower MAT score indicates a better accuracy and a higher COV score indicates a better diversity for the generative model. In many cases, δ is set as 0.5 Å for QM9 and 1.25 Å for GEOM-Drugs. Ganea et al. [252] further propose recall metrics, MAT-R and COV-R, to measure the number of correctly predicted conformers, as well as the precision metrics, MAT-P and COV-P, to measure the number of generated structures of high quality.

Mansimov et al. [253] proposed to leverage a VAE framework with MPNN to directly predict the atomic coordinates from 2D graphs, which does not conserve the SE(3)-equivariance. To this end, the following works instead predict invariant geometric attributes like interatomic distances and torsional angles. Simm et al. [254] propose graph distance geometry (GraphDG) following the VAE framework to predict the interatomic distances. Conditional graph continuous flow (CGCF) by Xu et al. [255] predicts the distance matrix via a flow-based model and optimizes generated conformations via a Markov chain Monte Carlo (MCMC) process and an energy-based tilting model (ETM). Both GraphDG and CGCF utilize the distance geometry (DG) method [256] to search atomic coordinates from the predicted distance matrix. ConfVAE by Xu et al. [257] improves the VAE pipeline which learns to encode molecule graphs into latent space and computes the 3D conformations as a principled bilevel optimization problem. However, these methods predict the distances and conformations separately where the errors in predicted distances accumulate in conformation calculations. In some cases, the predicted distance matrix may even fail to preserve valid 3D conformations. To avoid the issues caused by the two-stage strategy, ConfGF by Shi et al. [250] and DGSM by Luo et al. [258] directly estimates the gradient fields of the logarithm density of atomic coordinates (pseudo force fields) via a score-based generative method. The former relies on static molecule graphs to predict conformations as previous methods, while the latter deploys a dynamic graph construction that better models long-range interactions. Further, GeoDiff by Xu et al. [259] leverages the denoising diffusion probabilistic model (DDPM), a diffusion generative model, to generate 3D conformations via reversing a diffusion process. To keep the SE(3)-equivariance, ConfGF, DGSM, and GeoDiff leverage equivariant GNNs to make predictions of atomic coordinates. Besides interatomic distances and atomic positions, recent works have investigated to predict torsional angles in conformation generation. Ganea et al. propose GeoMol [252] which predicts local 3D structures and ensembles the local structures by predicting the torsion angles. Jing et al. [260] introduce torsional diffusion, a diffusion generative model that formal-

izes molecular conformation generation in the space of torsional angles. TorsionNet by Gogineni et al. [261] presents the conformation generation as a reinforcement learning problem to sample torsion at each time step.

In summary, various generative models, including VAE, GAN, flow-based model, and score-based/diffusion model, have been implemented for a molecular generation. The fragment-based generation defines the generative process as sequentially adding atoms, bonds, or motifs to build the graph. Such a framework, incorporated with RL or autoregressive models, makes the process tractable. RL provides a suitable framework for molecular optimization. Recently works that utilize a flow-based model to generate molecules autoregressively have also been a success. The molecule-based generation, on the other hand, generates the whole molecule simultaneously. Within this category, VAE-based generative models are widely used and explicitly learn a meaningful latent space where molecular search and optimization can be easily conducted. A few works leverage GAN models for molecular generation, however, it is not as widely used as VAE, due to potentially unstable training and hard to manipulate the generation. Flow-based models can also be leveraged for the molecule-based generation which generates node features, bond features, and/or adjacency matrices simultaneously. Recently, the development of score-based or diffusion models in image generation has also inspired works in molecular generation. Moreover, recent research is paying more attention to 3D molecular structure generations beyond 2D graphs. Since 3D structures play an essential role in various molecular applications. Notably, there are also massively works that build molecular generative models on string-based representations (e.g., SMILES) [179, 262, 263]. However, this is not the focus of this chapter, and compared to string-based representations, molecule graph generation with GNNs provides more topological and geometric information that is vital in molecular science.

2.3.5 Others

Synthesis Planning and Retrosynthesis Prediction Synthesis planning aims at determining the synthesis path of a chemical compound from available starting materials through a series of chemical reactions [264]. Retrosynthetic prediction, formalized by Corey [265], refers to the reverse problem: to select suitable disconnections recursively given the desired product [266]. The major challenge in the computational retrosynthetic analysis is the combinatorial exploration space of chemical reactions for target compounds. There are two major approaches: template-based and template-free methods. The former matches the target molecule to chemical reaction rules to yield one or multiple candidate precursors. The latter directly predicts the reactants from target products. Recently, several works have investigated template-free or semi template-free retrosynthesis prediction via GNNs. USPTO-50k [267] which contains 50k atom-mapped reactions with 10 reaction classes is the most widely used benchmark. G2G proposed by Shi et al. [268] learns to translate a target molecule into a set of reactant molecule graphs. G2G contains two steps: (1)

edit prediction that splits the target molecule into multiple synthons via predicting the reaction centers, and (2) synthon completion that transforms the synthons into reactant graphs. Such a workflow has been leveraged by many works. Sun et al. [269] propose to formalize such a retrosynthesis prediction problem as an energy-based model (EBM). Somnath et al. [270] introduce leaving graphs from a precomputed vocabulary for synthon completion, which greatly reduces the complexity of synthon generation. Lin et al. [271] further present G2GT, which leverages recent graph transformer (i.e., Graphormer [29]) in template-free retrosynthesis prediction. Han et al. [272] propose GNN-Retro that combines GNN with the A* search algorithm.

Molecular Knowledge Graph Knowledge graphs (KGs) are graphical structured knowledge bases where each node models a knowledge entity and each edge models the relations between entities [4]. KGs are a powerful tool that incorporates multiple data sources. In molecular science, KGs have been leveraged to represent biomedical knowledge. Specifically, nodes may model various entities, including drugs, protein targets, diseases, side effects, pathways, etc. Such biomedical KGs can be leveraged to predict drug-drug interactions (DDIs), which aims at predicting the outcomes of combined use of two or more drugs, like side effects, adverse reactions, or even toxicity [273–275]. Sources of DDI knowledge include DrugBank [122], KEGG [276], BioSNAP [277], PharmGKB [278], Bio2RDF [279], etc. In DDI KG, nodes model drugs and edges model interactions, thus DDI prediction can be treated as link prediction tasks. GNNs have since made progress on DDI prediction based on KGs. Feng et al. [280] implement GCN [21] on drug-drug interaction KG to predict potential DDIs. Yu et al. [281] introduce SumGNN which utilizes a local subgraph in KG around drug pairs instead of the entire KG to obtain information. Multiple works investigate leveraging multi-modal data to enhance DDI predictions. Lyu et al. [282] propose MDNN containing KG and heterogeneous pathways to obtain multi-modal representations. Lin et al. [283] propose an end-to-end framework built upon GNNs which captures potential neighborhoods of drugs by leveraging their relations in KG. Zhang et al. [284] propose MKGE which combines KG embedding with molecular structure features for DDI predictions. He et al. [285] propose to fuse the topological information from molecular graphs and the interaction information in the form of SMILES. Some works directly build GNNs on molecular graphs of drug pairs to predict DDIs [286–288] instead of relying on KGs. Besides DDI KGs, other KGs have also been investigated like protein-protein interaction KGs [289, 290] and chemical reaction networks [291]. Zitnik et al. [292] propose Decagon based on graph autoencoder to predict side effects of drug pairs and build a large multi-modal KG including protein-protein, drug-protein, and drug-drug interactions.

Biomolecules Though the focus of this chapter lies in small organic molecules. It is worth mentioning that GNNs have also been leveraged to learn expressive representations of large biomolecules (i.e., proteins, RNAs, DNAs, etc.). AlphaFold [293], which is a milestone in protein structure prediction, utilizes attention and message-passing mechanisms. Following this, multiple works have probed improving protein structure prediction from amino acid sequences [294–296]. Many works

have also developed GNN-based models in various protein applications, including protein function prediction [297, 298], protein-compound interactions [299, 300] and protein design [246, 301].

2.4 Self-supervised Learning on Molecule Graphs

The previous section reviews various GNN architectures with expressive message-passing operations for molecular systems. However, a well-designed neural network does not guarantee to perform well, as the performance of the model is also closely related to the quantity and quality of data. In supervised learning, the ML models require instance-label pairs to learn representations. Labeling data can be expensive and time-consuming, and acquiring labels in molecular science takes even more effort. Measuring the attributes and properties of molecules depends on complicated wet-lab experiments and/or expensive simulations. Thus, labeled molecular data is far from sufficient in many applications. Besides, molecular labels are usually noisy since different experimental or simulation environments can lead to significant fluctuations. Moreover, the chemical space of potential molecules is magnificent. The number of potential pharmacologically active molecules is estimated to be in the order of 10^{23}–10^{60} [302]. For the GNN trained on limited and noisy data in a supervised learning manner, it is of great challenge to generalize novel molecules from the huge chemical space. Despite the limited labeled molecular data, the number of molecules in publicly available databases is growing fast in recent years. Such databases include ZINC [66], PubChem [84], Enamine REAL library, etc. A question may be asked: can one make use of the large unlabeled data to improve ML models?

Following the insight, self-supervised learning (SSL) [303] is proposed to utilize the large unlabelled data and learn better representations that are generalizable to various challenging applications. As the name indicates, SSL trains the ML models via the supervisory signals from the data itself, which usually utilizes the underlying structure in the data. There are two major categories of SSL: (1) non-contrastive learning methods, (2) contrastive learning methods. Many non-contrastive learning methods rely on generative models that learn meaningful representations by recovering the data from partially observed input. Examples in computer vision include colorization of grayscaled images [304], prediction of rotation of images [305], and reconstruction of partially masked image [306, 307]. In natural language processing (NLP), self-supervised pre-training via prediction of masked tokens has been a standard technique in deploying large language models [308, 309]. On the other hand, contrastive learning (CL) methods, like MoCo [310] and SimCLR [311], learns representations by attracting the representations of the positive pairs together and pushing representations of the negative pairs away, where positive pairs are perturbed instances from the same data while negative samples are instances from different data [312]. For a query representation $\boldsymbol{q}$ and a batch of $N + 1$ keys containing only one key $\boldsymbol{k}_+$ from a positive instance, the information noise-contrastive estimation (InfoNCE) loss, a popular type of contrastive loss, is given in Eq. 2.35.

$$\ell = -\log \frac{\exp\left((\boldsymbol{q}^\top \boldsymbol{k}_+)/\tau\right)}{\sum_{i=0}^{N} \exp\left((\boldsymbol{q}^\top \boldsymbol{k}_i)/\tau\right)}, \tag{2.35}$$

where τ is the temperature parameter. Such a loss can be considered as a $(N+1)$-way classifier to classify $\boldsymbol{q}$ as its positive instance $\boldsymbol{k}_+$. CL has been a success in a wide variety applications [313–315]. Following the objective of contrasting representation in CL, researchers design non-contrastive methods that only require pulling together representations of positive pairs, like BYOL [316], SimSiam [317], and VICReg [318], which simply the framework.

Since proposed, SSL has been introduced to learning molecular representations from large unlabeled data with GNNs [319]. As shown in Fig. 2.4, SSL on molecule graphs has two steps. In the pre-training step, the GNN is trained on large unlabeled molecular data via self-supervised learning tasks, where the GNN parameters are optimized to encode expressive molecular representations. In the downstream application step, the pre-trained GNN is fine-tuned or used to directly extract the learned representations for various molecular applications. By this means, the GNN model digs into the molecule graphs and learns better representations that can perform well even on tasks with noisy and limited data. The following paragraphs will enumerate works that investigate different SSL techniques on molecule graphs.

Non-contrastive SSL methods are first introduced to learn representations from 2D molecule graphs. Most of such molecular SSL methods are based on predicting the probability or attribute of a masked or partially observed substructure occurring within a molecule [320]. Hu et al. [43] introduce pre-training tasks at the levels of both individual nodes and the whole graphs. The node-level pre-training contains two self-supervised tasks: context prediction which uses subgraphs to predict their surrounding graph structures, and masked attribute prediction which pre-trains GNNs through predicting the attributes of masked nodes/edges. While the graph-

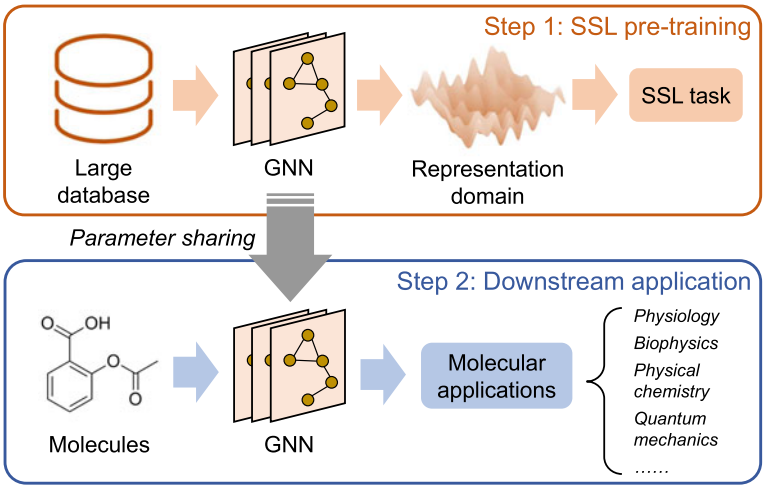


Fig. 2.4 Illustration of self-supervised learning pre-training and fine-tuning on downstream tasks

level pre-training is based on supervised learning of domain-specific graph labels, which is still constrained by the availability of molecular labels. Rong et al. [46] further introduce contextual property prediction which predicts masked subgraphs and graph-level motif prediction which predicts the containing functional groups within a molecule as two SSL tasks. This work also adapts a transformer-based model to learn representations from molecule graphs. Zhang et al. [321] also design motif-level self-supervised tasks that leverage the BRICS algorithm to construct motif trees and pre-trains GNNs by generating molecular graphs motif-by-motif. At each step of pre-training, the model first predicts the topology which determines whether a node has extra motifs to be connected to, and then predicts the motif label from motif vocabulary. He et al. [322] extends the idea of predicting masked attributes even further by leveraging information in chemical reactions. The work named masked molecule modeling (MMM) adapts Graphormer [29] to take in molecule graphs in a chemical reaction, the GNN model is then trained to reconstruct randomly masked atoms, motifs, and bonds within the reaction. Besides, Liu et al. [323] leverage N-gram graph representations to embed molecule graphs in an unsupervised manner that assembles the node features in short walks. The graph embedding method requires no training and can be used for different ML models. Sun et al. [324] introduce InfoGraphn which learns presentations by maximizing the mutual information between the graph-level embedding and multi-level substructure embedding.

Apart from 2D graphs, recent works have probed SSL strategies to include 3D geometric information. To acquire 3D conformations for pre-training, some works [325–327] uses efficient Merck molecular force field (MMFF94) function in RDKit[1] to simulation 3D atomic positions. While other methods [328–331] leverage datasets containing 3D molecular conformations at equilibrium calculated by more precise yet expensive density functional theory (DFT), including QM9, GEOM, PCQM4Mv2, and Molecule3D. Chen et al. [332] design an algebraic graph-assisted bidirectional transformer method that fuses representations from the algebraic graph and transformer. The algebraic graph embeds 3D stereochemical information into graph invariants and the bidirectional transformer embeds representations from SMILES [38] language. The fused representations can be incorporated with a variety of machine learning algorithms, including tree-based models and DNNs, for various molecular property predictions. Fang et al. [325] propose a geometry-enhanced molecular representation learning method (GEM) that learns representations through predicting bond lengths, bond angles, and interatomic distances. It also introduces an auxiliary graph-level SSL task that predicts molecular fingerprints, i.e., ECFP and MACCS keys. Zhou et al. [327] present masked atom prediction and 3D position denoising including pair-distance prediction and direct coordinate prediction in the framework named Uni-Mol. The input molecule is perturbed by masking some atoms and adding noise to 3D coordinates randomly. Uni-Mol also adapts a transformer-based model with invariant spatial positional encoding to embed geometric information. With the recent development of equivariant GNNs on 3D molecules, researchers have explored SSL strategies that perform to improve the expressiveness of such models. Zaidi et

[1] https://www.rdkit.org/.

al. [330], Liu et al. [331], and Jiao et al. [333] have demonstrated the effectiveness of position denoising as self-supervised pre-training on equivariant GNNs. The three works share the similar idea that adds noise to the 3D atomic positions at equilibrium and let the GNN predict the noise [330, 333], or predict the original interatomic distances [331]. Wang et al. [334] extend such denoising pre-training to non-equilibrium molecular systems. They have also shown that the positional denoising SSL objective is equivalent to the force field prediction from equilibrium structures. This sheds a light on leveraging equivariant GNNs for a wider range of applications like predicting potential energy surfaces and molecular dynamics simulations.

Inspired by the success of CL, CL-based self-supervised methods have been introduced to molecular systems. Wang et al. [335] propose MolCLR, a CL framework for molecular representation learning with GNNs. To create contrastive pairs, MolCLR implements three molecule graph augmentation strategies: atom masking, bond deletion, and subgraph removal. A positive pair contains graphs augmented from the molecule while a negative pair contains those augmented from different molecules. MolCLR adapts the normalized temperature-scaled cross entropy (NT-Xent) loss from SimCLR [311], which is a modification of the InfoNCE loss. Given a batch of N molecules, each molecule generates two augmented graphs, and the NT-Xent loss for a positive pair (i, j) in the batch of $2N$ graphs is given in Eq. 2.36.

$$\ell_{i,j} = -\log \frac{\exp((\boldsymbol{z}_i^\top \boldsymbol{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{\{k \neq i\}} \exp((\boldsymbol{z}_i^\top \boldsymbol{z}_k)/\tau)}, \tag{2.36}$$

where $\mathbb{1}_{\{\cdot\}}$ is an indicator function, and $\boldsymbol{z}_i, \boldsymbol{z}_j$ are normalized representations of the positive pair. Zhang et al. [336] further propose MICRO-Graph which extracts informative subgraphs from molecules via GNNs and performs contrastive pre-training on sampled subgraphs learned by GNNs. Similar to non-contrastive learning methods, CL has also gone beyond 2D molecule graphs and applies multi-view CL pre-training. Liu et al. [328], Stärk et al. [329], and Li et al. [326] propose GraphMVP, 3D infomax, and GeomGCL, respectively. All three methods conduct contrastive pre-training between 2D topological graphs and 3D geometric structures to learn molecular representations with 3D information embedded. Namely, 2D and 3D views from a molecule compose a positive pair while others are negative pairs. GraphMVP also utilizes a generative model in the representation domain as an auxiliary SSL task. Besides, Zhu et al. [337] develop a dual-view CL framework that contrasts between SMILES strings and 2D molecule graphs, where the former is encoded by a transformer and the latter is encoded by GNN. Further, MEMO from Zhu et al. [338] presents a multi-view CL that leverages four molecular representations, including 2D topology, 3D geometry, SMILES string, and fingerprint. Some works have sought to fuse chemical domain knowledge in molecular CL models to learn better representations. Following the insight, Wang et al. [339] introduce iMolCLR to improve previous CL frameworks in two aspects: (1) mitigating faulty negative instances via considering molecular similarities in NT-Xent loss, and (2) fragment-level contrasting between substructures decomposed via the BRICS algorithm. Fang et al.

[340] propose a molecule cluster strategy that utilizes functional groups and fingerprints similarity to select similar molecules as positive pairs and dissimilar molecules as negative pairs in contrastive training. The work also includes functional group embeddings as an input to GNN. MoCL by Sun et al. [341] leverages local-level and global-level in contrastive learning on molecule graphs. Specifically, the local-level domain knowledge augments the molecule graph by replacing a substructure with a bioisostere which perturbs the instance while maintaining the chemical properties. The global-level domain knowledge introduces a loss that minimizes the difference between the similarity of learned representations and ECFP similarity between two molecules. Fang et al. [342] propose knowledge-enhanced CL that includes graph augmentations based on the chemical element knowledge graph as well as the know-aware message-passing functions. In another model named CoSP, Gao et al. [343] leverage information besides small organic molecules to learn better representations. CoSP adapts a cross-domain CL framework that generates contrastive pairs from ligand molecules and protein binding pockets. It also introduces ChemInfoNCE loss to reduce the negative sampling bias through a chemical similarity-enhanced negative ligand sampling strategy.

2.5 Conclusion

Recent years have witnessed an immense growth of graph neural networks in molecular sciences. In this survey, we review the generic message-passing framework for building graph neural networks. Graph neural network architectures designed for small organic molecules are introduced. Works on learning representations from 2D topological graphs as well as recent efforts in leveraging 3D geometric information are covered. Also, we try to provide a comprehensive overview of benchmarks, metrics, and graph neural network models in various molecular applications, including property predictions, molecular scoring, and docking, molecular optimization and generation, molecular dynamics, synthesis planning, molecular knowledge graph, etc. The last section epitomizes molecular self-supervised learning with graph neural networks. Graph neural networks have been successfully implemented to solve a wide variety of molecular tasks. However, the acquisition of high-quality labeled molecular data is still expensive and time-consuming. One direction that is worth exploring is to introduce domain knowledge to molecular graph neural networks for solving challenging problems with limited or noisy data.

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Chapter 3
Voxelized Representations of Atomic Systems for Machine Learning Applications



Matthew C. Barry, Satish Kumar, and Surya R. Kalidindi

Abstract The behavior of a material or chemical system is determined by complex physical phenomena taking place over a hierarchy of length scales. Atomic systems are typically considered to be the smallest relevant length scale in materials modeling and thus form the basis of hierarchical multiscale materials modeling. This chapter presents a detailed overview of voxelized representations of the atomic structure that can be used in machine learning (ML) applications not only at the atomic length scale, but also as the foundation of a hierarchical multiscale materials modeling framework. We first present a theoretical development of the continuous microstructure function as a mathematical definition of material structure. We then show that the mathematical definition of the discrete (i.e., voxelized) microstructure function is consistent for any length scale. Furthermore, we show that the microstructure function (both discrete and continuous) can be used both to quantify the local material structure directly by systematically selecting the reference axis of the material system and to statistically quantify the global material structure using n-point spatial correlation functions. We then present practical methods for computing microstructure functions and numerically quantifying the material structure. Finally, we present the voxelized atomic structure (VASt) framework for developing both ML-based interatomic potentials and ML models supporting materials discovery for desired effective properties.

Keywords Materials informatics · Structure–property relationship · Homogenization · Machine learning-based interatomic potential · Density functional theory · Atomistic modeling · High entropy alloy

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3.1 Introduction

Atomic systems are fundamental to understanding and modeling the physics of material and chemical systems. Atomic simulation methods such as density functional theory (DFT) and molecular dynamics (MD) allow us to elucidate the complex physical processes occurring in material/chemical systems, and to compute numerous fundamental physical properties. However, there are many limitations in physics-driven atomic simulation methods. First-principles methods such as DFT are highly accurate but impractical for physics involving more than a few hundred atoms (i.e., length scales $\gtrsim$ 1–2 nm) and time scales $\gtrsim$ 100 ps due to their high computational cost and poor scaling with the number of atoms. Empirical potentials, on the other hand, are significantly faster than first-principal methods and can scale to much larger atomic systems, but are less accurate, difficult to develop, and lack transferability. As a result, one is often forced to compromise between accuracy and computational cost.

Machine learning (ML) approaches offer new opportunities to access the accuracy of first-principles methods without the computational limitations. There are two direct use cases for ML in the context of atomic systems: materials/compound discovery [1–13] and ML-based interatomic potentials (MLIPs) [14–30]. In materials discovery, ML approaches are used to predict a set of desired effective physical properties of a material directly from its atomic structure, bypassing the computationally expensive physics simulations completely. On the other hand, MLIPs are used to drive MD simulations by predicting the energy and interatomic forces of an atomic system at each timestep. Because materials discovery and MLIPs aim to accomplish two fundamentally different tasks, their ML approaches require different representations and featurizations of the atomic structure. In particular, the input features to ML models for materials discovery must capture a global representation of the atomic structure, whereas the input features to MLIPs must capture the local atomic structure in the neighborhood of a given atom. However, although materials discovery and MLIPs require different representations and featurizations of the atomic structure, these representations and featurizations should be derived from a single, consistent mathematical definition of atomic structure.

As previously mentioned, atomic systems can provide substantial insight into the molecular-scale properties and fundamental physics of material and chemical systems. However, the full material response is determined by the material structure over a hierarchy of length scales, and only certain aspects of the material behavior can be determined by the atomic structure. Furthermore, atomic systems do not fit neatly into a single length scale, but rather span two length scales—the smaller covering full atomic systems at the first-principles level and local atomic neighborhoods up to atomic interaction cutoff distances, and the larger covering full atomic systems at the MD length scale. Thus, although many ML approaches have been developed for atomic systems—and notable successes have been reported in both materials

discovery [1–13] and MLIPs [14–30]—the scope of existing approaches is inherently limited by their failure to consider the atomic structure in the context of hierarchical multiscale materials modeling.

This chapter presents a detailed framework for developing structure–property relationships with atomic systems that is also consistent with the greater goal of hierarchical multiscale materials modeling. We first present a rigorous mathematical definition of material structure that is consistent across all length scales, and from which we can derive both a local and global representation of the material structure. We then show that from this definition of material structure, the global material structure can be quantified statistically using spatial correlations in the form of n-point spatial correlation functions and the local material structure can be quantified directly by systematically selecting the reference axis of the material system. We then outline practical considerations for numerically computing and quantifying the material structure in the context of structure–property relationships. Finally, we review the voxelized atomic structure (VASt) framework for developing structure–property relationships with atomic systems at the first-principles length scale [14]. We demonstrate that the VASt framework for MLIPs can be used to efficiently and accurately predict the thermal conductivity of silicon over a range of hydrostatic strain using the Boltzmann transport equation (BTE). We also demonstrate that the VASt framework for compound discovery can be used to accurately predict the bulk modulus of any chemical composition of AlNbTiZr refractory high entropy alloy (RHEA) using less than 25 training samples.

3.2 Mathematical Formulation of Material Structure

Before we mathematically define the material structure, it is important to first understand fully the specific properties this definition should satisfy. As mentioned previously, the mathematical definition of material structure should be consistent across all length scales and allow the material structure to be quantified both globally and locally. In addition to these requirements, this definition should be an injective mapping (i.e., unique material structures are uniquely defined), invariant under rotations and translations of the material structure, and invariant to internal permutations of equivalent features.

At the most abstract level, the structure of a given material is a mapping of material local states, $\boldsymbol{h}$, over a continuous spatial domain, $\Omega \subset \mathbb{R}^3$. The local state at a point $\boldsymbol{x} \in \Omega$ is the sequence $\boldsymbol{h} = (h_1, \ldots, h_n)$ of material structure attributes, h_i, needed to specify completely the local material response at that point. Each material structure attribute can take on a set of possible values, H_i, which can be either discrete or continuous. The set of all possible local states, $\boldsymbol{H} = \{(h_1, \ldots, h_n) \mid h_i \in H_i, i = 1, \ldots, n\}$, is called the local state space. Thus, given a material with spatial domain, Ω, and local state space, $\boldsymbol{H}$, the material structure is defined as the mapping

$$m\colon \Omega \times \boldsymbol{H} \to \mathbb{R}_{\geq 0}, \tag{3.1}$$

where m is called the microstructure function and $m(\boldsymbol{x}, \boldsymbol{h})$ defines a suitable measure such as the probability density of finding the local state $\boldsymbol{h} \in \boldsymbol{H}$ at the spatial location $\boldsymbol{x} \in \Omega$ [31–37].

The mapping given by Eq. (3.1) satisfies all the previously identified properties that a mathematical definition of material structure must satisfy. It is independent of length scale since $m(\boldsymbol{x}, \boldsymbol{h})$ is defined at every point, $\boldsymbol{x}$, in the continuous domain, Ω. It can quantify the global material structure by choosing Ω to be either the entire material structure or a subset of the entire material structure that is statistically equivalent to the entire material structure, called a representative volume element (RVE). It can also quantify the local material structure at a point $\boldsymbol{x}_0 \in \Omega$ by restricting the spatial domain of $m(\boldsymbol{x}, \boldsymbol{h})$ to the closed ball in Ω centered at $\boldsymbol{x}_0$ with radius $r > 0$, defined as $B_r(\boldsymbol{x}_0) = \{\boldsymbol{x} \in \Omega \,|\, \|\boldsymbol{x} - \boldsymbol{x}_0\| \leq r\}$. Furthermore, the mapping given by Eq. (3.1) is clearly injective since two microstructure functions, m_i and m_j, are equivalent if and only if $m_i(\boldsymbol{x}, \boldsymbol{h}) = m_j(\boldsymbol{x}, \boldsymbol{h})$ for every $\boldsymbol{x} \in \Omega$ and $\boldsymbol{h} \in \boldsymbol{H}$. Similarly, it is invariant to internal permutations of equivalent features since two features located at spatial regions $U_i, U_j \subseteq \Omega$ are equivalent if and only if $U_j = U_i + \boldsymbol{r}$ for some $\boldsymbol{r} \in \mathbb{R}^3$ (i.e., U_j is a translation of U_i) and $m(\boldsymbol{x}, \boldsymbol{h}) = m(\boldsymbol{x} + \boldsymbol{r}, \boldsymbol{h})$ for every $\boldsymbol{x} \in U_i$ and $\boldsymbol{h} \in \boldsymbol{H}$. Finally, although the mapping given by Eq. (3.1) is not inherently rotationally and translationally invariant, we can impose rotational and translational invariance by procedurally defining the origin and orientation of the reference axis of the spatial domain, Ω.

In practice, we cannot experimentally measure or numerically compute the material local state on a continuous spatial domain. Instead, the local state at a given point is measured as an average over some finite volume—henceforth referred to as a voxel—determined by the length scale of the experimental characterization technique or numerical simulation. Thus, we employ a discrete version of the microstructure function, defined as the mapping

$$m\colon \boldsymbol{S} \times \boldsymbol{H} \to \mathbb{R}_{\geq 0}, \tag{3.2}$$

where $\boldsymbol{S} = \{(s_1, s_2, s_3) \,|\, s_i \in [0, 1, \ldots, S_i - 1],\ i = 1, 2, 3\}$ is the set of all voxel indices, $\boldsymbol{s} = (s_1, s_2, s_3)$, defining the discretized three-dimensional domain of the material structure, and $S_1, S_2, S_3 \in \mathbb{N}$ are the number of voxels along each dimension. Similar to the continuous case, for the discrete microstructure function we define $m[\boldsymbol{s}, \boldsymbol{h}]$ as the probability density of finding the local state $\boldsymbol{h} \in \boldsymbol{H}$ at the voxel with index $\boldsymbol{s} \in \boldsymbol{S}$. However, unlike the continuous case, the local state $\boldsymbol{h} \in \boldsymbol{H}$ at the voxel $\boldsymbol{s} \in \boldsymbol{S}$ is the sequence $\boldsymbol{h} = (h_1, \ldots, h_n)$ of material structure attributes, h_i, needed to specify completely the local material response *at the length scale of the voxel*, and is defined as the average local state over the spatial region $Q_{\boldsymbol{s}} \subseteq \Omega$ covered by the voxel,

$$\boldsymbol{h}(\boldsymbol{s}) = \frac{1}{\text{vol}(Q_s)} \int_{Q_s} \boldsymbol{h}(\boldsymbol{x}) d\boldsymbol{x}. \tag{3.3}$$

The discrete microstructure function given by Eq. (3.2) satisfies the same properties as the continuous microstructure function. Although the discrete microstructure function is now dependent on length scale, this dependency is accounted for in the definition of the discrete spatial domain and local state space. Thus, for any discrete spatial domain and local state space, the mathematical definition of material structure given by Eq. (3.2) remains consistent, regardless of the voxel length scale. Similar to the continuous microstructure function, the discrete microstructure function can quantify the global material structure by choosing $\boldsymbol{S}$ to be a discretization of the spatial domain corresponding to the entire material structure or an RVE. It can also quantify the local material structure at a point $\boldsymbol{x}_0 \in \Omega$ by restricting the spatial domain of $m[\boldsymbol{s}, \boldsymbol{h}]$ to a discretized cube centered at $\boldsymbol{x}_0$ with side length $L > 0$, where L is an integer multiple of the voxel side length, l. Furthermore, the mapping given by Eq. (3.2) is injective since two discrete microstructure functions, m_i and m_j, are equivalent if and only if $m_i[\boldsymbol{s}, \boldsymbol{h}] = m_j[\boldsymbol{s}, \boldsymbol{h}]$ for every $\boldsymbol{s} \in \mathbf{S}$ and $\boldsymbol{h} \in \boldsymbol{H}$. Similarly, it is invariant to internal permutations of equivalent features since two features located at $\boldsymbol{G}_i, \boldsymbol{G}_j \subseteq \mathbf{S}$ are equivalent if and only if $\boldsymbol{G}_j = \boldsymbol{G}_i + \boldsymbol{t}$ for some $\boldsymbol{t} \in \mathbb{Z}^3$ (i.e., $\boldsymbol{G}_j$ is a translation of $\boldsymbol{G}_i$) and $m[\boldsymbol{s}, \boldsymbol{h}] = m[\boldsymbol{s} + \boldsymbol{t}, \boldsymbol{h}]$ for every $\boldsymbol{s} \in \boldsymbol{G}_i$ and $\boldsymbol{h} \in \boldsymbol{H}$. Finally, although the mapping given by Eq. (3.2) is not inherently rotationally and translationally invariant, we can impose rotational and translational invariance by procedurally defining the origin and orientation of the reference axis of the discrete spatial domain.

3.3 Mathematical Quantification of the Microstructure

We have shown that the microstructure function can mathematically define both the local and global material structure. However, suitable descriptors (or features) need to be extracted from the microstructure function for building high-fidelity structure–property relationships. Before we can determine the best descriptors of the material structure for building structure–property relationships, we must first understand fully what exactly we are trying to quantify.

In structure–property relationships, we are not necessarily interested in the material structure directly, but rather the spatial features underlying the material structure. At the most abstract level, a spatial feature of a material structure is some measure of a set of one or more material local states in a specific spatial configuration. A quantification of the material structure is thus given by some set of spatial features, called a feature vector. The feature vector is a projection of the material structure onto a metric space—called the feature space—on which the metric between two feature vectors quantifies some notion of distance (i.e., similarity/difference) between the two corresponding material structures. Therefore, the mathematical representation of

the material structure must be defined such that the resulting feature space quantifies material structures as similar/different in a manner that is consistent with the given structure–property relationship.

In cases where our interest lies in predicting the material response at a point $\boldsymbol{x}_0 \in \Omega$, we are only interested in the spatial positioning of the material local states relative to the point $\boldsymbol{x}_0$. This implies that the distance between two material structures is determined locally. Thus, the spatial features underlying the material structure can be uniquely and completely quantified directly by the local microstructure function. In certain applications, however, we may know from physics that as the distance from a feature to the point of interest, $\boldsymbol{x}_0$, increases, its effect on the material response decreases. To account for this, the material structure is represented by the damped local microstructure function

$$\eta(\boldsymbol{x}, \boldsymbol{h}) = m(\boldsymbol{x}, \boldsymbol{h}) f_d(\boldsymbol{x}), \tag{3.4}$$

where $m\colon B_r(\boldsymbol{x}_0) \times \boldsymbol{H} \to \mathbb{R}_{\geq 0}$ is the local microstructure function and $f_d(\boldsymbol{x})$ is a damping function included to reflect the potential lower importance of features further away from the point of interest, $\boldsymbol{x}_0$ [14]. In this notation, the spatial variable $\boldsymbol{x}$ is defined with an origin at $\boldsymbol{x}_0$. The damping function can also be used to ensure that $\eta(\boldsymbol{x}, \boldsymbol{h})$ is smooth on the boundary, $\partial B_r(\boldsymbol{x}_0)$, by choosing $f_d(\boldsymbol{x}) \in C^\infty(B_r(\boldsymbol{x}_0))$ such that

$$f_d(\boldsymbol{x}) = 0 \text{ and } f_d'(\boldsymbol{x}) = 0 \text{ for } \|\boldsymbol{x}\| \geq r. \tag{3.5}$$

When $f_d(\boldsymbol{x}) \equiv 1$, the damped local microstructure function is equivalent to the local microstructure function (i.e., the importance of the features is independent of their distance from the point of interest). A discrete version of the damped local microstructure function is given by

$$\eta[\boldsymbol{s}, \boldsymbol{h}] = m[\boldsymbol{s}, \boldsymbol{h}] f_d(\boldsymbol{s} \cdot l), \tag{3.6}$$

where $m\colon C_L(\boldsymbol{x}_0, l) \times \boldsymbol{H} \to \mathbb{R}_{\geq 0}$ is the discrete local microstructure function defined on the discretized cube centered at $\boldsymbol{x}_0$ with side length $L > 0$ whose voxel indices are given by the set $C_L(\boldsymbol{x}_0, l)$. We choose the voxel side length, l, such that the number of voxels discretizing a cube side, $N = L/l$, is odd and define $C_L(\boldsymbol{x}_0, l) = \left\{(s_1, s_2, s_3) \mid s_i \in \left[-\frac{N-1}{2}, \ldots 0, \ldots, \frac{N-1}{2}\right],\ i = 1, 2, 3\right\}$ so that the voxel indexed by $\boldsymbol{s}_0 = (0, 0, 0) \in C_L(\boldsymbol{x}_0, l)$ is centered on $\boldsymbol{x}_0$ and $\|\boldsymbol{s} \cdot l\|$ is the distance from $\boldsymbol{x}_0$ to the voxel indexed by $\boldsymbol{s} \in C_L(\boldsymbol{x}_0, l)$.

In cases where our interest lies in predicting the global response (i.e., property) of a material structure (i.e., homogenization), we are interested in the spatial positioning of two or more material local states relative to each other. Since this type of spatial feature can be present at multiple locations throughout the material structure, the intensity (i.e., signal value) of a given spatial feature must be some quantification of the frequency with which it occurs throughout the material structure. This implies that the distance between two material structures is determined globally, and therefore

that the material structure must be quantified statistically. A complete statistical representation of the material structure is given by the n-point spatial correlation function

$$f_n(\boldsymbol{h}_1, \ldots, \boldsymbol{h}_n, \boldsymbol{r}_1 = 0, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_n) = \frac{1}{|\Omega_{\boldsymbol{r}}|} \int_{\Omega_{\boldsymbol{r}}} \prod_{i=1}^{n} m(\boldsymbol{x} + \boldsymbol{r}_i, \boldsymbol{h}_i) d\boldsymbol{x}, \tag{3.7}$$

where $\Omega_{\boldsymbol{r}} = \bigcap_{i=1}^{n} \Omega_{\boldsymbol{r}_i}$ and $\Omega_{\boldsymbol{r}_i} = \{\boldsymbol{x} \in \Omega \,|\, \boldsymbol{x} + \boldsymbol{r}_i \in \Omega\}$ and $|\Omega_{\boldsymbol{r}}|$ is the measure of the set $\Omega_{\boldsymbol{r}}$. Physically, $f_n(\boldsymbol{h}_1, \ldots, \boldsymbol{h}_n, \boldsymbol{r}_1 = 0, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_n)$ is the probability that the local state $\boldsymbol{h}_1$ is positioned relative to the local states $\boldsymbol{h}_2, \ldots, \boldsymbol{h}_n$ by the vectors $\boldsymbol{r}_2, \ldots, \boldsymbol{r}_n$, respectively [31, 37–44]. Thus, as required, $f_n(\boldsymbol{h}_1, \ldots, \boldsymbol{h}_n, \boldsymbol{r}_1 = 0, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_n)$ quantifies the frequency with which a given spatial feature (i.e., a spatial positioning of two or more material local states relative to each other) occurs throughout the material structure. As a result, the feature vector representation of the material structure given by the n-point spatial correlation function is a statistical representation of the global material structure and the distance between two material structures in this feature space is not sensitive to local fluctuations in their respective material structures. A discrete version of the n-point spatial correlation function is given by

$$f_n[\boldsymbol{h}_1, \ldots, \boldsymbol{h}_n, \boldsymbol{t}_1 = 0, \boldsymbol{t}_2, \ldots, \boldsymbol{t}_n] = \frac{1}{|\boldsymbol{S}_{\boldsymbol{t}}|} \sum_{\boldsymbol{s} \in \boldsymbol{S}_{\boldsymbol{t}}} \prod_{i=1}^{n} m[\boldsymbol{s} + \boldsymbol{t}_i, \boldsymbol{h}_i], \tag{3.8}$$

where $\mathbf{S}_{\boldsymbol{t}} = \bigcap_{i=1}^{n} \boldsymbol{S}_{\boldsymbol{t}_i}$ and $\mathbf{S}_{\boldsymbol{t}_i} = \{\boldsymbol{s} \in \mathbf{S} \,|\, \boldsymbol{s} + \boldsymbol{t}_i \in \mathbf{S}\}$ and $|\mathbf{S}_{\boldsymbol{t}}|$ is the cardinality of the set $\boldsymbol{S}_{\boldsymbol{t}}$. In practice, it is typically sufficient to consider only the two-point spatial correlation function

$$f_2[\boldsymbol{h}_1, \boldsymbol{h}_2, \boldsymbol{t}] = \frac{1}{|\boldsymbol{S}_{\boldsymbol{t}}|} \sum_{\boldsymbol{s} \in \boldsymbol{S}_{\boldsymbol{t}}} m[\boldsymbol{s}, \boldsymbol{h}_1] m[\boldsymbol{s} + \boldsymbol{t}, \boldsymbol{h}_2]. \tag{3.9}$$

When $\boldsymbol{S}$ discretizes the spatial domain of an RVE on which the microstructure function is periodic, we have $\boldsymbol{S}_{\boldsymbol{t}} = \boldsymbol{S}$ and Eq. (3.9) is a discrete cross-correlation and can be written in terms of discrete Fourier transforms as

$$f_2[\boldsymbol{h}_1, \boldsymbol{h}_2, \boldsymbol{t}] = \frac{1}{|\boldsymbol{S}|} \mathcal{F}^{-1}\left[M_{\boldsymbol{k}}^{\boldsymbol{h}_1 *} M_{\boldsymbol{k}}^{\boldsymbol{h}_2}\right], \tag{3.10}$$

where $\mathcal{F}^{-1}$ denotes the inverse discrete Fourier transform, $M_{\boldsymbol{k}}^{\boldsymbol{h}}$ is the discrete Fourier transform of $m[\boldsymbol{s}, \boldsymbol{h}]$, and $M_{\boldsymbol{k}}^{\boldsymbol{h}*}$ is the complex conjugate of $M_{\boldsymbol{k}}^{\boldsymbol{h}}$. The advantage of defining the discrete two-point spatial correlation function in the form of Eq. (3.10) is that it can take advantage of computationally efficient fast Fourier transform algorithms.

3.4 Numerical Implementation of Microstructure Descriptors

In order to have meaningful feature space representations of the material structure for building structure–property relationships, the microstructure function must be defined on a standardized discrete spatial domain using a consistent, standard voxel length. However, in many cases, it is impossible to experimentally measure or numerically compute the material local states for each material structure in a given set of material structures all on a standardized domain. In crystalline atomic systems, for example, the spatial domain is determined by the lattice vectors and thus for even a small number of such systems, it is impossible to find a (reasonable) voxel length for which every lattice can be represented by an integer number of voxels. Therefore, before the feature space representation of a given material structure is computed, its microstructure function must first be approximated on a standardized spatial domain.

Before we can approximate the microstructure function on a standardized discrete spatial domain, we must first determine a procedure for standardizing a nonstandard discrete spatial domain. For simplicity, throughout this section we will assume that all discrete spatial domains (standardized and nonstandardized) are cubes. Given a nonstandardized discrete spatial domain defined by the set of voxel indices, $\boldsymbol{S}$, with side length, L, and voxel length, l, we define the standardized discrete spatial domain, $\boldsymbol{\Sigma}$, as the closest approximation whose side length, Λ, is an integer multiple of the standard voxel length, λ (see Fig. 3.1a). Thus, depending on the definition of "closest" approximation, Λ can be defined as either

$$\Lambda^{(1)} = \lambda \cdot \left\lfloor \frac{L}{\lambda} + 0.5 \right\rfloor \quad \text{or} \quad \Lambda^{(2)} = \lambda \cdot \left\lceil \frac{L}{\lambda} \right\rceil, \tag{3.11}$$

where $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$ are the floor and ceiling function, respectively [3]. These definitions are related to L by

$$L - 0.5\lambda \leq \Lambda^{(1)} \leq L + 0.5\lambda \quad \text{and} \quad L \leq \Lambda^{(2)} \leq L + \lambda. \tag{3.12}$$

Thus

$$\left|\Lambda^{(1)} - L\right| \leq \left|\Lambda^{(2)} - L\right|, \tag{3.13}$$

so $\Lambda^{(1)}$ will always be the closest approximation to L by distance, but has the potential to "destroy" information near the boundary of the nonstandard discrete spatial domain when $\Lambda^{(1)} < L$. On the other hand, $\Lambda^{(2)}$ is a worse approximation than $\Lambda^{(1)}$ by distance whenever $\Lambda^{(2)} > L + 0.5\lambda$, but will never destroy information since the material volume covered by the standardized discrete spatial

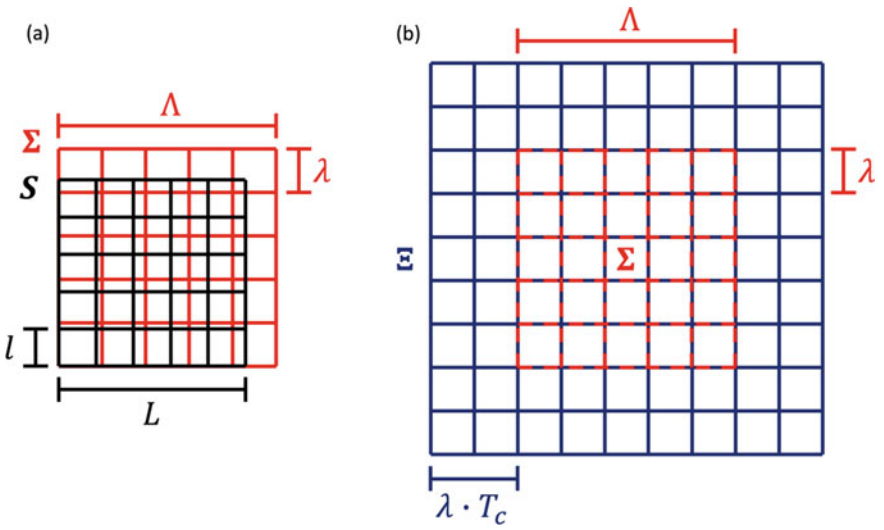


Fig. 3.1 **a** A nonstandard discrete spatial domain (black) with voxel indices, $\boldsymbol{S}$, side length, L, and voxel length, l, and the corresponding standardized discrete spatial domain (red) with voxel indices, $\boldsymbol{\Sigma}$, side length, Λ, and standard voxel length, λ. **b** The standardized discrete spatial domain (red) and corresponding extended standardized discrete spatial domain (blue) with voxel indices, $\boldsymbol{\Xi}$, and $T_c = \lceil R_c/\lambda \rceil$

domain will always contain the material volume covered by the original, nonstandard discrete spatial domain. Given Λ, we define the standardized voxel indices as $\boldsymbol{\Sigma} = \{(\sigma_1, \sigma_2, \sigma_3) \,|\, \sigma_i \in [0, 1, \ldots, \Sigma - 1], i = 1, 2, 3\}$, where $\Sigma = \Lambda/\lambda$ is the number of standard voxels that discretize each side.

Once the standardized discrete spatial domain is defined, the next step is to approximate the microstructure function on this domain. The appropriate method for approximating the microstructure function on the standardized discrete spatial domain depends on the topology of the nonstandard discrete spatial domain and the physical properties of the material local states under consideration. Typically, the material local states are field variables whose values are experimentally measured or numerically computed at each voxel. In these cases, the microstructure function must be approximated on the standardized discrete spatial domain using some form of interpolation. When the topology of the nonstandard discrete spatial domain is an RVE on which the microstructure function is periodic, such as in crystalline atomic systems, a natural choice is trigonometric interpolation [45, 46]. In this approach, a continuous microstructure function is approximated from the original discrete microstructure function defined on the nonstandard discrete spatial domain in the form of the trigonometric polynomial

$$\tilde{m}(\boldsymbol{x}=(x_1,x_2,x_3),\boldsymbol{h})=\frac{1}{|\boldsymbol{S}|}\sum_{k_1=-\frac{S}{2}+1}^{\frac{S}{2}}\sum_{k_2=-\frac{S}{2}+1}^{\frac{S}{2}}\sum_{k_3=-\frac{S}{2}+1}^{\frac{S}{2}}\hat{m}_{k_1k_2k_3}(\boldsymbol{h})e^{\frac{2\pi ik_1x_1}{L}}e^{\frac{2\pi ik_2x_2}{L}}e^{\frac{2\pi ik_3x_3}{L}}, \tag{3.14}$$

where $\hat{m}_{k_1k_2k_3}$ are the discrete Fourier coefficients given by the discrete Fourier transform of the original discrete microstructure function, defined as

$$\hat{m}_{k_1k_2k_3}(\boldsymbol{h})=\sum_{s_1=0}^{S-1}\sum_{s_2=0}^{S-1}\sum_{s_3=0}^{S-1}m_{s_1s_2s_3}(\boldsymbol{h})e^{-\frac{2\pi is_1k_1}{S}}e^{-\frac{2\pi is_2k_2}{S}}e^{-\frac{2\pi is_3k_3}{S}}. \tag{3.15}$$

The approximated microstructure function given by Eq. (3.14) is a continuous function that is defined at every point, $\boldsymbol{x}$, in the continuous spatial domain covered by the nonstandard discrete spatial domain. Thus, the value of the approximated microstructure function at a standardized voxel, $\boldsymbol{\sigma} \in \boldsymbol{\Sigma}$, is given by evaluating $\tilde{m}(\boldsymbol{x},\boldsymbol{h})$ for $\boldsymbol{x}=\boldsymbol{\sigma}\cdot\lambda$. Furthermore, since $\tilde{m}(\boldsymbol{x},\boldsymbol{h})$ is periodic on the nonstandard spatial domain, it can be evaluated for $\|\boldsymbol{x}\|>L$. Finally, we note that when $\boldsymbol{\Sigma}=\boldsymbol{S}$, Eq. (3.14) is simply the inverse discrete Fourier transform of $\hat{m}$ and $\tilde{m}[\boldsymbol{\sigma},\boldsymbol{h}]=m[\boldsymbol{s},\boldsymbol{h}]$.

Although the approximated microstructure function will no longer be periodic on the standardized discrete spatial domain, one can still approximate the periodic two-point spatial correlation function by suitably extending the standardized discrete spatial domain on which the microstructure function is approximated. Typically, we are not necessarily interested in the complete set of spatial correlations, but rather the spatial correlations up to some cutoff distance, $R_c>0$. In this case, we define the extended standardized discrete spatial domain by the set $\boldsymbol{\Xi}=\{(\xi_1,\xi_2,\xi_3)\,|\,\xi_i\in[-T_c,\ldots,0,\ldots,\Sigma-1+T_c],i=1,2,3\}$, where $T_c=\lceil R_c/\lambda\rceil$ is the minimum number of voxels such that $T_c\lambda\geq R_c$. Thus, $\boldsymbol{\Sigma}\subseteq\boldsymbol{\Xi}$ and the two-point spatial correlation function is given by

$$f_2[\boldsymbol{h}_1,\boldsymbol{h}_2,\boldsymbol{t}]=\frac{1}{|\boldsymbol{\Sigma}|}\sum_{\boldsymbol{\xi}\in\boldsymbol{\Xi}}\tilde{m}\big[\boldsymbol{\xi},\boldsymbol{h}_1\big]\chi_{\boldsymbol{\Sigma}}(\boldsymbol{\xi})\tilde{m}\big[\boldsymbol{\xi}+\boldsymbol{t},\boldsymbol{h}_2\big], \tag{3.16}$$

where $\chi_{\boldsymbol{\Sigma}}(\boldsymbol{\xi})$ is the indicator function. For the standardized discrete spatial domain shown in Fig. 3.1a, the corresponding extended standardized discrete spatial domain is shown in Fig. 3.1b. The product $\tilde{m}\big[\boldsymbol{\xi},\boldsymbol{h}_1\big]\chi_{\boldsymbol{\Sigma}}(\boldsymbol{\xi})$ is equivalent to zero-padding a T_c voxel thick border around the discrete microstructure function defined on $\boldsymbol{\Sigma}$. The discrete Fourier transform based formulation of the two-point spatial correlation function is then given by

$$f_2[\boldsymbol{h}_1,\boldsymbol{h}_2,\boldsymbol{t}]=\frac{1}{|\boldsymbol{\Sigma}|}\mathcal{F}^{-1}\Big[\mathcal{F}\big(\tilde{m}\big[\boldsymbol{\xi},\boldsymbol{h}_1\big]\chi_{\boldsymbol{\Sigma}}(\boldsymbol{\xi})\big)^*\mathcal{F}\big(\tilde{m}\big[\boldsymbol{\xi},\boldsymbol{h}_2\big]\big)\Big]. \tag{3.17}$$

Hence, one can still take advantage of the computationally efficient discrete Fourier transform based formulation of the two-point spatial correlation function.

3.5 The Voxelized Atomic Structure Framework

The voxelized atomic structure (VASt) framework is a comprehensive and highly generalizable approach for developing structure–property relationships using first-principles computations [14]. The VASt framework can be used to develop structure–property relationships in the form of MLIPs (i.e., VASt potentials) [14] and homogenization models for predicting effective physical properties of materials (i.e., VASt compound discovery). The goal of the VASt framework is to represent the true atomic structure as it is physically defined in the context of a given structure–property relationship. In particular, the VASt framework aims to minimize the introduction of nonphysical parameters and the use of *ad hoc* feature engineering when defining the atomic structure. In the context of MLIPs, this is achieved by representing the neighborhood of an atom as a discrete local microstructure function whose values represent the volume fraction of each chemical species within a voxel and directly utilizing this representation as the input to a convolutional neural network (CNN) for implicit feature engineering. In the context of homogenization, this is achieved by representing the global atomic structure by the two-point spatial correlations of its charge density field, projecting this representation to a low-dimensional feature space using principal component analysis (PCA), and developing a reduced-order probabilistic model using Gaussian process regression (GPR).

3.5.1 Voxelized Atomic Structure Potentials

Machine learning-based interatomic potentials have recently attracted great interest for their potential to predict the interatomic forces and energy of atomic systems with near first-principles accuracy at a fraction of the computational cost [14–30]. MLIPs are particularly valuable for studying the physics of complex atomic systems in extreme environments (i.e., extreme high/low temperature or externally applied strain), where current empirical interatomic potentials may fail and new empirical interatomic potentials may be difficult to develop due to the complexity of the underlying material physics in such harsh conditions [14, 47].

Machine learning-based interatomic potentials are a specific type of structure–property relationship in which some representation of the atomic structure about the neighborhood of an atom is connected to its interatomic force and/or energy. In the context of MLIPs, the local atomic structure of an atom of interest is fully defined by the relative positions and chemical species of the surrounding atom center

coordinates. Thus, the local state space is given by the set $\boldsymbol{Z} = \{Z_1, \ldots, Z_z\}$ of unique chemical species present in the atomic system. Hence, the local atomic structure in a neighborhood of radius R_c of an atom whose center is located at $\boldsymbol{\tau}_i$ is given by the mapping $\varphi(\boldsymbol{x}, Z)$: $B_{R_c}(\boldsymbol{\tau}_i) \times \boldsymbol{Z} \rightarrow \{0, 1\}$ defined as

$$\varphi(\boldsymbol{x}, Z) = \sum_{\boldsymbol{\tau} \in \mathcal{S}^Z} \delta(\boldsymbol{\tau} - \boldsymbol{x}), \tag{3.18}$$

where $\mathcal{S}^Z$ is the set of rotationally, translationally, and permutationally invariant atom center coordinates of chemical species Z defined relative to the atom of interest (including the atom of interest). Although one could take Eq. (3.18) directly as the local microstructure function, there are two issues. From a ML perspective, the use of delta functions can lead to numerical instabilities in which small perturbations of the atomic positions may result in large changes in model predictions. From a physics perspective, the delta function is a nonphysical representation of a real atom, which we know occupies a finite volume. Therefore, we define a physics-based representation of an atom in the form of the raised cosine distribution

$$\nu(\boldsymbol{x}, Z) = \begin{cases} \frac{1}{2R_a^Z}\left[1 + \cos\left(\frac{\|\boldsymbol{x}\|}{R_a^Z}\pi\right)\right], & \|\boldsymbol{x}\| \leq R_a^Z \\ 0, & \|\boldsymbol{x}\| > R_a^Z \end{cases}, \tag{3.19}$$

where R_a^Z is the atomic radius of element Z. We choose Eq. (3.19) because it smoothly approaches zero at R_a^Z and does not introduce any nonphysical parameters. The local microstructure function is then given by the convolution of $\varphi(\boldsymbol{x}, Z)$ and $\nu(\boldsymbol{x}, Z)$, defined as

$$m(\boldsymbol{x}, Z) = \sum_{\boldsymbol{\tau} \in \mathcal{S}^Z} \frac{1}{2R_a^Z}\left[1 + \cos\left(\frac{\|\boldsymbol{\tau} - \boldsymbol{x}\|}{R_a^Z}\pi\right)\right] H\left(R_a^Z - \|\boldsymbol{\tau} - \boldsymbol{x}\|\right), \tag{3.20}$$

where H is the Heaviside step function. The damped local microstructure function is then given by

$$\eta(\boldsymbol{x}, Z) = m(\boldsymbol{x}, Z) f_d(\boldsymbol{x}). \tag{3.21}$$

We choose $f_d(\boldsymbol{x})$ to be the normalized raised cosine distribution

$$f_d(\boldsymbol{x}) = \begin{cases} \frac{1}{2}\left[1 + \cos\left(\frac{\|\boldsymbol{x}\|}{R_c}\pi\right)\right], & \|\boldsymbol{x}\| \leq R_c \\ 0, & \|\boldsymbol{x}\| > R_c \end{cases}, \tag{3.22}$$

because it smoothly approaches zero at R_c. This ensures that $\eta(\boldsymbol{x}, Z)$ is smooth as atoms transition across the boundary of the atomic neighborhood, $\partial B_{R_c}(\boldsymbol{\tau}_i)$.

Equations (3.18)–(3.22) are depicted graphically in Fig. 3.2. The VASt representation of the local atomic structure is the discrete damped local microstructure function given by the discrete version of Eq. (3.21). Thus, the VASt representation of the local atomic structure is equivalent to a digital 3D image, where, instead of quantifying the values of [red, blue, green] within a voxel, a feature vector quantifies the volume fraction of each chemical species present in a voxel. As a result, the VASt representation of the local atomic structure is an ideal descriptor to serve as the input to a CNN. In this context, the calibrated convolutional layers of the CNN act as a nonlinear mapping from the VASt representation of the local atomic structure to a low-dimensional feature space from which the target property is predicted. Additionally, since each input channel of the CNN accounts for the contribution of a single chemical species to the spatial domain of the local atomic structure, the VASt framework for MLIPs naturally allows for a direct, parameter free representation of multielement systems. Furthermore, the VASt representation of the local atomic structure, given by the discrete version of Eq. (3.21), does not invoke any nonphysical parameters other than the cutoff distance and resolution (i.e., voxel length). Since the true local atomic structure is directly input to the CNN, all feature engineering is done implicitly by the CNN, allowing VASt potentials to implicitly learn the complex spatial relationships and multibody interactions that govern the physics of atomic systems.

In order to demonstrate the VASt framework for MLIPs, we developed a VASt potential capable of efficiently and accurately computing the thermal conductivity of crystalline silicon subjected to any hydrostatic strain in a range of − 0.035 to 0.035. For this study, the training dataset consisted of 6000 configurations of the 216-atom 3 × 3 × 3 silicon supercell. Unique atomic configurations were obtained by linearly distributing the lattice vectors of the supercells over a strain range of − 0.05 to 0.05 and displacing each atom in the supercell by a randomly oriented vector whose magnitude is generated from a truncated exponential distribution with a maximum possible displacement of 0.20 Å. A 12 Å cutoff distance was selected for the damped local microstructure function in order to capture long-range phonon interactions that can play a significant role in thermal transport. The spatial domain was discretized into 97 × 97 × 97 voxels, resulting in voxel length of ~ 0.25 Å.

The VASt potential is used to compute thermal conductivity with the BTE by predicting the interatomic forces required to compute the second- and third-order interatomic force constants (IFCs) via the finite displacement method [48]. The thermal conductivities predicted using the VASt potential ($\boldsymbol{k}_{\mathrm{VASt}}$) are shown in Fig. 3.3, along with the thermal conductivity values obtained using IFCs computed with DFT. For the thermal conductivities computed using the VASt potential, the second-order IFCs are computed using the typical 0.01 Å finite displacement distance, whereas the third-order IFCs are computed using a larger 0.15 Å finite displacement distance in order to reduce their sensitivity to noise. To verify that the larger third-order finite displacement distance did not introduce significant error, we computed the thermal conductivity using third-order IFCs obtained from DFT with

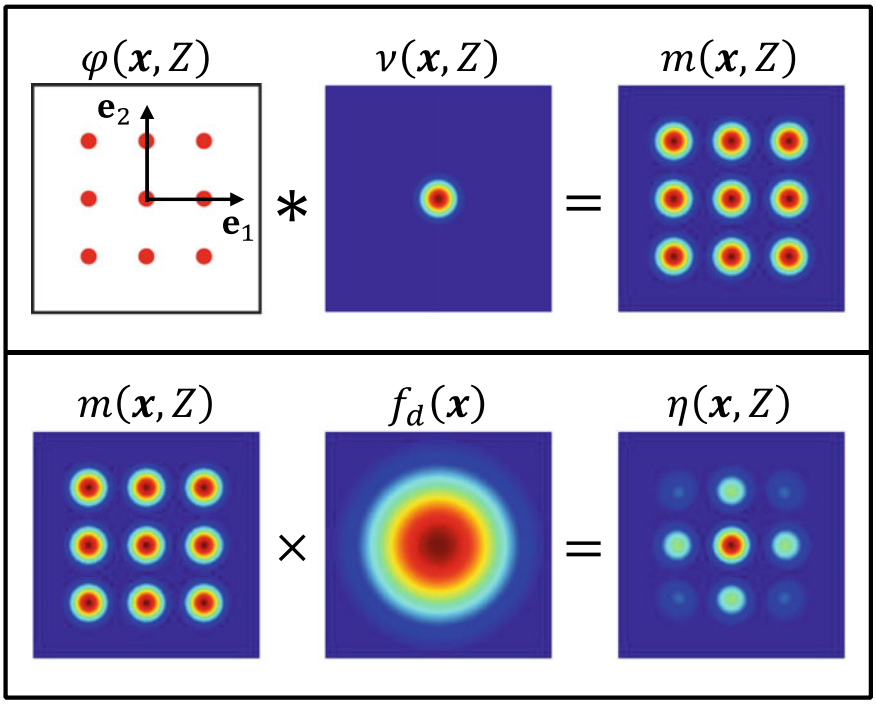


Fig. 3.2 A visual depiction of the damped local microstructure function constructed via Eqs. (3.18)–(3.22). (Top) The atom center coordinates of chemical species Z in the neighborhood of the atom of interest (including the atom of interest), $\varphi(\boldsymbol{x}, Z)$, are convolved with the raised cosine distribution representation of the atom volume, $\nu(\boldsymbol{x}, Z)$, to obtain the local microstructure function, $m(\boldsymbol{x}, Z)$. (Bottom) The local microstructure function is multiplied by the damping function, $f_d(\boldsymbol{x})$, to obtain the damped local microstructure function, $\eta(\boldsymbol{x}, Z)$. Reprinted with permission from Ref. [14]. Copyright 2020 American Chemical Society

both the typical 0.01 Å finite displacement distance ($\boldsymbol{k}_{\text{ref}}$) and the larger 0.15 Å finite displacement distance $\left(\boldsymbol{k}_{0.15\text{ Å}}\right)$ and found that the average and maximum percent errors between the $\boldsymbol{k}_{\text{ref}}$ and $\boldsymbol{k}_{0.15\text{ Å}}$ are ~ 7.6% and ~ 10.5%, respectively.

The $\boldsymbol{k}_{\text{VASt}}$ were found to be in excellent agreement with the and $\boldsymbol{k}_{0.15\text{ Å}}$—the average and maximum percent error over the full strain range considered was ~ 5.0% and ~ 10.9%, respectively. Additionally, the VASt-predicted thermal conductivities accurately modeled the trend of thermal conductivity remaining relatively constant from compression to equilibrium and then decreasing from equilibrium to tension. The accuracy of the thermal conductivities predicted from the VASt potential was also found to be a significant improvement over those computed using MD with the Tersoff empirical potential [49], which have an average and maximum error of ~ 48.1% and ~ 60.7%, respectively, for hydrostatic strain ranging from − 0.03 to 0.03 and incorrectly show thermal conductivity as monotonically decreasing from compression to tension [50]. Furthermore, compared to the experimentally measured

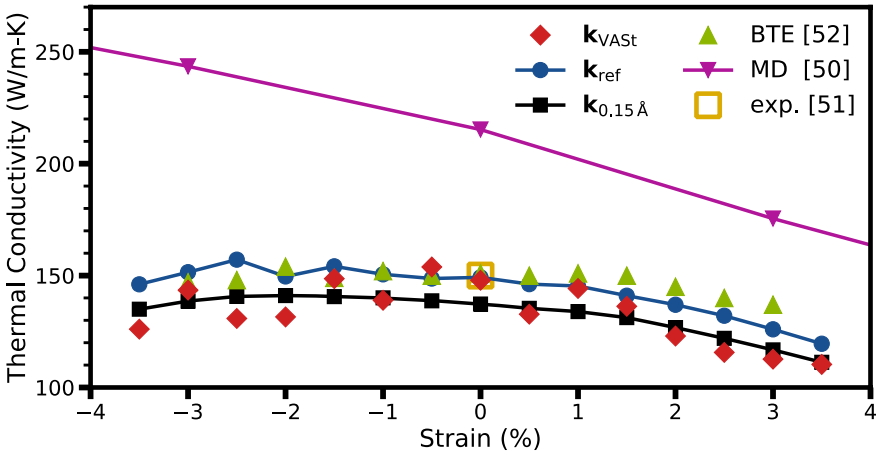


Fig. 3.3 The thermal conductivity of silicon at 300 K as a function of hydrostatic strain. The $\boldsymbol{k}_{\mathrm{VASt}}$ are computed with the BTE using IFCs obtained from interatomic forces predicted by our VASt potential, while the $\boldsymbol{k}_{\mathrm{ref}}$ and and $\boldsymbol{k}_{0.15\,\mathrm{Å}}$ are computed with the BTE using IFCs obtained from DFT. The $\boldsymbol{k}_{\mathrm{ref}}$ are computed using the typical 0.01 Å finite displacement distance for the third-order IFCs, while the $\boldsymbol{k}_{\mathrm{VASt}}$ and and $\boldsymbol{k}_{0.15\,\mathrm{Å}}$ are computed using a larger 0.15 Å finite displacement distance for the third-order IFCs. The thermal conductivities computed in Ref. [52] using the BTE with IFCs obtained from DFT and computed in Ref. [50] using MD with the Tersoff empirical interatomic potential [49] are also shown along with the experimentally measured thermal conductivity of unstrained silicon [51]. Reprinted with permission from Ref. [14]. Copyright 2020 American Chemical Society

thermal conductivity of unstrained silicon (~ 150 W/mK) [51] the percent error of the VASt-predicted thermal conductivity (147.9 W/mK) was only ~ 1.4%, whereas the percent error of the MD-computed thermal conductivity (~ 215 W/mK) was ~ 43.5%. The results of this study demonstrate that VASt potentials can be used to accurately model complex physical phenomena in material systems in extreme environments such as under externally applied strain.

3.5.2 Voxelized Atomic Structure Discovery

Materials discovery is a form of the homogenization problem in which the global measures of the material internal structure are related to a set of one or more (effective) physical properties. As previously mentioned, in the VASt framework for compound discovery the atomic structure is represented by its charge density field. The charge density field provides a complete representation of the atomic structure that is fully determined by physics and therefore does not require any *ad hoc* feature engineering. It also implicitly embeds the chemical species information present in the atomic system, thereby reducing a high-dimensional feature space of possible atomic attributes needed to differentiate between chemical species to a single local

state variable—charge density. As a result, structure–property relationships that use the charge density field to represent the atomic structure do not depend explicitly on the chemical species present in the atomic systems, and therefore are capable of extrapolating to atomic systems containing chemical species not present in the training dataset.

Physically, the charge density field of an atomic system is a continuous function $\rho\colon \Omega \to \mathbb{R}_{\geq 0}$, where $\Omega \subset \mathbb{R}^3$ is the continuous spatial domain of the atomic system. As previously mentioned, the charge density field representation of the atomic structure has only a single local state variable—charge density. Therefore, in the VASt framework for compound discovery, the microstructure function is fully defined by the charge density field, and we can take $m(\boldsymbol{x}, \boldsymbol{h}) = \rho(\boldsymbol{x})$. Of course, in practice the numerically computed charge density field is a discrete function $\rho\colon \boldsymbol{S} \to \mathbb{R}_{\geq 0}$, where $\boldsymbol{S}$ is the discretized spatial domain of the atomic system. Since a given set of crystalline atomic systems cannot naturally be defined on a standardized discrete spatial domain, we approximate the discrete charge density field on a standardized discrete spatial domain using trigonometric interpolation as outlined in Sect. 3.4. The discrete two-point spatial correlations are then given by

$$f_2[\boldsymbol{t}] = \frac{1}{|\boldsymbol{\Sigma}|} \sum_{\boldsymbol{\xi} \in \boldsymbol{\Xi}} \left[\tilde{\rho}\left[\boldsymbol{\xi}\right]^{0.5} \chi_{\boldsymbol{\Sigma}}(\boldsymbol{\xi})\right] \tilde{\rho}\left[\boldsymbol{\xi} + \boldsymbol{t}\right]^{0.5}, \tag{3.23}$$

where $\boldsymbol{\Xi}$, $\boldsymbol{\Sigma}$, and $\chi_{\boldsymbol{\Sigma}}(\boldsymbol{\xi})$ are as defined in Sect. 3.4. In the definition of the two-point spatial correlations of the charge density field, we choose to use the square root of the charge density field so that

$$f_2[\boldsymbol{t} = (0, 0, 0)] = \frac{1}{|\boldsymbol{\Sigma}|} \sum_{\boldsymbol{\sigma} \in \boldsymbol{\Sigma}} \tilde{\rho}[\boldsymbol{\sigma}], \tag{3.24}$$

which is just the average charge density in a voxel.

The two-point spatial correlation function is a high-dimensional representation of the charge density field. Although this representation can be used directly as the input to a CNN, achieving a highly accurate CNN requires a substantial quantity of training data. In materials discovery applications, the computational cost of computing the desired properties using physics-based simulations is very high and therefore, it is typically not feasible to generate a large enough training dataset to achieve a high accuracy CNN. As a result, it is necessary to project the high-dimensional two-point spatial correlation function representation of the charge density field to a low-dimensional representation that can be used as the input to a non-CNN ML model.

Given a representative set of charge density fields, we obtain a salient low-dimensional representation of the charge density field by applying principal component analysis (PCA) to the corresponding set of two-point spatial correlations. PCA is a linear, distance-preserving transformation from the original feature space to a new Fourier basis defined such that the basis vectors are aligned with the directions of

maximum variance in the dataset. There are two main advantages of PCA. First, there is typically a strong decay in the amount of variance captured by each successive basis vector. As a result, the dataset can often be approximated on a low-dimensional subspace of the full PC space with minimal information loss. Second, since the PC space is a Fourier basis, any feature vector inside the convex hull of the dataset in PC space corresponds to a valid microstructure (beyond those in the original dataset). This is crucial for reverse engineering new microstructures with tailored properties.

The low-dimensional representation of the charge density field is used as the input to a Gaussian process regression (GPR) model. GPR is a nonparametric, kernel-based ML method for developing probabilistic models using Bayesian inference. In GPR, the mapping from the input (i.e., feature vector) to the output (i.e., target/property) is modeled as a joint multivariate Gaussian distribution called a Gaussian process. The output of the Gaussian process is defined completely by a mean and covariance. As a result, GPR formally treats uncertainty in the model predictions since it not only estimates the expected value of the output, but also its distribution. For VASt homogenization models, the covariance matrix of the GPR model is given by the automatic relevance determination squared exponential (ARD-SE) [53] kernel function

$$k(\boldsymbol{x}, \boldsymbol{x}') = \sigma_s^2 \exp\left[-\frac{1}{2} \sum_{d=1}^{D} \frac{\left(x_d - x_d'\right)^2}{l_d^2} \right] + \sigma_n^2 \delta_{xx'}. \tag{3.25}$$

Typically, the computational cost of obtaining the structure of a sample (i.e., charge density field) is orders of magnitude lower than the computational cost of computing the ground-truth value of its target property. Therefore, we may assume to have access to a large set of candidate structures, but that our initial training dataset is limited to a small subset of these candidate structures for which the corresponding ground-truth properties have been computed. Thus, given this initial training dataset, we not only wish to train a GPR model, but also to identify which of the remaining candidate structures we should compute the ground-truth property of next so that the improvement to the GPR model is maximized. We accomplish this by using the uncertainty quantification provided by GPR to implement an active learning procedure based on Bayesian experiment design. First, we train a GPR model using the initial training dataset of structures with corresponding ground-truth computed properties. Once the GPR model is trained, the candidate structure that maximizes the information gain to the GPR model is obtained by evaluating an acquisition function. The ground-truth property of this candidate structure is then computed, and the sample is added to the training dataset. This process is repeated until the accuracy of the GPR model converges.

To demonstrate the capabilities of the VASt framework for compound discovery, we develop a VASt structure–property relationship capable of predicting the bulk modulus of AlNbTiZr RHEAs for any chemical composition. To sufficiently cover the composition space, we consider the set $\mathcal{C} = \left\{ \mathrm{Al}_\alpha \mathrm{Nb}_\beta \mathrm{Ti}_\gamma \mathrm{Zr}_\delta \,|\, \alpha, \beta, \gamma, \delta \in [0, 4, \ldots, 128] \text{ and } \alpha + \beta + \gamma + \delta = 128 \right\}$ of

possible 128-atom compositions. For each composition, we first obtain the special quasirandom structure (SQS) corresponding to the perfect BCC atomic structure [54–58]. We then obtain the ground state atomic structure (i.e., atomic positions and lattice constant) by fitting a Birch–Murnaghan Equation of State (EOS) [59, 60]. The ground-truth bulk modulus is also obtained from the EOS. Using the procedures outlined above, we then compute the two-point spatial correlations of the charge density field corresponding to the ground state atomic structure. Here, we use a standard voxel size of 0.2 Å for the standardized charge density fields and a 6.0 Å cutoff distance for the two-point spatial correlations. We choose 6.0 Å for the cutoff distance since it is sufficiently large to guarantee that the spatial correlations span at least the volume of a hypothetical BCC unit cell for all compositions. The two-point spatial correlations of elemental Nb and equiatomic AlNbTiZr are shown in Fig. 3.4a, b, respectively. We then apply PCA to the set of two-point spatial correlations. The dataset is plotted as a function of its first 2 PCs in Fig. 3.5a, b. In Fig. 3.5a, the dataset is colored by the chemical species present in the atomic systems. The projection of the dataset onto PC space closely approximates a tetrahedron whose vertices correspond to the 4 elemental systems (i.e., Al, Nb, TI, Zr), edges correspond to compositions of the 6 binary systems (i.e., AlNb, AlTi, AlZr, NbTi, NbZr, and TiZr), faces correspond to compositions of the 4 ternary systems (i.e., AlNbTi, AlNbZr, AlTiZr, and NbTiZr), and interior corresponds to the compositions of the quaternary system (i.e., AlNbTiZr). In Fig. 3.5b, the dataset is colored by the DFT-computed bulk modulus. As expected, Nb has the highest bulk modulus, followed by Ti, then Zr, and finally Al.

The results of the Bayesian experiment design active learning procedure for training the GPR model is shown in Fig. 3.6a. In this study, the first 3 PCs are used as the input to the GPR model and the initial training dataset for the active learning process consists of only the 4 elemental systems (i.e., Al, Nb, Ti, Zr). The results show that the accuracy of the VASt homogenization model converges to a mean absolute percentage error below 2% with the addition of fewer than 20 candidate charge density fields (out of a total of 6541 candidate samples) to the original 4 elemental

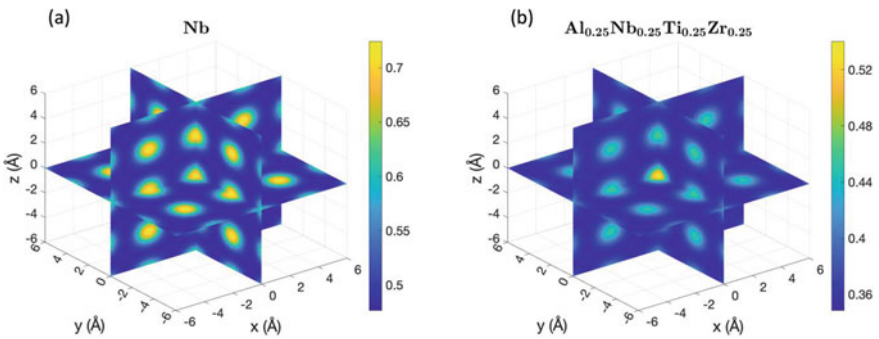


Fig. 3.4 **a** The two-point spatial correlation function of BCC niobium. **b** The two-point spatial correlation function of equiatomic BCC AlNbTiZr SQS

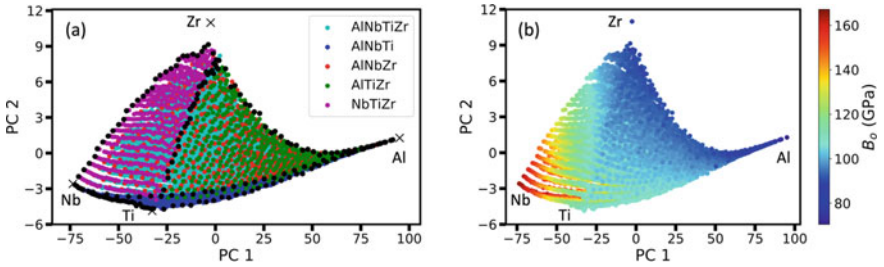


Fig. 3.5 The dataset of AlNbTiZr RHEA compositions as represented by the first 2 PCs and colored by **a** the chemical species present in the atomic system and **b** the DFT-computed bulk modulus

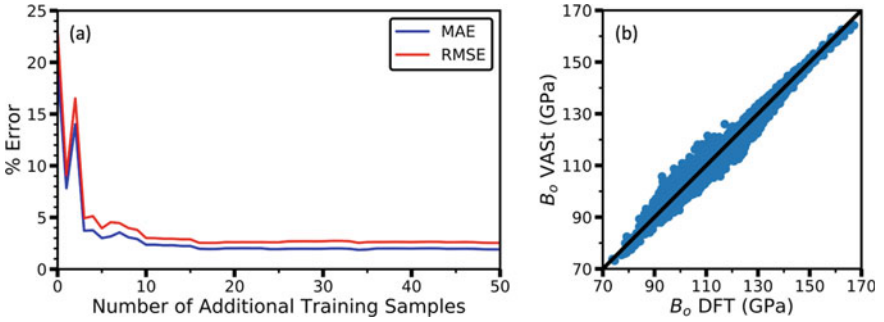


Fig. 3.6 **a** The convergence of the VASt homogenization model with respect to the number of candidate samples added to the initial training dataset. Here, the initial training dataset consists of only the 4 elemental systems (i.e., Al, Nb, Ti, Zr). **b** A parity plot of the DFT-computed and VASt-predicted bulk moduli of the AlNbTiZr RHEA compositions in the validation dataset. The VASt predictions are from the first GPR model in the active learning process to reach a mean absolute percentage error of $< 2\%$

systems. The predicted values of bulk modulus obtained from the first GPR model to reach $< 2\%$ error are compared to the DFT-computed values in Fig. 3.6b. The results of this study demonstrate that the VASt framework for compound discovery is capable of predicting the physical properties of complex atomic systems using a limited number of training samples.

3.6 Conclusion

In this chapter, we present a detailed overview of voxelized representations of atomic structure for building high-fidelity structure–property relationships in the context of hierarchical multiscale materials modeling. We first presented the continuous microstructure function as a mathematically rigorous definition of material structure that is independent of length scale. We then showed that the equivalent mathematical

definition for the discrete (i.e., voxelized) microstructure function was consistent across all length scales. We then showed that the microstructure function could be used to quantify the local material structure directly by systematically selecting the reference axis of the material system and that it could be used to quantify the global material structure statistically using n-point spatial correlation functions. We then outlined practical considerations for numerically computing and quantifying microstructure functions in the context of developing structure–property relationships. Finally, we presented the VASt framework for developing structure–property relationships using first-principles computations. We outlined the VASt framework for developing MLIPs and demonstrated that a VASt MLIP could be used to accurately predict the thermal conductivity of silicon over a range of hydrostatic strain. We then outlined the VASt framework for compound discovery and demonstrated that a VASt homogenization model could accurately predict the bulk modulus of AlNbTiZr RHEAs for all possible chemical compositions using fewer than 25 training samples.

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Chapter 4
Development of Exchange-Correlation Functionals Assisted by Machine Learning



Ryo Nagai and Ryosuke Akashi

Abstract With the recent rapid progress in the machine-learning (ML), there have emerged a new approach using the ML methods for developing the exchange-correlation functionals of density functional theory. In this chapter, we review how the ML tools are used for this and the performances achieved recently. It is revealed that the ML, not being opposed to the analytical methods, complements human intuition and advances the development of the first-principles calculation with desired accuracy.

Keywords Density functional theory · Electronic structure · Machine learning · Neural network

4.1 Introduction

A holy grail in the materials science and chemistry is the general theoretical method to emulate chemical reactions. For this purpose, we need to evaluate the energy of electrons distributed over the ionic configuration, with the accuracy of chemical reaction, typically the order of 1 kcal/mol. We already know what equation to solve—the electronic Schrödinger equation

$$H = \sum_i \left[-\frac{\nabla_i^2}{2} + V_{\text{ion}}(\mathbf{r}_i) \right] + \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \tag{4.1}$$

$$H\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots). \tag{4.2}$$

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Here, i and j are the indices of electrons and V_{ion} denotes the one-body potential due to the distribution of ions. Arguably this is the theory of everything for electrons (apart from the relativistic, ion vibrational and external perturbation effects). But this equation is too difficult to solve exactly for realistic many-electron systems because its computational cost is of order of $\exp(N)$, with N representing the system size. An alternative yet exact route to the first-principles electronic structure calculation is the density functional theory (DFT) [1, 2]. This theory allows one to recast the original many-body problem to a set of one-body self-consistent equations, Kohn–Sham (KS) equations, as introduced later.

The self-consistent potential entering the KS equations, the exchange-correlation potential V_{xc} is exactly defined, but in a way that requires computational cost equivalent to the exact solution of the Schrödinger equation. It, therefore, has to be approximated for practical use. The systematic way to improve the approximate formula, however, has been lacking. People have constructed the approximate forms of V_{xc} by assembling exactly computable asymptotic behaviors of the electronic systems, but, because of the lack of systematic derivation, they have had to rely on both human intuition and empiricism to decide how to integrate them into a unified formula.

Recently, a new trend has emerged in the development of V_{xc}: Applying machine learning (ML) to V_{xc}. The ML methods enable us to implement a numerically calculable function that imitates some other function that is difficult to run on the computer. Typically the ML targets human functions like pattern recognition, while it can also be utilized to make a computational shortcut, by building an approximate model that accurately reproduces any other numerically demanding formula, algorithm, procedure, etc. An idea thus arose, that we may construct calculable models of V_{xc} by machine-learning numerous specific calculable cases. This may seem to be an extremely empirical approach that opposes the standard theoretical one. Here we are revealing that, in reality, the ML is capable of complementing the human, by providing tools to explore models that well represent nonideal systems where any analytical approach does not apply, and accelerating the development of approximate V_{xc}.

In this book chapter, we review recent studies of formulating approximate forms V_{xc} for the KS equations using the ML. In Sect. 4.2, we append a pedagogical review of the foundation of DFT to clarify what to machine-learn. In Sect. 4.3 we introduce the basic concept of applying ML to V_{xc}, explain the approaches, and make a survey of the accuracy achieved by the ML approach. Section 4.4 is devoted to the summary and future perspectives.

4.2 Basics of Density Functional Theory

4.2.1 Hohenberg–Kohn Theorem

The heart of the DFT is the concept that the charge density in the ground state,

$$n(\mathbf{r}) = \int d\mathbf{r}_2 d\mathbf{r}_3 \ldots |\Psi_{\text{GS}}(\mathbf{r}, \mathbf{r}_2, \mathbf{r}_3, \ldots)|^2, \tag{4.3}$$

determines all the physical properties of the quantum system. This has been formalized as the two Hohenberg–Kohn theorems [1]. According to the Schrödinger Eq. (4.2), when the ionic potential distribution V_{ion} is given, the ground state wavefunction Ψ and charge density n is unambiguously determined. Hohenberg and Kohn have proved the converse—when n is given, V_{ion} that reproduces the given n in the ground state is unambiguously determined (up to an arbitrary constant). Consequently, Ψ_{GS}, and everything defined with Ψ_{GS}, are determined via the Schrödinger equation. The related quantity f is thus regarded as a functional of n, $f[n]$. (Hereafter we express the dependence as a functional of n as $[n]$.) The other theorem is the variational principle. Define the functional $F[n]$ as

$$F[n] = \langle \Psi_{\text{GS}}[n] | \left\{ \sum_i \left(-\frac{\nabla_i^2}{2} \right) + \sum_{i \neq j} \frac{1}{2|\mathbf{r}_i - \mathbf{r}_j|} \right\} | \Psi_{\text{GS}}[n] \rangle \tag{4.4}$$

When an external potential V_{ion} is given, the following inequality holds

$$E[n] \equiv F[n] + \int d\mathbf{r} n(\mathbf{r}) V_{\text{ion}}(\mathbf{r}) \geq E[n_0], \tag{4.5}$$

where n_0 satisfies $V_{\text{ion}} = V_{\text{ion}}[n_0]$. The equality holds only when $n = n_0$. This theorem yields that the ground state charge density at a given ionic potential can be obtained by solving the variational problem [3, 4]. With these theorems, the solution of the Schrödinger equation for the ground state can be recast to optimization of a scalar function on $\mathbf{R}^3$ space, n. The latter procedure can be executed with a surprisingly cheap computational cost, typically of order $O(N)$, if an explicit form of F as a functional of n is available. In this procedure, we do not have to explicitly treat the wave character of the electron. Although this approach has been pursued as orbital-free- (OF-) DFT, a more practical method has become the standard, which incorporates the concept of orbital for convenience as detailed below.

4.2.2 Kohn–Sham Equation

The roadblock for the practical OF-DFT is that the kinetic energy, which largely reflects the wave character of electrons and is difficult to represent as a functional of n, accounts for a large fraction of the total energy. While the development of the functional form of the kinetic energy has long been conducted, an alternative approach has been initiated by Kohn and Sham [2]. In their theory, an auxiliary "non-interacting" system, reproducing n in the interacting ground state, is formulated

$$\left[-\frac{\nabla^2}{2} + V_{\text{eff}}[n](\mathbf{r}) \right] \varphi_a(\mathbf{r}) = \varepsilon_a \varphi_a(\mathbf{r}), \tag{4.6}$$

where the charge density is given by

$$n(\mathbf{r}) = \sum_a \theta(\mu - \varepsilon_a)|\varphi_a(\mathbf{r})|^2 \equiv \sum_{a:\text{occ.}} |\varphi_a(\mathbf{r})|^2. \tag{4.7}$$

Here and hereafter a labels the Kohn–Sham eigenstate $(\varepsilon_a, \varphi_a)$. θ is the step function $\theta(x) = 1\,(x > 0); 0\,(x < 0)$. Summation $\sum_{a:\text{occ.}}$ runs over the occupied states, whose eigenvalues are smaller than a chemical potential μ. The effective potential has a form

$$V_{\text{eff}}[n](\mathbf{r}) = V_{\text{ion}}(\mathbf{r}) + \int d\mathbf{r}' \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + V_{\text{xc}}[n](\mathbf{r}). \tag{4.8}$$

The second (Hartree) term represents the classical Coulomb potential effect and the third, the exchange-correlation potential, includes the remaining interaction effects. Equation (4.6), the Kohn–Sham equation has n on the left hand side and has to be therefore solved self-consistently.

V_{xc} is defined by

$$V_{\text{xc}}[n](\mathbf{r}) = \frac{\delta E_{\text{xc}}[n]}{\delta n(\mathbf{r})} \tag{4.9}$$

with $E_{\text{xc}}[n]$ being the quantity named exchange-correlation (XC) energy. $E_{\text{xc}}[n]$ is defined with the following equality

$$E_{\text{GS}}[n] = T_{\text{s}}[n] + \int d\mathbf{r} V_{\text{ion}}[n](\mathbf{r}) n(\mathbf{r}) + E_{\text{H}}[n] + E_{\text{xc}}[n]. \tag{4.10}$$

$E_{\text{GS}}[n]$ is the ground-state energy as a functional of n. $T_{\text{s}}[n]$ is the noninteracting kinetic energy defined by

$$T_{\text{s}}[n] = \min_{|\Phi|^2 = n} \langle \Phi | \sum_i \left[-\frac{\nabla_i^2}{2} \right] | \Phi \rangle, \tag{4.11}$$

where Φ is the Slater determinant reproducing the given n as its squared norm. $E_{\text{H}}[n]$ is the Hartree energy

$$E_{\text{H}}[n] = \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' \frac{n(\mathbf{r}) n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \tag{4.12}$$

The exchange-correlation energy E_{xc} is defined as the residual part of the total energy, which is also a functional of n. Note that the interacting ground-state energy $E_{\text{GS}}[n]$ can be rewritten as

$$E_{\text{GS}}[n] = \sum_{i:\text{occ.}} \varepsilon_i + E_{\text{H}}[n] + E_{\text{xc}}[n] - \int d\mathbf{r} n(\mathbf{r}) V_{\text{xc}}[n](\mathbf{r}). \tag{4.13}$$

We can thus calculate the ground-state charge density and total energy using Eqs. (4.6) and (4.13), provided that a specific expression of $E_{\rm xc}$ is available.

4.2.3 Approximating the Exchange-Correlation Functional

Once the exact formula of $E_{\rm xc}[n]$ is obtained, apparently we no longer have to solve the Schrödinger equation for calculating the ground state charge density and total energy. However, such a formula is currently unavailable. Because the definition of $E_{\rm xc}[n]$ involves $E_{\rm GS}[n]$, which is the exact ground-state energy obtained through the Schrödinger equation, its analytical formula is probably impossible to write in an efficiently calculable form.

Practical use of the KS equation is realized by constructing calculable approximate forms of $E_{\rm xc}[n]$. For this goal we introduce the exchange-correlation energy density $\varepsilon_{\rm xc}$ by

$$E_{\rm xc}[n] = \int d\mathbf{r} n(\mathbf{r}) \varepsilon_{\rm xc}[n](\mathbf{r}). \tag{4.14}$$

The most useful approximation is the semi-local approximation [1], in which we simplify that $\varepsilon_{\rm xc}[n](\mathbf{r})$ depends only on the charge density and its derivatives at $\mathbf{r}$. Within this approximation, $\varepsilon_{\rm xc}[n](\mathbf{r}) \simeq \varepsilon_{\rm xc}(n(\mathbf{r}))$ gives the local density approximation (LDA). This approximation becomes exact in the uniform electron gas limit as $n(\mathbf{r}) = n =$ constant. Vosko et al. [5] and Perdew and Zunger [6] referred to the accurate calculation of $\varepsilon_{\rm xc}$ in the homogeneous electron gas obtained by the diffusion Monte Carlo method [7] and fitted their model functions. The model functions VWN and PZ, after the authors' names are used in practice today. The next order correction within the semilocal approximation is efficiently included by adding a variable $|\nabla n(\mathbf{r})|/n(\mathbf{r})$; this is called generalized gradient approximation (GGA). The GGA-PBE functional [8] is a famous model functional within this level. To construct this model functional, one refers to rigorous formulas that have to be satisfied by $\varepsilon_{\rm xc}$; asymptotic equalities and inequalities in some limits like the uniform density limit, and scaling relations. The model functional is constructed with a minimal number of parameters and those are determined so that the model complies with those formulas.

Further improvement has been conducted by considering the higher-order derivatives of $n(\mathbf{r})$ and explicitly including the KS orbitals. The modern meta-GGA [9–12] and hybrid functionals [13, 14] have been proposed in this way. For example, the hybrid functionals incorporate some fraction of the exact exchange energy formulated by the KS orbitals

$$E_{\rm x}^{\rm EXX} = -\frac{1}{2} \sum_{ab}^{\rm occ.} \iint d\mathbf{r} d\mathbf{r}' \varphi_a^*(\mathbf{r}) \varphi_b^*(\mathbf{r}') \frac{1}{|\mathbf{r}-\mathbf{r}'|} \varphi_a(\mathbf{r}') \varphi_b(\mathbf{r}). \tag{4.15}$$

See e.g. [15, 16] for a review. Precisely speaking, the approximations of this class is beyond the original KS theory in that the corresponding exchange-correlation potential is non-local: Namely, the matrix element $[V_{\rm xc}[n]]_{\mathbf{r},\mathbf{r}'}$ is not proportional to $\delta(\mathbf{r}-\mathbf{r}')$. Such extension is in fact justifiable as the generalized Kohn–Sham theory [17].

The goal of the development of the DFT (for the condensed matter) is a unified computational scheme that achieves accuracy at the energy scale of chemical reactions. The target accuracy of scale ~ 1 kcal/mol is called chemical accuracy. The levels of approximations for DFT toward this goal have been classified as steps in "Jacob's ladder" [18], where the respective rungs correspond to the LDA, GGA, metaGGA, hybrid, and those depending on the unoccupied KS orbitals, in ascending order to the chemical accuracy world.

4.2.4 Exact Exchange-Correlation Functional

The celebrated statement that the exact xc functional has been elusive needs some disclaimer. Actually, we already have both well-defined formulas of $E_{\rm xc}$ and $V_{\rm xc}$ as functionals of n and numerical algorithm to evaluate them. The problem is that we do not have a method that is computationally cheaper than directly solving the many-body Schrödinger equation. Here we review some numerically executable exact formalisms for the functionals, which would be useful, and are already in use somehow, for developing the ML functionals.

$V_{\rm xc}$, as a functional of a given charge distribution n, is determined by Eq. (4.10) and derivative Eq. (4.9). On the other hand, the one-to-one correspondence between $V_{\rm ion}[n]$ and n allows us to calculate $V_{\rm xc}$ by solving inversion problems [19, 20] numerically. Here we remember that, in the non-interacting system (Eq. 4.6), the one-to-one correspondence holds between $V_{\rm eff}[n]$ and n, and, in DFT, $V_{\rm ion}[n]$ is not an input but a quantity derived from the given n. Wagner and coauthors [21] have demonstrated the numerical algorithm strictly conforming to this input-output relation. They solved the inverse problems of the KS (Eq. 4.6) and many-body Schrödinger equations (Eq. 4.2) to determine $V_{\rm eff}[n]$ and $V_{\rm ion}[n]$, respectively. Subtracting the latter from the former yields $V_{\rm H}[n]+V_{\rm xc}[n]$. This algorithm, involving the solution of the Schrödinger equation, is not computationally advantageous but was useful for proving the converging nature of the exact KS self-consistent equations [22].

If the ionic positions and exactly calculated n are available, we can derive $V_{\rm xc}$ by only the inversion of the KS equation thanks to the one-to-one correspondence. This procedure is called KS inversion [19]. The calculation cost of the KS inversion is in principle as same as the forward solution of the KS equation, but its numerical convergence has remained a cumbersome problem, especially in three dimensions [23].

An analytical expression is also available for $E_{\rm xc}$, that is called analytically continued fluctuation dissipation formula [24]

$$E_{\mathrm{xc}} = E_{\mathrm{x}}^{\mathrm{EXX}} - \frac{1}{2}\int_0^1 d\lambda \int \frac{d\omega}{2\pi} \int d\mathbf{r}d\mathbf{r}' \frac{\chi_\lambda[n](\mathbf{r},\mathbf{r}',i\omega) - \chi_0[n](\mathbf{r},\mathbf{r}',i\omega)}{|\mathbf{r}-\mathbf{r}'|}. \tag{4.16}$$

Here, parameter λ defines the system

$$H_\lambda[n] = \sum_i \left[-\frac{\nabla_i^2}{2} + V_{\mathrm{ion}}(\mathbf{r}_i) + V_\lambda[n](\mathbf{r}_i)\right] + \frac{\lambda}{2}\sum_{i\neq j}\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \tag{4.17}$$

and V_λ is defined to be the potential that reproduces the ground state density n, identical to that of $H = H_{\lambda=1}$ and therefore $V_{\lambda=1} = 0$ and $V_{\lambda=0} = V_{\mathrm{eff}} - V_{\mathrm{ion}}$. χ_λ denotes the density-density response function calculated for the ground state of H_λ, that is dependent on imaginary frequency $i\omega$. The λ integral is taken with the variable n fixed. Here we notice that, for evaluating χ_λ, we need to solve the interacting Schrödinger equation for various λ. An approximation has been derived by, for example, approximating χ_λ to the independent-particle polarization expressed by the KS orbitals [25]. The random phase approximation [25, 26], which belongs to the fifth step of the Jacob's ladder, can describe the dispersion interactions like the van der Waals one, which propagate between the intermediate excited states.

4.3 Development of ML-Based XC Functionals

4.3.1 ML to DFT: Concepts and Pioneers

From the theoretical structure of DFT seen above, we read its significant features. (i) Once the exact form of functional, $F[n]$, $E_{\mathrm{xc}}[n]$ or $V_{\mathrm{xc}}[n](\mathbf{r})$, is formulated, we can calculate the ground state charge density and energy apparently without using the Schrödinger equation. (ii) The functionals are well defined as functionals of the charge density n through rigid procedures, but not as an explicit analytical functional of n. (iii) The procedures involve a step of solving the Schrödinger equation, because of which it is almost impossible to faithfully execute for general many-electron systems.

Here we notice that the DFT formalism imposes a crucial limitation to the functionals as input-output relations; vector in, vector out. Namely, any specific distribution of n can be numerically described as a vector (for example, $(n(\mathbf{r}_1), n(\mathbf{r}_2), \ldots)$ with arbitrary discretization in real space). Letting a specific vector n through the functional yields a definite scalar E_{xc} or vector $(V_{\mathrm{xc}}(\mathbf{r}_1), V_{\mathrm{xc}}(\mathbf{r}_2), \ldots)$. The previously proposed functional forms, referring to the numerically exact uniform electron gas case and several asymptotes, are regarded to be approximations that converge to the exact functionals in limited spaces formed by the n vector. In their constructions, model functionals are prepared and the few parameters are determined so that the asymptotes are satisfied.

From the above view, one can conceive an approach that is a straightforward but extreme extension of the standard ones; prepare an extremely flexible model functional and determine numerous parameters so that the function becomes exact in maximally large region in the n space. Here, machine learning (ML) provides us with useful models which can implement any kind of vector-to-vector function. The neural network (NN), the ML model we focus on in this chapter, is defined by repetitive linear and non-linear operations

$$\mathbf{x} \to \cdots \phi\left(W_2 \phi\left(W_1 \mathbf{x} + \mathbf{b}_1\right) + \mathbf{b}_2\right) \cdots . \tag{4.18}$$

Here, $W_1, W_2, \ldots$ and $\mathbf{b}_1, \mathbf{b}_2, \ldots$ are parameter matrices and vectors, respectively, having arbitrary intermediate dimensions. Non-linear transformation ϕ acts on each vector component. It has been proved that, by increasing the intermediate dimensions, the NN can reproduce any continuous functions up to arbitrary accuracy; this is known as the universal approximation theorem [27, 28]. This theorem ensures that the NN, though not being the exact functional, can approximate the exact one up to arbitrary accuracy.

The idea of approximating the exact functional $V_{\mathrm{xc}}[n]$ with an NN has been first exemplified by Tozer et al. [29] in 1996. They calculated n for some atoms with an accurate wave function method, derived V_{xc} by the KS inversion method, and optimized a few-parameter NN as a mapping from $n(\mathbf{r})$ to $V_{\mathrm{xc}}(\mathbf{r})$ so that it well reproduces the example $n - V_{\mathrm{xc}}$ pairs (training). Their procedure can be scaled up straightforwardly, to the modern machine-learning approaches to the DFT functional.

As a pioneering work, a specific mention also proceeds to a work by Zheng and coauthors in 2004 [30]. They proposed to make the parameters in a hybrid functional, B3LYP [31, 32], system-dependent. They implemented the mapping from the atomic species and positions to the B3LYP parameters by the NN and trained it referring to experimentally observed atomization energy data of molecules.

The surge of the ML application to DFT in the recent decade seems to have been prompted by a work by Snyder and coauthors [33]. In a one-dimensional model system, they implemented the kinetic energy functional $T_{\mathrm{s}}[n]$ (Eq. 4.11) by the Gaussian kernel regression, for possible execution of the OF-DFT. This model form does not need the model parameter optimization but needs to store the n-$F[n]$ pairs with various system parameters for application to untrained cases. In this and their subsequent works [34, 35] the Gaussian kernel regression model was used, though the NN is more frequently used in later studies.

The basic concepts of more recent ML applications to the DFT problems have already been put forward in those predecessors. Below we summarize the various ML applications to the task of calculating the electronic states on the basis of DFT.

4.3.2 Levels of ML Applications

The electronic properties are determined ab initio by atomic positions through the Schrödinger equation. The ML scheme can in principle be applied to every step in this procedure. Firstly, the whole procedure itself is well defined as a mapping from atomic positions (+ number of electrons) to electronic properties. One may therefore apply the ML to skip the solution of the Schrödinger equation by modeling the atomic position-to-energy mapping. Methods of this kind have been extensively explored for efficient molecular dynamics with the electronic effects considered (e.g., NN potential in molecular dynamics [36]). Although such an ML model, if efficiently trained, enables us to calculate the electronic properties with the cheapest computational cost, the model would be less transferable since it becomes atomic-species-dependent. We do not review the applications of this kind in detail in this chapter.

The fundamental mappings in DFT are independent of atomic species—the charge density n does not regard which atomic positions it distributes. Brockherde and coauthors [37] implemented ML models for the mapping from V_{ion} to n, whose one-to-one correspondence has been proved as the Hohenberg–Kohn theorem [1]. Moreno et al. [38] and Denner et al. [39] attempted implementation of some of such functionals; n to ground-state wave function and two-particle correlation function.

Successful implementation of the direct mapping from n to universal functional $F[n]$ paves the way to the OF-DFT. This is one of the most active fields for the ML application from the beginning of this decade [33, 34, 37, 40, 41]. In OF-DFT, the ground state energy is calculated by directly minimizing Eq. (4.5) with respect to n without solving the KS equations. The main issue is to approximate the kinetic energy as a functional of n, which constitutes the largest fraction of the total energy.

Another approach adopts the KS scheme and put more importance on accuracy than on calculation speed. At the expense of the computational cost of solving the KS equation, this approach takes advantage of treating the kinetic energy exactly as an operator on the (KS) wavefunction. This gives us two major benefits. First, the largest part of the total energy is treated exactly, on top of which the ML model can learn precise n dependence of the remainder. Second, since the form of kinetic energy operator, $\frac{-1}{2}\nabla^2$, prefers smooth eigenstate. (Remember the solution of quantum harmonic oscillator problem. The eigenvalue with frequently oscillating eigenstate is larger than that of smooth ones.) This property of the kinetic energy operator suppress spurious oscillation in φ and n coming from error in ML-predicted V_{xc} [42]. A typical ML application studies in this level constructs the mapping from n to E_{xc} or $V_{\text{xc}} \equiv \delta E_{\text{xc}}/\delta n$. Using the trained ML model as $E_{\text{xc}}[n]$ or $V_{\text{xc}}[n]$, they solve the KS equation in untrained systems in the same way as existing analytical XC functionals. The following sections mainly introduces the methods used in this approach.

4.3.3 How to Train the ML Models

There is a common strategy for training whatever functionals. We first prepare "accurate" examples of input $\{n^{(t)}\}$ (t: index of training data) and output value(s) of functional(s) for specific systems, $\{f^{(t)} \equiv f[n^{(t)}]\}$. We define a loss function by the difference between $\{f^{(t)}\}$ and the values of the same quantity approximately evaluated by the use of ML, $\{f_{\mathrm{ML}}^{(t)}\}$, and optimize the ML model so that the loss function is minimized. The method for generating the "accurate" data should be numerically exact, as the approximate solvers for the training data limit the accuracy of the ML model.

The training data of exchange-correlation potential as a functional of n can be calculated by the Kohn–Sham inversion technique. Notably, the formula of $V_{\mathrm{xc}}[n]$ is enough for evaluating the total energy of system by the KS equation, even though the total energy formula Eq. (4.13) seemingly needs $E_{\mathrm{xc}}[n]$, not only $V_{\mathrm{xc}}[n]$. This can be made possible by correcting the constant term of $V_{\mathrm{xc}}[n^{(t)}]$, at the end of the inversion, by using the KS eigenvalues $\varepsilon_a^{(t)}$ and energy $E_{\mathrm{GS}}^{(t)}$ as [42]

$$V_{\mathrm{xc}}\left[n^{(t)}\right] \to V_{\mathrm{xc}}\left[n^{(t)}\right] + \frac{1}{M}\left(E_{\mathrm{GS}}^{(t)} - \sum_{a:\mathrm{occ.}} \varepsilon_a^{(t)}\right) \tag{4.19}$$

with M being the number of electrons. With this treatment, the total energy is given by the simple sum of the eigenvalues of the KS equations with the corrected constant. This is a practical usage of the Levy–Zahariev formalism [43], where we introduce a scalar functional $c[n]$ in Eq. (4.13) as

$$E_{\mathrm{GS}} = \sum_{a:\mathrm{occ.}} (\varepsilon_a + c[n]) + E_{\mathrm{H}} + E_{\mathrm{xc}} - \int d\mathbf{r} n(\mathbf{r}) \{V_{\mathrm{xc}}[n](\mathbf{r}) + c[n]\} \tag{4.20}$$

and define $c[n]$ so that the latter terms vanish.

Although learning the n to V_{xc} mapping as explained above is the straightforward way to apply ML to the KS equation, there are variations in the way to model the functionals. Here we classify the modeling methods used in the recent studies, though these methods are often used in combination.

4.3.3.1 Learning $n \to V_{\mathrm{xc}}$ Mapping

Most generally, $V_{\mathrm{xc}}[n]$ is a functional relating the entire distribution of n to the entire potential. Nagai and coauthors demonstrated the training of $V_{\mathrm{xc}}[n]$ in a one dimensional two spinless electron system [42]. The fully nonlocal form is little transferable since it does not apply to the cases with different space grids (range, spacing, etc.). Moreover, such a form requires large amount of training data because a single set of model parameters gives us only a single set of training pairs.

These two problems are solved by limiting the model form so that the potential value depends only on the near distribution of n (e.g., semilocal approximation introduced in Sect. 4.2.3). First, a single global distribution of n has various $n(\mathbf{r})$ values at the respective $\mathbf{r}$ points, which enables us to train the model with smaller amount of training data. Second, such a near region model is applicable to systems with different sizes. Schmidt and coauthors implemented a model form of the exchange-correlation energy density $\varepsilon_{\mathrm{xc}}[n](\mathbf{r})$ that depend on density within a near region of $\mathbf{r}$ for a 1D model [44]. Ryabov et al. showed that a similar near-region model can learn nonuniform systems to reproduce the traditional LDA-PZ and GGA-PBE functionals in three dimensions [45], which is a sort of re-invention of the semilocal approximation.

4.3.3.2 Learning $n \rightarrow E$ Mapping

Another direct way to train the functional is to use data of density and physical/chemical quantities such as atomization energy or ionization potential. In this strategy, E_{xc} rather than V_{xc} is conveniently modeled as a functional of n (and $\varphi[n]$). The physical/chemical quantities are prepared by any database of experimental values or highly-accurate wavefunction theory such as coupled-cluster theory [46]. The loss function is then defined with the difference of the reference quantity $|Q^{(t)} - Q^{\mathrm{ML}}[n^{(t)}]|$ for reference pairs $(n^{(t)}, Q^{(t)})$. The training dataset must include a large number of materials to train the ML parameters adequately since the reference physical quantities are integrated values, which can only be obtained once per material.

Dick and Fernandez-Serra [47] built a model of Behler–Parrinello form [36] to correct the difference between energies calculated with a computationally cheap method ("baseline") and an accurate one: The charge density is translated to atomic descriptors and each energy correction per atom is modeled as a function of the descriptors. The reference charge densities were generated with the KS calculation with the GGA-PBE [8] functional, whereas the accurate reference energy was calculated using the CCSD(T) method.

Kirkpatrick et al. [48] performed the training of ML XC functional along the above strategy. They modeled ΔE_{xc}, the difference in E_{xc} between the reactant and product as a functional of semilocal values of n and partial exchange energy, and trained it with 1161 chemical reactions. They collected the reference pairs $(n^{(i)}, \Delta E_{\mathrm{xc}}^{(i)})$ using the KS equation with the B3LYP [13] functional for n and CCSD(T) for ΔE_{xc}, respectively. The training data pool was supplemented with 570 (504) atomic systems with fractional charges (spins) (see later Sect. 4.3.4.1) and 5150 data of the uniform electron gas. Their study (and some studies introduced below) uses the quantities defined as the energy difference between two states, such as the atomization energy or the ionization potential, in their training procedure. The absolute total energy is not directly used as training data since it shows large error between DFT and the exact wavefunction theory; for example, the difference in total energies between those with PBE and the full configuration-interaction method is a few Hartree's. In

contrast, because of the error cancellation, the errors to quantities defined as the energy difference become far smaller (a few kcal/mol $\sim$ 0.01 Hartree). Since the accuracy for those quantities are actually more important in practical use, it is an effective strategy to put them into the loss function.

4.3.3.3 Implicit Training Using KS Equations

One can also define the loss function by the difference between the accurately calculated n and n derived through the self-consistent KS equation with the ML exchange-correlation potential included. The model parameters are optimized so that the two densities agree. Unlike that mentioned in Sect. 4.3.3.1, this approach does not concern the numerically demanding KS inversion but is actually equivalent to that referring to V_{xc}. This is because, to the HK theorem (4.2), V_{xc} reproducing the exact n is exact and unique. A primary drawback of this method was that the gradient of the loss function with respect to the model parameters was unavailable. Nagai and coauthors [49] defined the loss function by the differences in n and reaction energies and used a non-gradient stochastic method for updating the model parameters, which took large computational cost for the optimization. This problem has been solved by implementing the KS equation in differential programming. This idea has been first demonstrated in a 1D model system by Li and coauthors [50]. Kasim and Vinko [51] and Dick and Fernandez-Serra [52] implemented a single NN for the molecules in 3D, where the derivatives iteratively back-propagates through the KS cycle. The differential programming, however, requires a large amount of memory, which makes it difficult to train on large dataset.

4.3.4 Imposing Physical Constraints

Various analytic formulas obeyed by the exact E_{xc} have been revealed [53–57]; asymptotic expansion forms, inequalities, scaling relations, and so on. The uniform electron gas limit [5, 58] is in particular important for the functionals to be accurate in solid systems [59]. Ideally, the approximate functionals should comply with all such known formulas. Controlling the analytical structure of ML models as such is, however, not straightforward because of their flexibility. In this section, we review methods to impose physical constraints on ML-based XC functionals, which have been found to result in better accuracy and transferability.

4.3.4.1 Learning Data Satisfying Physical Constraints

The most direct way to impose physical constraints is to include data satisfying the physical constraints in the training data set. Kirkpatrick et al. [48] and Gedeon et al. [60] included training data in systems with fractional number of electrons and

fractional spins into the training dataset, to make the XC functional comply to the ideal linear dependence [61]; namely, the ground-state total energy of the M-electron system $E(M)$ is exactly linear for M being non-integer. This linearity condition is important for accurately treating the charge-transfer properties, such as ionization potential and bandgap, and bond stretching [62, 63], but most existing XC functionals do not satisfy it. Kirkpatrick et al. and Gedeon et al. generated data of a noninteger number of electrons by interpolating those of integer number of electrons. They trained ML-based functionals with some non-locality (see descriptions in Sects. 4.3.3.1 and 4.3.3.2) using those data, and their XC functional well reproduced the behavior of total energy of fractional electrons.

4.3.4.2 Designing ML Architectures

Some physical constraints represented by the limiting values, asymptotes and inequalities can be equipped by the NN models analytically. An example is the spin-scaling relationship for the spin-dependent exchange energy [64]

$$E_{\mathrm{x}}\left[n_{\uparrow}, n_{\downarrow}\right] = \frac{1}{2}\left[E_{\mathrm{x}}[2n_{\uparrow}] + E_{\mathrm{x}}[2n_{\downarrow}]\right], \tag{4.21}$$

which may be applied to any model exchange functionals.

Nagai et al. has proposed a way to analytically impose a certain class of asymptotic constraints [65]: Assume that we want to impose an asymptotic constraints that the output of the ML function $f(\mathbf{x})$ should converge to f_0 at $\mathbf{x} \to \mathbf{x_0}$. This constraints can be imposed by a post-processing operation $\hat{\theta}_{\{\mathbf{x}_0, f_0\}}$ defined as follows:

$$\hat{\theta}_{\{\mathbf{x}_0, f_0\}}[f](\mathbf{x}) = f(\mathbf{x}) - f(\mathbf{x}_0) + f_0. \tag{4.22}$$

Even if we have multiple asymptotic constraints, where f converges to $f_0^{(1)}, f_0^{(2)}, \ldots f_0^{(N_c)}$ at $\mathbf{x}_0^{(1)}, \mathbf{x}_0^{(2)}, \ldots, \mathbf{x}_0^{(N_c)}$, we can impose them simultaneously by combining Eq. (4.22) in a way like the Lagrange interpolation:

$$\hat{\Theta}_{c=1,\ldots N_c}[f](\mathbf{x}) = \sum_{c=1}^{N_c} \prod_{c' \neq c} \frac{l_{c,c'}(\mathbf{x} - \mathbf{x}_{c'})}{l_{c,c'}(\mathbf{x}_c - \mathbf{x}_{c'})} \hat{\theta}_{\{x_0^{(c)}, f_0^{(c)}\}}[f](\mathbf{x}) \tag{4.23}$$

$l_{c,c'}(c, c' = 1, 2, \ldots N_c)$ are functions satisfying $l_{c,c'}(\mathbf{x}) = 0$ at $\mathbf{x} = 0$ and $l_{c,c'}(\mathbf{x}) \to 1$ at $|\mathbf{x}| \to \infty$. They suppress oscillations coming from the high-dimensional polynomial in Eq. (4.23).

Most of the physical constraints are asymptotic-type, and thus they constructed an ML-based XC functional satisfying most of known physical constraints for the meta-GGA approximation. In their results, imposing physical constraints improved not only accuracy but also computational stability. They used dataset consisting of molecular systems for the training. The SCF calculation using the XC functional

constructed without physical constraints as in Nagai et al. [49] did not converge for some solid systems (metals, semiconductors), while that with physical constraints converged in the same speed as the conventional analytical functionals. Imposing physical constraints in this way is expected to regularize functional forms in limiting regimes of the n space, which cannot be efficiently learned with specific materials data, and improves transferability.

In conjunction with the linearity condition, the exact E_{xc} shows a discontinuity in the derivative of E_{xc} with respect to the total number of electrons M at the integer M [61]:

$$\frac{\partial E_{\mathrm{xc}}}{\partial M}(M+0) - \frac{\partial E_{\mathrm{xc}}}{\partial M}(M-0) \equiv \Delta_{\mathrm{xc}} \neq 0 \tag{4.24}$$

Gedeon et al. [60] applied a function of form $|\sin \pi M|$ so that the derivative discontinuity Δ_{xc} is efficiently represented. They used this in combination with the aforementioned training data design to reproduce the ideal behavior.

The constraints represented by inequalities can be well enforced by using bounded non-linear transformation. Dick and Fernandez-Serra [52] employed a function of form $\frac{a}{1-(a-1)\exp(-x)} - 1$ with a being a constant to impose the Lieb–Oxford bound for E_{x} [66], which defines lower bounds of exchange-correlation energy.

4.3.4.3 Designing the Loss Function

The ML model is optimized by minimizing the loss function defined by the difference between the values calculated by the ML model and those in the reference data. By adding extra penalty terms to that, we can loosely enforce the constraints to the model. Pokharel et al. [67] introduced a penalty term of form $\propto \mathrm{ReLu}(f(x) - a)$, with $\mathrm{ReLu}(x)$ being the rectified linear unit $\mathrm{ReLu}(x) = \max(0, x)$, so that the model approximately satisfies an inequality condition $f(x) \leq a$. Another interesting approach has been put forward by Cuerrier et al. [68]. They modeled the negative-definite exchange-correlation hole by neural networks, defined an "entropy" with that, and targeted it as the loss. Although there is no guarantee that their maximum entropy principle is physical, this method makes the model smooth and likely enhances the transferability.

4.3.4.4 Compressing Input Parameter Space with Physical Constraints to Improve Transferability

The scaling condition relates the values of density functionals between densities in different length scales. By changing the length scale by a uniform factor γ, the charge density is transformed by

$$n(\mathbf{r}) \to n_{\gamma}(\mathbf{r}) \equiv \gamma^{d} n(\gamma \mathbf{r}) \tag{4.25}$$

with d being the spatial dimension. Exact kinetic energy functional (Eq. 4.11), for example, is then transformed by

$$T_s[n] \to T_s[n_\gamma] \equiv \gamma^2 T_s[n]. \tag{4.26}$$

Hollingsworth et al. studied the use of coordinate-scaling condition to improve transferability in one-dimensional models [69]. They first trained the ML-based kinetic energy functional using dataset consisting of densities in a specific scale. When they applied the trained model to data in a different scale, the scaling parameter was adapted to the value of the training data by Eq. (4.25). The output of ML $T_s[n]$ is then converted to that of original density scale by Eq. (4.26). The resulting accuracy For a certain model (Hooke's atom) this method yielded better accuracy than directly predicting $T_s[n]$ in different scales, but did not for the other (stretched H_2); the accuracy depends on whether the scaling relation is of primary relevance for the system.

4.3.5 Performance of ML-Based XC Functionals

Here we quote some state-of-the-art results accomplished by the ML-based exchange-correlation functionals. Nagai et al. demonstrated the construction of NN-based meta-GGA functional using dataset including atomization energy and density of 3 molecules [49] (see Sect. 4.3.3.3 for their training scheme). Its mean absolute error (MAE) for atomization energy of 147 molecules was 4.7 kcal/mol, while that of conventional meta-GGA functional with the best performance (M06-L [70]) was 5.2 kcal/mol. Figure 4.1 exhibits the accuracy of energy-related quantities and density. Later, they improved the accuracy by incorporating physical constraints into ML model architecture (see Sect. 4.3.4.2): its MAE for the same 144 molecules was 3.6 kcal/mol [65].

Li et al. implemented differentiable KS cycle and performed training with solving KS cycle for 1D molecules [50]. Figure 4.2 exhibits predicted effective potential v_s ($= V_{\text{eff}}$ in Eq. 4.6) and the density n obtained from solving KS equation on v_s. The results well reproduce the exact v_s and n, and we can see that the XC functional trained using density and energy in the loss function (Fig. 4.2a) had better accuracy than that using only energy (Fig. 4.2b). Kasim et al. implemented the training using differentiable KS equations on 3D molecules [51]. They used 4 molecules each for training and validation, and MAE for 104 molecules of their NN-based GGA functional was 7.4 kcal/mol while that of GGA-PBE [8] was 16.5 kcal/mol. Dick et al. also demonstrated NN-based meta-GGA XC functional using differentiable KS equation [52]. They used AE, IP, barrier height (BH), and density of tens of molecular systems as the training dataset. Table 4.1 compares their functional and standard conventional functionals by the MAE for hundreds of molecular systems. See Sect. 4.3.3.3 for their differential implementation of the KS equations.

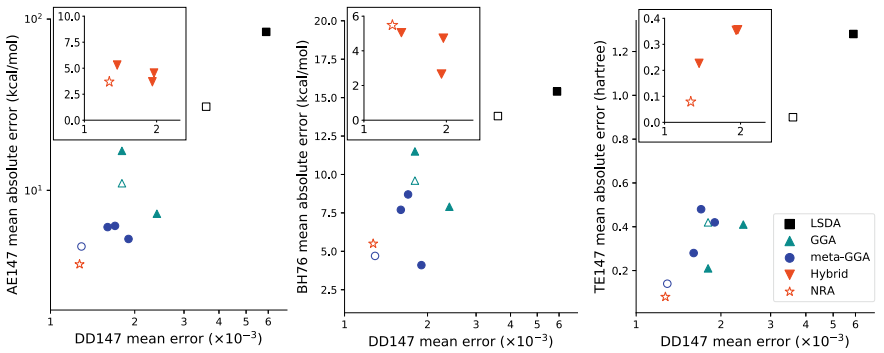


Fig. 4.1 Relations between accuracy of density and that of atomization energy, density distribution, total energy, cited from Ref. [49]. The panels represent mean absolute error (MAE) for atomization energy of 147 molecules (AE147), barrier height of 76 chemical reactions (BH76), and total energy of 147 molecules (TE147) against MAE for density distribution of 147 molecules (DD147). The closed and open markers represent the accuracy of existing and the NN-based functionals, respectively. Legend "NRA" represents an improved approximation with a non-local density descriptor, constructed using NN

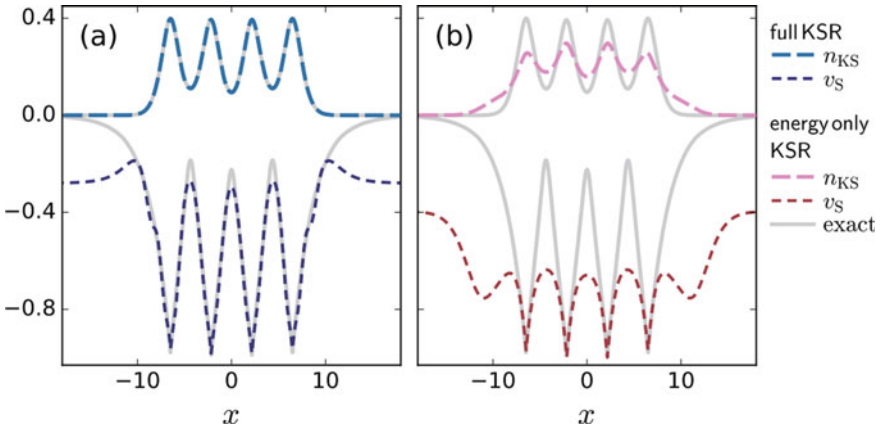


Fig. 4.2 Density and KS potential of H_4 from neural XC functionals trained with **a** energy and density loss and **b** energy loss only [50]. This figure was reprinted from Li et al. [50], under the Creative Commons Attribution 4.0 International License (CC BY 4.0)

As mentioned in the previous chapter, Kirkpatrick et al. included fractional electron/spin systems in their training dataset (see Sects. 4.3.3.2 and 4.3.4.2 for their scheme). Furthermore, they adopted local HF features, which are defined as range-separated exact exchange energy density

$$e^{\sigma}_{\omega\mathrm{HF}}(\mathbf{r}) = -\frac{1}{2}\sum_{ab:\mathrm{occ.}} \varphi^*_a(\mathbf{r})\varphi_b(\mathbf{r}) \int d\mathbf{r}' \frac{\mathrm{erf}(\omega|\mathbf{r}-\mathbf{r}'|)}{|\mathbf{r}-\mathbf{r}'|} \varphi^*_b(\mathbf{r}')\varphi_a(\mathbf{r}'), \qquad (4.27)$$

Table 4.1 Mean absolute errors of atomization energies (AE) for the W4-11 dataset, barrier heights (BH) for BH76, decomposition energies (DE) for MB16-43, and densities (ϵ_n) for G2/97, cited from Ref. [52, 71]

	AE (kcal/mol)	BH (kcal/mol)	DE (kcal/mol)	WTMAD-2 (kcal/mol)	$\epsilon_n \times 10^3$ (–)
RPBE [72]	8.3	9.0	50.8	10.5	8.8
B97 [73]	4.7	7.3	36.1	8.6	7.0
OLYP [74]	9.9	8.5	29.0	8.5	10.1
revPBE [75]	7.6	8.3	27.1	8.4	9.4
M06L [70]	4.4	3.9	63.3	8.6	9.4
revTPSS [76]	5.7	8.9	36.7	8.4	7.9
SCAN [12]	4.1	7.8	17.8	8.0	6.2
xc-diff	3.5	6.5	22.7	7.3	5.2
PBE0 [14]	3.7	5.0	15.9	6.4	5.7
B3LYP [13]	3.4	5.7	24.8	6.5	8.3
M05-2X [77]	4.0	1.7	26.3	4.6	7.5
ωB97X-V [78]	2.8	1.8	32.5	4.1	5.0

"xc-diff" represents meta-GGA NN-based XC functional proposed by Ref. [52]. Adapted with permission from Dick and Fernandez-Serra [52]. Copyright (2021) by the American Physical Society

with $\omega = 0.4$ and ∞. Those features are expected to include information on non-local interaction [48]. For their training, they used 1161 data including atomization, ionization, electron affinity, and intermolecular binding energies of small molecules, 1074 data represented fractional electron and fractional spin systems of atoms. In one of the tests they performed, the MAE was 1.66 kcal/mol for the QM9 dataset [79] containing 133,857 chemical reactions, while the MAE of an existing hybrid functional called ωB97X was 2.13.

A factor that contributes to the high transferability seen in those results is the use of local descriptors of electron density. When applying a constructed XC functional to a material, the value of XC energy density can be well predicted if similar local density distribution is included in the training dataset, regardless of the kinds of ions. We, therefore, do not need to have exactly the same elements of the target material in the training dataset.

Another factor contributing to the transferability is a mathematical property of the KS equation. When an ML model is applied to local density distribution outside the region of the training dataset, there would be error in predicted XC energy density. As a result, there would occur spurious fluctuations in the XC potential. However, the kinetic energy term in the Kohn–Sham equation (Eq. 4.6) has an effect of penalizing the non-smooth solution, and thus we can obtain smooth KS orbitals (detailed discussions are in Ref. [42]).

4.4 Summary and Future Perspectives

In this chapter, we introduced methods for constructing ML-based XC functionals for the Kohn–Sham equation. ML models such as NNs are trained with highly accurate electronic structure data of real materials to construct the XC functional. Although data-driven techniques of ML enable construction without physical constraints, the inclusion of physical constraints improves the accuracy and computational stability. There exist some choices in the way of the training. One can directly use the pair of electron density and energy as data, while incorporating the self-consistent cycle of the KS equations into the training procedure is also an effective way to ensure the numerical stability of constructed XC functional. The trained ML model can be used by substituting it into KS equation for any material. Since the electron density, which exists in common in any materials, is input of ML model, once an XC functional is well-trained, it would yield accurate predictions for various materials. In that sense, ML techniques are expected to be effective for the improvement of DFT.

In some studies, ML-based XC functionals have outperformed the standard ones constructed by humans. By using flexible ML models and using data from real materials for the training, the ML-based XC functional is expected to learn the essential features of the electronic interaction effects which cannot be represented by existing physical theories.

The goal of this field is to realize a versatile XC functional that is universally accurate for any materials. The ML models are far more flexible than the conventional analytical models, and this feature can be exploited for a highly transferable XC functional. However, at present, using data of small molecules is the mainstream, while learning with data of large materials has not been realized. It will be important to establish a method to utilize data of experiments as well as to expand the highly accurate electronic structure data set for various materials.

Meanwhile, specializing in a particular system would also be a practical application [47]. For the systems whose electronic features are far from typical ones used in the existing XC functionals, i.e. the uniform electron system, the conventional constructions would fail (e.g. localized d-electron). The ML-based construction is also expected to be effective for that cases because it can systematically extract the essence of electronic behavior from the dataset.

The construction of a functional by ML may be useful for understanding physics. For example, by limiting the types of materials used in training and evaluating the performances of constructed ML-based XC functionals for different types of materials, the similarity in electronic structures of the two systems can be analyzed. It would also be possible to investigate which physical constraint contributes most to the accuracy by comparing differently constrained functionals in the way introduced in Sect. 4.3.4.

In this chapter, we introduced the construction of density functionals for ground state electron systems, but ML can be applied to other systems, such as time-dependent DFT [80] or classical DFT for solvent systems [81]. Through the accumu-

lation of studies for those problems, we expect that a systematic way to complement unknown functionals in more general variational problems (e.g., Ref. [82]) would be established.

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Chapter 5
Machine-Learning for Static and Dynamic Electronic Structure Theory



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Abstract Machine learning has emerged as a powerful technique for processing large and complex datasets. Recently it has been utilized for both improving the accuracy and accelerating the computational speed of electronic structure theory. In this chapter, we provide the theoretical background of both density functional theory, the most widely used electronic structure method, and machine learning on a generally accessible level. We provide a brief overview of the most impactful results in recent times. We, further, showcase how machine learning is used to advance static and dynamic electronic structure calculations with concrete examples. This chapter highlights that fusing concepts of machine learning and density functional theory holds the promise to greatly advance electronic structure calculations enabling unprecedented applications for in-silico materials discovery and the search for novel chemical reaction pathways.

Keywords Materials science · Electronic structure · Density functional theory · Machine learning · Neural networks

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5.1 Introduction

Progress in chemistry and materials science is driven with increasing importance by computational studies based on electronic structure theory calculations. They enable the understanding of matter on the microscopic level of quantum mechanics, in principle, from first principles without any empirical input. High-performance computing has fueled the usefulness of electronic structure calculations and has enabled large-scale computational studies. However, the ever-increasing demand for accurate first-principles data renders even the most efficient simulation codes infeasible.

On the other hand, the use of data-driven machine-learning methods has grown rapidly across a number of research fields. Such methods are increasingly gaining importance, as they are utilized to accelerate, replace, or improve traditional electronic structure theory methods.

In recent years combining machine-learning techniques with electronic structure theory has emerged as an active area of research. By far the largest application domain is to leverage machine learning for predicting chemical and material properties. Prominent applications include the search for novel catalysts for chemical reactions [2], novel materials [3, 4] including unexplored materials for photovoltaic applications [5]. Machine learning techniques have also been used for predicting elastic and structural properties [4, 6], quantum chemical properties such as polarizabilities [7] and bond dissociation energies [8], transport and diffusion properties [9–11], surface energies [12], and magnetic properties [13].

Another application space is the development of interatomic potentials for machine-learning based mapping of atomic positions to potential energy surface, such as neural network potentials [14–18], Gaussian approximation potentials [19], the related spectral neighborhood analysis potential [20, 21] and moment tensor potentials [22, 23].

Finally, tackling the electronic structure problem has been attempted in terms of machine-learning driven kinetic energy [24, 25] and exchange-correlation approximations [26, 27], density of states predictions [28], electron density predictions [29], and local density of states predictions [30, 31].

Furthermore, machine learning and density functional theory have also been combined as components of other workflows, such as the generation of molecules with specific properties [32, 33], as part of elaborate statistical frameworks [34] and to predict hyperparameters for stable ab-initio calculations [35].

In this chapter, we discuss the utility of combining density functional theory, the most widely used electronic structure method, with machine learning. To that end, we first provide a concise introduction to electronic structure theory with an emphasis on density functional theory in Sect. 5.2.1. This is complemented by an introduction to the basic concepts of machine learning in Sect. 5.2.2. This forms a solid basis for showcasing surrogate modeling workflows based on machine learning in Sect. 5.3. We illustrate two concrete examples of how machine learning is utilized to advance electronic structure calculations. In the first example, we illustrate a

workflow based on neural networks that replaces time-independent density functional theory calculations. In the second example, we use very recently developed physics-informed neural networks to accelerate the simulation of electron dynamics.

5.2 Theoretical Background

5.2.1 Electronic Structure Theory

In the following, we provide a short overview of the basic formalism. We thereby focus on density functional theory which is the most popular electronic structure method due to its balance of accuracy and computational cost. Note that we work within atomic units throughout, where $\hbar = m_e = e^2 = 1$, such that energies are expressed in Hartrees and length in Bohr radii.

5.2.1.1 Exact Framework

We begin with a brief outline of the electronic structure problem in the most common theoretical context which is non-relativistic quantum mechanics. The focus is to solve the problem of N_e electrons with collective coordinates $\underline{\boldsymbol{r}} = \{\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{N_e}\}$ that are coupled to N_i ions with collective coordinates $\underline{\boldsymbol{R}} = \{\boldsymbol{R}_1, \ldots, \boldsymbol{R}_{N_i}\}$, where $\boldsymbol{r}_j \in \mathbb{R}^3$ refers to the position of the jth electron, while $\boldsymbol{R}_\alpha \in \mathbb{R}^3$ denotes the position of the αth ion of mass M_α and charge Z_α. Note that for sake of brevity we do not take spins into account. The physics in this framework is governed by the Hamiltonian

$$\hat{H} = \hat{T}^i(\underline{\boldsymbol{R}}) + \hat{T}^e(\underline{\boldsymbol{r}}) + \hat{V}^{ii}(\underline{\boldsymbol{R}}) + \hat{V}^{ei}(\underline{\boldsymbol{r}}, \underline{\boldsymbol{R}}) + \hat{V}^{ee}(\underline{\boldsymbol{r}}), \tag{5.1}$$

where

$$\hat{T}^i(\underline{\boldsymbol{R}}) = \sum_{\alpha}^{N_i} -\frac{\nabla_\alpha^2}{2M_\alpha} \tag{5.2}$$

denotes the kinetic energy of the ions and

$$\hat{T}^e(\underline{\boldsymbol{r}}) = \sum_{j}^{N_e} -\frac{\nabla_j^2}{2} \tag{5.3}$$

the kinetic energy of the electrons, while

$$\hat{V}^{ii}(\underline{\boldsymbol{R}}) = \sum_{\alpha}^{N_i} \sum_{\beta \neq \alpha}^{N_i} \frac{Z_\alpha Z_\beta}{2|\boldsymbol{R}_\alpha - \boldsymbol{R}_\beta|} \tag{5.4}$$

represents the ion–ion interaction,

$$\hat{V}^{ei}(\underline{\boldsymbol{r}};\underline{\boldsymbol{R}}) = -\sum_{j}^{N_e}\sum_{\alpha}^{N_i}\frac{Z_\alpha}{|\boldsymbol{r}_j - \boldsymbol{R}_\alpha|} \tag{5.5}$$

the electron–ion interaction, and

$$\hat{V}^{ee}(\underline{\boldsymbol{r}}) = \sum_{j}^{N_e}\sum_{k\neq j}^{N_e}\frac{1}{2|\boldsymbol{r}_j - \boldsymbol{r}_k|} \tag{5.6}$$

the electron–electron interaction.

The coupled electron-ion problem is an area of active research and novel methodologies are being developed continuously [36–42].

In most cases, the Born–Oppenheimer approximation [43] is applied. By separating the coupled electron-ion problem, the computational complexity of the Hamiltonian in Eq. (5.1) is reduced. This separation is feasible because the inverse dependence of the kinetic energy on the particle mass results in a far smaller kinetic energy for the ions than for the electrons. This approximation is valid as long as the motion of the ions happens on a much larger time scale than the motion of the electrons. The resulting Born–Oppenheimer Hamiltonian reads [44]

$$\hat{H}^{BO}(\underline{\boldsymbol{r}};\underline{\boldsymbol{R}}) = \hat{T}^{e}(\underline{\boldsymbol{r}}) + \hat{V}^{ee}(\underline{\boldsymbol{r}}) + \hat{V}^{ei}(\underline{\boldsymbol{r}};\underline{\boldsymbol{R}}) + E^{ii}(\underline{\boldsymbol{R}}), \tag{5.7}$$

where the ion–ion interaction turns into a constant shift in energy E^{ii}, because it only depends on the coordinates $\underline{\boldsymbol{R}}$.

Zero Temperature

At temperature $\tau = 0$, the time-independent, electronic Schrödinger equation

$$\hat{H}^{BO}(\underline{\boldsymbol{r}};\underline{\boldsymbol{R}})\Psi(\underline{\boldsymbol{r}};\underline{\boldsymbol{R}}) = E\Psi(\underline{\boldsymbol{r}};\underline{\boldsymbol{R}}) \tag{5.8}$$

provides the basis for computing the electronic structure of matter. Most importantly, the lowest eigenvalue of this equation is labeled the ground-state energy E_0. It provides the basis for calculating various chemical and materials properties including molecular or crystal structures [45], charge densities [46], cohesive energies [47], elastic properties [48], vibrational properties [49], magnetic properties [50, 51], dielectric susceptibilities [52], magnetic susceptibilities [53], phase transitions [54], HOMO/LUMO energies (highest-occupied molecular orbital, lowest unoccupied molecular orbital) [55–57], bond dissociations [58], enthalpies of formation [58], ionization potentials [59], electron affinities [59], band gaps [60, 61], and the equation of state [62].

Non-zero Temperature

Temperature plays an important role in a considerable class of problems, particularly in materials science. For example, in liquid metals, the energy scale due to the

electronic temperature becomes similar to the first electronic excitation energy. Under the assumption that the electrons find their thermal equilibrium on a time scale small compared to the motion of the ions, the Born–Oppenheimer approximation still holds. However, the electronic temperature needs to be taken into account explicitly.

Hence, the zero-temperature framework is extended in terms of an *ensemble* of quantum mechanical states that describe a local thermodynamical equilibrium of a system. Usually, the grand canonical ensemble is employed [63, 64].

The central quantity is the grand-canonical operator

$$\hat{\Omega} = \hat{H}^{BO} - \tau \hat{S} - \mu \hat{N}, \tag{5.9}$$

where $\hat{S} = -k_{\mathrm{B}} \ln \hat{\Gamma}$ the entropy operator, $\hat{N}$ the particle number operator, and k_{B} the Boltzmann constant. Using an orthonormal set of N_e-particle states $\Psi_{N_e,m}$, one defines the statistical density operator as

$$\hat{\Gamma} = \sum_{N_e} \sum_{m} w_{N_e,m} \left|\Psi_{N_e,m}\right\rangle \left\langle\Psi_{N_e,m}\right| \tag{5.10}$$

by drawing on statistical weights $w_{N_e,m}$ restricted to satisfy the normalization condition $\sum_{N_e,m} w_{N_e,m} = 1$. Observables of an operator $\hat{O}$ are then calculated

$$O[\hat{\Gamma}] = Tr\{\hat{\Gamma}\hat{O}\}, \tag{5.11}$$

in terms of the statistical density operator.

As is apparent from Eq. (5.9), the entropy and particle number operators are coupled to the Hamiltonian via the chemical potential μ and the temperature τ. In analogy to Eq. (5.8), the energy is calculated using the grand-canonical potential

$$\Omega[\hat{\Gamma}] = E - \tau S - \mu N. \tag{5.12}$$

Based on this, thermodynamical equilibrium is defined as the statistical density operator for which the grand-canonical potential is minimized [65]. The minimizing statistical density operator is referred to as grand-canonical statistical operator [63].

5.2.1.2 Density Functional Theory and Beyond

The multitude of practical electronic structure methods employs different approximations to overcome the underlying complexity of the problem. Consequently, they differ in terms of accuracy and computational cost. In the following, we give a concise overview of these methods with an emphasis on density functional theory. In the following, we always work within the Born–Oppenheimer approximation. We take the parametric dependence on $\underline{\boldsymbol{R}}$ as granted and, therefore, suppress denoting it.

Density functional theory is a very popular method for carrying out electronic structure calculations, as it balances acceptable accuracy with reasonable computational cost. The central quantity is the electronic density $n(\boldsymbol{r})$. The Hohenberg–Kohn theorems [66] guarantee a one-to-one correspondence between the electronic density and the external potential, e.g., the electron-ion potential $\hat{V}^{ei}(\underline{\boldsymbol{r}})$. This means every property of interest can be determined as a functional of the density, most prominently, the ground-state energy. In moving to a description of the system based on the electronic density, one moves away from the need to deal with terms directly dependent on the collective electronic positions $\underline{\boldsymbol{r}}$ in favor of quantities solely depending on the position in real space, denoted by $\boldsymbol{r}$. This applies to the single-particle wave functions and mean-field potentials as well, as illustrated below.

Density Functional Theory at Zero Temperature
Practical density functional theory calculations usually rely on the Kohn–Sham ansatz [67]. Here, the system of interacting electrons is replaced by an auxiliary system of non-interacting electrons constrained to reproduce the electronic density of the interacting system. This is achieved in terms of the Kohn–Sham equations

$$\left[-\frac{1}{2}\nabla^2 + v_s(\boldsymbol{r})\right]\phi_j(\boldsymbol{r}) = \epsilon_j \phi_j(\boldsymbol{r}), \tag{5.13}$$

a set of N_e one-particle Schrödinger equations, where $v_s(\boldsymbol{r})$ denotes the Kohn–Sham potential $v_s(\boldsymbol{r})$. It is calculated self-consistently and includes both the electron–ion interaction and the electron–electron interaction, the latter in terms of a mean-field description. The electronic density is obtained from the Kohn–Sham orbitals $\phi_j(\boldsymbol{r})$ as

$$n(\boldsymbol{r}) = \sum_j |\phi_j(\boldsymbol{r})|^2. \tag{5.14}$$

The total energy functional is expressed within the Kohn–Sham formalism as

$$E[n] = T_S[n] + E_H[n] + E_{XC}[n] + E^{ei}[n] + E^{ii}, \tag{5.15}$$

where $T_S[n]$ is the kinetic energy of the Kohn–Sham system, $E_H[n]$ the Hartree energy, i.e., the energy contribution from the electrostatic interaction of the density with itself, E^{ii} the energy contribution from the ion–ion interaction, $E^{ei}[n]$ the energy contribution from the electron–ion interaction, and $E_{XC}[n]$ the exchange-correlation energy. The main motivation of the Kohn–Sham ansatz is that the interacting kinetic energy $T[n]$, for which no exact density functional expression is known, can be approximated by $T_S[n]$ as

$$T[n] \approx T_S[n] = \sum_{j=1}^{N_e} \int d\boldsymbol{r}\, \phi_j^*(\boldsymbol{r}) \left(-\frac{1}{2}\nabla^2\right) \phi_j(\boldsymbol{r}). \tag{5.16}$$

Given that the form of $E_{\mathrm{XC}}[n]$ was known, it is ensured that density functional theory calculations yield *in principle* the exact result. In practice, however, $E_{\mathrm{XC}}[n]$ has to be approximated. The fundamental approximation to $E_{\mathrm{XC}}[n]$ is the local density approximation (LDA) [67] which is parametrized from quantum Monte Carlo data of the uniform electron gas [68]. Incorporating the density gradients leads to generalized gradient approximations (GGA) such as the PBE functional [69]. Including further constraints and quantities (e.g., the gradient of the Kohn–Sham orbitals) leads to functionals such as meta-GGAs (e.g., SCAN [70]) and hybrid-GGAs which include a fraction of Hartree–Fock exchange (e.g., B3LYP [71, 72]). A comprehensive comparison of exchange-correlation functionals is given in Refs. [60, 73].

The definition of the Kohn–Sham potential is obtained by the aid of the variational principle, i.e., that the ground-state energy is stationary for small variations of the density $\delta n(\boldsymbol{r})$. This leads to the expression

$$v_{\mathrm{S}}(\boldsymbol{r}) = v^{ei}(\boldsymbol{r}) + \frac{\delta E_{\mathrm{H}}[n]}{\delta n(\boldsymbol{r})} + \frac{\delta E_{\mathrm{XC}}[n]}{\delta n(\boldsymbol{r})}. \tag{5.17}$$

Density Functional Theory at Non-zero Temperature

The previous statements also hold true for non-zero temperatures in terms of $\hat{\Gamma}$ [63, 74–76]. In analogy to the zero-temperature case, the Kohn–Sham equations become

$$\left[-\frac{1}{2}\nabla^2 + v_{\mathrm{S}}^{\tau}(\boldsymbol{r})\right]\phi_j(\boldsymbol{r}) = \epsilon_j^{\tau}\phi_j(\boldsymbol{r}) \quad \{\tau>0\}, \tag{5.18}$$

where now both the Kohn–Sham potential and the Kohn–Sham eigenvalues depend on the temperature. Similarly, the electronic density is obtained from the Kohn–Sham orbitals $\phi_j(\boldsymbol{r})$ as

$$n^{\tau>0}(\boldsymbol{r}) = \sum_j f^{\tau}(\epsilon_j^{\tau})\,|\phi_j(\boldsymbol{r})|^2 \tag{5.19}$$

with f^{τ} denoting the Fermi–Dirac distribution due to which the electronic density becomes temperature dependent. Likewise, the free energy functional becomes

$$A = T_{\mathrm{S}}[n] - k_{\mathrm{B}}\tau S_{\mathrm{S}}[n] + E_{\mathrm{H}}[n] + E_{\mathrm{XC}}^{\tau}[n] + E^{ei}[n] + E^{ii}. \tag{5.20}$$

In this expression, the electronic entropy S_{S} is evaluated as a functional of the Kohn–Sham eigenvalues, similar to T_{S}. While the explicit temperature dependence of the exchange-correlation energy has been neglected in most applications, the recent interest in warm dense matter [77] has given rise to functionals that address the explicit temperature dependence of the exchange-correlation energy [76, 78–81].

Coupling Density Functional Theory to Molecular Dynamics

In classical molecular dynamics (MD) simulations, the dynamics of the ions is modeled with classical mechanics. This requires the calculation of an interatomic potential (IAP), which determines how the ions in the system interact and propagate in

time. These potentials represent a logical connection point for density functional theory calculations. The coupled dynamics of electrons and ions within the Born–Oppenheimer approximation is commonly realized by combining classical MD with density functional theory calculations. This is achieved either by performing density functional theory calculations during each time step of an MD simulation (i.e., so-called ab-initio MD) or by using calculation data to tailor IAPs for later use. In their simplest form, these IAPs are expressed as pair potentials, such as the Lennard–Jones potential [82] which has a simple functional form with two tuneable parameters. More sophisticated approaches such as the embedded atom model (EAM) [83] account for pair-wise interaction not just between ions, but also between ions and the electron gas present in metals. These effectively constitute many-body potentials and require the adjustment of a wider range of parameters. Other examples of such many-body potentials are given in Refs. [83–91]. Recently, machine learning has given rise to a new class of IAPs that are constructed using data-driven algorithms (see Sect. 5.2.2). Notable examples include ANI-1 [92], GAP [19], BLAST [93], HIP-NN [94], SchNet [95], SNAP [20], DPMD [96], and the AGNI machine-learning force fields [97]. These approaches differ in their technical aspects, e.g., in machine learning algorithms or choice of descriptor vectors, but strive for the same overall goal.

Time-Dependent Density Functional Theory

A notable extension of density functional theory is time-dependent density functional theory, which is a formally exact representation of the electronic time-dependent Schrödinger equation in terms of a single-particle picture. It thus enables simulating ultra-fast electron dynamics in relatively large molecules and materials when exposed to time-dependent external fields (such as laser pulses) at high accuracy with reasonable computational overhead.

Instead of solving the time-dependent Schrödinger equation, time-dependent density functional theory calculations rely on solving the time-dependent Kohn–Sham equations

$$\left[-\frac{1}{2}\nabla^2 + v_\mathrm{S}(\boldsymbol{r},t)\right]\phi_j(\boldsymbol{r},t) = i\frac{\partial\phi_j(\boldsymbol{r},t)}{\partial t}. \tag{5.21}$$

Analogous to the HK theorems [66], the Runge–Gross theorem [98] states that there is a unique one-to-one correspondence between the time-dependent electron density $n(\boldsymbol{r},t)$ and the external potential. Similar to the static case, the time-dependent electron density is obtained from the Kohn–Sham orbitals as

$$n(\boldsymbol{r},t) = \sum_j |\phi_j(\boldsymbol{r},t)|^2. \tag{5.22}$$

In time-dependent density functional theory codes [99–101], the time-dependent Kohn–Sham equations are solved numerically. A review of numerical techniques for solving time-dependent Kohn–Sham equations can be found in [102, 103]. Since these equations are solved iteratively until convergence is achieved in the self-

consistency calculations, they represent a significant fraction of the computational cost in the time-dependent density functional theory workflow, especially for high-resolution multi-atom systems.

Practical Calculations

While the preceding paragraphs outline the general theory that enables density functional theory simulations, practical calculations have to address a few technical problems to allow for the efficient use of specialized density functional theory software. These include the implementation of basis sets, self-consistency iterations, density mixing, exchange and correlation functionals, and parallelization strategies, just to name a few. Long-lasting developmental efforts in the field have led to reliable, numerically stable, and efficient algorithms used in a spectrum of software frameworks. Here we will give only a short overview of those technical aspects of density functional theory calculations relevant to the findings presented within this work.

Very generally, density functional theory calculations can be categorized into two types of applications. Firstly, the simulation of bulk materials in the material science and condensed-matter physics communities, and secondly the treatment of (isolated) molecules and compounds in the quantum chemistry community. One technical issue one has to overcome in either scenario is the representation of the Kohn–Sham wave functions in a way that makes practical computer simulations possible. In both cases, this is done in terms of linear combinations of basis functions $\mu_l(\boldsymbol{r})$ as

$$\phi_j(\boldsymbol{r}) = \sum_l c_{lj}\mu_l(\boldsymbol{r}), \tag{5.23}$$

with appropriate expansion coefficients c_{lj}. The choice of basis functions differs between the two domains, with quantum chemists usually employing basis sets constructed via atomic wave functions, which allow for efficient treatment of molecules.

A different type of basis set is needed for the simulation of bulk materials, on which we shall focus throughout the rest of this chapter. The overall idea of these calculations is to deduce the bulk properties of a material by simulating only a small portion of it, which is assumed to repeat infinitely in either direction, i.e., periodic boundary conditions are assumed. This is feasible since the materials in question exhibit a periodic crystal structure, even if said structure may be slightly perturbed as one moves to higher temperatures. Bloch's theorem [104] states that in such a case the eigenfunctions of the Hamiltonian take the form of a plane wave p with phase $\boldsymbol{k}$, given as

$$p(\boldsymbol{r}) = e^{i\boldsymbol{k}\boldsymbol{r}}, \tag{5.24}$$

multiplied by a periodic function $u(\boldsymbol{r})$. Therefore, one employs plane waves as basis functions. To avoid the impracticality of working with infinitely many plane waves (as theory technically dictates), in practice, only a subset of plane waves is included in the basis set. Since every plane wave corresponds to a kinetic energy, this is done by only including plane waves below a certain kinetic energy cutoff.

However, since the plane waves in Eq. (5.24) are phase-dependent, integrals over real space involving wave functions expressed by such plane waves now require integration over the $\boldsymbol{k}$-space. In practice, this is done via a summation over a concrete number of values for $\boldsymbol{k}$, often referred to as $\boldsymbol{k}$-points or $\boldsymbol{k}$-grid. Conceptually, larger $\boldsymbol{k}$-grids produce more accurate results, but drastically deteriorate the efficiency of the simulation.

The selection of an appropriate size of the basis set (i.e., the determination of the cutoff energy for the plane waves) and the determination of an appropriate set of $\boldsymbol{k}$-points is one of the fundamental tasks to be solved when performing condensed-matter density functional theory simulations. They are usually realized by performing a convergence analysis, i.e., performing calculations with differently sized basis sets and $\boldsymbol{k}$-grids until the calculated observable (usually the total free energy) does not vary below a certain threshold between individual calculations. For further reading, we point the reader to Ref. [105].

Beyond Density Functional Theory

In contrast to density functional theory, wave function methods center on the many-particle wave function $\Psi(\underline{\boldsymbol{r}})$. The simplest method is the Hartree–Fock theory, which constructs a single Slater determinant by using a system of auxiliary wave functions (similar to the Kohn–Sham ansatz). Hartree–Fock calculations neglect electronic correlation by construction, limiting the predictive capabilities of the method [106]. Thus, several post-Hartree–Fock methods have been developed that incorporate electronic correlation explicitly, such as the coupled-cluster method (CC/CCSD(T)) [107–112], configuration interaction (CI) [113–123], and Møller–Plesset perturbation theory (MP) [124]. These methods generally outperform density functional theory calculations in terms of accuracy [125] but do so at a much higher computational cost. They are often employed in quantum chemistry, where one is interested in moderately sized molecules rather than extended systems.

Electronic structure calculations can also be carried out using Monte-Carlo methods, i.e., methods that use repeated, randomized sampling. They portray some obvious advantages, such as a generally trivial parallelizability, good scalability, and a formally exact quantum mechanical treatment of the electronic structure problem. There exists a variety of methods that are based on Monte Carlo calculations, such as variational Monte Carlo [126], fixed-node diffusion Monte Carlo [127] or path-integral Monte Carlo [128, 129], useful for $\tau > 0$.

5.2.2 Machine Learning

Machine learning (ML) is a form of computational pattern recognition which has had a profound impact on multiple fields over the past decade. Advances in GPU hardware coupled with the development of high-level frameworks have led to many different applications in various fields. Some of these applications include autonomous driving [130], protein structure prediction [131], high-energy particle-physics data

analysis [132], generative language models [133], and materials discovery [134]. In this section, a brief background on machine learning is given, as well as some specifics relevant to the most common models in materials science applications.

5.2.2.1 Introduction

While there is no standard definition of machine learning, it is commonly defined as the study of computer algorithms that improve automatically through experience [135]. Depending on the nature of the model and its relationship with data, machine learning is loosely divided into the following categories.

Supervised Learning
Supervised learning refers to the class of problems where some prediction has to be made with respect to the ground truth. A model is trained on training data and its accuracy is tested by comparing its predictions with already known labels. The labels can be continuous real numbers (regression) or discrete states (classification). Deep learning [136] is a subset of supervised learning that has gained prominence in the last decade because of its effectiveness in mapping complex datasets.

Unsupervised Learning and Generative Models
When the dataset consists of only input data without ground truth, unsupervised learning algorithms are used. These algorithms are used for dimensionality reduction, to understand some underlying structure in the dataset, and to generate synthetic data with a similar distribution as the dataset even when such distribution would be complex, non-trivial, and intractable analytically.

Reinforcement Learning
In contrast to supervised and unsupervised learning, there is no pre-existing data set in reinforcement learning (RL) problems. The model, referred to as an agent for RL problems, has a set of actions that it can perform, which changes the state of the environment. Depending on the state, a score is assigned using the scoring function. The goal of the agent is to maximize this score. In complex environments, the state space may be too large to be enumerated as a discrete matrix. A neural network may be used to learn the mapping between the environment, action, and score. This is called deep reinforcement learning [137].

Machine learning models can be further divided into parametric and non-parametric models. When the model involves a finite set of learning parameters $\underline{\boldsymbol{W}}$, the model is said to be parametric [138]. These parameters are not set explicitly and are learned iteratively through the learning process. Models without a finite set of parameters are non-parametric. In non-parametric models, the set of weights $\underline{\boldsymbol{W}}$ is generally infinite.

Some examples of machine learning models include *supervised parametric models* (linear regression, neural networks), *unsupervised parametric models* (Gaussian mixture models, generative adversarial networks [139]), *supervised non-parametric models* (Gaussian Process Regression (GPR) [140], Decision Trees (DT) [141], Support Vector Machines (SVM) [142]), and *unsupervised non-parametric models* (k-means algorithm) [143].

Learning models can also be combined into ensembles. Using multiple learning models reduces overfitting and improves generalizability. Examples of ensemble learning techniques are Random Forests [144] and Gradient Boosting (GB) [145].

5.2.2.2 The Learning Process

Supervised parametric learning problems generally follow a common structure. For a dataset, a model with parameters $\boldsymbol{w}$ and $\boldsymbol{b}$ is to be fitted such that the cost function J is minimized. The components of the learning process are described broadly.

Dataset
The dataset consists of N samples of dimensionality D_x, $\underline{\boldsymbol{X}} = \{\boldsymbol{x_1}, \boldsymbol{x_2}, \ldots, \boldsymbol{x_N}\}$, where $\boldsymbol{x}_i \in \mathbb{R}^{D_x}$, and labels of dimensionality D_y, $\underline{\boldsymbol{Y}} = \{\boldsymbol{y_1}, \boldsymbol{y_2}, \ldots, \boldsymbol{y_N}\}$, where $y_i \in \mathbb{R}^{D_y}$. This dataset is split into training and test datasets. The *training data* is a subset of the dataset that is used to train the model. Using the known data points $\underline{\boldsymbol{X}}_{train}$ and the corresponding labels $\underline{\boldsymbol{Y}}_{train}$, the model is adjusted to find the best possible mapping in the training set. Once trained, a model is only useful if it can be generalized to previously unseen data. This performance is quantified by using the model to predict labels for unobserved data for which true labels are known. The *testing data* is used to measure the performance of the model by comparing predicted labels with true labels.

Descriptors
The ML models act on inputs in real space $\mathbb{R}$ but the raw data might not necessarily belong to $\mathbb{R}$, or the raw data might include much more information than needed for the task, thus making ML computation costly. In both cases, domain knowledge can be used to process the input data into tractable forms that are more suitable representations of the underlying problem for the machine learning task. The process of designing new representations for raw input data is called feature engineering and the resultant representations are called features or descriptors [146].

Model
The learned mapping between input data $\underline{\boldsymbol{X}}$ and labels $\underline{\boldsymbol{Y}}$ is represented by the model f. It is a function $f : \mathbb{R}^{D_x} \mapsto \mathbb{R}_{\frown}$. This mapping depends on parameters $\boldsymbol{w}$ and $\boldsymbol{b}$ of the model. Once trained on the known labels, the model is used to make predictions for previously unseen data (inference).

Loss Function
The loss or cost function, denoted by L or J, is a quantitative measure of the penalty due to error. The cost function changes as the parameters $\boldsymbol{w}$ and $\boldsymbol{b}$ are adjusted. The average loss value over the training set is defined as empirical risk. During the training phase, the parameters are adjusted to minimize the empirical risk. The difference between loss during training and loss during testing is referred to as generalizability error. Since only training data is observed by the model, the parameters might be tuned such that it performs well on the training set (low empirical risk) but performs badly on unobserved data (high generalizability error). This is called overfitting. When

the model does not capture the complexity of the underlying function sufficiently, it is called underfitting. Commonly used loss functions are mean squared error and cross-entropy loss [138].

Optimisation Scheme
Adjusting parameters in response to the cost function is governed by the optimization scheme. This is often done by gradient descent algorithms [147]. The parameters are adjusted iteratively according to the gradient of the loss function with respect to the parameters

$$\boldsymbol{w}_{t+1} = \boldsymbol{w}_t - \eta \nabla J. \tag{5.25}$$

Here, the learning rate η is a parameter that controls the rate of change of model parameters.

Hyperparameters
The parameters used in the learning process that are set beforehand are called hyperparameters. This is in contrast to the model parameters which are iteratively updated during the learning process. The choice of hyperparameters is important because model performance can be considerably impacted by it. Examples of hyperparameters are the number of iterations in the training process, convergence criteria for the loss function, the learning rate for gradient descent, and the architecture of a neural network.

5.2.2.3 Supervised Learning Models

A brief description of commonly used supervised learning models in materials science is given in the following.

Linear Regression
Linear regression [148] is one of the most used algorithms for regression tasks. Given the training dataset of N points, $(\underline{\boldsymbol{X}}_{train}, \boldsymbol{y}_{train})$ consisting of labeled data, regression is used to determine the mapping between input $\boldsymbol{x}_i$ and prediction y_i^*. For illustration, we use scalar dependent variables $y_i^* \in \mathbb{R}^1$ with independent variables $\boldsymbol{x}_i \in \mathbb{R}^{D\times 1}$. Given an input $\boldsymbol{x}_i$, the prediction y_i is defined as

$$y_i = \boldsymbol{x}_i^T \boldsymbol{w} + b_i, \tag{5.26}$$

where weights $\boldsymbol{w} \in \mathbb{R}^{D\times 1}$ and bias $b_i \in \mathbb{R}^{1\times 1}$ are the learnt parameters of the model. These equations can be stacked as

$$\boldsymbol{y} = \boldsymbol{X}^T \boldsymbol{w} + \boldsymbol{b}, \tag{5.27}$$

where $\boldsymbol{y} \in \mathbb{R}^{N\times 1}$, $\boldsymbol{X} \in \mathbb{R}^{D\times N}$, $\boldsymbol{w} \in \mathbb{R}^{D\times 1}$, $\boldsymbol{b} \in \mathbb{R}^{N\times 1}$. The analytical solution for the optimal value of parameters $\boldsymbol{w}^*$ is given by

$$\boldsymbol{w}^* = \left(\mathbf{X}\mathbf{X}^T\right)^{-1}\mathbf{X}\boldsymbol{y}^*. \tag{5.28}$$

However, the computational complexity for calculating $\boldsymbol{w}^*$ is $\mathcal{O}(D^3)$ which becomes intractable for large data sets with high dimensions. For such intractable cases, the parameters are determined by iteratively minimizing the least squares cost function

$$J(\boldsymbol{w}) = \frac{1}{2}\sum_{i=1}^{N}\|y_i - y_i^*\|^2. \tag{5.29}$$

If the number and range of the parameters are ill-suited for the dataset, the model might reach a sub-optimal state due to overfitting or underfitting. To overcome this, a regularisation term is added to the cost function [138]

$$J(\boldsymbol{w}) = \frac{1}{N}\sum_{i=1}^{N}\|y_i - y_i^*\|^2 + \lambda\|\boldsymbol{w}\|_p \tag{5.30}$$

with a hyperparameter λ. When $p = 1$ (i.e., the absolute sum) is used, the model is called Least Absolute Shrinkage and Selection Operator (LASSO) [149]. With $p = 2$ (i.e., the sum of squares), the model is referred to as ridge regression [150]. Sure Independence Screening and Sparsifying Operators (SISSO) [151] is an extension to LASSO with better performance for feature selection. Kernel Ridge Regression (KRR) [152] is a non-parametric realisation of ridge regression. Instead of learned parameters $\boldsymbol{w}$ and $\boldsymbol{b}$, a kernel function measuring the distance between different data points is used.

Implementing a linear regression model is a relatively simple task and provides interpretable results. However, this technique is not suitable for classification problems and for modeling complex non-linear relationships in datasets that have a high degree of dimensionality.

When classification is needed, logistic regression [153] can be applied. Here, the input data is mapped to discrete labels by applying a sigmoid function on the linear regression model.

Neural Networks

Neural networks (NNs) are a promising approach to regression due to the *universal approximation theorem* [154] that shows that any function can be modeled by a sufficiently sophisticated NN. Deep learning has been widely applied to problems in different domains including image classification [155], linguistics [156], biomedical engineering [157], and cosmology [158]. However, neural networks also have some challenges in determining their hyperparameters and training routines that avoid overfitting.

Introduction

A neural network is essentially a set of nested linear regression functions with non-linear activation functions. The base unit of a neural network is a perceptron or artificial neuron [159]. For data point $\boldsymbol{x}$, the perceptron is defined as

$$f = \sigma(\boldsymbol{x}^T \boldsymbol{W} + \boldsymbol{b}), \tag{5.31}$$

where σ is a non linear activation function, $\boldsymbol{W}$ consists of the weights, and $\boldsymbol{b}$ denotes the bias. Note that the dimensions of $\boldsymbol{W}$ and $\boldsymbol{b}$ are chosen according to the desired network architecture. When nested together in layers, the model is known as a multilayer perceptron or NN. For layer j we have

$$\begin{aligned} f^{(j)} = \sigma^{(j-1)}\Bigg(\sigma^{(j-2)}\Bigg(\cdots\sigma^{(2)}\Bigg(\sigma^{(1)}\Big(\boldsymbol{x}^T\boldsymbol{W}^{(1)} + b^{(1)}\Big)^T \boldsymbol{W}^{(2)} \\ + b^{(2)}\Bigg)^T \cdots\Bigg)^T \boldsymbol{W}^{(j-2)} + b^{(j-2)}\Bigg)^T \boldsymbol{W}^{(j-1)} + b^{(j-1)}. \end{aligned} \tag{5.32}$$

The first layer, which accepts the input samples, is called the input layer, and the final layer is called the output layer. The layers in between are referred to as hidden layers. A layer is defined by the number of its neurons which is often denoted as width. A NN is defined by the number and structure of hidden layers. A network with a large number of hidden layers is called a deep neural network (DNN). This gives rise to the term deep learning [136].

Design of a Neural Network

The design of neural networks depends on the learning task. The components of a neural network are:

Activation function: The choice of activation function affects the performance and output of NNs [160]. Commonly used activation functions are Tanh and ReLU [161].

Loss function: The learning objective is quantified by the loss function. Sophisticated loss functions can be designed to achieve multiple objectives [162].

Optimisation scheme: Standard gradient descent can be a bottleneck for training DNNs with large datasets [147]. Stochastic gradient descent [163] with mini-batches is used to accelerate the training process. Learning rate and momentum [164] terms are used to reduce overfitting. These parameters can be tuned throughout training with sophisticated optimisation schemes like AdaGrad [165], ADAM [166] and BFGS [167, 168]. The gradients with respect to weights are calculated numerically using backpropagation [147].

Architectures

The operations performed by the neurons can be configured for each layer. The following are some of the most commonly used architectures:

Fully Connected Neural Network (FCN): Each neuron in a layer of FCNs is connected to every neuron in the next layer. Also called feed-forward NNs, FCNs are the earliest and simplest NN architecture [169].

Convolutional Neural Network (CNN): CNNs are useful for tasks with two-dimensional or three-dimensional data such as image classification [170], optical character recognition [171], and medical image analysis [172]. Popular CNN models include UNet [173] and AlexNet [174].
Graph Neural Network (GNN): Instead of a vector input, GNNs [175] can be used with data points in the form of arbitrary graph structures. This is helpful for many-body problems in cosmology [176] and molecular dynamics [177].
Recurrent Neural Network (RNN): RNNs are used for tasks that involve sequential or temporal evolution such as language translation [178], time series prediction [179], and text generation [133]. Long Term-Short Memory (LSTM) [180] is a popular RNN architecture.
Physics-Informed Neural Network (PINN): PINNs are used to solve partial differential equations. This is done by incorporating initial and boundary conditions in the loss term [181].

NNs are effective at modeling complex non-linear functions and can be generalized well to new data [182]. Shortcomings of NNs are that large training datasets are required for practically useful results and that the black-box models lack explainability and interpretability [183].

Gaussian Process Regression
Gaussian Process Regression (GPR) is a powerful non-parametric supervised learning technique with robust uncertainty quantification. Here, a Gaussian random variable $X \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ is defined as having the distribution:

$$P(\boldsymbol{x}; \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{\frac{D}{2}} |\Sigma|} e^{-\frac{1}{2}((\boldsymbol{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}))} \tag{5.33}$$

where $\boldsymbol{\mu}$ is the mean, Σ is the covariance matrix, and D is the dimension of the data.

A Gaussian process (GP) is a collection of random variables, any finite number of which have a joint Gaussian distribution [140]. It can be thought of as a distribution over functions. By constraining the probability distribution at training points, a learned function can be drawn over the test points. Instead of learning any learning parameters $\boldsymbol{w}$, the inference is done by constructing the appropriate covariance matrix and drawing functions from it. Let $\underline{\boldsymbol{X}^*}$ be the set of training points, $\boldsymbol{y}^*$ be the set of training labels, and $\underline{\boldsymbol{X}}$ be the set of test points. For a simple noise-free case, prediction functions for y can be drawn from the distribution

$$\begin{aligned} \boldsymbol{y} \sim \mathcal{N}(&\mathbf{K}\left(\underline{\boldsymbol{X}}, \underline{\boldsymbol{X}^*}\right) \mathbf{K}(\underline{\boldsymbol{X}^*}, \underline{\boldsymbol{X}^*})^{-1} \boldsymbol{y}^*, \\ &\mathbf{K}\left(\underline{\boldsymbol{X}}, \underline{\boldsymbol{X}}\right) - \mathbf{K}\left(\underline{\boldsymbol{X}}, \underline{\boldsymbol{X}^*}\right) \mathbf{K}(\underline{\boldsymbol{X}^*}, \underline{\boldsymbol{X}^*})^{-1} \mathbf{K}\left(\underline{\boldsymbol{X}^*}, \underline{\boldsymbol{X}}\right)\Big), \end{aligned} \tag{5.34}$$

where $\mathbf{K}_{i,j} = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$. The kernel function k defines the covariance between any two vectors $\boldsymbol{x_i}, \boldsymbol{x_j}$ in the dataset. Some simple and commonly used kernel functions are squared exponential and periodic kernels [184]. The choice of k is an important design decision and depends on the nature of the dataset. With appropriate kernels and

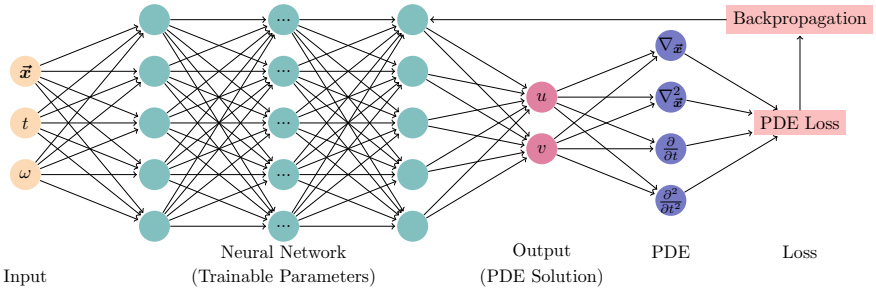


Fig. 5.1 Architecture of physics-informed neural networks

under certain conditions, GPs are equivalent to other machine learning methods such as neural networks [185] and Kernel Ridge Regression [140]. GPs provide in-built uncertainty quantification and are effective for small datasets. However, the inference step involves matrix inversion which scales as $\mathcal{O}(N^3)$ for explicit inversion. This makes GPs computationally expensive for larger datasets. Approximate inference to increase the scalability of GPs is an active area of research [186] and tractability has been greatly increased for sparse matrices with approaches such as Subset of Regressors [187].

Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) are constructed by encoding the constraints posed by a given differential equation and its boundary conditions into the loss function of a Fully Connected Network (Fig. 5.1). This constraint guides the network to approximate the solution of the differential equation. For a system f, with solution $u(\underline{\boldsymbol{x}}, t)$, governed by the following equation

$$f(u) := u_t + \mathcal{N}[u; \lambda], \underline{\boldsymbol{x}} \in \Omega, t \in [0, T] \tag{5.35}$$

$$f(u) = 0 \tag{5.36}$$

where $\mathcal{N}[u; \lambda]$ is a differential operator parameterised by $\lambda, \Omega \in \mathbb{R}^{\mathbb{D}}, \underline{\boldsymbol{x}} = (x_1, x_2, \ldots, x_d)$
with boundary conditions

$$\mathcal{B}(u, \underline{\boldsymbol{x}}, t) = 0 \quad \text{on} \quad \partial\Omega \tag{5.37}$$

and initial conditions

$$\mathcal{T}(u, \underline{\boldsymbol{x}}, t) = 0 \quad \text{at} \quad t = 0 \tag{5.38}$$

A neural network $u_{net} : \mathbb{R}^{D+1} \mapsto \mathbb{R}^1$ is constructed as a surrogate model for the true solution u,

$$f_{net} = f(u_{net}) \tag{5.39}$$

The constraints imposed by the system are encoded in the loss term L for neural network optimization

$$L = L_f + L_{BC} + L_{IC} \tag{5.40}$$

where L_f denotes the error in the solution within the interior points of the system, enforcing Eq. (5.35). This error is calculated for N_f collocation points.

$$L_f = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| f_{net} \left(\underline{\boldsymbol{x}}_f^i, t_f^i \right) \right|^2 \tag{5.41}$$

$$L_{BC} = \frac{1}{N_{BC}} \sum_{i=1}^{N_{BC}} \left| u \left(\underline{\boldsymbol{x}}_{BC}^i, t_{BC}^i \right) - u^i \right|^2 \tag{5.42}$$

$$L_{IC} = \frac{1}{N_{IC}} \sum_{i=1}^{N_{IC}} \left| u \left(\underline{\boldsymbol{x}}_{IC}^i, t_{IC}^i \right) - u^i \right|^2 \tag{5.43}$$

L_{BC} and L_{IC} represent the constraints imposed by the boundary and initial conditions, calculated on a set of N_{BC} boundary points and N_{IC} initial points respectively, with u_i being the ground truth. The different sampling domains are visualized in Fig. 5.2. The sampling of these collocation points depends on the problem domain. For smooth functions, uniform sampling can be used. For a problem with sharp discontinuities, other sampling techniques such as Halton sampling [188] may be used to incorporate more information from the discontinuities.

Once sufficiently trained, the network can be used as a solver for the PDE, potentially for a range of parameters λ [189].

5.2.2.4 Further Resources and Frameworks

Details on machine learning are covered in comprehensive textbooks such as Refs. [190–192]. A particular focus on deep learning is provided in Refs. [193, 194], while a historical overview can be found in Ref. [195]. The applications of machine learning in the natural sciences is the central theme in Refs. [196, 197].

Scikit-learn [198] is a popular python framework for regression and non-deep learning models. PyTorch [199] and TensorFlow [200] are commonly used deep learning frameworks with high level abstractions PyTorch-Lightning [201] and Keras [202], respectively. George [203] and GPyTorch [204] can be used for GPR with support for approximate inference.

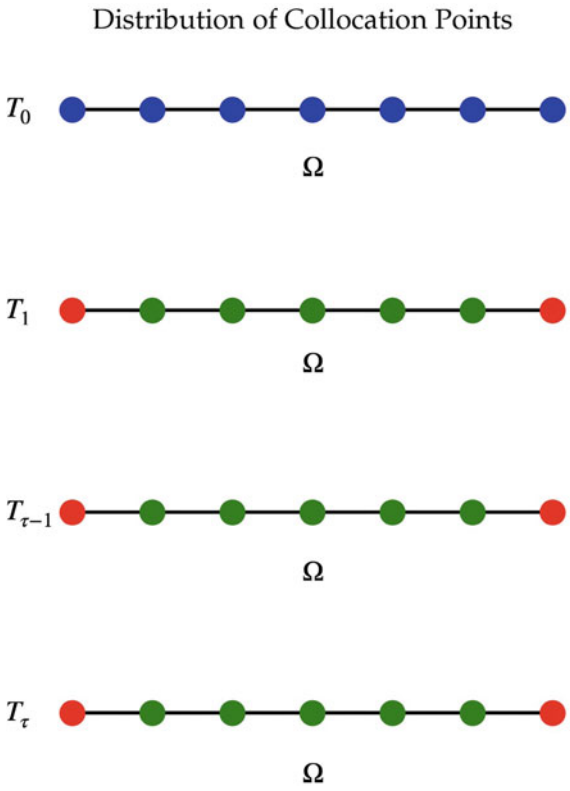


Fig. 5.2 Distribution of collocation points for calculating the loss function in Eq. (5.40)

5.3 Surrogate Modeling Workflows

5.3.1 Learning the Local Density of States with Neural Networks

Learning quantities that carry information on the electronic structure improves the performance of machine-learning-based electronic structure workflows [205]. Despite pioneering efforts to build density functional theory surrogate workflows based on electronic structure quantities [29, 206], an ML framework that captures a range of electronic structure observables is still subject to ongoing research. Problems in identifying such a scheme arise both from a theoretical point of view, i.e., which quantities to target, as well as from an ML perspective, as the quantities in question can be difficult to treat computationally.

The local density of states (LDOS) holds promise as a universal target for such workflows. Since the LDOS characterizes the energy spectrum around discrete points in space, models constructed around it are *in principle* independent of system size and can be used to evaluate important quantities of interest. Consequently, the LDOS

has been investigated as the basis for such surrogate models in Refs. [30, 31]. In the following, an overview of both theory and application of the workflow presented in Ref. [31] as implemented in the *Materials Learning Algorithms* package (MALA) [207] will be given.

5.3.1.1 Theory

The main obstacle within building density functional theory surrogate models lies in the fact that the Kohn–Sham total free energy functional defined by Eq. (5.20) is only implicitly a functional of the density. The kinetic energy $T_S[n]$ and the electronic entropy $S_S[n]$ are explicit functionals of the Kohn–Sham wave functions and energy eigenvalues, respectively. While this does not violate the fundamental notion of density functional theory that any system is determined solely by its (ground-state) density, as wave functions and energy eigenvalues can be regarded as functionals of the electronic density, it complicates the development of machine learning applications—simply approximating the electronic density does not suffice.

The LDOS, defined via the Kohn–Sham wave functions and eigenvalues as

$$d(\epsilon, \boldsymbol{r}) = \sum_j |\phi_j(\boldsymbol{r})|^2 \delta(\epsilon - \epsilon_j^\tau), \tag{5.44}$$

closes this gap by providing information in both the spatial as well as the energy domain. For each point in real space, it gives a probability distribution in the energy domain for the electronic states to be occupied. As such, it holds spatially resolved information about the electronic structure of a system. It is important to note that while Eq. (5.44) is formally correct and will be used for analytical derivations, it cannot be treated computationally in practice due to two reasons. Firstly, as discussed in Sect. 5.2.1.2, within the plane wave density functional theory framework we operate in, integration of the real space is performed as summation in $\boldsymbol{k}$-space, it follows

$$d(\epsilon, \boldsymbol{r}) = N_k^{-1} \sum_k^{N_k} \sum_j^{N_e} |\phi_{jk}(\boldsymbol{r})|^2 \delta(\epsilon - \epsilon_{jk}^\tau), \tag{5.45}$$

where N_k is the number of $\boldsymbol{k}$-points.

Secondly, the δ-distribution in Eqs. (5.44) and (5.45) cannot be treated explicitly by a computer simulation, since it is defined as having infinitesimal small width. Practically, it is approximated by a Gaussian function

$$\tilde{\delta}(\epsilon - \epsilon_j^\tau) = \frac{\exp\left[(-(\epsilon - \epsilon_j^\tau)^2/\sigma^2)\right]}{\sqrt{\sigma}}, \tag{5.46}$$

of width σ.

The LDOS can be used to construct powerful ML workflows since it is related to the electronic density

$$n(\boldsymbol{r}) = \sum_j f_j^\tau |\phi_j(\boldsymbol{r})|^2 = \int d\epsilon \; f^\tau(\epsilon) d(\epsilon, \boldsymbol{r}) \tag{5.47}$$

and the density of states

$$D(\epsilon) = \sum_j \delta(\epsilon - \epsilon_j^\tau) = \int d\boldsymbol{r} \; d(\epsilon, \boldsymbol{r}), \tag{5.48}$$

where $f_j^\tau = f^\tau(\epsilon_j^\tau)$ and $f^\tau(\epsilon)$ denotes the Fermi–Dirac distribution. The importance of the electronic density has already been discussed in Sect. 5.2.1, via Eq. (5.47) any LDOS based workflows gains access to all density based contributions in Eq. (5.20).

The density of states $D(\epsilon)$ on the other hand, which holds the same information on the distribution of electronic states *per-system* the LDOS holds *per-point*, can be used to calculate remaining energy distributions. Namely, it yields the band energy E_b through

$$E_b = \sum_j f_j^\tau \epsilon_j^\tau = \int d\epsilon \; f^\tau(\epsilon) \epsilon D(\epsilon), \tag{5.49}$$

and the band energy can be shown to yield $T_\mathrm{S}[\phi_j]$ by

$$T_\mathrm{S}[\phi_j] = E_b[D] - \int d\boldsymbol{r} \; v_\mathrm{S}^\tau(\boldsymbol{r}) n(\boldsymbol{r}). \tag{5.50}$$

In doing so, one obtains a representation of the kinetic energy that can be calculated without direct access to the Kohn–Sham wave functions. Likewise, it is straightforward to express the S_S in terms of D, it follows

$$S_\mathrm{S}(\epsilon_j^\tau) = -\sum_j \left[f_j^\tau \ln f_j^\tau + (1 - f_j^\tau) \ln(1 - f_j^\tau) \right], \tag{5.51}$$

$$S_\mathrm{S}[D] = -\int d\epsilon \; \left(f^\tau(\epsilon) \ln\left[f^\tau(\epsilon) \right] + \left[1 - f^\tau(\epsilon) \right] \ln\left[1 - f^\tau(\epsilon) \right] \right) D(\epsilon). \tag{5.52}$$

Drawing on this framework, we can finally express Eq. (5.20) as

$$\begin{aligned} A[d] = {} & E_b[D[d]] - k_\mathrm{B} \tau S_\mathrm{S}[D[d]] - E_\mathrm{H}[n[d]] \\ & + E_\mathrm{XC}^\tau[n[d]] - \int d\boldsymbol{r} \; v_\mathrm{XC}^\tau(\boldsymbol{r}) n[d](\boldsymbol{r}). \end{aligned} \tag{5.53}$$

This equation is the basis for the machine learning surrogate workflow discussed here. It allows direct, analytical access to the total free energy of any system, provided the LDOS can be approximated by an ML model to a sufficient degree. Via numerical

or analytical derivation, it also gives the atomic forces. Approximating the LDOS is realized by a feed-forward NN in a *per-grid-point* manner, i.e., for each point $\boldsymbol{r}$ in real space, a mapping of

$$d(\epsilon, \boldsymbol{r}) = M(B(j, \boldsymbol{r}))[\boldsymbol{\lambda}], \tag{5.54}$$

is performed, where $M[\boldsymbol{\lambda}]$ is a NN characterized by a set of hyperparameters $\boldsymbol{\lambda}$. In Eq. (5.54), $B(j, \boldsymbol{r})$ denote *descriptors* utilized to encode information at point $\boldsymbol{r}$ from which this mapping can be performed, while j denotes the feature dimension of these descriptors, similar to how the LDOS gives a vector in the ϵ-domain for each point in space. Simply put, Eq. (5.54) maps one vector (in the j-domain) to another one (in the ϵ-domain) per point in space, $\boldsymbol{r}$.

One is in principle not restricted to a specific type of descriptor, although MALA currently employs descriptors based on the Spectral Neighborhood Analysis Potential (SNAP) [20, 21, 208, 209], consequently named SNAP descriptors. As further outlined in Ref. [31], SNAP descriptors for any point $\boldsymbol{r}$ are calculated by placing each point at the origin and thereafter calculating the local atomic density $\rho(\boldsymbol{r})$ around this point via

$$\rho(\boldsymbol{r}) = \delta(\boldsymbol{0}) + \sum_{r_k < R_{\text{cut}}} f_c(|\boldsymbol{r}_k|, R_{\text{cut}}^{v_k}) w_{v_k} \delta(\boldsymbol{r}_k). \tag{5.55}$$

This local atomic density incorporates information about neighboring atoms k of species v_k, located at $\boldsymbol{r}_k$, and assigns a species-dependent weight w_{v_k}, within a cutoff distance $R_{\text{cut}}^{v_k}$. f_c denotes a switching function, which guarantees that contributions to $\rho(\boldsymbol{r})$ smoothly go to zero for $r_k = R_{\text{cut}}^{v_k}$. To construct $B(j, \boldsymbol{r})$ from $\rho(\boldsymbol{r})$, one then performs a coordinate transformation, a calculation of hyperspherical harmonic functions, and an expansion in Fourier coefficients; the details of which will not be discussed here but are given in Ref. [31]. For increased performance, $B(j, \boldsymbol{r})$ only includes these transformed and expanded components up to a degree defined by the hyperparameter J_{max}, thus J_{max} determines the dimensionality of $B(j, \boldsymbol{r})$. For example, when $J_{\text{max}} = 5$, it follows that any descriptor vector has 91 entries in the j-domain. How both $R_{\text{cut}}^{v_k}$ and J_{max} are determined practically will be discussed in the next section. The equations outlined here give rise to the workflow shown in Fig. 5.3, drawing on the LAMMPS [210], Quantum ESPRESSO [211–213] and PyTorch [199] codes.

5.3.1.2 Application and Results

To demonstrate the LDOS-based workflow implemented in MALA, a simple test system of 128 beryllium atoms at room temperature (298 K) and ambient mass density (1.896 g/cc) will be used. The data discussed here has been released as part of Ref. [214].

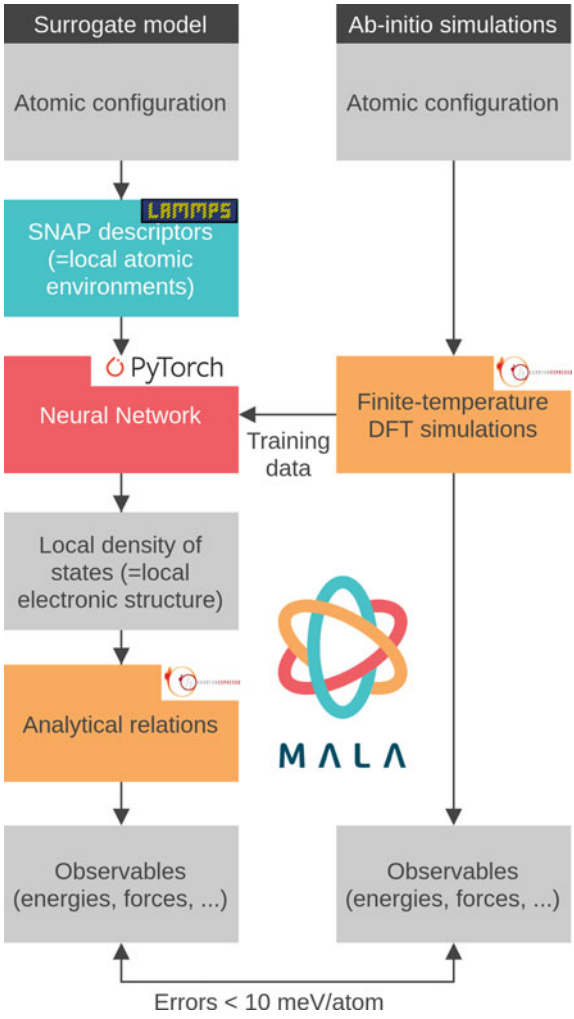


Fig. 5.3 General workflow of MALA

Snapshot Generation

MALA workflows typically require a small number of atomic configurations, for which the LDOS has been sampled accurately. The atomic configurations, also referred to as snapshots, need to be representative of the chosen temperature and mass density. Thus, the first step in creating a surrogate model is to perform an ab-initio MD simulation. In the present case, this was done using the Quantum ESPRESSO code with the Berendsen thermostat [215], approximating the canonical ensemble [216]. Density functional theory calculations were carried out at the Γ-point using an ultrasoft pseudopotential [217, 218], the PBE exchange-correlation functional [69] and a plane-wave basis set with a cutoff energy of 40 Ry. Care has to be taken to ensure that atomic configurations are only sampled after the system has been prop-

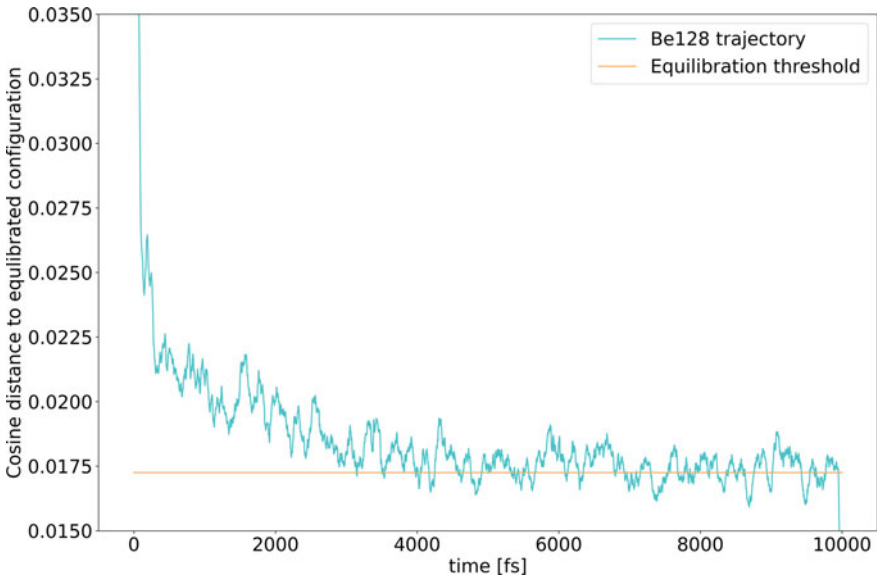


Fig. 5.4 Equilibration for a configuration of 128 beryllium atoms. The equilibration threshold was reached after approximately 4200 time steps

erly equilibrated. This is achieved by calculating an equilibration threshold from the radial distribution functions (RDF) of individual snapshots of the trajectory. Given that the very last snapshot of the trajectory can be considered equilibrated (which can be confirmed via visual inspection of the RDF), the cosine distance between the RDFs of the last and all other snapshots can be calculated. The average cosine distance over the last 10% of snapshots is then taken as the equilibration threshold, and should the distance fall below this value for a reasonable amount of snapshots, the system can be considered equilibrated. This analysis is shown in Fig. 5.4. To further ensure that the snapshots sampled from a trajectory are sufficiently diverse, snapshots are not sampled at random. Rather, beginning with the first equilibrated snapshot, a new snapshot is only considered for sampling given that it has a sufficiently large distance from the last sampled snapshot. Multiple distance metrics are possible here, in this case, the average euclidean distance between each atom in a snapshot to the atom closest to it in another snapshot was employed. Based on this sampling, 20 snapshots have been selected.

Density Functional Theory and LDOS Calculation

The accuracy and efficiency of MALA surrogate models are chiefly dependent on the quality of the LDOS used. "Quality" in this context relates to two aspects, first a minimal error when evaluating A_{tot} via the LDOS based Eq. (5.53) compared to the density and wave functions based Eq. (5.20) and secondly a smooth LDOS, i.e., one that is free of unphysical oscillations. Ensuring the latter does not only ease the machine learning procedure itself by approximating a smooth rather than a noisy

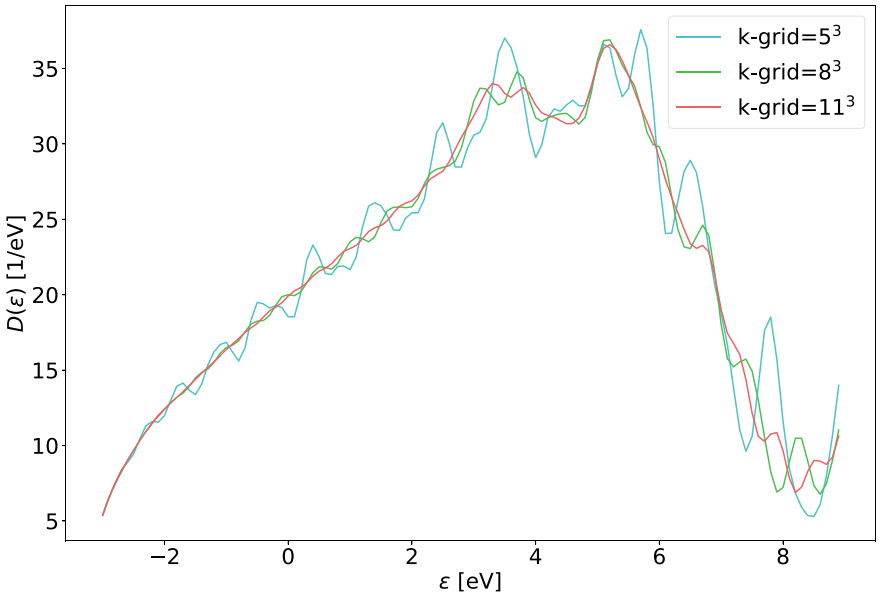


Fig. 5.5 Comparison of DOS for different $\boldsymbol{k}$-grids. Unphysical oscillations can be seen for small $\boldsymbol{k}$-grids, demanding high-fidelity calculations for this type of workflow

function but further makes analysis of the resulting DOS easier. As first discussed in Ref. [31] for an aluminum system, adjusting the LDOS in this regard is only truly possible via adjusting the $\boldsymbol{k}$-grid. Choosing σ in Eq. (5.46) too large may lead to a smooth LDOS by applying very broad Gaussian functions to the wave functions, yet such an LDOS is not physically correct, as confirmed by energy comparisons. These results have been confirmed for beryllium in [219]. We are thus simply left with the task of choosing an appropriate σ, in this case adhering to Ref. [31] with $\sigma = 0.2$ eV and determining the appropriate $\boldsymbol{k}$-grid. This is done by comparing energies and the DOS illustrated in Fig. 5.5. In combination with the energy comparisons outlined above, we arrive at an optimal $\boldsymbol{k}$-grid of $11 \times 11 \times 11$ $\boldsymbol{k}$-points. Using this $\boldsymbol{k}$-grid, density functional theory calculations were performed consistent with the calculation parameters for the ab-initio MD calculations. Thereafter, the LDOS was sampled from resulting wave functions and energy eigenvalues on an energy grid of 250 values with an energy spacing of 0.1 eV.

SNAP Descriptor Calculation

Based on the snapshots determined in the first step of the workflow, we have to calculate the SNAP descriptors. As mentioned above, this is done using the LAMMPS code. From a user perspective, the calculation itself is trivial; however, one needs to determine the correct $R_{\text{cut}}^{\nu_k}$ and J_{max}. These values determine how well the SNAP descriptors capture local environments, and if they are chosen wrong, the machine learning model will never be able to correctly learn the data provided. As shown in

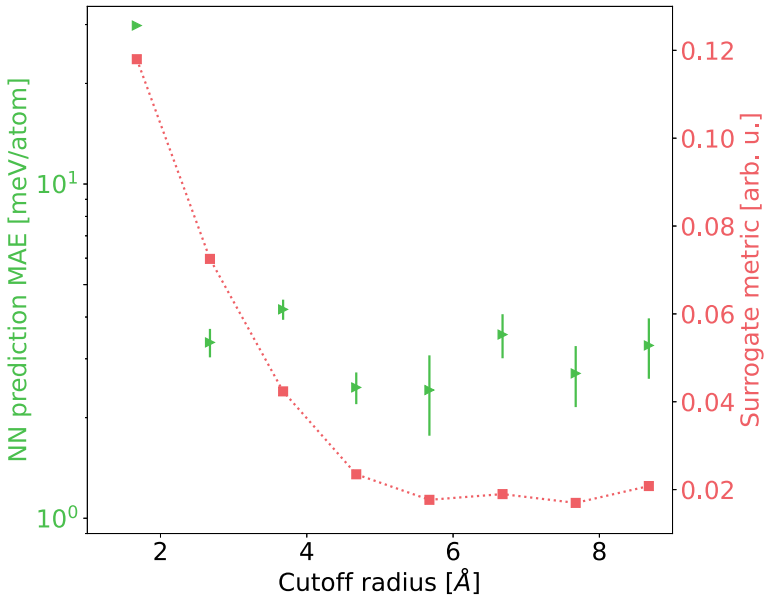


Fig. 5.6 Correlation of ACSD and network prediction errors (mean absolute errors on the test set) for different SNAP descriptor hyperparameters, as taken from Ref. [219]. Network prediction errors were measured over five networks with the architecture described in Ref. [31] and different initializations. For each data point, five networks have been trained and evaluated, with the antennas indicating the spread over network initializations

Ref. [219], it is not necessary to determine their ideal values through a costly and training-based hyperparameter optimization. Rather, they can be determined through a surrogate metric called the Average Cosine Similarity Distance (ACSD), which measures how drastically the similarity distributions of given LDOS and SNAP data differ. To put it more intuitively, small values of the ACSD indicate that similar SNAP vectors correspond to similar LDOS vectors. As shown in Fig. 5.6 this metric corresponds well to actual network prediction accuracies, yet can be calculated in a matter of minutes on a single CPU. Drawing on this we determine the optimal SNAP descriptor parameters for our system to be $R_{\text{cut}}^{v_k} = 4.676$ Å and $J_{\text{max}} = 5$. Technically, lower values J_{max} would suffice, as Ref. [219] suggests, yet for reasons of compatibility with prior aluminum models, we will stick with this slightly larger value.

Model Selection

As with any machine learning application, the accuracy and efficiency of MALA surrogate models depend chiefly on the architecture and tuning of the used machine learning model, in this case, a feed-forward NN. Different hyperparameter optimization techniques can be employed, from relying on intuition, to direct search [220, 221], advanced hyperparameter optimization libraries like Optuna [222], or experimental approaches such as Neural Architecture Search Without Training (NASWOT)

[223]. Reference [219] gives an overview of hyperparameter optimization techniques available within MALA and shows how training-free hyperparameter optimization such as NASWOT can be employed to identify optimal network architectures. For this small example, we choose the well-tested network architecture discussed in Ref. [31], an NN with four hidden layers of which the first three contain 800 neurons and the last layer contains 250 neurons.

Model Training and Testing
With data and model architecture in place, a surrogate model can be trained. Since the beryllium cells chosen in this example are smaller than the aluminum data used in Ref. [31], we can expect a larger number of snapshots to be necessary for comparable performance. Thus, for the training of the model, four atomic snapshots are used. One atomic snapshot is used as validation data, i.e., for monitoring model performance throughout training. Training is performed on a standard GPU using the PyTorch framework implemented in MALA. Five separate NNs were trained, to assess the performance of the training routine itself, using the ADAM [166] optimizer. Convergence was achieved in 86–154 epochs. The remaining 14 snapshots not used during the training procedure are used to confirm the model performance as a test set. As can be seen in Fig. 5.7, the resulting model is able to predict the total energy of all 14 beryllium snapshots very accurately. Apart from total energies, electronic density and DOS are important quantities usually calculated via density functional theory. Figure 5.8 illustrates how MALA predicts these quantities accurately as well. As mentioned above, it is advantageous to train the same network multiple times using different network initializations, to quantify the robustness of the model. Here we choose to train and test the model a total of five times. The mean absolute error (MAE) across all 14 configurations and network initializations amounts to 2.45 meV/atom, with a standard deviation of 0.58 meV/atom, thus confirming that our model is suitable to be used as a surrogate model for beryllium.

5.3.2 Physics-Informed Neural Networks

Physics-Informed Neural Networks [224, 225] are a new class of machine learning algorithms for the solution of partial differential equations. This is achieved by incorporating structured prior information derived from physical laws into the learning algorithm. In this section, we discuss the theory behind PINNs and demonstrate the utility of PINNs with a proof-of-concept workflow for solving the non-relativistic time-dependent Schrödinger equation of interacting and spinless particles.

Since PINNs can be used to solve systems of arbitrary resolutions once they are trained and generalize well over different parameter spaces [189], they might be used to accelerate the solution of time-dependent Kohn–Sham equations. In the next section, we provide a proof-of-concept workflow for solving a simple system using PINNs.

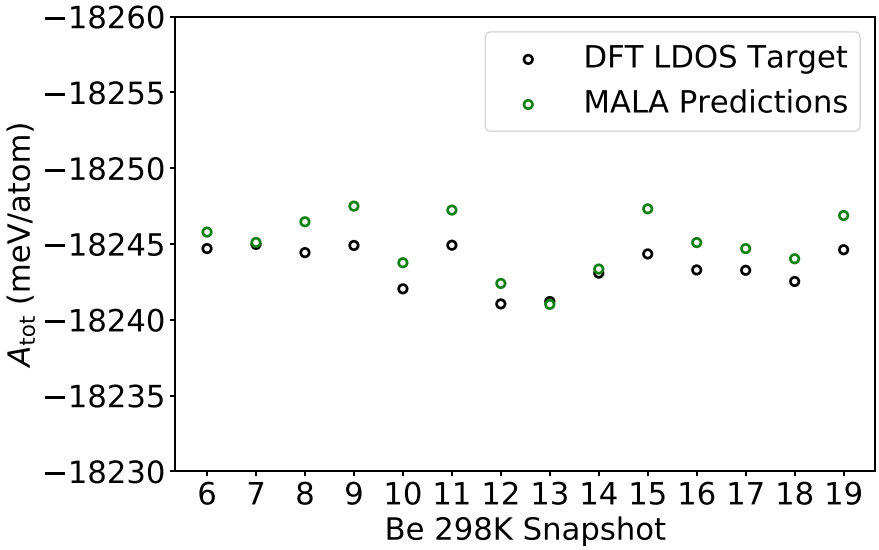


Fig. 5.7 Comparison of density functional theory calculated and machine learning predicted total energies across the test set of 14 beryllium snapshots

Workflow

The workflow for surrogate modeling using PINNs is demonstrated for the solution of the time-dependent Schrödinger equation

$$i\frac{\partial\phi(\boldsymbol{r},t)}{\partial t} - \hat{H}\phi(\boldsymbol{r},t) = 0. \tag{5.56}$$

As a prototypical example, we take the isotropic quantum harmonic oscillator whose Hamiltonian in three-dimensional space reads

$$\hat{H} = -\frac{1}{2}\nabla^2 + \frac{\omega^2}{2}|\boldsymbol{r}|^2, \tag{5.57}$$

where ω denotes its frequency. By separation of variables $\phi_{klm} = \phi_k(x)\,\phi_l(y)\,\phi_m(z)$, the Hamiltonian can be decomposed into a sum of Hamiltonian for each dimension in space

$$\hat{H} = \hat{H}_x + \hat{H}_y + \hat{H}_z. \tag{5.58}$$

For the sake of this demonstration, we will hence consider only the one-dimensional problem with Hamiltonian

$$\hat{H}_x = -\frac{1}{2}\frac{\partial^2}{\partial x^2} + \frac{\omega^2}{2}x^2. \tag{5.59}$$

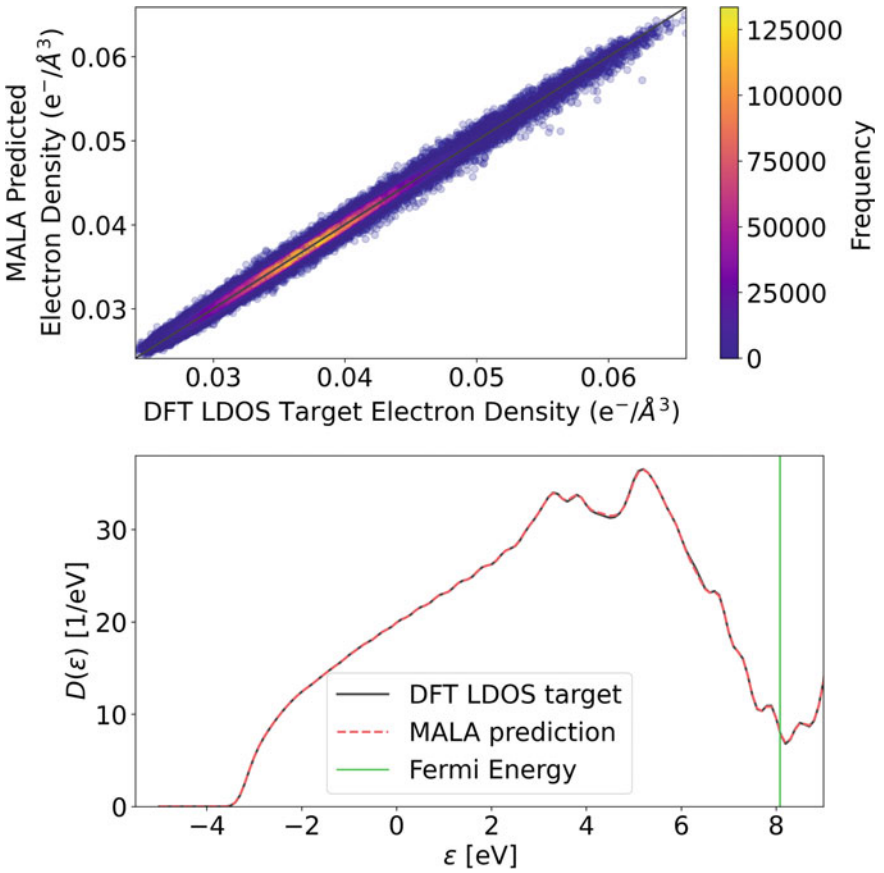


Fig. 5.8 Top: comparison of predicted and actual electronic density. Bottom: comparison of predicted and actual DOS

The eigenstates and eigenvalues of this one-dimensional harmonic oscillator are known analytically. They are

$$\phi_j(x) = (2^j j!)^{-1/2} H_j(\sqrt{\omega}x)\, \phi_0(x) \tag{5.60}$$

$$\epsilon_j = \omega(j + 1/2),\ j \in \mathbb{N}, \tag{5.61}$$

where H_j denotes the Hermite polynomials and

$$\phi_0(x) = \left(\frac{\omega}{\pi}\right)^{1/4} e^{-\omega x^2/2}. \tag{5.62}$$

A simple form of quantum dynamics is achieved by a superimposing these eigenstates. The superposition of two eigenstates

$$\phi_{m,n}(x,t) = \frac{1}{\sqrt{2}}\left[e^{-i\epsilon_m t}\phi_m(x) + e^{-i\epsilon_n t}\phi_n(x)\right] \tag{5.63}$$

yields a solution of the time-dependent Schrödinger equation, where the entire dynamics takes place due to the phases $\exp\left(-i\epsilon_j t\right)$.

Since the neural network is constrained to $\mathbb{R}$, the complex-valued solution can be represented as

$$\phi(x,t) = u + iv, \tag{5.64}$$

where $u = \mathrm{Re}(\phi)$ is the real part of the solution and $v = \mathrm{Im}(\phi)$ its imaginary part. The time-dependent Schrödinger equation can then be written as

$$\left(-\frac{\partial v}{\partial t} + \frac{1}{2}\frac{\partial^2 u}{\partial x^2} - \frac{\omega^2}{2}x^2\right) + i\left(\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial^2 v}{\partial x^2} - \frac{\omega^2}{2}x^2\right) = 0. \tag{5.65}$$

Training

Training data is generated on a high-resolution grid from the analytical solution described above. The surrogate model $f_{net} : \mathbb{R}^2 \mapsto \mathbb{R}^2$ is constructed, with inputs (x,t) and outputs (u,v). The loss function L is given by

$$L = L_f + L_{BC} + L_{IC}, \tag{5.66}$$

where

$$L_{BC} = \frac{1}{N_{BC}}\sum_{i=1}^{N_{BC}}\left(\left|u\left(x_{BC}^i, t_{BC}^i\right) - u^i\right|^2 + \left|u\left(x_{BC}^i, t_{BC}^i\right) - v^i\right|^2\right) \tag{5.67}$$

$$L_{IC} = \frac{1}{N_{IC}}\sum_{i=1}^{N_{IC}}\left(\left|u\left(x_{IC}^i, t_{IC}^i\right) - u^i\right|^2 + \left|v\left(x_{IC}^i, t_{IC}^i\right) - v^i\right|^2\right) \tag{5.68}$$

$$L_f = \frac{1}{N_f}\sum_{i=1}^{N_f}\left|f_{net}\left(x_f^i, t_f^i\right)\right|^2. \tag{5.69}$$

At each training step, the loss function is calculated on N_f collocation points, sampled randomly from the real-space grid.

Training Considerations

Architecture

The architecture and associated hyperparameters have a significant impact on the performance of the PINN. The learning model architecture can be customized depending on the nature of the domain. For example, CNNs, RNNs, and GNNs can be used for spatial, temporal, and interacting problems respectively. For the one-dimensional quantum harmonic oscillator workflow, we use an FCN with 7 layers of 10 neurons each. The choice of activation functions is constrained by the fact that they have to

be $(n+1)$ times differentiable for n-order PDEs, because the gradient of the loss function Eq. (5.66) is needed for backpropagation. Since the governing equation is of second order, we use tanh activation functions because they are thrice-differentiable. Optimizer selection is important for convergence of the learner and to avoid solutions at local minima. It has been shown that a combination of Adam [166] at early training and L-BFGS [168] for later stages has been effective for solving a variety of PDEs through PINNs [189]. We use the Adam optimizer only given the relatively simple nature of this problem.

Collocation Points
The accuracy of the NN increases with an increase in the density of collocation points. However, the computational resources required also increase exponentially with the increase in density of points, plotted in Fig. 5.9. This can lead to training bottlenecks, especially for high-dimensional systems. A trade-off has to be made between the desired accuracy and the number of collocation points because training the NN on a large number of points may lead to overfitting and adversely affect generalizability. The distribution of points can also be customized according to the problem domain. The collocation point density could be increased around areas of sharp discontinuities to capture more information about the domain. For the one-dimensional quantum harmonic oscillator, we use 1000 points per unit length of the domain interior with uniform random sampling.

Results
The domains for the quantum harmonic oscillator are: $x \in [-\pi, \pi], t \in [0, 2\pi]$ with fixed boundary conditions

$$u(-\pi, t) = 0 \qquad v(-\pi, t) = 0, \tag{5.70}$$
$$u(\pi, t) = 0 \qquad v(\pi, t) = 0, \tag{5.71}$$

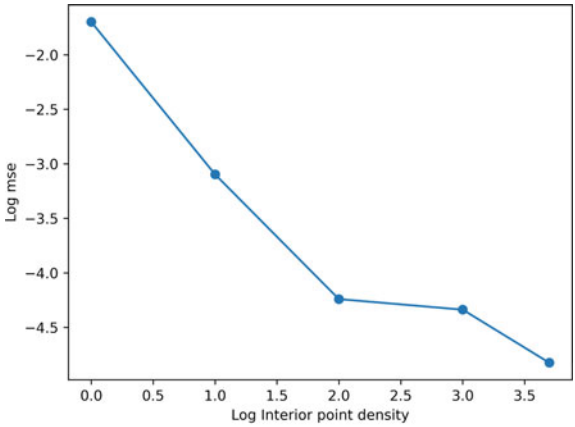
and initial conditions

$$\begin{aligned} u(x, 0) &= Re(\phi_{m,n}(x, 0)) \qquad v(-\pi, t) = Im(\phi_{m,n}(x, 0)), \\ &= \phi_{m,n}(x, 0) \qquad = 0. \end{aligned} \tag{5.72}$$

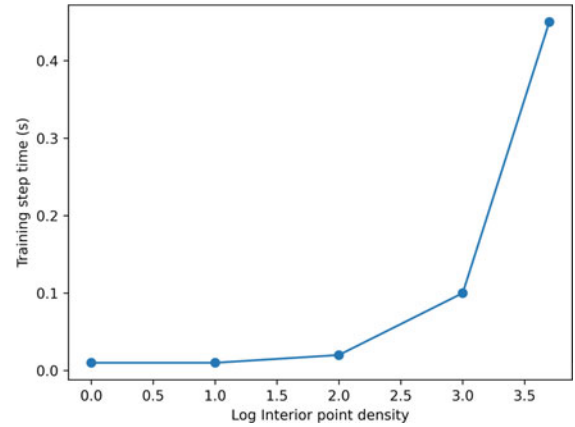
The following results were generated using TensorFlow [200] and the Modulus framework (v21.06) [226]. We show the utility of PINNs across various eigenstates and the frequency which is the primary parameter of the differential equation.

Simple Case
The initial surrogate model was created for a one-dimensional quantum harmonic oscillator with fixed parameter $\omega = 1$. The initial state is chosen as a superposition of the ground and the first excited state

$$\phi_{0,1}(x, t) = \frac{1}{\sqrt{2}} \left(e^{(-i\epsilon_0 t)} \phi_0(x) + e^{(-i\epsilon_1 t)} \phi_1(x) \right), \tag{5.73}$$



(a) Log Collocation Point Density vs Log MSE



(b) Log Collocation Point Density vs Time per training step

Fig. 5.9 Performance (**a**) and training time (**b**) with respect to the collocation point density

where the analytical solution is

$$\phi_{0,1}(x,t) = \frac{1}{\sqrt{2}}\sqrt[4]{\frac{\omega}{\pi}}\exp\left(-\frac{\omega x^2}{2}\right)\left(\exp\left(-i\frac{\omega}{2}t\right)+\exp\left(-i\frac{3\omega}{2}t\right)\sqrt{2\omega}x\right). \tag{5.74}$$

The PINN solver accurately yields the solution as shown in Fig. 5.10. The error is quantified as Mean Squared Error (MSE) between true and predicted values. The left column shows the true value for u, v, and the probability density $|\phi|^2$. The center column shows the corresponding predicted PINN solution. The right column shows the difference between the true values and the prediction. The MSEs are in the order of 10^{-5}. In addition, Fig. 5.11 illustrates the true (solid) and predicted (dashed) probability density at various time steps.

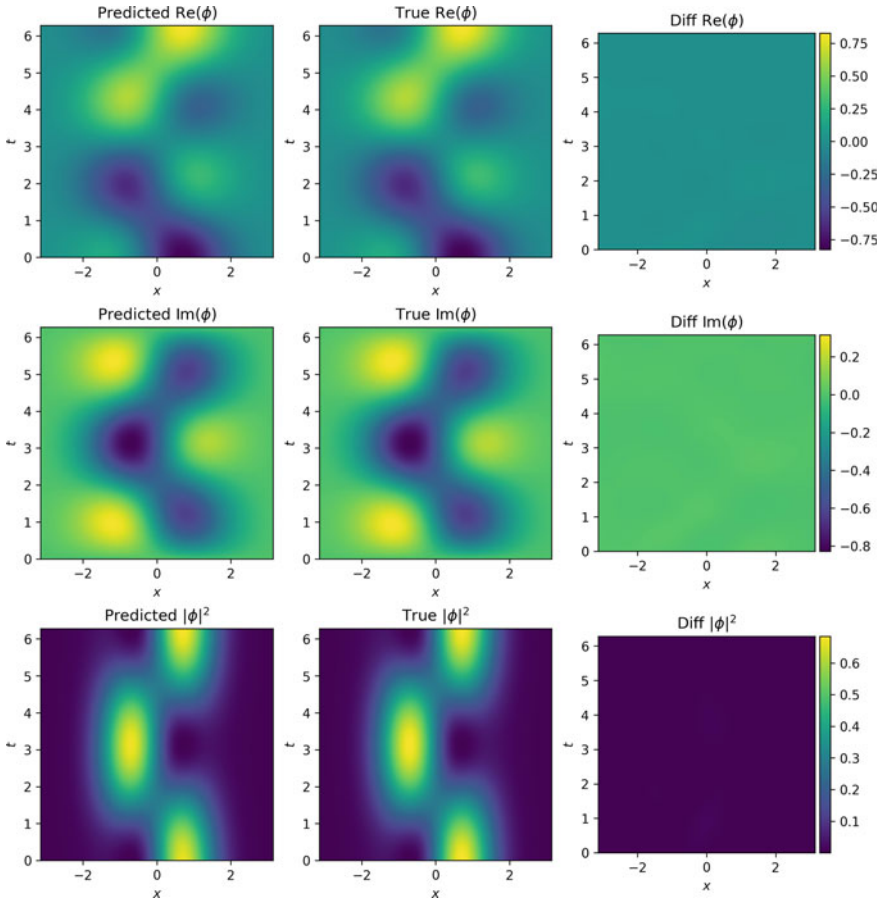


Fig. 5.10 Electron dynamics snapshots in the $x - t$ plane for true and predicted values for $\phi_{0,1}$ with $\omega = 1.0$. $MSE_u = 1.60\text{e}{-5}$, $MSE_v = 1.37\text{e}{-5}$. Here "True" value refers to the ground truth, "Predicted" value refers to the PINN predicted value, and "Diff" is the difference (True − Predicted)

Generalizability

To demonstrate the interpolative performance over a parameter range, the PINN was trained on a range of ω values. As in the previous example, the initial $\phi_{0,1}(x, t)$ was chosen. The neural network was trained on a fixed set of values $\omega \in [0.75, 2.0]$. After training, the inference was carried out for parameters $\omega \in [0.25, 2.25]$. Good performance was observed on previously unseen points within the training dataset range (interpolation) but accuracy diminished somewhat as the values of ω deviated from the training range during inference (extrapolation), as shown in Fig. 5.12. This can be attributed to the change in the width of the wave function relative to our fixed space domain Eq. (5.71). As ω decreases, the width increases, and the fixed boundary condition no longer holds true in the domain. As ω increases, the width

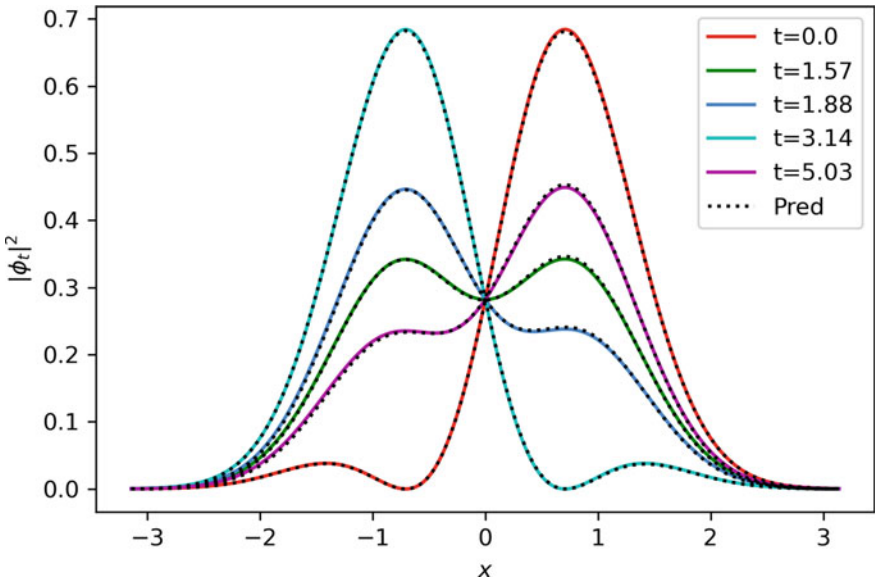


Fig. 5.11 Probability density $|\phi_t|^2$ at various time steps for $\phi_{0,1}$ with $\omega = 1.0$. Bold colors denote ground truth, dotted black line denotes the corresponding predicted density

decreases, and the solution becomes zero-valued across most of the domain with a sharp non-zero wave function in the center. The structure of the narrow wave function is difficult to be captured by PINN trained on a fixed domain as the network overfits to zero-valued points. This problem could be alleviated by using dynamic domains while training PINNs, such that the domain adjusts to the width of the wave function.

Higher Excited States

Finally, we investigate the time evolution of higher excited states. Here, the FCN-PINN was trained on an initial state

$$\phi_{0,3}(x,t) = \frac{1}{\sqrt{2}}\left(e^{(-iE_0t)}\phi_0(x) + e^{(-iE_3t)}\phi_3(x)\right). \tag{5.75}$$

The analytical solution in this case is

$$\phi_{0,3}(x,t) = \frac{1}{\sqrt{2}}\sqrt[4]{\frac{\omega}{\pi}}\exp\left(-\frac{\omega x^2}{2}\right) \left(\exp\left(-i\frac{\omega}{2}t\right) + \exp\left(-i\frac{7\omega}{2}t\right)\frac{1}{\sqrt{3}}\left(2\sqrt{\omega^3}x^3 - 3\sqrt{\omega}x\right)\right). \tag{5.76}$$

As is evident in Fig. 5.13, the predicted solution falsely converged to the static ground state ϕ_0 as time evolves. This incorrect time propagation is also reflected in the spatially and temporally resolved error which is in the order of 10^{-1}. To allevi-

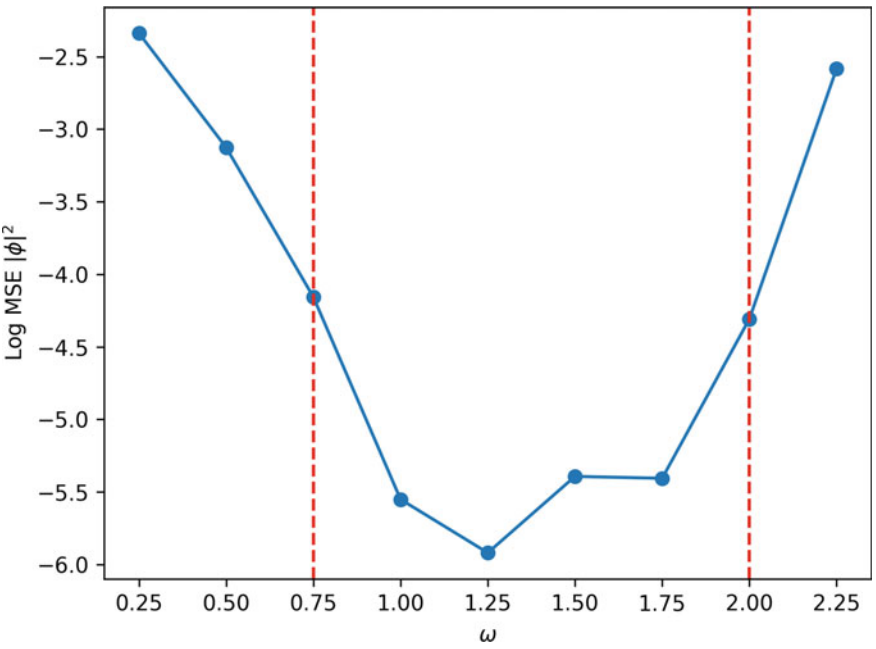


Fig. 5.12 PINN performance (log MSE) for various values of ω. Blue dots represent ω values used for inference. Training was done on systems with a set of random ω values within the region denoted by red dashed lines

ate this incorrect time propagation we consider RNNs which are NN architectures more suited to time series and time-dependent systems. We construct a PINN based on an RNN architecture with Gated Recurrent Units (GRUs) [227]. As illustrated in Fig. 5.14, the RNN-PINN performs much better, capturing the structure of the solution over time with an MSE in the order of 10^{-3}.

5.4 Conclusion

In this chapter, we have introduced the general background of density functional theory and machine learning. Based on this, we exemplified the use of machine learning to accelerate electronic structure calculations for both static and dynamic properties.

For static electronic structures, the MALA surrogate models give direct access to band energy, total energy, the density of states, and electronic densities, and can as such be used for a number of investigations themselves. Here, we have shown how such models can be used to accurately reproduce the electronic structure of beryllium, while Ref. [31] shows that MALA surrogate models are capable of correctly

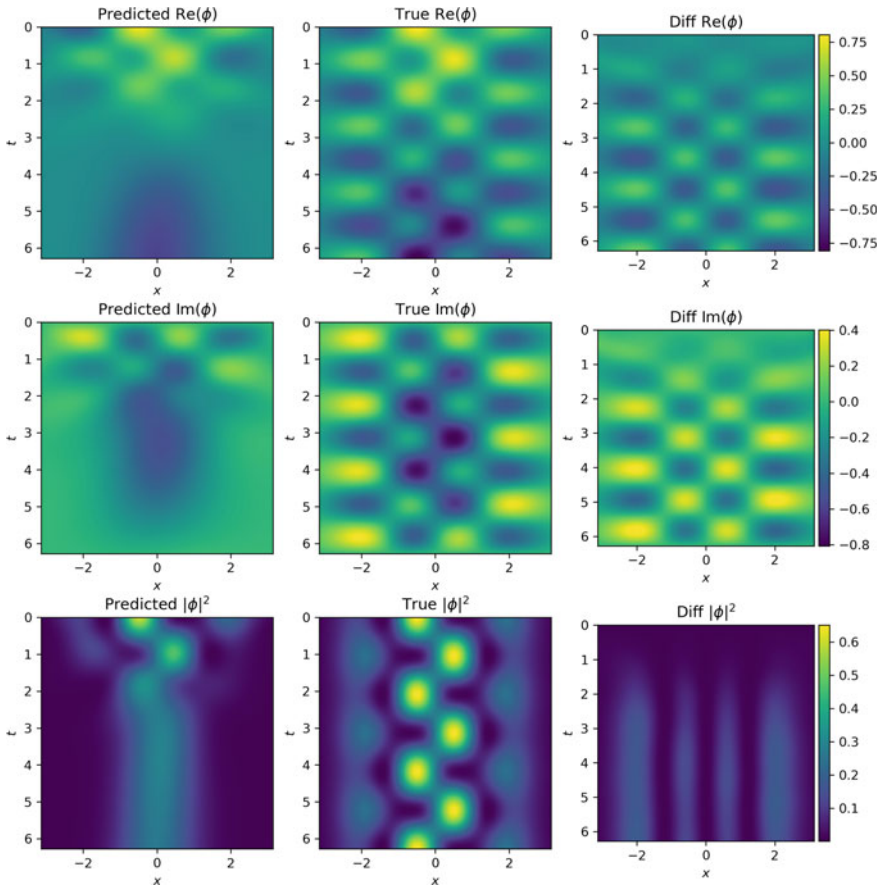


Fig. 5.13 Electron dynamics snapshots for $\phi_{0,3}$ with $\omega = 1.0$ in the $x - t$ plane obtained with an FCN-PINN. $MSE_u = 2.76\text{e}{-1}$, $MSE_v = 2.77\text{e}{-1}$

capturing the electronic structure of multiple types of configurations, in this case, liquid and solid aluminum. Coupling MALA to classical Monte-Carlo simulations thus holds the promise of investigating melting curves or phase transition. Likewise, the implementation of atomic forces into MALA, which is a current research project, will enable large-scale MD simulations. In all of the aforementioned applications, *size transfer* is implied, i.e., training a model on small numbers of atoms (e.g., 128 atoms) and then running inference on larger numbers of atoms, such as tens or even hundreds of thousands of atoms. Ongoing work and preliminary results indicate this is possible [228].

Regarding the dynamics of the electronic structure, PINNs can act as surrogate models. The primary advantage of PINNs over traditional numerical solvers is that they are mesh free and can be used to perform simulations at arbitrary resolutions once sufficiently trained. Depending upon the training regiment, PINNs can be gen-

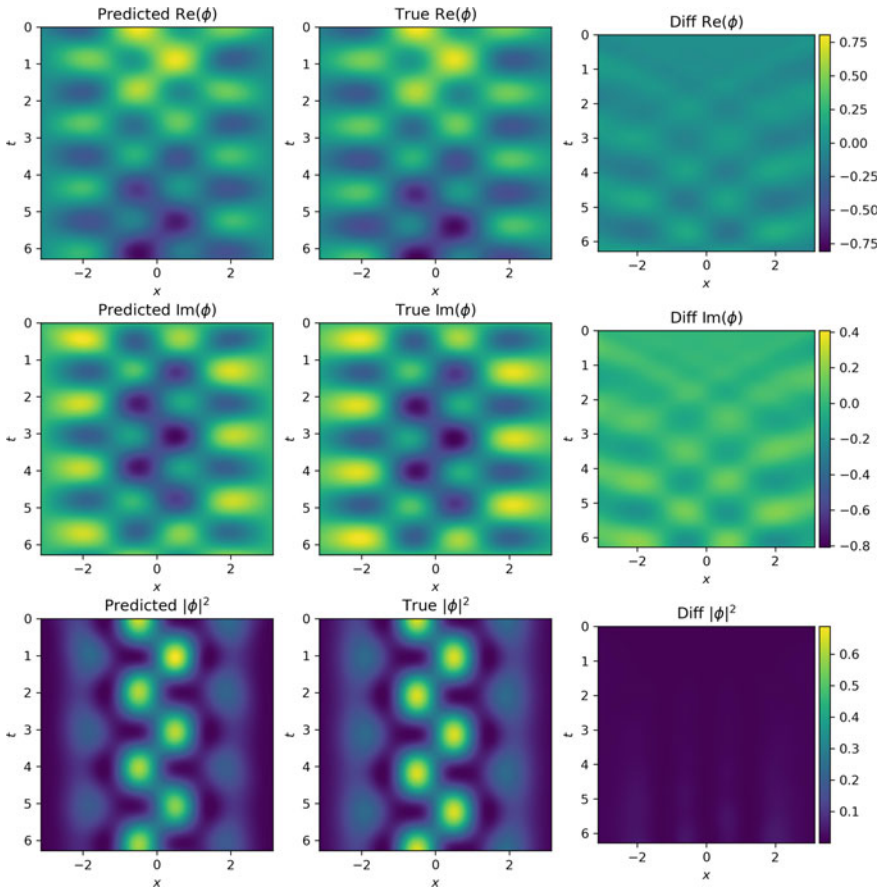


Fig. 5.14 Electron dynamics snapshots for $\phi_{0,3}$ with $\omega = 1.0$ in the $x - t$ plane obtained with an RNN-PINN. $MSE_u = 4.54\text{e}{-3}$, $MSE_v = 5.00\text{e}{-3}$

eralized to a range of the PDE parameter space. Their mesh-free nature and generalizability can be utilized for accelerating simulations of electron dynamics in the context of time-dependent density functional theory. The time-dependent Kohn–Sham equations in Eq. (5.21) are solved through iterative numerical algorithms which represent a significant computational cost [103]. A PINN framework would enable on-the-fly modeling of the electronic response properties of laser-excited or shock-compressed samples in various scattering experiments that are conducted at photon sources around the globe. This would enable fast simulations that generalize well over the input parameters of the experimental setup. Given the effectiveness of PINNs for the time-dependent Schrödinger equation, applying it in the context of the time-dependent Kohn–Sham equations seems very promising.

Recent work has shown the feasibility of accelerating electronic structure calculations with machine learning. The time has come to put these developments into action by developing production-level simulation codes for accelerated calculations. This will require efforts towards uncertainty quantification, active learning approaches, scalable machine learning, and automated artificial intelligence. Paired with fast-paced developments in machine-learning techniques, the future of artificial intelligence driven electronic structure calculations is bright. Previously unattainable applications in terms of length and time scales are now within reach.

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Chapter 6
Data Quality, Data Sampling and Data Fitting: A Tutorial Guide for Constructing Full-Dimensional Accurate Potential Energy Surfaces (PESs) of Molecules and Reactions



Jun Li and Yang Liu

Abstract Molecular dynamic properties, including spectra, collision energy transfer, kinetics and dynamics, are largely determined by the system's potential energy surface (PES), whose performance is essentially determined by data quality, data sampling and data fitting. Here, we provide careful discussions and considerations on these three key factors with ample interpretative examples, centering on accuracy, efficiency, and generality. Briefly, for a molecular system, a sufficiently large number of data points are sampled and calculated at some accurate electronic structure level. Then the PES can be fitted to a specific function, such as the permutational invariant polynomial-neural network (PIP-NN) method, which has been successfully applied to nearly 60 neutral or charged molecular systems with up to 8 atoms. For more and more complicated molecular systems, a NN based Δ-machine learning method is proposed to efficiently obtain high-level electronic energies of nuclear configurations in all dynamically relevant regions based on ample direct low-cost low-level calculations. Relevant perspectives and improvements are provided. Finally, a checklist is proposed to train and report a full-dimensional accurate PES, which is essential for publishers, authors, referees, readers, and users for reproducing, evaluating, and guiding.

Keywords Potential energy surface · Permutational invariant polynomial-neural network · Coupled cluster method · Δ-machine learning · Molecular reaction dynamics · Data sampling · Data quality

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6.1 Introduction

For a molecular system, the lighter electron moves much faster than the relatively much heavier nucleus. Therefore, for any nuclear motion, the electrons can adjust their positions instantaneously. The electronic and nuclear motions are thus reasonably assumed to be decoupled and can be treated separately. This so-called Born–Oppenheimer (BO) approximation consequently leads to the concept of the potential energy, which is consisted of the electronic energy and the nucleus-nucleus repulsion at a fixed nuclear position [1]. Varying nuclear coordinates within some ranges results the potential energy surface (PES), which is a hypersurface that describes the correlation between the energy and nuclear configuration in the system [2].

There exist some key points on the PES of a molecular system. Generally, for a reaction, these key points include reactants, intermediates, products, and transition states (TSs). All of them are known as stationary points. However, the TS corresponds to a configuration point that has one and only one imaginary vibrational frequency. All vibrational frequencies are real for other stationary points, namely, reactants, intermediates and products. The intrinsic reaction coordinate (IRC) analysis [3] of the TS can determine an minimum energy path (MEP) that connects the desired reactants and products. Properties of these stationary points (geometries, energies, and vibrational frequencies, etc.) and those along the MEPs are important and necessarily only required to predict the reaction selectivity, competition, and reactivity within the framework of the transition state theory (TST) [4], or Rice-Ramsperger-Kassel-Marcus (RRKM) theory [5, 6]. Despite powerful, efficient, and successful in many cases, these statistical theories may fail and nonstatistical dynamic effects have to be taken into account [7]. For instance, the minimum along the reaction coordinate may be not passed by in a reactive trajectory if a flat potential energy region, following a rate-limiting transition state, leads to multiple products [8]. Dynamic effect may also occur if the PES involves the so-called "bifurcating energy surface": the rate-limiting TS was followed by another TS that interconverts products, and the reactants can lead to different products through the same rate-limiting TS [9]. Besides, products can be reached via the so-called roaming mechanism without through any conventional tight TS. Such a roaming mechanism is again largely determined by the potential energy topology [10–12]. Further, it has been well recognized that different forms of energy (translations, vibrations, and rotations) may possess disparate influences on the detailed scattering dynamics. Indeed, the PES is a key factor that is related to molecular equilibrium structures, interactions, various spectra, energy transfer, reaction kinetics, and dynamics. Different research targets may have distinct requirements for the PES.

Theoretically, the potential energy values of any configuration spanned in chemically relevant regions for a target molecular system can be determined by various electronic structure methods. With potential energies of ample configurations for a specified molecule or molecular reaction system, fitting or interpolating techniques are employed to construct a multi-dimensional function of nuclear configurations. This multi-dimensional function is called the PES of the target system. The PES

discussed in this work is restricted to be full-dimensional and accurate (all were based on the CCSD(T) or MRCI calculations), although reduced-dimensional PESs [13, 14] may be efficient and reliable in some scattering or spectroscopy computations [15, 16]. Note that BO approximation may fail at where energies of different electronic states are close even degenerate. For such cases, the concepts of the BO approximation and the PESs remain useful. Nonadiabatically coupled PESs can be derived in a diabatic representation with the coupling given as the off-diagonal elements of a potential energy matrix. Fitting PESs of different electronic states and their couplings in a diabatic potential energy matrix is not discussed further in this work. Relevant excellent reviews and work are available [17–32].

Then various properties can be determined by running either classical or quantum dynamic calculations on the PES of the system. Consequently, although PES can't be measured directly by experiments, its performance significantly affects the dynamic outcome. As stressed in our previous review article, "an inaccurate PES is not expected to yield correct results no matter how accurate the dynamical method might be. … the lack of accurate global PESs represented the main stumbling block in our understanding of reaction dynamics" [33]. The energies and its derivatives can be also determined on the fly [34, 35]. This so-called direct dynamics approach is popular for exploring the potential landscape of a complex system without the need to construct a full-dimensional PES. However, the calculation can't be done at a high-level electronic structure level. It is also only appropriate for simulating fast processes by (quasi-)classical dynamics, as the wave packet is delocalized. Therefore, various quantum effects may be hard to be considered in direct dynamics simulations. Further, the statistical errors are generally large, because it's unrealistic to perform large number of direct dynamical trajectory calculations due to the high computational cost. Figure 6.1 displays the contour plot (with other coordinates relaxed) of the PES for a typical 1,3-dipolar cycloaddition reaction $N_2O + C_2H_2 \rightarrow$ oxadiazole as a function of the approaching distance R and the N_2O bending angle θ, whose definitions are also depicted in the same figure. Starting from the initial reactants, two reactants approach each other to hit the repulsive wall, and finally arrive at the product via a narrow region around the tight TS, as illustrated by three representative trajectories on the contour in solid lines [36] However, the direct dynamics on the B3LYP/6-31G(d) surface, starting from the TS with additional 0.6 kcal mol^{-1} energy deposited into the reaction coordinate, showed that the reaction took place mildly without hitting the repulsive wall but following the MEP [37]. Such a typical reactive trajectory is shown in Fig. 6.1 by the gray (short dot) line.

Significant computing costs can be saved by first constructing a high-quality PES based on accurate but expensive ab initio points covering all dynamically relevant regions. However, it is not an easy task to obtain reliable full-dimensional PESs, whose topology is complicated with reactants and products, saddle points and intermediates. As discussed above, a set of molecular configurations need be sampled first and their electronic energies (and/or forces) are computed at an appropriate level of theory. For a molecular system with N_{atom} atoms ($N_{\text{atom}} \geq 3$), the full-dimensional PES is dependent on $3N_{\text{atom}} - 6$ intramolecular coordinates. The sampling size

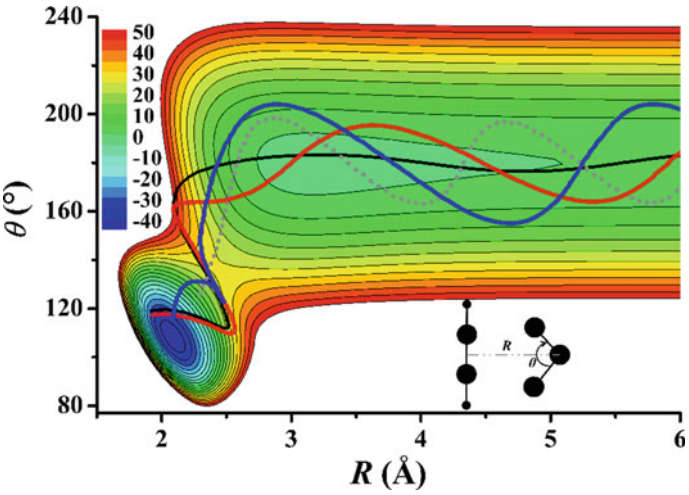


Fig. 6.1 Contour plot of the $N_2O + C_2H_2 \rightarrow$ oxadiazole reaction on the full-dimensional PIP-NN PES (energy in kcal mol^{-1}) as a function of R and θ, with R being the distance from the central N atom to the midpoint of the C–C bond, θ the bending angle NN′O. Other coordinates are relaxed. Three representative trajectories on the PES (starting from the asymptotic reactants) are shown in solid lines [36]. The gray (short dot) line corresponds to a direct dynamical trajectory starting from the TS [37]. Reprinted with permission from Ref. [36]. Copyright American Chemical Society

increases remarkably with the size of the system. Thanks to the huge development in electronic structure theory and the rapid progress in computing power, it is now straightforward to calculate reliable energies of hundreds of thousands of configurations for a molecular system with a reasonable size. There are many electronic structure methods of varying accuracies and computing speeds, such as the Hartree–Fock method, the Moller–Plesset (MPn) perturbation methods [38, 39], and more sophisticated electron correlation methods, including the highly accurate coupled cluster method [e.g. CCSD(T)] [40], the configuration interaction method [41], the complete-active-space self-consistent field (CASSCF) [42–44] or its variants, CASPT2, CASPT3, and the multi-reference configuration interaction method (MRCI) [44–46]. Generally, with a bigger basis set and more sophisticated post-Hartree–Fock electron correlation treatment, the result is generally more reliable. Recently, Dawes and his coworkers have reviewed recent developments in single- and multi-reference electronic structure methods that are used to obtain accurate energies for constructing global molecular PESs [47]. Nonetheless, one has to perform comprehensive tests to choose an appropriate electronic structure method for good and consistent description of all sampling data with acceptable calculation cost.

The next step is to sample sufficiently a large number of points in the configuration space. The sampling data points should have sufficient wide ranges with appropriate sampling densities in both the configuration and energy spaces for specified research targets. This is because that with any method to represent the PES, the performance of the PES can't be guaranteed for regions where no data is sampled. Generating

configurations on a random or regular grid is an efficient and convenient way for small molecules (under four atoms) with only a few degrees of freedom. However, this sampling approach becomes unfeasible for larger systems due to the exponential growth of the number of data points with the system size. For a specified system, we need to consider regions where the quantum/classical reaction dynamics simulations would access. Running classical dynamical simulations at some low-level methods, such as density functional theory (DFT), is efficient to sample points in the dynamically related regions. Note that the sampling is related to a specified research target. For instance, some regions are classically forbidden, but maybe accessed in quantum dynamic (QD) simulations. Various techniques, such as active learning and concurrent learning, were developed to select the most and probably the smallest representative samples effectively [48, 49].

Then the sampling data points, which are represented as a set of coordinates and its associated reliable electronic energies, are used to obtain an analytical function of the nuclear degrees of freedom, namely, the PES, whose functional form can't be derived generally based on physical considerations. Many forms of analytical functions have been suggested to represent the PES and summarized in the review by Jiang et al. [1]. Among them, the permutation invariant polynomial (PIP) method is powerful to represent the full-dimensional PESs with a reasonable accuracy and exact invariant property with respect to the permutation of like atoms [50, 51]. Neural network (NN) approach, a nonlinear regression method, is a universal approximator as it can in principle approximate any unknown multidimensional function with arbitrary accuracy by learning from a set of known data, namely, the sampling data points. Many NN methods have demonstrated excellent fitting performance in representing PESs of various systems in the gas phase [2], and large condensed systems from bulk solids via surfaces and interfaces to solutions [52, 53]. Indeed, many approaches have been now available for the representation of the PES, such as Gaussian approximation potentials (GAP) [54, 55], symmetric gradient domain machine learning (sGDML) [56], SchNet [57], PhysNet [58], gradient-domain machine learning (GDML) [59], Gaussian Process Regression (GPR) [60], PIP [50, 51], NN [53, 61], and NN with PIP inputs [1, 62, 63]. Please refer to these reviews and literatures cited therein for further detailed descriptions and discussions.

From the discussion above, it is clearly that developing any PES is consisted of the following three components: sampling data, determining their electronic energies, and fitting them to a specific function. This is essentially similar to machine learning (ML), which enables the computers to address problems by learning from data [64]. Apparently, data locates at the core position: no data, no learning; bad data leads to "wrong" learning, namely, unreliable PESs. In this Chapter, we will restrict the discussion to data quality, data sampling and data fitting for the PES development of gas-phase molecular systems, reactive or non-reactive. It is not our purpose to provide a comprehensive and rigorous review of all relevant investigations. Instead, we provide a tutorial guide for developing full-dimensional accurate PESs of small molecular systems based on a large number of datasets, whose electronic energies are computed at the CCSD(T) or MRCI level.

6.2 Data Quality

Learning can't happen without data [64] and unreliable data result in unreliable PESs and predictions. For developing a full-dimensional reliable PES of any chemical system, the first step is to obtain high-quality energies efficiently for large amounts of configurations. Then the essential question becomes how to select a suitable method for obtaining accurate electronic energies for the system and the research target.

6.2.1 How to Choose a Suitable Electronic Structure Method

No matter which method is used to represent the data (the configuration and its electronic energy), the accuracy of the PES reaches at most the accuracy level of the sampling data. Therefore, it is essential to obtain high-quality electronic energies for the sampling data in any molecular system. For practical reasons, one has to balance the calculation cost and accuracy, as hundreds of thousands of configurations are often computed for developing a reliable full-dimensional PES. For small-size molecular systems, generally coupled cluster [40] and configuration interaction theories [65] associated with reasonable size of basis sets are employed as they can effectively provide reliable electronic energies of configurations spanned in the dynamically relevant regions. Particularly, the coupled cluster approach with single, double, and perturbative triple excitations, CCSD(T), is well recognized as the gold standard for quantum chemistry, as it can provide results of chemical accuracy for a medium sized molecular system at a reasonable calculation cost [40].

For a specific molecular system, ample relevant experimental and theoretical investigations have been often reported for the equilibrium geometries and harmonic frequencies of reactants, products and intermediates, reaction energies or enthalpies, reaction profile, as well as kinetics that is related to the barrier height and the microscopic reaction mechanism. Therefore, the properties of all stationary points (geometries, energies, harmonic frequencies) can be evaluated first by various electronic structure methods. A quick comparison can be made to guide the choice of the electronic structure method.

If related information is rare in the literature, benchmark calculations are carried out to assess possible errors of various approximations employed in solving the exact electronic Schrodinger equation. Pioneering chemists have proposed several composite protocols to effectively obtain highly accurate electronic energies, such as the Gaussian-n theory [66–68], the Weizmann-n approach [69, 70], the complete basis set (CBS) model [71–73], the focal point analysis (FPA) [74, 75], and the High accuracy Extrapolated Ab initio Thermochemistry (HEAT) protocol [76–78]. In these methods, the accuracy can be systematically achieved by including various basis set extrapolation procedures, higher-order correlations, and some small corrections (core electron correlations, relativistic, spin–orbit, basis set superposition effect,

diagonal Born–Oppenheimer corrections, etc.). Most were tacitly neglected in theoretical calculations [79, 80]. As demonstrated in the $F + H_2O \rightarrow HF + OH$ reaction, the FPA and HEAT calculated barrier heights are about 2 kcal/mol lower [81] than that predicted at the MRCI with Davidson correction, MRCI+Q/aug-cc-pVTZ, level [82]. Following FPA, [74, 75] the geometries of stationary points were firstly determined by the all-electron (AE) UCCSD(T)-F12 method [83–85] with the basis set cc-pCVTZ-F12 [86], which takes the core-electron correlation effect into account. Then a series of single-point energy calculations are carried out to obtain the benchmark FPA or HEAT energy, namely,

$$E_{FPA} = E(FC\text{-}CCSDT(Q))/CBS + \Delta_{core} + \Delta_{rel} + \Delta_{core} + \Delta_{SO} + \Delta_{DBOC} \quad (6.1)$$

In Eq. (6.1), the complete basis set (CBS) limit is extrapolated according to results calculated at the aug-cc-pVXZ (X = 5 and 6) basis sets. The electron correlation contribution beyond UCCSD(T) is obtained by running UCCSDT/aug-cc-pVTZ and UCCSDT(Q)/aug-cc-pVTZ calculations. The core-electron correlation effect, Δ_{core}, is taken as the difference between AE and frozen core (FC) UCCSD(T)/aug-cc-pCVQZ energies. The scalar relativistic effect, Δ_{rel}, is computed by the AE Douglas-Kroll-Hess UCCSD(T)/aug-cc-pCVQZ calculation (DKH2). The spin–orbit correction, Δ_{SO}, can be taken from available experimental measurements or can be calculated. The diagonal Born–Oppenheimer correction, Δ_{DBOC}, determined by the Hartree–Fock calculation, has been demonstrated sufficiently accurate [76]. Sub-chemical accuracy is generally expected by FPA or HEAT, or similar composite approaches [87, 88].

The IRC analysis can provide important information along the reaction coordinate. The potential energy curve along the IRC is expected to significantly affect the reaction kinetics and dynamics [89]. Along the IRC, the reactants, products or intermediates, and transition states are connected. Accordingly, the electronic structure information is altered significantly. It is useful and efficient to test the performance of the electronic structure approach on the potential energy curve along the IRC. For instance, the reactant asymptote $F(^2P_u) + H_2O(X^1A_1)$ has three nearly degenerate lowest-lying electronic states, while there are two nearly degenerate lowest-lying electronic states in the product asymptote $HF(X^1\Sigma^+) + OH(X^2\Pi)$ [90]. From reactants to products, at least three low-lying electronic states are therefore needed to be considered in the electronic structure calculation: the two lowest electronic states correlate adiabatically to both reactants and products and one excited state in the reactant side correlates to $HF(X^1\Sigma^+) + OH(A^2\Sigma^+)$, namely, ground state HF and excited state OH [90]. In addition, the situation is even more complicated near the TS by charge transfer states [91]. Non-adiabatic effect may occur via derivative, Renner-Teller, and spin–orbit coupling, which are useful to rationalize the non-adiabatic transitions observed in Nesbitt's experiments [90, 92, 93]. Therefore, multi-reference methods, such as CASPT2 or MRCI, are suggested to provide reliable descriptions for all regions. Indeed, the tests by Deskevich, Nesbitt, and Werner showed that even the three state-averaged CASSCF (3-state-SA-CASSCF) results discontinuities in the IRC potential energy curve, which can be made much smoother

by including more electronic states, like 5-state-SA-CASSCF. However, artificial non-degeneracies were introduced in both reactant and product asymptotes with equal weightings for all states in the CASSCF calculations. Finally, the dynamically weighted SA-CASSCF (DW-SA-CASSCF) calculation was proposed to eliminate the discontinuities and the artificial nondegeneracies [91]. Indeed, the MRCI calculation, followed by 5-state-DW-SA-CASSCF computation [94], was used to develop the first full-dimensional ab initio based PESs for reactions F/Cl + $H_2O \rightarrow$ HF/HCl + OH [82, 95]. As expected, the near triple degeneracy is clearly shown in the pre-reaction well (F•••H_2O or Cl•••H_2O) with the double degeneracy in the post-reaction well (HO•••HF or HO•••HCl). As shown in Fig. 6.2a, the weight of state 3 reduces smoothly from the reactant side to the product side for the Cl + H_2O reaction. The weight of state 4 is significant near the TS and negligible in both reactant and product sides. It also shows that state 5 has a relatively small contribution along the IRC. Thus, it is sufficiently accurate to consider only the first five lowest lying electronic states for F/Cl + H_2O reactions [82, 95].

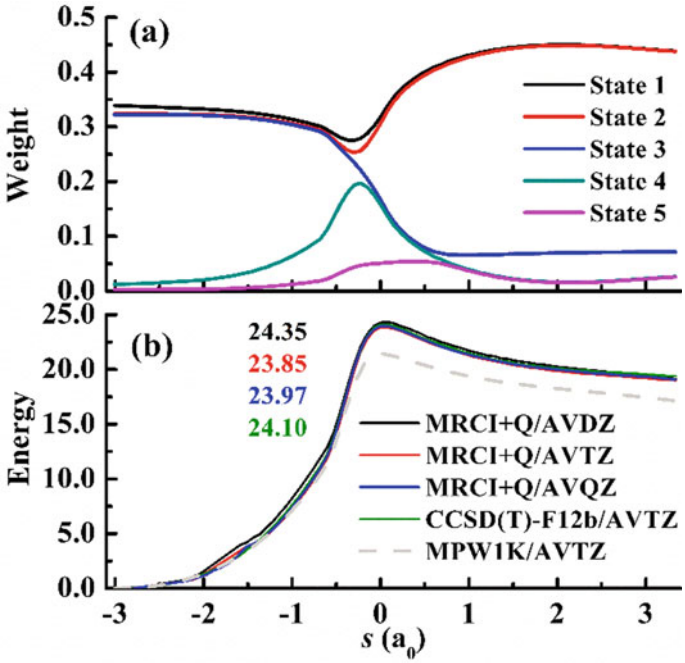


Fig. 6.2 **a** The weights of the lowest-lying five electronic states as a function of the reaction coordinate in the 5-state-DWCASSCF(15e,10o)/aug-cc-pVTZ calculations, and **b** comparison of the potential energies along the IRC at various levels and basis sets using DFT MEP geometries for the Cl + $H_2O \rightarrow$ HCl + OH reaction [95]. Adapted with permission from Ref. [95]. Copyright 2013 AIP

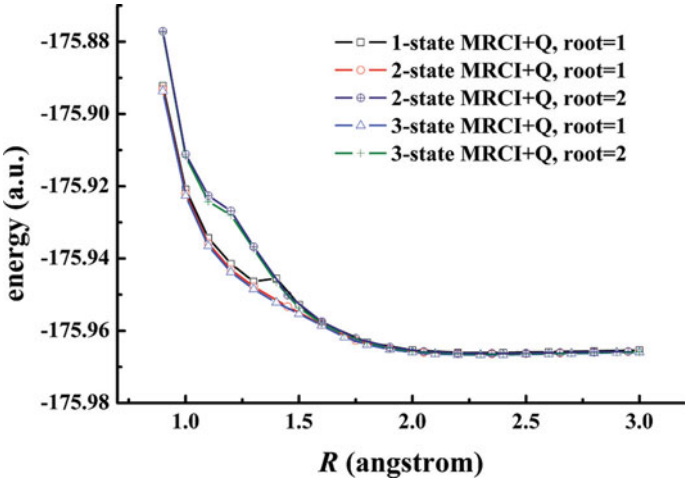


Fig. 6.3 Comparison of potential energies along the F•••H_2O distance by using one, two, and three-state MRCI methods based on the 5-state-DW-SA-CASSCF scheme. It is shown that one-state MRCI may converge to a wrong root, while two-state or three-state MRCI calculations give essentially the same results [82]. Adapted with permission from Ref. [82]. Copyright 2012 AIP

Another useful and efficient test is to plot some one-dimensional potentials in the entrance or exit channels. For instance, the potential energy changes along the center of mass distance between reactants or between products at different orientations. As shown in Fig. 6.3, for F and H_2O approaching each other, the 1-state MRCI calculation with the Davidson correction (1-state-MRCI+Q) may converge to a wrong root in certain regions with degenerate electronic states, thus leading to a non-smooth potential. While both 2- and 3-state MRCI schemes predicted essentially identical and smooth potentials with appropriate behavior. Hence, a 2-state MRCI scheme with the rotated reference Davidson correction (2-state-MRCI+Q_{rot}) was selected if only the ground electronic state is focused [82].

6.2.2 Single-Reference Versus Multi-reference Methods

It is noted that the interpolating methods do not guarantee performance in regions that are lacking data points. Thus, a large number of configurations should be sampled to cover all dynamically related regions for developing a full-dimensional PES of a chemical system. Their electronic energies are required to be computed by a suitable electronic structure method, which can be single- or multi-reference wavefunction methods. Note that most sampled points are far away from the stationary points and may encounter the convergence issue in the Hartree–Fock step, particularly for regions of multireference nature, such as bond-breaking regions. Even successfully converged, the Hartree–Fock calculation may converge to a wrong state and the

resulting energies might be not consistent. Multi-reference methods become the natural choice. As pointed out by Dawes and Ndengué in their review paper, "in a multireference PES, an inconsistency in a preliminary HF calculation may not be reflected in the final MRCI energy, as long as the CASSCF reference 'recovers' and converges correctly" [47]. Existing example PESs based on MRCI or CASPT2 calculations include HO_2 [96], O_3 [97], F/Cl + H_2O [82, 95], N_4 [98], OH + O_2 [99], CH_4 [100], etc.

Note a lot of effort needs be paid to select a suitable active space for all dynamical regions. Sometimes a small active space may be suitable. For instance, in the F + C_2H_6 reaction, a minimal active space of 5 electrons in 3 spatial 2p-like orbitals was used in the MRCI-F12+Q computations [101]. With such a small active space, various issues such as orbital switching may happen throughout the configuration space. Indeed, issues still exist even with full valence electrons included in the active space. Several procedures have been proposed and tested [47], although none of them are completely satisfactory. The central idea of these procedures is to keep the active space consistent everywhere in all dynamical regions.

Multi-reference methods are generally much more expensive than the single-reference method, CCSD(T). Even worse, the predicted results of the stationary points by the multi-reference methods are generally less accurate than those calculated by the single-reference methods. As discussed above for the F + H_2O reaction, the 2-state-MRCI+Q_{rot}/aug-cc-pVTZ calculation predicted a barrier height of 3.83 kcal mol^{-1}, while the UCCSD(T)-F12b/aug-cc-pVTZ result was 1.93 kcal mol^{-1} [82], much close to the FPA or HEAT values [81]. For the OH + $HO_2 \rightarrow H_2O + O_2$ reaction, a single point energy calculation required about 4 h by the SA-CASSCF (20e, 14o)/cc-pVDZ method on the Intel Xeon CPU E5-2680 v3 @ 2.50 GHz. The CASSCF estimated reaction energy is − 45.3 kcal mol^{-1}, about 25 kcal/mol higher than the experimental value − 70.96 kcal mol^{-1}. In addition, the resulting potentials along the MEP are not smooth, as shown in Fig. 6.4a. While the UCCSD(T)-F12a/aug-cc-pVTZ computation predicted a reaction energy of − 70.57 kcal mol^{-1} and only about 0.5–1.5 h were required for a single point energy calculation [102].

Unlike the CCSD(T) method, MRCI even with Q_{rot} does not have the size consistency and extensivity properties and it is prohibitively expensive to capture high-order correlations within the multireference realm. The simple and efficient external scaling strategy, proposed by Brown and Truhlar [103] and Ramachandran and Peterson [104], was used to bring the reaction energies and barriers of the F + H_2O reaction closer to the benchmark values or experiment while retaining the PES topology without introducing discontinuities [81, 105]. Simply, the final electronic energy by 2-state-DW-MRCI+Q_{rot}/aug-cc-pVTZ was scaled according to the following formula,

$$E_{scaled} = E_{DW\text{-}CAS} + \alpha\left(E_{MRCI+Q} - E_{DW\text{-}CAS}\right) \tag{6.2}$$

As shown, the external correlation energy, $E_{MRCI+Q} - E_{DW\text{-}CAS}$, is adjusted by the constant α. No external correlation energy corresponds to $\alpha = 0$ and full MRCI+Q_{rot} correlation energy corresponds to $\alpha = 1$. For $\alpha > 1$, additional correlation energy can

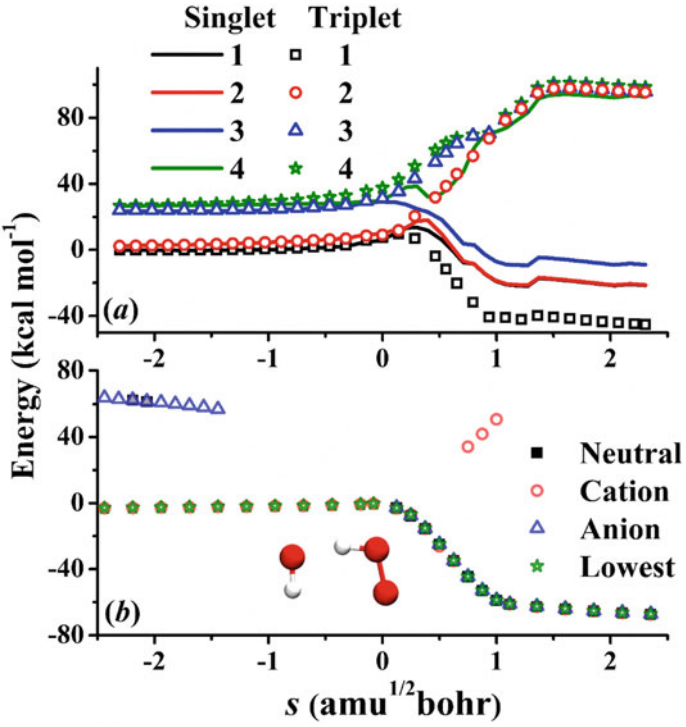


Fig. 6.4 **a** The SA-CASSCF(20e, 14o)/cc-pVDZ calculated energies of the low-lying electronic singlet and triplet states as a function of the reaction coordinate for the $OH + HO_2 \rightarrow H_2O + O_2$ reaction. **b** The CCSD(T)-F12a/aug-cc-pVTZ calculated potentials with the initial guess from the corresponding neutral, cation, anion species, and the lowest among them [102]. Reproduced from Ref. [102] with permission from the Royal Society of Chemistry

be recovered, which enlarged the active space to some extent. The value of α was then empirically determined by some known experimental and benchmark theoretical values. For $F + H_2O$, α was determined to be 1.135 by reproducing the FPA barrier height [81, 105]. As shown in Fig. 6.5, the potential energies along the MEP at the 2-state-MRCI+Q_{rot}/aug-cc-pVTZ (labeled as ab initio) were smoothly scaled and the scaled ab initio and the scaled PES are in good agreement [105]. Overall, the MRCI calculations are expensive, limited to small-size systems, and not as accurate as the gold standard CCSD(T). Even challenging, it is difficult to select a suitable active space that is appropriate for all dynamical regions.

Most existing full-dimensional accurate PESs are based on ample CCSD(T) calculated points. The explicitly correlated CCSD(T), CCSD(T)-F12, converges faster with respect to the size of the basis set [83, 84] than the conventional CCSD(T) calculations. CCSD(T)-F12/aug-cc-pVTZ has been widely [106] used to obtain accurate electronic energies of numerous configurations for constructing PESs of reactive and non-reactive systems due to its efficiency and accuracy. However, many issues may

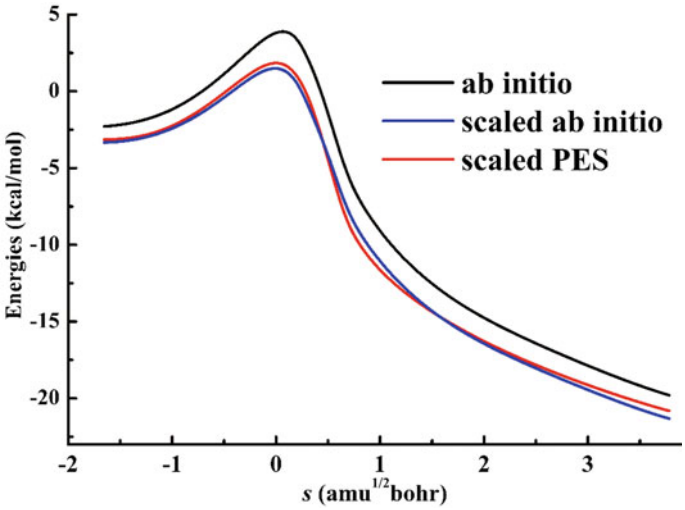


Fig. 6.5 The potential energies from various resources along the MEP of the $F + H_2O$ are compared [105]. See details in the text. Adapted with permission from Ref. [105]. Copyright 2013 AIP

encounter, as dynamical simulations often reach regions where different electronic states have small gaps in energy. For these multi-reference regions, the CCSD(T) and the Hartree–Fock calculation are very likely to converge to a wrong electronic state or the Hartree–Fock calculation can't successfully converge. If an incorrect high-state Hartree–Fock solution was obtained, the calculated properties are not reliable. Many useful options have been suggested and tested [47]. The Molpro manual [107] provides some useful solutions, such as using a small but reasonable basis set (cc-pVDZ, def2-SVP, aug-cc-pVDZ, etc.) firstly, adding a level shift in the HF calculation step, using initial guesses from its corresponding close-shell cation or anion, using initial guesses from DFT calculations with a small basis set, using initial guesses from its neighboring configuration, etc. If all fail, the robust MCSCF method can be launched. Dawes and Ndengué have discussed various options and gave some suggestions in their review [47]. Vaucher and Reiher proposed a fast and automated approach to detect and cure incorrect orbital convergence by the randomized orbital perturbation method [108].

The reaction $OH + HO_2 \rightarrow H_2O + O_2$ is of multi-reference in nature. Using initial guesses of the neutral, anion and cation species as the spin-restricted Hartree–Fock calculations, the following CCSD(T)-F12a/aug-cc-pVTZ calculations always predict some strange points of high energies along the MEP of this reaction, as shown in Fig. 6.4b. To make the PES smooth and reliable, for each configuration, three individual Hartree–Fock calculations were calculated with initial guesses of cationic, neutral, and anionic species, respectively. The lowest Hartree–Fock solution for the

neutral species was then employed in the CCSD(T)-F12a calculation. As shown in Fig. 6.4b, the potential along the MEP is smooth and the resulting exothermicity is also in good agreement with experiment [102].

For calculating ample points of reactive PESs, Győri and Czakó recently reported a pragmatic method, called ManyHF, which can automatically get better, lower-energy Hartree–Fock solutions by carrying out an empirically determined set of preliminary SCF calculations to generate multiple initial guesses. Tests for $CH_3OH + OH$, $C_2H_6 + F$, and $CH_3NH_2 + Cl$ showed that their method can make the following CCSD(T) energies reliably [109].

The electronic structure changes significantly along the $OH + SO \rightarrow H + SO_2$ reaction. The explicitly correlated multi-reference configuration interaction (MRCI-F12) [110–112] with the basis sets aug-cc-pVDZ and aug-cc-pVTZ revealed that for the entrance channel, the ground electronic state corresponds to $OH(^2\Pi)$ and $SO(^3\Sigma^-)$ with three singly occupied orbitals, while the remaining regions corresponds to a doublet state with one singly occupied orbital. Thus, the UCCSD(T)-F12a/aug-cc-pVTZ calculations for a doublet state correspond to $OH(^2\Pi) + SO(a^1\Delta)$, 22.7 kcal mol^{-1} higher than the ground electronic state $OH(X^2\Pi) + SO(X^3\Sigma^-)$, which should be well described by UCCSD(T)-F12a/aug-cc-pVTZ for a quartet state [113, 114]. The correlation is depicted in Fig. 6.6 [114]. Practically, for those points with the distance $R_{HO\cdots SO}$ larger than 2.0 Å, UCCSD(T)-F12a/aug-cc-pVTZ calculation is done for a quartet and doublet state, respectively. Then the lower energy is taken as its true ground-state electronic energy. One can see that clearly, a smooth and correct potential curve is obtained.

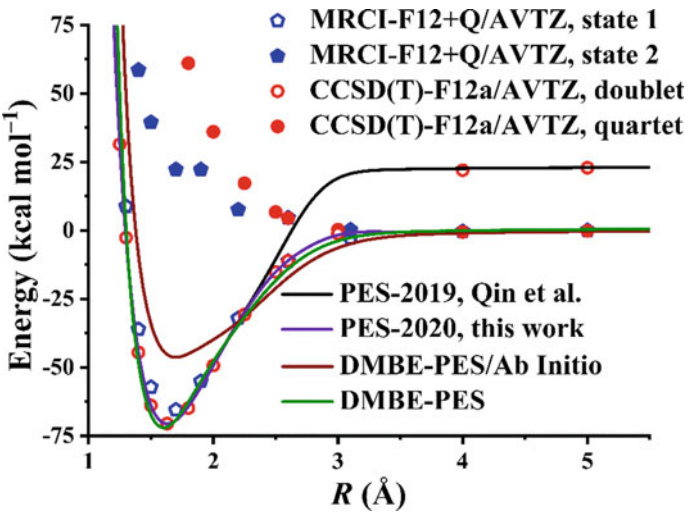


Fig. 6.6 Potential energy dependence along $R_{HO\cdots SO}$ at various levels of theory with other coordinates fixed at the HOSO equilibrium [114]. Reproduced from Ref. [114] with permission from the Royal Society of Chemistry

For PESs, reactive or nonreactive, the consistency of the electronic energy throughout the dynamical region is crucial. Generally, one electronic structure method is used for all sampling points, for a specified system. Additionally, detailed scrutiny and settings are carried out to make sure that the resulting energies are consistent everywhere. For some systems, different electronic structure methods are used for different regions. For instance, for describing the collision induced dissociation correctly, $H_2 + H_2O \rightarrow H + H + H_2O$ or $H_2 + H + OH$, most points are calculated at the level of CCSD(T)-F12a/aug-cc-pVTZ and minor points are calculated at the MRCI-F12+Q_{rot}/aug-cc-pVTZ level. Both electronic structure methods yield essentially identical potential energy curves along the H–H distance, $r_{HH} \leq 4$ Å, as shown in Fig. 6.7a, or along the H–OH distance, $r_{OH} \leq 2$ Å, as shown in Fig. 6.7b. For $r_{OH} > 2$ Å, the single-reference method CCSD(T)-F12a/aug-cc-pVTZ predicts a wrong behavior and the MRCI-F12+Q_{rot}/aug-cc-pVTZ method is used to obtain energies of points in this region. The combining of single and multireference methods for different regions is expected to provide accurate and consistent energies efficiently [115].

The accuracy of various density functional theory (DFT) methods needs be evaluated for developing full-dimensional reliable PESs. Schaefer and coworkers evaluated the performance of 49 popular DFT methods for relative energies of the stationary points of the $F + H_2O$ reaction. Their calculations showed that all predictions by the DFT methods were significantly different from the benchmark values calculated by CCSD(T)/V5Z [116]. Interestingly, the double hybrid XYG3 functional [117] has demonstrated accuracy and efficiency in describing several hydrogen abstractions, compared to the calculations at the CCSD(T) level [118]. Encouragingly, the PES of the $H + CH_4$ reaction, developed by the XYG3 functional, yielded QD results that were close to those on the highly accurate but expensive PES [119] based on CCSD(T)-F12/aug-cc-pVTZ calculations [118]. Significantly computational costs can be saved by DFT, whose general applicability, however, needs further careful evaluation. Czako and coworkers developed 20 analytical PESs for the $F^- + CH_3I$ S_N2 and proton-abstraction reactions with the energies computed at the levels of HF, MP2, several density function theories(DFT), coupled-cluster based methods associated with the basis sets, aug-cc-pVDZ and aug-cc-pVTZ. Quasiclassical trajectories (QCTs) were carried out on these PESs to reveal the influence of the electronic structure levels on the dynamic results [120].

6.2.3 Composite Methods

For each sampled point, the same electronic structure computing procedure is generally adopted to obtain consistent and well-behaved energies throughout the dynamically related regions. However, due to the steep computational scaling of the highly accurate electronic structure methods, full dimensional PESs for small molecules with 50–100 electrons were developed by the CCSD(T) or MRCI calculations. Various high-level composite methods were adopted to determine the electronic

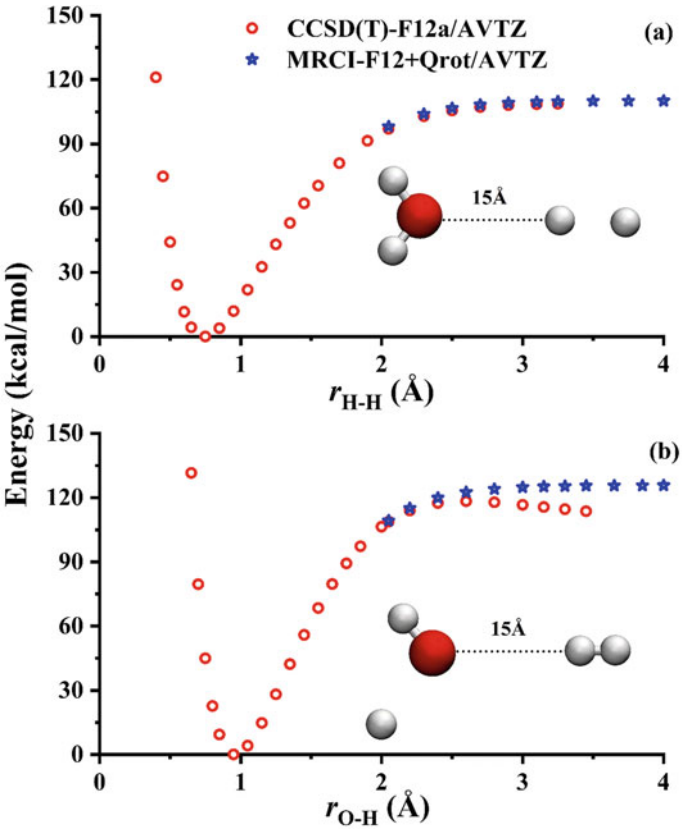


Fig. 6.7 **a** Potential energy as a function of the bond length H–H for $H_2 \cdots H_2O$, the ab initio energies are calculated at the level of CCSD(T)-F12a/aug-cc-pVTZ; **b** Potential energy as a function of the bond length O–H for $H_2 \cdots H_2O$, the ab initio energies before 2 Å are calculated by CCSD(T)-F12a/aug-cc-pVTZ and MRCI-F12+Q_{rot}/aug-cc-pVTZ is employed after 2 Å. Energies in kcal/mol are all relative to the reactant asymptote $H_2 + H_2O$. Reproduced from Ref. [115] with permission from the Royal Society of Chemistry

energies efficiently, particularly for complicated systems. For the $F + H_2O$ reaction, as mentioned above, the external scaling scheme was used to obtain highly accurate energies efficiently [81, 105]. The $F^- + CH_3CH_2Cl$ reaction system has 44 electrons with 21 degrees of freedom. The final energies of the sampled 35,474 points were determined by a composite strategy: CCSD(T)-F12b/aug-cc-pVDZ + (MP2-F12/aug-cc-pVTZ − MP2-F12/aug-cc-pVDZ). The explicitly correlated CCSD(T)-F12b method was expected to provide a reliable description of electron correlation while the energy difference between MP2-F12/aug-cc-pVTZ and MP2-F12/aug-cc-pVDZ was computationally less expensive to recover the basis set effects beyond aug-cc-pVDZ for CCSD(T)-F12b [121]. Similarly for the $Cl + C_2H_6$ reaction, the electronic

energies were computed at the same composite level of theory, with additional spin-orbit corrections in the reactant region being determined at the level of MRCI+Q/aug-cc-pVDZ [122].

For the $OH^- + CH_3I/CH_3F$ reaction, Tasi et al. found that CCSD(T) calculations yielded too negative energies for certain geometries, for which the following strategy was used to obtain their reliable electronic energies: CCSD-F12b/aug-cc-pVTZ + (BCCD(T)/aug-cc-pVDZ − BCCD/aug-cc-pVDZ) [123, 124]. BCCD denotes the Brueckner coupled cluster doubles (BCCD), and BCCD(T) for BCCD with perturbative triples (BCCD(T)) methods [125].

6.3 Data Sampling

The calculating cost increases with the sampling size. The PES performance and the target dynamic results can be converged with the number of the sampling points. Namely, some targets can't be improved by a new PES that is fitted with more points. Hence, data sampling is also very important. It is not expected to make reliable predictions for out-of-sample data. The sampling range determines essentially the applicability of the PES range, which should cover sufficiently large regions. As mentioned above, generating configurations on a random or regular grid becomes unfeasible for larger systems. Still, the grid approach can be employed for sampling with some degrees of freedom fixed. For instance, regular grids can be used to sample configurations along the two reactive bond lengths, r_{XH} and r_{HY}, in the hydrogen abstraction reaction $X + HY \rightarrow X{\bullet\bullet\bullet}H{\bullet\bullet\bullet}Y \rightarrow HX + Y$, with other coordinates fixed.

For polyatomic molecular systems, the low-level direct dynamics can be helpful to obtain a preliminary data set, which can be used to fit a preliminary PES. Then various tests can be carried out to further explore the configuration space. If trajectories propagate to regions that lack points, the chemical structures may collapse and adopt unphysically short interatomic distances or "explode". The PES can be improved by adding points in these regions.

The vibrational, rotational, and translational motions at the initial reactant asymptote can be sampled according to the Boltzmann distribution at some specified temperature, for instance, 3000 K, which is sufficiently high to access all dynamically relevant regions. Thermal kinetics often require the temperature as high as 3000 K, particularly for combustion. In most cases, the resulting sampling is appropriate.

The research target is related to the sampling. If only reactivity initialized from the ground ro-vibrational reactants is focused at different collision energies, the sampling range is generally narrower than that by the thermal sampling of each degree of freedom at 3000 K. If only the forward reaction is focused, the configurations in the product channels may be not well sampled and the corresponding PES is generally not appropriate for the reverse. For ro-vibrational spectra of the simplest Criegee intermediate, CH_2OO [126–128], and its isomer, dioxirane [129], only configurations around CH_2OO and dioxirane are needed. No configuration for the reactive channel such as $H_2CO + O$ and $HCO + OH$ is required. Similarly, for

the ro-vibrational spectra of the weakly interacted complexes, such as H_2O•••HCl [130–132] and H_2O•••CO [133, 134], sampling for other regions except the wells is not helpful for the research target [130, 135].

For low-temperature kinetics or low-energy dynamics, long-range regions between reactants or products need be well sampled for long-range interactions. For the H_2 + HCl inelastic scattering, a total of 42,417 points were selected to cover regions with the center-of-mass distance between H_2 and HCl up to 30 Å [136]. For the ion–molecule reaction, $H_2 + NH_2^- \leftrightarrow H^- + NH_3$, the sampling points cover a very large dynamical relevant configuration space. As shown in Fig. 6.8, there still exists weak interactions, even the distance between NH_2^- and H_2 or between NH_3 and H^- reaches 50 Å. Hence, 98,154 points were sampled in a large configuration space to provide reliable descriptions for both long-range and strong-interacted regions [137]. For efficiency and accuracy, it is often to represent the long-range interaction [82, 95, 136, 138, 139] in the multipolar expansion form [140].

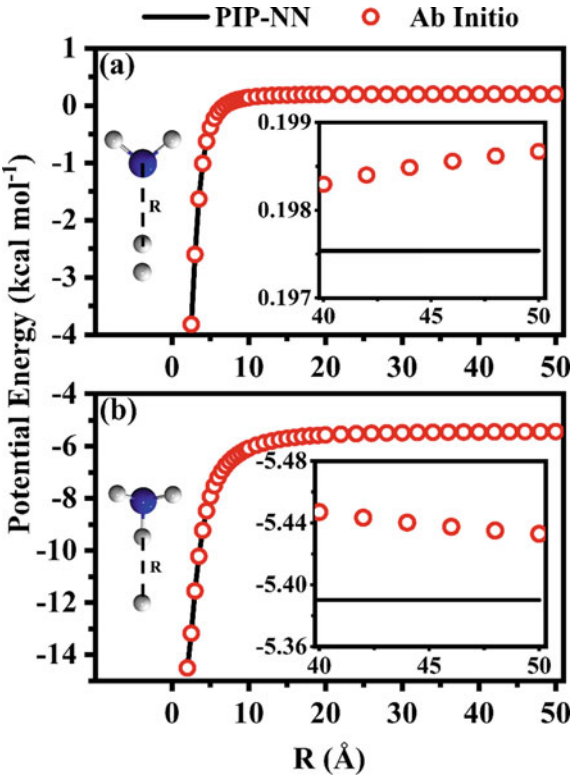


Fig. 6.8 One-dimensional cuts for the potential energy between NH_2^- and H_2, **a** from NH_2^-•••H_2 and between NH_3 and H^-, **b** from NH_3•••H^-. The energies in kcal mol^{-1} are relative to the reactant asymptote $NH_2^- + H_2$. The definitions of R in two panels are also shown [137]. Reproduced from Ref. [137] with permission from the Royal Society of Chemistry

For gases in shock tubes, shocked gases surrounding hypersonic Earth-re-entry vehicles, the chemical species may be highly excited in the ro-vibrational and translational degrees of freedom, resulting far-from-equilibrium energy populations. During hypersonic flight through Earth's atmosphere, the translational temperature of the air (most are N_2 and O_2) in front of the vehicle can reach 10,000 K and the internal vibrational energy can be close to the dissociation threshold [141]. Therefore, the sampling configurations for $N_2 + N_2$ and $N_2 + O_2$ cover fully dissociated regions for N + N + N + N or N + N + O + O, and the energies of the sampling points span up to 2000 kcal/mol [98, 142, 143]. For the plasma chemistry, such as CO (v) + CO (v′) [144] or CO_2 (v) + CO_2 (v′) [145], high internal vibrational excitations and low collision energies would sample the corresponding configurations.

The sampling is also relevant to the consequent nuclear simulation methods. With (quasi-)classical dynamics, some regions can't be accessed and configurations in these classically forbidden regions are not sampled. If quantum mechanics approach is used for the simulation, generally the sampling configurations are in wider regions. This is due to that the action of the potential energy operator onto the wave packet [146] is calculated by the discrete variable representation (DVR) [147] based on some specified coordinates [148]. Therefore, additional points, often with high energies and distorted configurations, should be added to make the potential matrix meaningful. Or else, lacking points in these classically forbidden regions may result "holes" [149] in the PESs and affect the quantum dynamics.

There is another way proposed by Behler [150] to explore the poorly sampled regions of the configuration space. This approach is based on the limited extrapolation capabilities of NN PES: if some regions were not covered well, different NN potentials would give different energy predictions for configurations in these regions. Namely, these regions are of high uncertainty. In detail, firstly, several different NN PESs are obtained by fitting the initial dataset and a large number of configurations are generated by trajectory calculations on one of these PESs. Then all fits are used to recalculate the energies of the newly generated structures, configurations that the fits give significantly different energy predictions are selected for additional ab initio calculations and added to the training set. Recently, Lin et al. proposed a trajectory-free active learning strategy, based on the property of NN potentials, to automatically sample points for the construction of PESs. Interestingly, this sampling method is self-driven to explore configuration space without the dynamic simulations. Besides, they introduce an energy-dependent weighting function to avoid adding some highly distorted configurations, which may significantly decrease the fitting accuracy [151]. Automated PES development tools have been available, such as the GROW package [152], the AUTOSURF package [153, 154], the PES-Learn software package [155], and the ROBOSURFER program [156].

Practically, the following steps need be considered for sampling:

(1) The reaction profile is primarily characterized by exploring the reactants, products, intermediates, and transition states connecting them. The reaction $F + H_2O \rightarrow HF + OH$ is exothermic, − 17.61 kcal mol^{-1}, with a barrier height of 1.93 kcal mol^{-1}. Other channels, such as HOF + H (89.9 kcal mol^{-1}) and FO

$+ H_2$ (87.1 kcal mol^{-1}), were too high in energy to be included in the sampling [82].

(2) Regions around the stationary points and along the MEPs are very important to affect the dynamics. The energies, geometries, harmonic frequencies of the stationary points, and the potentials along the MEPs should be well sampled. As shown in the inset of Fig. 6.9, the potential along the MEP of the OH + $HO_2 \rightarrow H_2O + O_2$ reaction is well reproduced by the PES, compared to the CCSD(T)-F12a/aug-cc-pVTZ calculations [102]. Other important and useful one-dimensional scans, such as approaching between reactants or products at various directions, are also suggested to be included in the sampling.

(3) Depict a plot with the sampling points projected onto the reactive coordinates. On the same figure, plot the potential contour along the reactive coordinates. To some extent, the comparison can indicate if the sampling covers the contour with a high cut-off energy. Figure 6.9 shows the contour along the two reactive bonds, r_{OH} ($HO'H \bullet\bullet\bullet O_2$) and $r_{O'H}$ ($HO' \bullet\bullet\bullet HO_2$), for the reaction OH + $HO_2 \rightarrow$ $H_2O + O_2$. The sampled points with two reactive bond distances up to 5 Å are also shown in the same figure. Clearly, the points cover sufficient wide regions with denser points sampled in the valleys with relatively low energies from reactants to products [102].

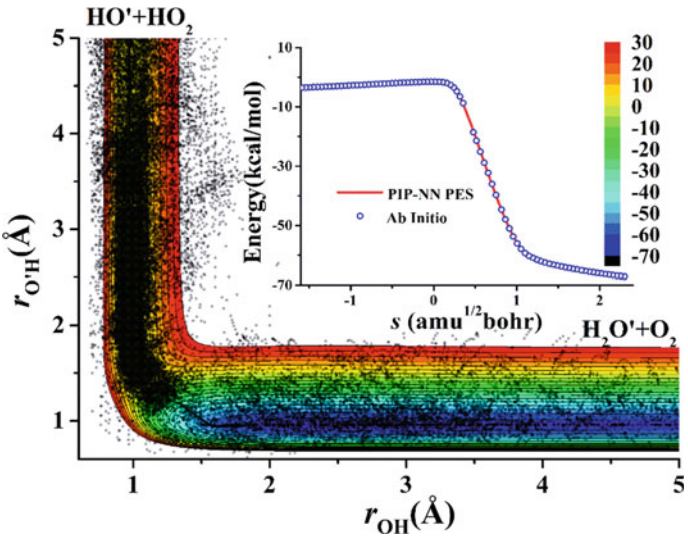


Fig. 6.9 Contour plot of the potential (energy in kcal mol^{-1}) for OH + $HO_2 \rightarrow$ TS1 $\rightarrow$ H_2O + O_2 as a function of the two reactive bond lengths. All other coordinates are fixed at the transition state, TS1. The sampled points with the two reactive bond lengths up to 5 Å are shown. The inset plot shows the corresponding potentials along the MEP on the PES and determined directly by ab initio calculation [102]. Reproduced from Ref. [102] with permission from the Royal Society of Chemistry

6.4 Data Fitting

The PES can then be obtained by learning from sufficiently a large number of configuration points of high-quality electronic energies. PIP-NN takes the PIPs as input layer of NN [1, 62, 63]. Therefore, it combines the advantages of the PIPs [50, 51] and the NNs [157, 158]. Namely, the invariant property with respect to the permutation of like elements is enforced by PIP and high fidelity can be obtained by NN. Please refer to Refs. [1, 62, 63] for detailed descriptions and discussions of the PIP-NN method, particularly for the translation/rotational/permutation symmetry adaptation, the maximum order of the PIPs used in NN.

Explicitly for NNs with two hidden layers, the PIP-NN function is expressed as,

$$V_{\text{PIP-NN}} = b_1^{(3)} + \sum_{k=1}^{K} \left(\omega_{1,k}^{(3)} \cdot f_2 \left(b_k^{(2)} + \sum_{j=1}^{J} \left(\omega_{k,j}^{(2)} \cdot f_1 \left(b_j^{(1)} + \sum_{i=1}^{I} \omega_{j,i}^{(1)} \cdot G_i \right) \right) \right) \right), \tag{6.3}$$

In Eq. (6.3), the input layer is consisted of the PIPs, namely,

$$G_i = \widehat{S} \prod_{n_1 < n_2}^{N} p_{n_1 n_2}^{l_{n_1 n_2}} \tag{6.4}$$

with the Morse-like variables $p_{n_1 n_2}$ of internuclear distances between atoms n_1 and n_2 ($r_{n_1 n_2}$),

$$p_{n_1 n_2} = \exp(-r_{n_1 n_2}/\lambda) \tag{6.5}$$

and the corresponding order, $l_{n_1 n_2}$.

The real parameter λ in Eq. (6.5) is generally not optimized but set as a universal constant for all internuclear distances. Note that the magnitude of λ is related to the potential changing along the distances between fragments in the reactant or product asymptotes. For neutral molecular systems without long-range interactions, λ is generally set as a number around 1.0 Å. For systems with long-range interactions, such as the ion–molecule reaction, a large number should be adopted for λ. Figure 6.10a clearly shows the changing trends of p along r at different λ. In addition, Fig. 6.10b–d shows the corresponding plots of p^n with $n = 2, 3, 4$, respectively. For the PES of the reaction $H_2 + NH_2^- \rightarrow H^- + NH_3$ [137], the corresponding fitting RMSE is 0.026 kcal/mol for $\lambda = 2.1$ Å. For $\lambda = 0.95$ Å, the RMSE is 0.060 kcal/mol. Both PESs are quite accurate, as shown in Fig. 6.11. However, they show quantitatively different changing trends of potentials for reactants $H_2 + NH_2^-$ or products $H^- + NH_3$ approaching each other, as shown in the inset plot of Fig. 6.11. Clearly, a smaller $\lambda = 0.95$ Å makes the potential converges earlier along R than the PES with a larger $\lambda = 2.1$ Å. The potential with a smaller λ deviates from the direct ab initio calculations by about 0.1–0.3 kcal/mol. The potential with a larger λ is in much

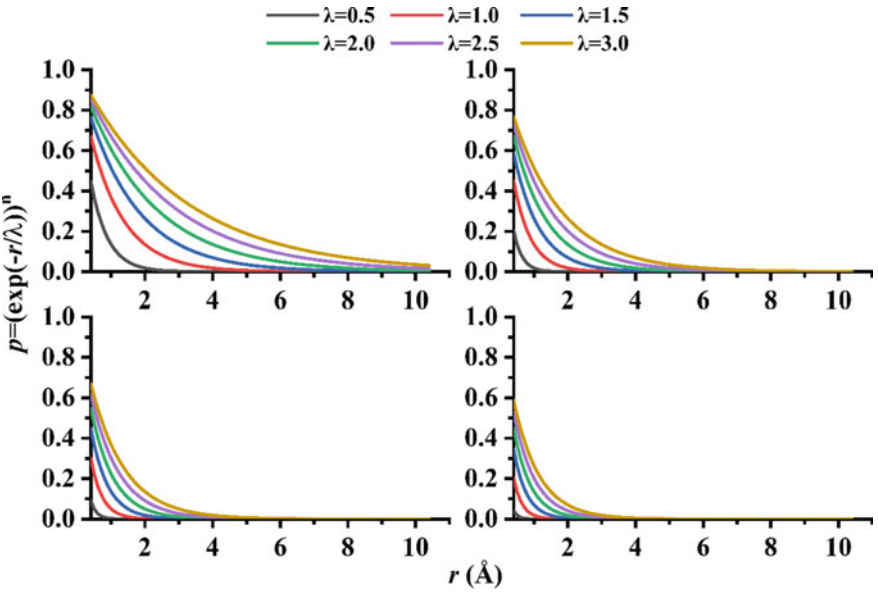


Fig. 6.10 The variable $p = [\exp(-r/\lambda)]^n$ as a function of the internuclear distance r at different λ and order n: **a** $n = 1$, **b** $n = 2$, **c** $n = 3$, **d** $n = 4$

better agreement with the ab initio calculations. The two PESs yield significantly different integral cross sections (ICSs) of the reaction $H_2 + NH_2^- \rightarrow H^- + NH_3$: at the collision energy of 1 kcal/mol, the ICSs are 19.6 Å^2 on the PES with $\lambda = 2.1$ Å, significantly higher than 11.0 Å^2 on the PES with $\lambda = 0.95$ Å. Therefore, scrutiny is always suggested.

The symmetrization operator, $\hat{S}$, in Eq. (6.4) contains all permutation operations between like atoms associated with the molecular system. For the $F + H_2O$ system, $\hat{S} = \hat{I} + \hat{S}_{12}$ with $\hat{I}$ being the identical operation and $\hat{S}_{12}$ being the permutation between the two hydrogen atoms if the atoms are labeled as H1H2F3O4 or H1H2O3F4. Consequently, the values of PIPs, $\{G_m\}$, are not changed with respect to any permutation of like atoms. Namely, the invariance property with respect to the permutation of like atoms is naturally and rigorously retained.

In Eq. (6.2), the number of the input PIPs is denoted as I. The number of neurons in the first and second hidden layers are denoted as J and L, respectively. The hyperbolic tangent function, $\tanh(x) = (e^{2x} - 1)/(e^{2x} + 1)$, is adopted as the nonlinear transfer functions for the two hidden layers, f_i $(i = 1, 2)$. $\omega_{j,i}^{(l)}$ are weights that connect the ith neuron of $(l - 1)$th layer and the jth neuron of the lth layer; $b_j^{(l)}$ denote the biases of the jth neuron of the lth layer. The root-mean square error (RMSE) is used to measure the performance of the fitting, which is defined as,

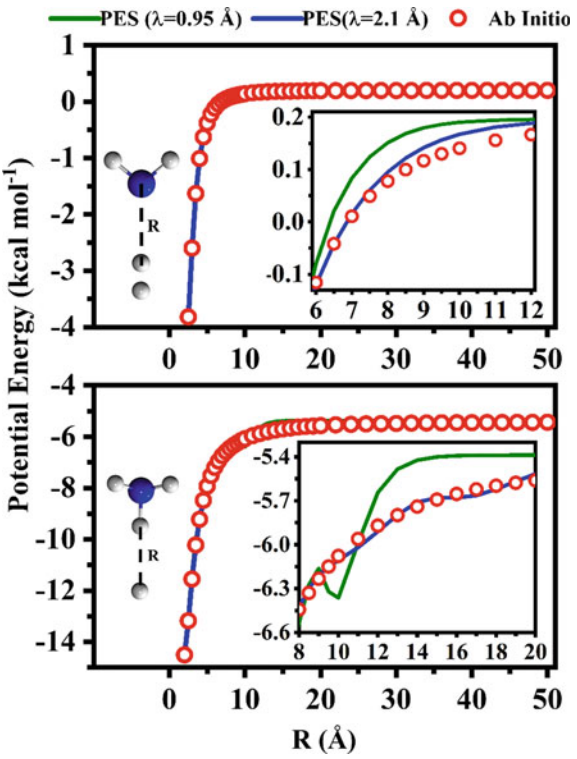


Fig. 6.11 Comparison of two different optimal PIP-NN PESs with different adjusting parameter λ along the distance between NH_2^- and H_2 (**a**) and between NH_3 and H^- (**b**) [137]. Reproduced from Ref. [137] with permission from the Royal Society of Chemistry

$$\mathrm{RMSE} = \sqrt{\sum_{i=1}^{N_{\mathrm{data}}} \left(E^i_{\mathrm{output}} - E^i_{\mathrm{target}}\right)^2 / N_{\mathrm{data}}}, \tag{6.6}$$

where E_{target} and E_{output} represent the fitted and input energies, respectively.

To obtain better optimal fitting, several different NN architectures with two hidden layers have been tested. For each NN architecture, 50–200 NN trainings were carried out. For each NN training, the sampling dataset was randomly divided into three parts, namely, the training (90%), validation (5%), and test (5%), to avoid the so-called "over-fitting" issue. In addition, the fitting was adopted only if all three sets (training, validation, and test) have similar RMSEs. Meanwhile, the maximum deviation was also considered for choosing the final optimal PES.

The permutation symmetry property is rigorously retained without the need to modify NNs. The size of the input PIPs, I, is determined by the maximum order of the PIPs, $M = \sum_{n_1<n_2}^{N} l_{n_1 n_2}$. As has been discussed in Refs. [1, 63], the lower limit of M can be determined using the Molien's series of the complete nuclear permutation and inversion (CNPI) group [50, 150, 159, 160] of the system [50, 161]. Practically, the required PIPs under the order of M can be obtained by the software, MAGMA [162], SINGULAR [163], and the monomial symmetrization approach (MSA) [164].

Full-dimensional accurate PESs of the following systems have been fitted by the PIP-NN method: F + HCl [165], H + Li_2 → Li + LiH [166], Si + H_2 → SiH + H [167], N + O_2 (doublet, quartet, and sextet states) [168], H + MgH → Mg + H_2 [169], O^+ + H_2 [170], S^+ + H_2 → SH^+ + H [171], K + H_2 → H + KH [172], H + HBr → H_2 + Br [173], N^+ + H_2 → NH^+ + H [174], Au^+ + H_2 → H + AuH^+ [175], OH + SO → H + SO_2 [113, 114], C + H_2O → CO + H_2/2H [176], F + H_2O → HF + OH [105, 138, 177], Cl + H_2O → HCl + OH [95], HCl + OH → Cl + H_2O [178], OH + CO → H + CO_2 [179], $HOCO^-$ [180], CO + CO [144], O + H_2O [138, 181], H + HO_2 [182], C_2H_2 [183], HF + HF [184], KRb + KRb → K_2 + Rb_2 [185], H + H_2S → H_2 + SH [186], NH^+ + H_2 → N + H_3^+/NH_2^+ + H [187], OH + O_2 → H + O_3/HO_2 + O [99], OH^+ + H_2 → H + H_2O^+ [188], H_2 + HF [189], NH_3 [31], H_2 + HCl [136], N_2 + HF [190], N_2 + HOC^+ → N_2H^+ + CO/N_2 + HCO [191], CH_2OO [126], dioxirane [129], O + C_2H_2 → H + HCCO/CO + CH_2 [192], CO + H_2O [135], C_2H_2 + Ne [193, 194], OH + H_2O → H_2O + OH [195], HCl + H_2O [130], OH + HO_2 → H_2O + O_2 [102], NH_4^+ [196], CH_4 [100], H + NH_3 → H_2 + NH_2 [197], H_2 + NH_2^- → NH_3 + H^- [137], H_2O^+ + H_2 → H_3O^+ + H [198], H_2CO + H_2 [199], Cl + CH_4 → HCl + CH_3 [200], CO_2 + CO_2 [145], HO_2 + HO_2 → H_2O_2 + O_2 [201], H + CH_4 → H_2 + CH_3 [119], N_2O + C_2H_2 → Oxadiazole [36], F + CH_3OH → HF + CH_3O/CH_2OH [202, 203], Cl + CH_3OH → HCl + CH_3O/CH_2OH [204], H + CH_3OH → H_2 + CH_3O/H_2 + CH_2OH/H_2O + CH_3 [205], OH + CH_4 → H_2O + CH_3 [206]. One can see that they include reactions, nonreactive collisions, and single molecule with or without apparent charges.

It is a two-edged sword to use the PIPs as the input layer of NN. The number of PIPs increases remarkably with the complexity of symmetry of the molecular system. For OH + CH_4 [206], F + CH_3OH [202], Cl + CH_3OH [204], H + CH_3OH [205], the input layers are consisted of 1331, 1104, 1104, and 1331 PIPs, respectively. For the 12-atom *N*-methyl acetamide (NMA), as discussed in Ref. [207], if only a reduced permutational symmetry is included, the PIPs contain 8040 terms at a maximum polynomial order of 3, and 95,965 at a maximum polynomial order of 4. One can use only the non-redundant PIPs as the input [186, 208], as many PIPs are not independent [164]. For the H + H_2S system, there are 9 non-redundant PIPs from 23 3th-order PIPs [186]. For a molecular system with a given symmetry, all invariant polynomials can be derived from a finite generating invariant polynomials [209], which are called fundamental invariants (FIs) by Prof. Zhang and his group [210]. Correspondingly, FIs can be used as the input of NN and the approach is fundamental invariant NN (FI-NN) [210]. The size of the FIs is much smaller than that of PIPs, making FI-NN more efficient in evaluation than PIP-NN [211, 212].

It is hard to apply the polynomial-based NN methods, PIP-NN, non-redundant PIP-NN, to a molecular system with about 10 atoms and above. This "10 atom limit" is broken by the fragmented PIP method, which relies on the fact that the Morse variables are close to zero at large internuclear distances and thus can be discarded [207, 213, 214]. It's hard to remove those basis polynomials with small Morse variables from the total basis as generating the full fitting basis is prohibitive for large systems. Alternatively, one can directly fragment the large target molecule

into a few smaller parts for which the PIP basis can be efficiently generated and then the union of these fragmented bases was used as the entire basis of the system. Details of the fragmented PIP method can be found in the literature [207, 213, 214]. The fragmented PIP method significantly decreases the sizes of PIPs, which can be adopted in the PIP-NN or FI-NN methods for fitting PESs of large complex molecules but awaits further validation in the future.

The high-dimensional NN (HD-NN) method, proposed by Behler and Parrinello in 2007 [61], is highly scalable for large molecular systems of up to thousands of atoms. PESs have been constructed by the HD-NN method for condensed systems with diverse types of interactions from covalent bonds to metallic bonding and dispersion interactions [52, 53]. For HD-NN, the entire energy of the molecular system sums all energy contributions from each atom, whose local atomic chemical environment, or "fingerprint", is characterized by many-body atom-centered symmetry functions [215]. The permutationally invariant properties are fulfilled by requiring that an identical NN (namely same NN structure with same fitting parameters) is used for the same element in the system. Atomic energy contributions are not physically meaningful, and the total electronic energies are used to obtain the optimal parameters of NNs [52, 53, 150, 215–218]. The HD-NN method has been adopted to represent PESs of OH + CO, H + H_2S, H + NH_3, H + CH_4, OH + CH_4, H + CH_3OH. Compared to one single NN in the PIP-NN, the number of NN in the HD-NN PESs is identical to N_{atom}. Although the resulting PESs have excellent performance, the evaluation of HD-NN is slower than PIP-NN or FI-NN for small molecules [205, 208, 219, 220]. However, HD-NN is very suitable for larger systems as it scales linearly with the number of atoms [205, 208, 219, 220].

6.5 NN Based Δ-ML Approach

For developing globally accurate full-dimensional PESs of large systems, the difficulties come from the large configuration space and the high cost to obtain high-quality electronic energies at the CCSD(T) or MRCI levels. Nowadays, the CCSD(T) calculation remains too computationally demanding for systems with more than 10 heavy atoms. Solutions include the dual-level Shepard interpolation method [221], the hierarchical construction scheme (HC) [222], the Δ-machine learning approach (Δ-ML) [223], or the transfer learning (TL) approach [224]. TL is based on the idea that knowledge gained from solving one problem can be applied to a different but related problem. Thus, an NN PES based on a large number of low-level points can be improved by slightly adjusting its parameters with a smaller number of high-level points.

The central idea of the dual-level Shepard interpolation method, the HC scheme, and the Δ-ML approach is that the PES of some target high-level quality (V_{HL}) was expressed as a sum of a PES fit (V_{LL}) to a large number of low-level ab initio calculated points and an energy difference surface ($\Delta V_{HL\text{-}LL}$) between high-level and low-level ab initio methods based on much fewer data points [221–223]. Namely,

$$V_{\mathrm{HL}} = V_{\mathrm{LL}} + \Delta V_{\mathrm{HL\text{-}LL}}. \tag{6.7}$$

In the HC work by Fu et al., the cubic spline method was used to construct V_{LL} and $\Delta V_{\mathrm{HL\text{-}LL}}$ for the PES of the tri-atom reaction F + H_2. For V_{LL}, 15,000 symmetry-unique points were sampled. For $\Delta V_{\mathrm{HL\text{-}LL}}$, only 1400 points were selected from 15,000 points [222]. Recently, Bowman and coworkers proposed a Δ-learning approach based on PIP [225, 226]. Correspondingly, V_{LL} and $\Delta V_{\mathrm{HL\text{-}LL}}$ are both fitted by the PIP. Tests for PESs of CH_4, H_3O^+, *N*-methylacetamide and acetylacetone have shown that significant computational cost can be saved since only a small number of sampling configurations are required to be calculated at the target high level [225, 226].

Similarly, V_{LL} and $\Delta V_{\mathrm{HL\text{-}LL}}$ can be also learned by the PIP-NN fitting. Note that the evaluation of Eq. (6.7) can be slow. We propose to employ the $\Delta V_{\mathrm{HL\text{-}LL}}$ PES, based on a much smaller number of points calculated directly at the target high-level ab initio method, to predict the difference energies of the remaining points that are not calculated at the high level. Then, all data are of high quality, and are fitted by the PIP-NN method. The evaluation speed of the final PIP-NN PES is thus not affected. Indeed, the new method, PIP-NN based Δ-ML approach, is essentially to obtain high quality energies of many data points.

In the PIP-based Δ-learning approach, the root mean square (RMS) bond difference between two configurations was used as the criterion to select points near the equilibrium of stationary points. This strategy can produce a subset that spans the same range of configurations as the original low-level dataset and the correction PES based on a small number of configurations correctly improves the key properties around the stationary points or along the MEP to a high level [225, 226]. In the PIP-NN based Δ-ML approach, we propose to search the poorly sampled regions of the configuration space according to the limited extrapolation capabilities of NN methods. The PIP-NN method can be replaced by other NN methods. Figure 6.12 shows the general flow chart for developing full-dimensional accurate PESs by the NN based Δ-ML approach. Please see also detailed discussions in Ref. [201].

This new PIP-NN based Δ-ML approach is then applied to develop the globally accurate full-dimensional PES of the self-reaction of HO_2 ($HO_2 + HO_2$) at its leading triplet electronic state. First, we constructed a low-level PES fit to 75,300 points at the UM06-2X/aug-cc-pVTZ level. UM06-2X/aug-cc-pVTZ [227] was chosen as the most properties of the stationary points have the smallest deviations, compared to those calculated by UCCSD(T)-F12a/aug-cc-pVTZ. This is beneficial for developing the correction PES as cheap as possible. The next step is to build a correction PES. 1500 points were first chosen from the DFT dataset and then their electronic energies were re-calculated at the UCCSD(T)-F12a/aug-cc-pVTZ level. Some of these points were around the stationary points, and the remaining were well dispersed along the MEP. Then several preliminary correction PESs were obtained by fitting the energy differences between high- and low-level energies of these points, four best fittings were chosen to calculate the average energy difference among them, namely,

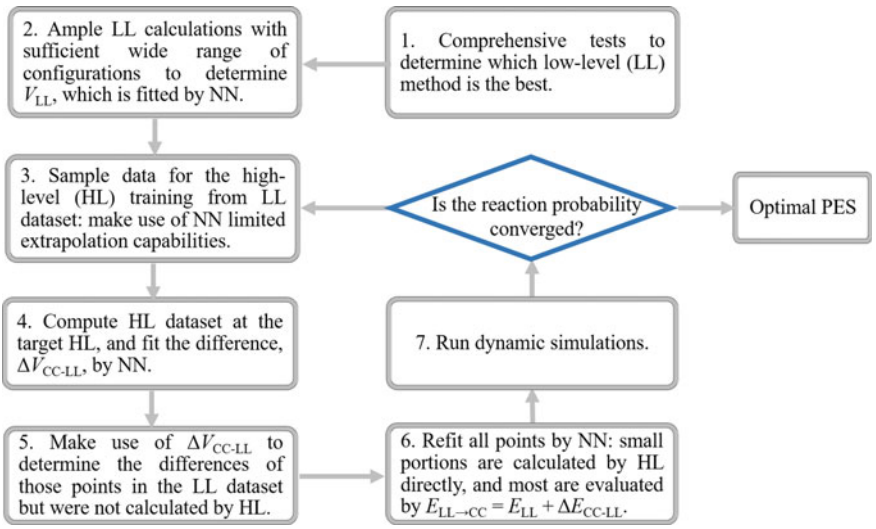


Fig. 6.12 The flow chart for developing accurate PESs by the NN based Δ-ML approach. Please see details in Ref. [201]

$$D = \frac{\sum_{i<j}^{n} \left| \Delta V^{i}_{\text{HL-LL}} - \Delta V^{j}_{\text{HL-LL}} \right|}{n \times (n-1)/2}, \tag{6.8}$$

where $\Delta V^{i}_{\text{HL-LL}}$ and $\Delta V^{j}_{\text{HL-LL}}$ are the ith and jth correction PES, respectively; n is the number of the selected best fittings, here $n = 4$. The energy differences of each remaining point in the DFT dataset were calculated. Then the first 300 points with the biggest energy deviations were added to the training dataset for the correction PES. Note that the generalized Euclidean distance $\chi(\{r_i\}) = \sqrt{\sum_i^{15} \left|\vec{r}_i - \vec{r}'_i\right|^2}$ defined in terms of the internuclear distances between two points, $\{\vec{r}_i\}$ and $\left\{\vec{r}'_i\right\}$, was used to exclude points that are close to each other. The procedure mentioned above was repeated to gradually improve the correction PES until the final correction PES converges. Only if the final correction PES correctly describes the topography of the energy differences between high- and low-level points, the final high-level PES can provide converged dynamical results and reproduce the key properties, such as geometries, frequencies, and energies of all stationary points, compared to the target high-level calculation. Finally, 10,700 points were selected and re-calculated at the UCCSD(T)-F12a/aug-cc-pVTZ level.

The distributions of the training dataset and the entire dataset are displayed in Fig. 6.13a, b, respectively, along the two reactive bond distances up to 15 Å. The insert plots show the details near TS1. By comparison of the two figures, one can see the training dataset has very similar but relatively sparse distributions to the entire dataset. In addition, much denser points exist around RC, TS1, and TS2 in both panels, which are necessary for the high-quality description around these regions.

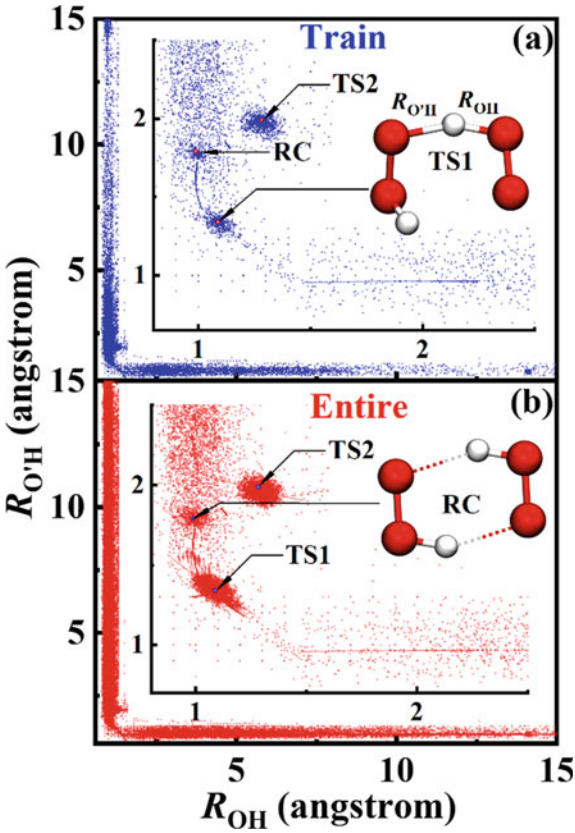


Fig. 6.13 Distribution of the training dataset for the correction PES (**a**) and entire dataset for the DFT PES (**b**) along the two reactive bond lengths up to 15 Å. The insert plots show the details near the transition state [201]. Adapted with permission from Ref. [201]. Copyright 2022 American Chemical Society

Figure 6.14 presents the contour plot for the dominant reaction channel $HO_2 + HO_2 \rightarrow H_2O_2 + O_2$ on the final high-level PES (denotes as PESH) along two reactive bonds (up to 4 Å for clarity) with other degrees of freedom relaxed. The regions for reactant asymptote, RC, TS1, and product asymptote are clearly shown. The potentials along the minimum energy path (MEP) on PESH and the low-level PES (denotes as PESL) and the corresponding energy difference between them are also shown in the inset of the same figure. As seen, PESL, PESH, and their differences provide accurate descriptions of the MEP and the corresponding energy difference between them compared to the direct electronic structure calculations. With the zero of the energy as the reactant asymptote, the energy difference is quite large near the TS region along the MEP.

For the computation time on the clusters with Intel Xeon CPU E5-2680 v3 @ 2.50 GHz processors, computing 75300 M06-2X/aug-cc-pVTZ points needed about 3140 h and the CPU time for 10700 UCCSD(T)-F12a/aug-cc-pVTZ points was 16,050 h. Consequently, building PESH only takes roughly 17% of CPU time for calculating 75300 UCCSD(T)-F12a/aug-cc-pVTZ points directly. The RMSEs for V_{LL}, $\Delta V_{HL\text{-}LL}$, and V_{HL} are 0.20, 0.13, and 0.17 kcal mol^{-1}, respectively. The NN

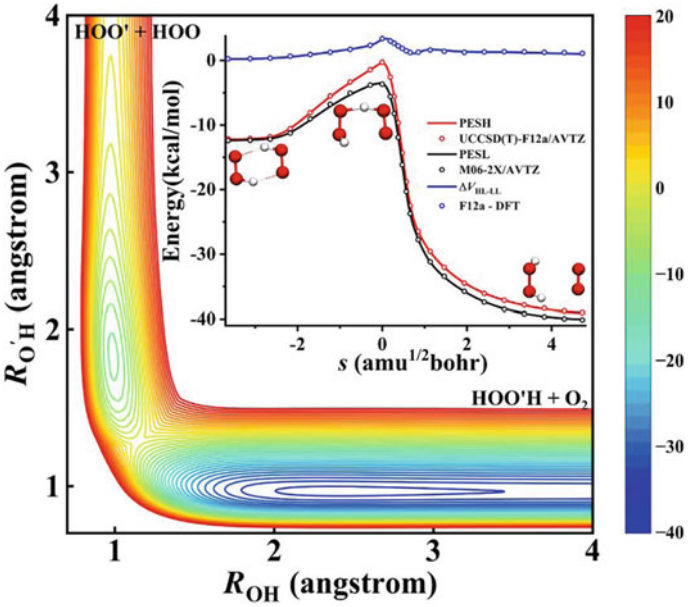


Fig. 6.14 Contour plot for the reaction $HO_2 + HO_2$ on PESH along the two reactive bonds $R_{O'H}$ and R_{OH} with other coordinates optimized. The energy is in kcal mol^{-1}. The inset plot shows the corresponding potential energies along MEPs on PESH and PESL, respectively, and determined directly by electronic structure computations. The energy differences between UCCSD(T)-F12a/aug-cc-pVTZ and DFT, and the corresponding energies on the correction PES $\Delta V_{HL\text{-}LL}$ along the MEP are shown in the plot [201]. Adapted with permission from Ref. [201]. Copyright 2022 American Chemical Society

based Δ-ML approach can be used to develop full-dimensional PESs of large molecular systems efficiently. In addition, one can improve previous PESs by adding more and more corrections gradually [222], which is invaluable for the science community of developing PESs and related fields [201].

6.6 Conclusions and Perspective

Potential energy surface (PES) of a molecular system is a function that represents the mapping correlation between the configuration and the energy. At a specified configuration, the PES function returns its energy. A given energy may correspond to many different configurations. Recent years have witnessed huge advances in developing full-dimensional accurate PESs based on tens even hundreds of thousands of configuration points, whose electronic energies are generally computed at the

levels of MRCI or CCSD(T). These advances in the PES construction are boosted by diverse factors, including the increasing computing power, the development in the algorithms of electronic structure methods, and advanced machine learning models.

In order to make a full-dimensional accurate PES of a molecular system, three components are needed to be considered: data quality, data sampling, and data fitting, indicating that the data locates at the core position. Indeed, data and artificial intelligence become an important pillar of an integrated strategy for studying complicated phenomena [228]. The data here refer to a set of configuration points and their associated electronic energies. These configurations are not only far from equilibrium, but also located in several regions of different stationary points, including reactants, products, transition states, and intermediates. The complicated electronic structure in different regions makes it difficult to get a high-quality dataset. Various issues have been discussed and reviewed for electronic structure calculations. Some useful tips and suggestions are given to obtain accurate and well-behaved energies without discontinuities and noises.

For a given research target, the sampled data should be representative enough with wide ranges, and as small as possible for efficiency. If the sampled configuration space is not enough, the prediction for regions lacking points is in principle not reliable. Data sampling is relevant to data quality. The DFT and the CCSD(T) PESs are different, and may require different samplings to represent them, although low-level direct dynamics or PESs can be used for preliminary sampling. Data sampling is also relevant to data fitting. The atomic coordinates are often transformed into variables to characterize the chemical environments of the system with appropriate symmetry properties. Indeed, the density and the representative of the sampling should be measured under the representation of the fitting, for instance, the PIP spanned space for PIP-NN, or the atom-centered symmetry function space for HD-NN. Data sampling is also relevant to the research target. Reaction dynamics reach much wider configurations than those only for ro-vibrational spectra simulations. Finally, data sampling is relevant to the simulation method. For instance, quantum dynamics require wider ranges than quasi-classical trajectory.

The PES is fitted by learning from the sampled data. Different representations of the coordinates, "fingerprints", or descriptors are designed. Particularly, the symmetry property can be considered in the input layer of the PIP-NN with no need to modify the NN implementation. For HD-NN, the symmetry property is retained by enforcing one identical NN to describe the energy contribution of the same chemical element. Then the exchange between like atoms does not affect the entire energy. With similar performance, the PIP-NN is faster in evaluation than the HD-NN. However, the increasing complexity of the PIP prevents the application of the PIP-NN for large molecular systems. The recently proposed fragment PIP or reduced-symmetry PIP has been applied to large systems and may be implemented into the PIP-NN or FI-NN in principle. HD-NN is very promising as it has no limit for molecular symmetry complexity.

However, the calculation cost in CCSD(T) or MRCI is prohibitively expensive for large molecular systems. The hierarchical construction scheme and the Δ-machine learning (Δ-ML) approach are useful to save the calculation cost. Both methods

express the PES of some target high-level quality (V_{HL}) as a sum of a PES fit (V_{LL}) to a large number of low-level points and an energy difference surface ($\Delta V_{\mathrm{HL\text{-}LL}}$) between high-level and low-level methods based on much fewer data points. We proposed to employ the PIP-NN method (or other NN methods) to fit V_{HL} and $\Delta V_{\mathrm{HL\text{-}LL}}$. More importantly, we took advantage of the limited extrapolating ability of the NN to get the sampling for the $\Delta V_{\mathrm{HL\text{-}LL}}$ effectively. In addition, $\Delta V_{\mathrm{HL\text{-}LL}}$ was used to make predictions for points that were not calculated directly at the high-level method. Finally, all data points were used to fit the final PES by PIP-NN. This PIP-NN based Δ-ML method was successfully applied to $HO_2 + HO_2$ and $OH + CH_3OH$ reactions.

Finally, we suggest a checklist for reporting PESs based on ample high-level data. It is our hope that this checklist will be helpful to provide a standard for training and reporting PESs. It will be also useful for authors, referees, readers, and users of PESs to reproduce the results, and to guide a critical evaluation on PESs. It might be useful for publishers as well since it can be considered submission guidelines for PES related papers [229]. This checklist is as follows and can be amended or improved in the future.

(1) The calculation detail of the electronic structures should be provided, including the software and its version, the computing platform, special settings such as convergence criteria, etc. An example input is suggested to be reported.
(2) Provide bad tests, such as data with discontinuities or convergence issues or data converged to a wrong state, and how to remove these issues or improve the data. These are very useful and important for the community of the field.
(3) Provide the ranges for the configuration space and the energy space, which are related to the applicability of the PES. It is also useful to check the detailed distributions of the sampling points in the configuration space or the energy space.
(4) Provide the complete configurations and their energies used for fitting. These currently high-level data can be employed as the "low-level" data for future improvements within the Δ-ML strategy. These high-level data are also useful to develop or benchmark other electronic structure methods, such as DFT.
(5) Provide the fitting error distributions to check where the large errors locate. Correspondingly, the energies of these configurations might be improved in the future.
(6) Provide some key evaluations of the final PES on reproducing the geometries, energies and harmonic frequencies of the stationary points, the potential along the MEP, potentials along the approaching distance between fragments, and potentials along the reactive coordinates.
(7) Provide the final PES subroutine publicly for the community, which has been required already by some peer-reviewed journals.

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Chapter 7
Machine Learning Applications in Chemical Kinetics and Thermochemistry



Lung-Yi Chen and Yi-Pei Li

Abstract Kinetic modeling can predict the performance of a reaction system and aids in understanding detailed reaction chemistry. However, high-fidelity reaction simulations require accurate thermodynamic and kinetic parameters of the involved species, which are not always available, especially for complex reaction networks containing hundreds or thousands of chemicals. This chapter aims to survey recent developments in machine learning in predicting molecular thermochemistry and kinetic properties, with an emphasis on advances in deep learning-based molecular property and reaction kinetics prediction. The pros and cons of commonly used conventional theoretical models are also discussed in this chapter.

Keywords Machine learning · Thermodynamic properties · Atomic fingerprints · Reaction kinetics · Transition state search

7.1 Introduction

Tremendous efforts have been made to improve reaction technology in the past decades due to increasing demands for new chemistry in many different fields. Nevertheless, developing and optimizing new reaction processes is often expensive due to the tedious and costly experimental processes, so it is highly desirable if computer simulations can be used to replace some of the many required experiments. In principle, with the detailed mechanism and the rate and equilibrium constants of each reaction, one can calculate the time evolution of the concentration of each chemical species and hence easily predict the performance of the reaction system, such as conversion and product distribution at different reaction conditions. However, there are still gaps in the predicting capacity of reaction simulations. One of the major challenges is the lack of accurate kinetic rates and molecular thermochemistry, which

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is typically used to calculate reverse reaction rates from the equilibrium constant [46]. In practice, complex reaction systems could involve thousands of species and reactions [101], many of which cannot be directly measured by experiments due to the limitations of the resolutions of experimental techniques. Therefore, the reported experimental data are usually limited, and obtaining those parameters by experiments may either be time-consuming or impractical.

This problem can be solved using quantum mechanical calculations, which allow one to directly compute the thermochemistry and kinetic parameters of specific reactions. However, because accurate quantum mechanical methods are computationally intensive, they are often not directly applicable to complex reaction systems containing thousands of reactions. Therefore, many empirical methods have been developed to estimate molecular thermochemistry and rate constants [32, 66, 79, 110, 144, 147]. For thermochemistry, one of the most widely used empirical methods is the Benson Group Increment Theory (BGIT) [12], which decomposes molecules into defined substructures with their corresponding group additivity values (GAVs). The BGIT method sums up the GAVs of the substructures present in the molecule to determine the molecular property and can successfully predict the thermochemistry of simple organic species. However, it does not work well for complex molecules such as polycyclic species with fused rings because the strain energies are usually not additive [4, 35]. Compared to thermodynamic properties, reaction rate constants are even more challenging to predict because they are determined not only by stable chemicals but also by transition state species. Due to the lack of accurate prediction methods, it is often assumed that the activation energy and pre-exponential factor of the Arrhenius equation can be roughly estimated using the parameters of other reactions within the same reaction class [74, 89].

Nowadays, with the rapid development of machine learning techniques, people have successfully constructed data-driven models to predict the thermodynamic [27, 45, 98, 174] and kinetic [22, 57, 77] properties of molecules. The major advantage of utilizing machine learning methods is that they can automatically identify trends and patterns in data and can be continuously improved when more data are available. This chapter offers a brief summary of recent progress and achievements of machine learning methods for kinetic modeling, with an emphasis on deep learning-based thermodynamic and kinetic property predictions. Comparisons between data-driven approaches and human-designed theoretical models are also provided in this chapter.

7.2 Predicting Molecular Thermodynamic Properties

Accurate prediction of molecular thermochemistry is important for modeling reaction systems with correct concentration distributions at equilibrium. This task can be regarded as the prediction of molecular properties, which has been extensively investigated by many different groups [59, 61, 139, 162]. Several methods have been proposed to transform molecules into a fixed-length representation that is suitable for conventional machine learning algorithms, such as artificial neural networks (ANNs).

These representations can generally be categorized as molecular and atomic fingerprints, which represent global information of molecules and local information of each atom, respectively. The details of molecular and atomic fingerprints are discussed in Sects. 7.2.1 and 7.2.2. Although data-driven methods have achieved great success in predicting molecular properties, training these models usually requires a considerable amount of data, which is not always available, especially in new research areas. Data scarcity is one of the major bottlenecks for applying machine learning in chemistry. So, two of the typical solutions for the data scarcity issue (transfer learning and active learning) are discussed in Sect. 7.2.3 of this chapter.

7.2.1 The Molecular Fingerprint Method

The concept of molecular fingerprint has existed for decades in quantitative structure–activity relationship (QSAR) analysis. This method has been widely used to estimate molecular properties [34, 152, 169] and has achieved great success in thermochemistry predictions [33, 37, 78, 137]. The molecular representation of QSAR is usually generated based on the chemical descriptors that represent constitutional, electrostatic, quantum-chemical, topological, geometrical, or fragment-based properties of the molecule. Because thousands of chemical descriptors can be used to construct molecular representations, a priori selection of descriptors based on domain knowledge [52, 53] is often required to reduce the reductant features [139].

The learned molecular fingerprint derived from deep learning encoders provides an appealing alternative for representing molecules. The learned fingerprint has recently become popular because it can automatically extract useful features from data without human intervention [162]. Typically, the learned molecular representation can be generated by encoding methods with three types of input: (i) 3D molecular structure, (ii) 2D molecular connectivity, and (iii) 1D SMILES representation. The pros and cons of these methods are discussed below.

The molecular representation generated from 3D structures can usually preserve the information of bond lengths, bond angles, cis–trans isomerism, stereoisomerism, and mutual orientation of atoms. Previous papers have proposed various model architectures to encode 3D molecular structures. For example, Schütt et al. developed the DTNN [132] and SchNet [133] models, both of which use a Gaussian basis to convert the pairwise interatomic distances into a 3D fingerprint. The Gaussian expansion functions allow the model to capture the interactions from different distance regimes and make the molecular representation rotational- and translational-invariant. Torng and Altman [149] and Kuzminykh et al. [88] applied 3D convolutional neural networks (3DCNNs) to generate 3D fingerprints for protein analysis and molecular property predictions, respectively. To be encoded by 3DCNNs, a molecule needs to be converted into a 3D image in which the voxel is the smallest unit representing an atom. However, since an atom only occupies a single voxel in the 3D image, most of the voxels contain no information, which leads to a data sparsity problem and affects the performance of the model. This issue can be resolved by

preprocessing the molecule by Gaussian smoothing and wave transformation so that the atoms will scatter in the 3D space, which leads to better learning of the interatomic interactions. Fang et al. proposed the GeoGNN model and demonstrated that it is possible to generate 3D molecular fingerprints using two graph-based encoders [41]. The first encoder is a standard graph encoder that regards the atoms and bonds as nodes and edges. On the other hand, the second encoder considers the bonds and the bond angles as nodes and edges, which contains the 3D information of bond length and angle. This novel architecture has enabled accurate molecular property predictions on various benchmarks [41].

The drawback of the 3D molecular representation is that the 3D structural information is not always available, and there is often more than one possible conformation for a given molecule, which results in various property values when different geometries are used. For predicting thermodynamic properties such as formation enthalpy, one can explore all the possible configurations and then calculate the Boltzmann weighted average [91] of each individual enthalpy of conformers [175] to derive the final enthalpy value. However, finding all the possible conformers is often challenging for large species, so this approach is not always applicable. This problem can be alleviated by the 1D- and 2D-input models (such as SMILES-based and graph-based encoding) because they sacrifice most of the 3D spatial information and consider only the connectivity between atoms. Moreover, previous works have shown that SMILES- and graph-based models can perform well on a broad range of tasks [82, 162], including thermodynamic property predictions [94, 121].

Despite the absence of 3D spatial information such as bond angles and bond lengths, 1D SMILES-based encoding provides an opportunity to fully exploit unlabeled data. SMILES stands for Simplified Molecular Input Line Entry System, a line notation for representing molecules [160]. The SMILES notation consists of a series of characters corresponding to the specification of atoms, bonds, branches, cyclic structures, disconnected structures, and aromaticity. The text-representation allows molecules to be handled by natural language processing (NLP). Recent advances in NLP [23] have enabled sequential models to extract meaningful information embedded in string representations via self-supervised pre-training [31, 39]. This technique has been successfully applied to SMILES-based molecular property predictions. For instance, Honda et al. [67] pre-trained a Transformer model [155] to output a SMILES string the same as the input. This pre-training process enables the model to learn the semantic rules of molecules to better construct the latent representation of molecules for the desired downstream tasks. Wang et al. and Chithrananda et al. also utilize the Transformer architecture and pre-train a SMILES-BERT model with large-scale unlabeled molecules on the masked SMILES recovery task [21, 156]. These self-supervised learning methods are particularly useful when the size of the training dataset is small. When there are enough training data, the benefits of self-supervised pre-training usually become less pronounced.

Another type of input for predicting molecular properties is the 2D molecular connectivity, which is often encoded by graph-based methods [24, 96, 145, 158, 166]. A molecule can be treated as a graph because atoms and bonds can be viewed as nodes and edges. Therefore, it is straightforward to use the graph convolutional

network (GCN) to learn the representation of molecules [173]. In addition, recent studies have shown that GCN models can be modified to handle the periodicity of crystal graphs and therefore can predict material properties such as band gap, Fermi energy, and formation energy [17, 130, 163]. One of the most popular and iconic GCN models designed for molecular property predictions is the Chemprop model proposed by Yang et al., which provides many user-friendly options to build GCN models for specific tasks conveniently [169]. In short, Chemprop uses a directed message passing neural network (D-MPNN) [49, 83] to iteratively update the states of the hidden nodes by collecting the information of their corresponding neighbor nodes and edges. The final molecular fingerprint is obtained by aggregating all the hidden node information with mean- or sum-pooling. This molecular fingerprint containing latent information on the molecular graph is then passed through the fully connected network to predict the molecular property of interest. For example, Chemprop has been implemented in the reaction mechanism generator (RMG) [46, 97], a widely used software for reaction simulation [81, 140, 172], to predict heat capacity, formation enthalpy, and entropy of molecules in complex reaction systems.

7.2.2 The Atomic Fingerprint Method

The molecular fingerprint methods discussed in the previous section aim to directly predict the property of the whole molecule. However, for extensive thermodynamic properties [18], it is possible to separately predict the property value of each molecular component and sum up these values to derive the molecular property. Indeed, the group contribution method has been widely used for predicting heat capacity, formation enthalpy, and entropy of molecules [25, 26, 47, 76, 108, 126]. However, the group contribution method requires tabulating the property values of chemical groups. Since the chemical groups are defined to describe a specific collection of molecules, the structural information that the model can perceive is limited by the dataset used to build the model. To expand the scope of applicability, one needs to define new groups or additional corrections, which requires expert knowledge and is not always straightforward, especially for molecules with complex structures, such as polycyclic species with fused rings [94].

Despite the limitation of the group contribution method, the concept of the additivity scheme has successfully been integrated with advanced machine learning models to predict extensive molecular properties. These machine learning methods employ various encoding techniques to generate atomic fingerprints for predicting the additivity values of each atom, which are summed together to obtain the property of the molecule. For example, the atom-centered symmetry function is one of the popular approaches that encode the pairwise interactions for the atoms [10]. Behler proposed using the radial and angular distribution functions to replace the 3D Cartesian coordinates so that the representations can be translational-, rotational-, and permutational-invariant [9]. Similar and improved encoding methods

have been proposed to encode the atomic fingerprint from 3D many-body interactions, including SOAP [6], wACSF [48], COMB [120], FCHL [40] and GMP [92]. It has been demonstrated that the improved atom-centered symmetry functions can perform well in predicting energy-related properties such as potential energy [171], atomization energy [129], and formation enthalpy [48]. Moreover, Pronobis et al. showed that one could slightly modify the architecture of an atomic fingerprint model to predict intensive properties [122]. For example, for a molecule with N atoms, the prediction can be formulated as $\hat{y} = \sum_{i=1}^{N} O(x_i)$ for extensive properties and $\hat{y} = \frac{1}{N} \sum_{i=1}^{N} O(x_i)$ for intensive properties, where O represents the output neural network and x_i represents the atomic fingerprint of atom i.

Similarly, the concept of atomic fingerprints can also be extended to 2D graph-based models [18, 159]. The hidden node state derived from the collection of the neighbor information can be viewed as the fingerprint of the atom [18]. Instead of aggregating the node vectors to derive a molecular representation, the latent node vectors can be fed to the fully connected network to calculate their corresponding atomic contribution values. And the sum of each atomic contribution derives the property of the whole molecule as shown in Fig. 7.1. Because the atomic fingerprints derived from graph-based encoding can preserve structural information, the property values of atoms are calculated with the effects of structural environments. In addition, because the atomic fingerprint model combines the advantages of the group contribution method and deep learning, it can be easily generalized to larger molecules not present in the training data [18].

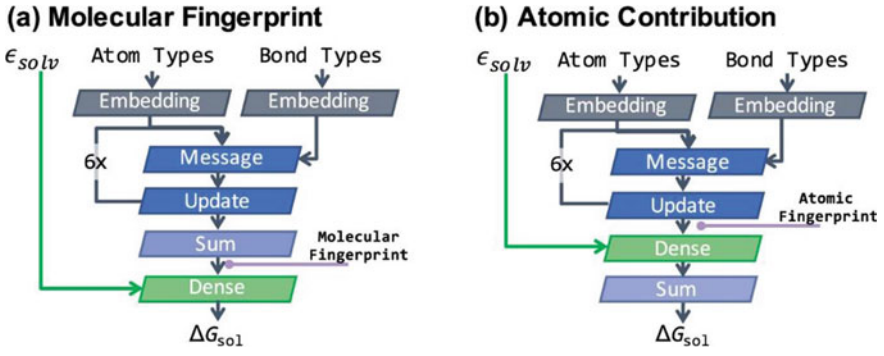


Fig. 7.1 The comparison between **a** the molecular fingerprint and **b** the atomic fingerprint methods. In the original paper, the models are designed to predict the solvation energy. To achieve this, the models take the dielectric constants of solvents (ϵ_{solv}) as inputs and feed them to the dense layer. Reprinted with permission from Ref. [159]. Copyright (2021) American Chemical Society

7.2.3 *Remedies for Data Scarcity: Transfer Learning and Active Learning*

Because accurate thermochemistry data are scarce, it is difficult to train a machine learning model solely using the data derived from experiments or high-level wave function-based ab initio calculations. One way to overcome this problem is to take advantage of density functional theory (DFT) methods that have low computational costs [58, 124]. The drawback of this approach is that the low-level data inevitably have larger errors, which will be inherited by the model when used for training. This issue can be alleviated using transfer learning (or knowledge transfer) [116, 161, 177], which is a machine learning technique that aims to improve the performance of a model when there is insufficient data in the domain of interest. Transfer learning can be done by training the model on a different but relevant domain with sufficient data before training the model in the domain of interest. For example, to build a machine learning model for accurate thermochemistry prediction, Grambow et al. [56] first trained a base model on the thermodynamic data calculated at the B3LYP level of theory (low level) and then fine-tuned the base model on a high-quality dataset composed of experimental data and data calculated with CCSD(T)-F12a (high level). Similar strategies have been applied in many different works [15, 57, 72, 75, 142, 165] to leverage low-quality but sufficient data to make the machine learning models more robust for real applications (Fig. 7.2).

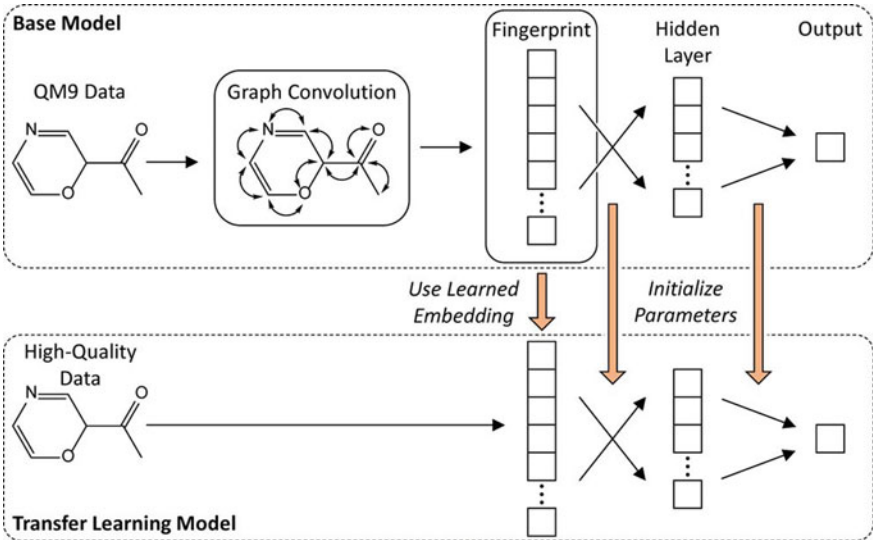


Fig. 7.2 Illustration of the transfer learning procedure. The based model is pre-trained on a low-quality dataset. Subsequently, the transfer learning model loads the weights of the base model and undergoes fine-tuning on a high-quality dataset. Reprinted with permission from Ref. [56]. Copyright (2019) American Chemical Society

Another possible solution to data scarcity is the active learning strategy, which reduces the amount of training data required by selectively collecting new data that can most effectively enhance the machine learning model [113, 151, 157, 164]. For example, Gubaev et al. propose a generalization of the D-optimality criterion that can be used for non-linear models as their query strategy [59]. The D-optimal training subset is determined by yielding the most linear rows in the Jacobi matrix, which is solved by the maxvol algorithm [55]. The authors illustrated that this method could reduce the training data required for high-accuracy enthalpy predictions. Active learning can also be carried out using an uncertainty-guided approach, which provides new training data for samples for which the predictions are potentially associated with significant uncertainty as judged by the current model [131, 148]. As discussed in the work of Hirschfeld et al. [65], there are four primary approaches for uncertainty quantification: ensemble-based methods [90], distance-based methods [127], union-based methods [70], and mean–variance estimation [112]. Some of these methods provide quantitative measurement of aleatoric or epistemic uncertainties, while others are only for indicating relative uncertainty. Generally, aleatoric uncertainty stems from the error of data, while epistemic uncertainty represents the shortage of knowledge of the model. Previous work has shown that it is more preferred to use epistemic uncertainty to guide active learning [111]. However, traditional ensemble-based epistemic uncertainty requires additional computational costs, which impedes the rapid screening of the chemical space. Recently, Soleimany et al. leveraged evidential deep learning to use one single model to replace ensemble models [143]. The evidential deep learning algorithm uses evidential priors with the Gaussian likelihood function and regularizes incorrect evidence penalty, which makes the models learn both aleatoric and epistemic uncertainty simultaneously [3, 136]. Hence, this method can help to avoid the high computational cost associated with exploring the unknown chemical space using epistemic uncertainty-guided active learning.

7.3 Predicting Rate Constants

7.3.1 Conventional Methods to Calculate Rate Constants

It is often assumed that the reactions forming the same type of transition state species tend to have similar activation energies and pre-exponential factors of the Arrhenius equation. For example, Dean [29] analyzed a number of addition and recombination reactions and obtained the generic pre-exponential factor and activation energy for these reaction types. By collecting kinetic data of reaction types of interest, one can quickly estimate the rate of similar reactions at specific conditions. The reaction mechanism generator (RMG) software [97] employs this idea to model combustion reactions by building a kinetics database containing various reaction types relevant to combustion systems. This model implemented in RMG is basically a hierarchical decision tree that aims to match a given reaction to the most similar reactions in the

database. The kinetic parameters of these similar reactions are then extracted and averaged to estimate the kinetic parameters of the given reaction. This algorithm relies on iteratively checking whether the reaction matches the preconstructed reaction templates, which are string representations specifying how the substructures of the species change in reactions, as illustrated in Fig. 7.3. The hierarchical decision tree provides a convenient and interpretable way to quickly estimate kinetic parameters and is particularly useful for the popular reaction types. Because the templates are defined to describe a specific reaction data set, the types of reactions that the decision tree can perceive are limited by the data set used.

The quantitative structure–activity relationship (QSAR) model [80, 123] has also been used to analyze the relationship between the kinetic properties and the physico-chemical descriptors of reactants and products [16, 109, 115, 127, 128, 167]. Hoonakker et al. proposed the condensed graph of reaction (CGR) encoding method that merges reactants and products into a single pseudomolecule [68]. The CGR

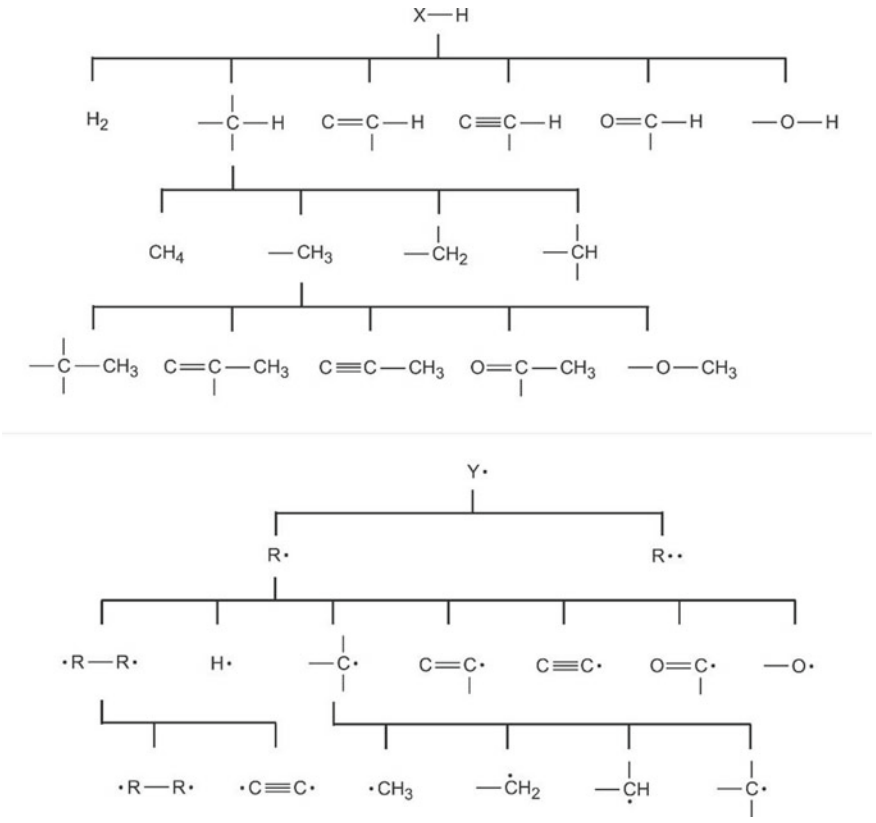


Fig. 7.3 Hierarchical decision trees for template mapping of an H-abstraction reaction. Reproduced with permission from Gao et al. [46], open access under a CC BY license (Creative Commons Attribution 4.0 International License)

method allows the combined molecular graph to have the superposition of atoms in the reactant, reagent, and product so that standard cheminformatics methods such as ISIDA fragment descriptors [154] and SiRMS simplex descriptors [87] can be applied to featurize the pseudomolecules. For example, Madzhidova et al. used CGR to establish support vector regression and random forest models to predict the rate constants of bimolecular elimination and nucleophilic substitution reactions [104, 105]. The QSAR approaches have also been used to analyze the cycloaddition [50, 125], hydroxyl radicals reaction [102], ozone-hydrogen peroxide oxidation [19], and radical C–H functionalization [93].

7.3.2 Neural Networks for Predicting Kinetic Parameters

With the emerging trends of deep learning, ANNs have also been applied to predict reaction rate constants [2, 22, 69]. The simple concept of using the multi-layer perceptron (MLP) to predict the logarithm of rate constants has been demonstrated for the aqueous OH reaction [36], OH radical reaction [2, 100], and tropospheric ozone degradation [42]. More recently, Komp et al. generated a large-scale database containing roughly 6.9 million reaction rate constants obtained from numerical integration of the transmission probability [86]. In this work, physical descriptors such as barrier height, width, and asymmetry serve as the inputs of the neural network, and the model is validated with the testing data of different reaction systems. These works require the user-chosen molecular descriptors as model inputs, so there is no standard representation that can be generalized to other types of reactions.

The more advanced deep learning architectures such as graph-based, SMILES-based and 3D-based inputs allow the model to automatically learn the implicit information from the molecular connectivity so that they can often generalize better than the conventional ANN models. For example, the recent work of Grambow et al. demonstrates that it is possible to expand the scope of applicability using a template-free graph-based model [57]. The authors constructed a D-MPNN model to predict the activation energy using the pair of reactant and product graphs as inputs. The architecture of this model is very similar to that of Chemprop [166] discussed in the previous subsection. The main difference is that both the molecular graphs of reactant and product are fed to D-MPNN to encode a reaction, as shown in Fig. 7.4. The hidden vector of each atom in the product is subtracted by that of the corresponding atom in the reactant to derive the atomic difference fingerprint. The difference fingerprints of each atom pair are passed through another neural network and then are summed to obtain the final reaction fingerprint to predict the activation energy. The idea of this model is similar to that of siamese neural networks [20], which are often used on pairs of atoms for similarity calculations. Because the difference fingerprints of the atoms far away from the reaction center have less contribution to the final reaction fingerprint, this model can focus on the change of structures near the reaction center and therefore can properly predict the activation energy required for the transformation. There are other successful applications of graph-based models in activation energy

predictions. For example, instead of using dual D-MPNN to encode the reactant and product graphs, Heid and Green [62] extend the default functions of the Chemprop package with CGR [28, 68, 114], which unites the atom and bond features of the reactant and product to form a single condensed molecular graph representing the reaction.

We note that the methods discussed above require the atom-mapping between the reactant and product atoms, which is not always available in reaction data sets. Therefore, automatic atom mapping algorithms have been developed, such as ReactionMap [44], MarvinJS [73], and RXNMapper [135]. Interested readers are referred to the review paper of Lin et al. [95].

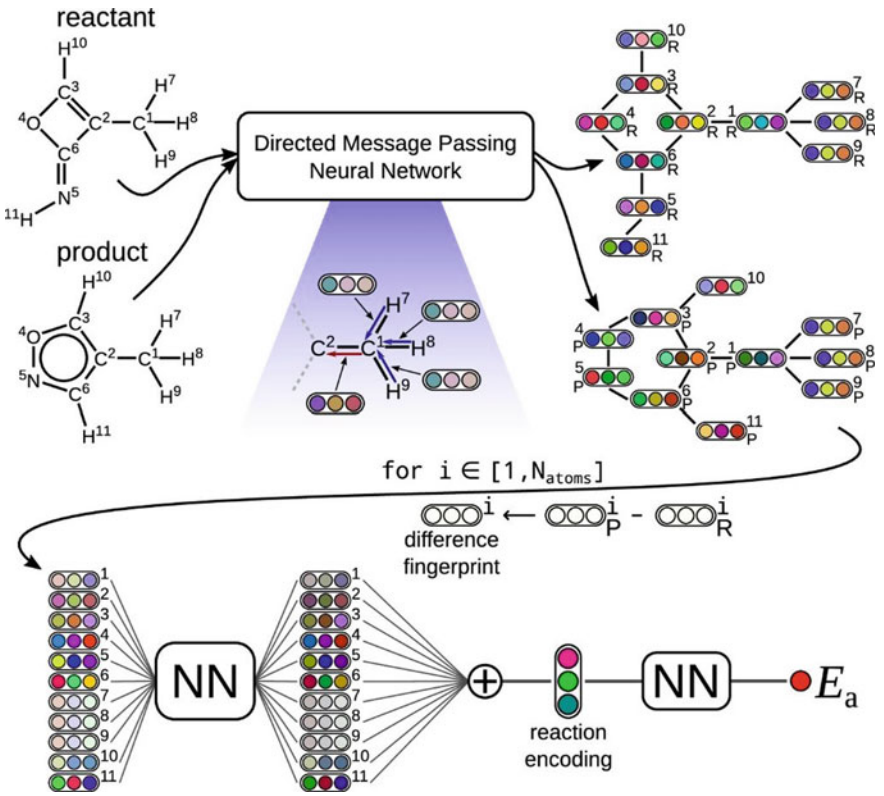


Fig. 7.4 Illustration of the model architecture developed by Grambow et al. For an atom-mapping reaction, reactant and product graphs are fed to the D-MPNN encoder to obtain the hidden atom states. The difference of each atom pair is calculated and passed through another neural network (NN). Finally, the summation of these difference fingerprints is used to predict the activation energy E_a of the reaction. Reprinted with permission from Ref. [57]. Copyright (2020) American Chemical Society

7.3.3 Machine Learning for Transition State Search

In addition to directly predicting kinetic parameters, machine learning techniques have also been applied to aid ab initio simulations for a more efficient transition state search. Finding the structure of transition state species is regarded as an important task because optimized transition state structures are required for reaction barrier and rate constant calculations. The simplest method for searching for transition states is a linear or quadratic interpolation between the three-dimensional reactant and product structures, which is fast but prone to failure in convergence. To address this issue, more advanced algorithms have been proposed, such as the freezing string method (FSM) [11, 107], growing string method (GSM) [54, 118], and nudged elastic band (NEB) method [63, 64, 71]. These algorithms aim to approximate the minimum energy path (MEP) that characterizes the reaction coordinates. The complete knowledge of MEP will guarantee that the transition state can be determined by calculating the highest energy point along this path [64]. However, these transition state search methods are computationally demanding because they need to compute gradients repeatedly using ab initio calculations. For multiple-step reaction pathways, the computational costs will grow exponentially with the increasing complexity of the reaction mechanism. Therefore, using ab initio calculations to generate kinetic parameters is seldom applied to complex systems containing hundreds or even thousands of reactions.

One way to alleviate this problem is to use machine learning models to mimic the ab initio potential energy surface [7, 10, 99, 119]. Since energies and gradients can be calculated rapidly using machine learning models, the search for transition states can be carried efficiently on the machine learning energy surface. Typically, NEB calculations require roughly hundreds of ab initio force calls to converge. Recent studies have shown that it is possible to reduce the number of ab initio force calls of NEB using Gaussian Process Regression (GPR) [30, 84, 85, 150]. The GPR is a popular, flexible, and non-parametric machine learning method that can provide uncertainty estimates based on Bayesian inference [13, 103, 134, 153]. The integration of GPR helps improve the sample efficiency on the potential energy surface. For example, Koistinen et al. introduce all-images-evaluated (AIE) and one-image-evaluated (OIE) algorithms for accelerating NEB calculations [85]. For both algorithms, one needs to provide the initial guess of the reaction path, which is often derived from linear synchronous transit (LST) [60], image dependent pair potential (IDPP) [141], or geodesic interpolation [176]. The main difference between AIE and OIE algorithms is how the training set of the Gaussian Process (GP) model is constructed. For the AIE method, the true energies and atomic forces of all the images in the initial path are evaluated and are used to train the initial GP model. The surrogate GP model serves as the approximate potential energy surface, in which the approximate force vectors of each image are calculated to determine the next iteration of intermediate images as shown in Fig. 7.5. Next, the true energies and atomic forces of the new images are evaluated and used to re-train the GP model until the NEB forces are converged. On the other hand, OIE does not evaluate the

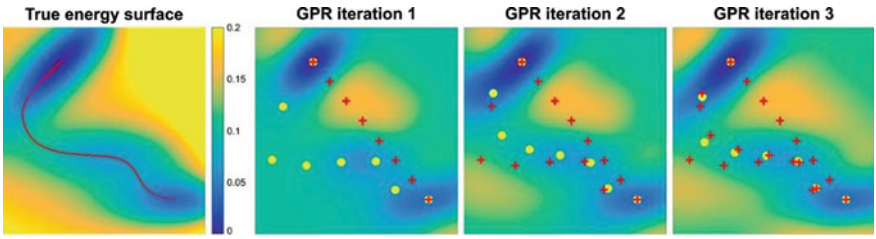


Fig. 7.5 Far left: Illustration of the two-dimensional Müller-Brown energy surface with the minimum energy path shown by the red curve. Three panels to the right: The first, second, and third iterative development of the approximate potential energy surface represented by the surrogate Gaussian Process model. The red plus signs represent the points whose energies and atomic forces are evaluated by ab initio calculations. The yellow disks show the intermediate images optimized by the nudged elastic band method and relaxed on the approximate potential energy surface. In each iteration, the energies of all the yellow disks are calculated to become the new training data for the Gaussian Process model (all-images-evaluated algorithm). Reprinted with permission from Ref. [85]. Copyright 2017 American Institute of Physics

true energies and forces of all the images during each iteration. Instead, as shown in Fig. 7.6, OIE only calculates the energy and force of the single image with the highest uncertainty predicted by the GP model. The concept of the OIE algorithm resembles the uncertainty-guided active learning, which continuously improves the model by adding the most unfamiliar data to the training set. Normally, this characteristic makes OIE require fewer ab initio force evaluations than the AIE algorithm [85].

An alternative approach to mitigate the computational cost of potential energy surface exploration is to employ reinforcement learning methods. Ahuja et al. [1] have successfully applied a reinforcement learning method to develop a model to produce a correction term for the quasi-Newton step calculated with the BFGS algorithm [14, 43, 51, 138], which enables the optimizer to complete geometry optimization in roughly 30% fewer steps compared to the pure BFGS. Zhang et al. designed the reinforcement learning game as a "shooting game" that aims to find out the correct paired reactant and product structures in the quantum mechanics/molecular mechanics (QM/MM) molecular dynamics (MD) trajectories [170]. Yoon et al. also developed a CatGym reinforcement learning environment to predict the kinetic pathways of the ternary $Ni_3Pd_3Au_2(111)$ alloy catalyst [168].

Recent studies have also shown that it is possible to utilize machine learning to predict the transition state structures directly [74, 120]. This approach provides an efficient way to generate initial guesses of the coordinates of the atoms of the transition state without the need to carry out any ab initio calculations. For example, Jackson et al. [72] used the three-dimensional rotation and translation-equivariant tensor field network (TSNet) [146] to predict the Cartesian coordinates of the gas-phase S_N2 reaction transition state structures. These authors demonstrated that only the 2D molecular graphs of reactants and products are needed for model inputs, which means that one does not even need to optimize the 3D structures of reactants and products. More recently, Pattanaik et al. [117] successfully generated the transition

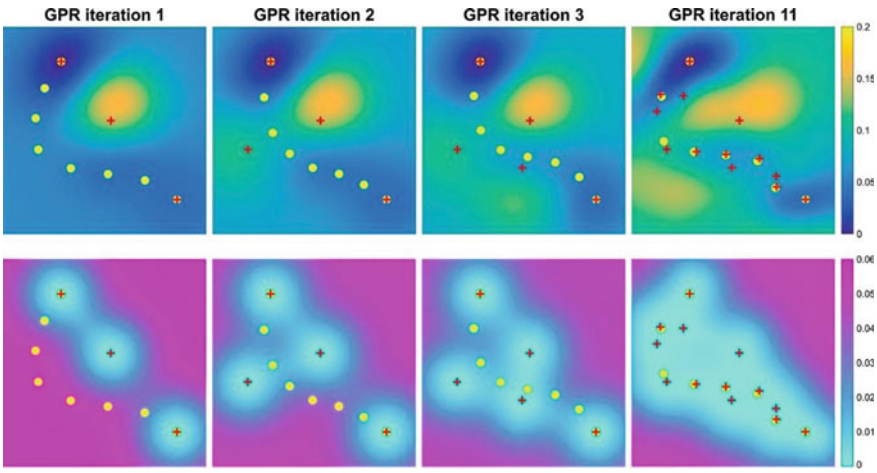


Fig. 7.6 Upper panel: Illustration of the two-dimensional Müller-Brown energy surface approximated by the Gaussian Process model. The red plus signs are the points whose energies and atomic forces serve as the training data of the Gaussian Process model. The yellow disks show the intermediate images optimized by the nudged elastic band method and relaxed on the approximate potential energy surface. Lower panel: The uncertainty estimated by the Gaussian Process model. In each iteration, the energy and force of only one intermediate image with the highest uncertainty are evaluated with ab initiocalculations (one-image-evaluated algorithm). Reprinted with permission from Ref. [85]. Copyright 2017 American Institute of Physics

state structures of isomerization reactions using a graph neural network (GNN) [8], which encodes the reaction graphs containing the atom features and the averaged bond features derived from the reactant and product. After graph convolutions, the embedding edge vectors are fed to another neural network to predict the interatomic distances (D_{ij}) of the transition state and the weight matrix (W) for the subsequent optimization. The predicted distance matrix is then reverted back to Cartesian coordinates by eigenvalue decomposition of the Gram matrix [5, 38]. Since the Cartesian coordinates (X_{init}) obtained from eigenvalue decomposition may not perfectly match the predicted distance matrix, the authors used non-linear square (NLS) optimization to further refine the coordinates. The framework of this transition state prediction model is illustrated in Fig. 7.7.

In addition to the methods mentioned above, Makoś et al. have also successfully used the conditional generative adversarial network (GAN) to generate the initial guess of transition state structure [106]. As shown in Fig. 7.8, the transition state GAN consists of a generator and a discriminator. The generator is a convolutional neural network that synthesizes fake transition state structures with the inputs of reactant and product Coulomb matrices, which are translational- and rotational-invariant representations of molecules commonly used in machine learning [129]. The real transition state data are combined with these fake data and fed to the discriminator,

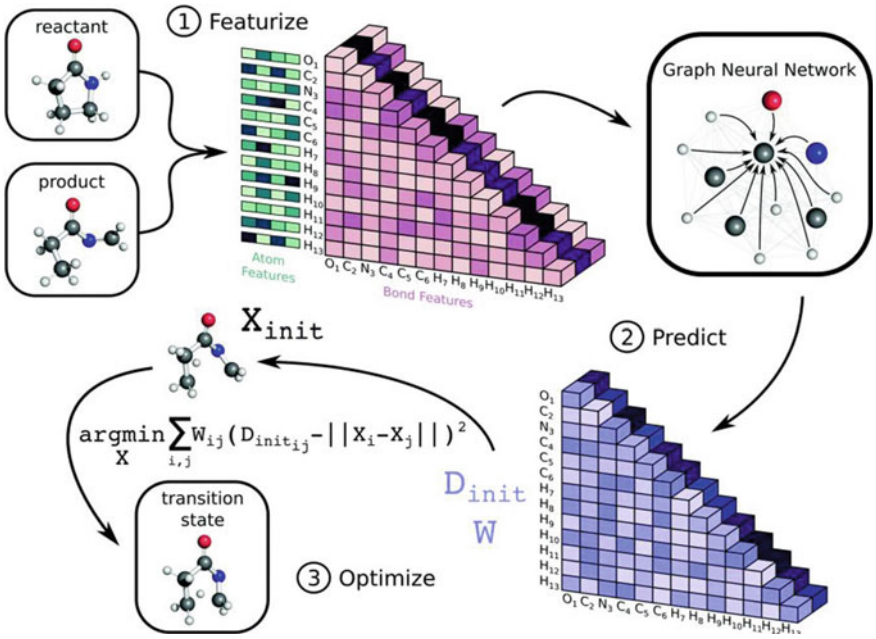


Fig. 7.7 Illustration of the generation of transition state structures. Reactant and product graphs are used as inputs to predict the distance and weight matrices of the transition state structure. Next, non-linear square optimization is performed to obtain the Cartesian coordinates of the transition state geometry. Reprinted with permission from Ref. [117] open access under a CC BY license (Creative Commons Attribution 3.0 International License)

a binary classifier that distinguishes between the fake and real transition states. For the generator, the training process is a competitive game against the discriminator because the generator has to produce the transition state structures closely resembling the real ones. Consequently, after training, a reliable initial guess of transition state structure can be obtained from this generative model.

7.4 Conclusions

High-fidelity kinetic modeling is essential for understanding detailed reaction chemistry and predicting the performance of a reaction system. However, the realism of reaction simulations is often affected by the accuracy of reaction rate constants and thermodynamic parameters, which determine the equilibrium constants and the reverse reaction rates. With the rapid development of machine learning techniques, people have successfully constructed data-driven models to predict thermodynamic and kinetic properties of molecules. Various approaches have been proposed to transform molecules into a fixed-length representation suitable for

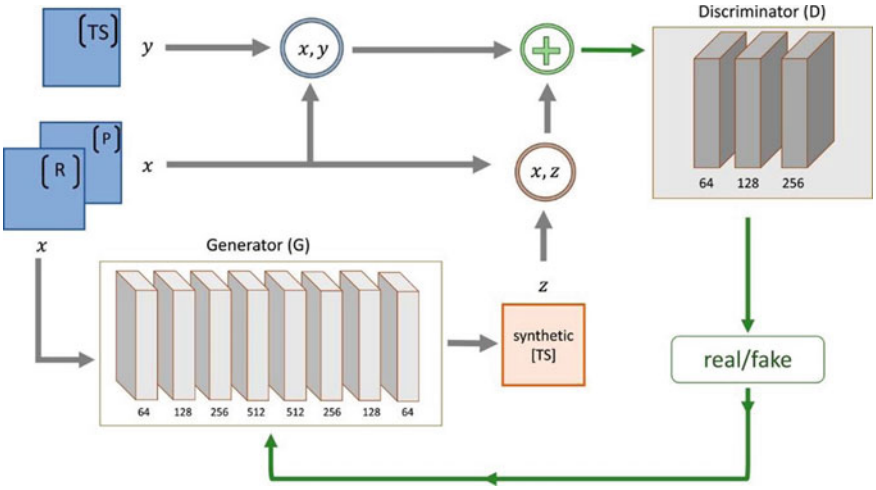


Fig. 7.8 Model architecture of the transition state GAN. The Coulomb matrices of reactant and product (x) are fed to the generator to synthesize data (z). The real pair (x, y) and fake pair (x, z) are concatenated by $\oplus$ and fed to the discriminator. Reprinted with permission from Ref [106]. Copyright 2021 American Institute of Physics

machine learning algorithms, including molecular descriptors, 3D molecular representations, SMILES-based, and graph-based encoding. For molecular thermochemistry predictions, SMILES- and graph-based methods without 3D structural information such as bond angles and bond lengths have been shown to perform well. Although adding 3D information can boost the model performance, the information of 3D structure is not always readily available, especially for applications in which one wishes to carry out high-throughput screening. For extensive thermodynamic properties such as formation enthalpy, entropy, and heat capacity, the atomic fingerprint method can often generalize better than the molecular fingerprint method, because the atomic fingerprint model combines machine learning with the concept of the additivity scheme.

Machine learning models have also been applied to predict kinetic parameters such as activation energies, which are directly related to the geometry of the transition state, and therefore can be inferred from the structures of the reactant and product. In addition, it is possible to use machine learning potential energy surfaces and reinforcement learning methods to facilitate geometry optimization or transition state search. Recent studies have also shown that one can use deep learning methods to predict the transition state structures without performing ab initio calculations. Though there is still room for improvement in the performance of these methods, they have shown great potential for improving conventional simulation protocols, and hence are expected to play a more critical role in reaction simulations in the future.

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Chapter 8
Synthesize in a Smart Way: A Brief Introduction to Intelligence and Automation in Organic Synthesis



Dian-Zhao Lin, Guichun Fang, and Kuangbiao Liao

Abstract In this chapter, we will give a brief introduction to intelligence and automation in organic synthesis. The chapter includes five main sections. In the first section, selected examples of computer-assisted retrosynthetic planning were placed and discussed. In the second section, we will discuss reaction outcome prediction models, which employ machine learning to establish models for reaction yield, reaction selectivity, reaction intermediate, reaction type, reaction product, and reaction condition prediction. A brief introduction to essential elements for the establishment of the reaction prediction model is also documented in this section. Related examples of machine learning-assisted reaction discovery and optimization will be presented in the third section, followed by exceptional representative cases of automated synthesis and laboratory robots. At the end of this chapter, several issues of machine learning and automation that chemists may be concerned about will be discoursed.

Keywords Artificial intelligence · Machine learning · Synthetic automation · Organic synthesis · Computer-assisted synthesis planning · Reaction prediction model

8.1 Introduction

Since the first organic molecule made from inorganic compound, urea, was synthesized in 1828 [1], organic chemistry has been a representation of an inefficient and labor-intensive process for two centuries. Synthetic chemists used to take months or years, to optimize reaction conditions, to discover new reactions, and to assemble natural product molecules. Thanks to the fourth industrial revolution (Industry 4.0)

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triggered by interconnected technologies, organic synthesis has also been stepping into Synthesis 4.0, a new era that unifies machines, automation, and artificial intelligence (AI) to deliver intelligent solutions to synthetic challenges. Drawing on technics from the biological field, synthetic chemists began to use high-throughput experimentation (HTE) for rapid multiple reaction parameters screening, large numbers of compound libraries parallel synthesis, as well as abundant reliable data acquirement [2–14] (vide infra). Taking advantage of synthetic automation and laboratory robotics, synthetic fellows are liberated from their day-after-day synthetic work [15–32]. Employing big data and machine learning (ML) [33], organic reactions are becoming predictable and synthetic folks will no longer discover their reaction by intuition or by chance [26, 30, 31, 34–38]. Organic synthesis is now walking on a digital and automated highway.

In this chapter, we would like to give a brief introduction to intelligence and automation in organic synthesis. Specifically, selected recent examples of computer-assisted synthesis planning (CASP), including retrosynthetic planning, reaction outcomes prediction model, as well as ML-assisted reaction discovery and optimization, will be presented. In addition, exceptional representative cases of automated synthesis and laboratory robots will be documented. Note that publications not mentioned in this chapter do not mean they are less critical, but because of the length limitation of this article. Finally, an outlook on issues of ML and automation that chemists may be concerned about will be discussed.

8.2 Retrosynthetic Planning

Training algorithms for computational multistep organic synthesis planning has been challenging for more than five decades. Since E. J. Corey, the Nobel Chemistry Prize laureate in 1990, conceived and formalized in the 1960s [39, 40], organic chemists and computer scientists have been exploring the pavement towards computer-assisted (retro-)synthesis planning [41] due to its attractive potential to search for "checkmate moves" in an ample chemical space initialing from a particular molecule [42–47]. Generally, algorithms for retrosynthetic planning are based on template libraries that involve matching generalized reaction rules that encode disconnections between atoms to predict reaction outcomes or candidate precursors. When a molecule structure is an input, the computer searches for the appropriate disconnection rules in the template library. The same procedure will continue in the sub-molecules until the fundamental molecules (e.g., readily available or no matched disconnection rules can be found) are obtained.

During the 1980s to 1990s, a decade after Logic and Heuristics Applied to Synthetic Analysis (LHASA), the first computational approach for suggesting synthetic pathways was developed [48], new algorithms for synthetic design and reaction prediction emerged rapidly. For example, Computer-Assisted Mechanistic Evaluation of Organic Reactions (CAMEO) was developed for reaction prediction by analysis of nucleophilic and electrophilic sites [49, 50]. EROS enabled to identify

of leading structures for retrosynthesis according to bond polarity, electronegativity, and the resonance effect [51]. SOPHIA was capable of predicting reaction outcomes with minimal user input [52, 53], while KOSP could be used to predict retrosynthetic targets [54]. These prediction tools suffered from apparent disadvantages, such as narrow application scope.

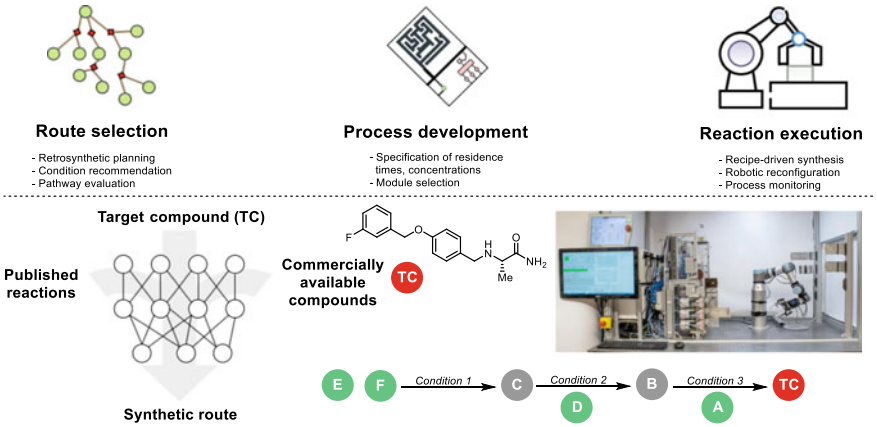
Segler and co-workers reported a knowledge-graph approach that used the concept of half-reactions to generate possible products in 2017 [55]. They soon reported a hybrid template neural-symbolic approach for both retrosynthesis and reaction prediction [56]. The algorithm could mimic human's intuitive decision-making to avoid reactivity conflicts. Compared with the rule-based expert systems, this approach was 150 times faster. However, this system did not take stereochemistry into account.

In 2018, Segler et al. used Monte Carlo tree search (MCTS) and symbolic AI to discover retrosynthetic routes for small organic molecules [57]. MCTS was combined with an expansion policy network that guided the search and a filter network to preselect the most promising retrosynthetic steps. At the same time, the deep neural works were trained on over 12.4 million known single-step chemical reactions that have ever been published in organic chemistry. In a double-blind AB test, the synthetic routes generated by AI were on average considered to be equivalent to reported literature routes by human chemists. Nonetheless, natural product synthesis was beyond this method's capabilities, while quantitatively predicting abilities in enantiomeric or diastereomeric ratios remained challenging.

In 2019, based on their previous work [58], Coley et al. described a comprehensive computer-augmented chemical synthesis strategy whose synthetic planning is proposed through millions of chemical publications and simulated in silico to maximize the possibility of success, followed by the execution by a modular continuous-flow platform which is automatically reconfigured by a robotic arm (Scheme 8.1) [59]. This development strategy, which is demonstrated by 15 drugs or drug-like compounds, enhanced the ability of chemists to perform synthesis while relieving them from reducing information collection and manual tasks.

In 2020, based on their early studies [60], Grzybowski et al. reported a new version of *Chematica* (or *Synthia*) [61]. The program's knowledge of organic chemistry and data-based artificial intelligence routines were augmented with causal relationships, which allowed it to "strategize" over multiple synthetic steps. A Turing-like test showed that the routes designed by such a hybrid AI-expert are principally indistinguishable from those designed by human chemists. Three computer-designed synthetic routes of complex natural products were successfully validated in the laboratory, demonstrating this expert-level automated synthetic planning program's potential in organic synthesis, especially total synthesis. Continued improvements to the reaction knowledge base and further code optimization for the system are still pending by the authors.

As mentioned above, most retrosynthetic planners base on template libraries. Corresponding reaction rules in such template libraries were basically codified manually, resulting in time-consuming, prediction impracticality [62], and reaction knowledge obsoletion [42, 63, 64]. Extracting transformations from reaction datasets is an



Scheme 8.1 Computer-augmented chemical synthesis strategy by Coley et al.

alternative to manually encoding reaction rules [65–67]. For example, Grzybowski et al. created a "Network of Organic Chemistry" by encoding all the reactions in the Beilstein database [68]. Synthetic pathways for any molecule similar enough to a molecule already in this database could be successfully drawn by searching this network [69]. Unfortunately, the algorithm still lacked the ability to predict unseen reaction outcomes [70, 71]. Additionally, such synthetic data collected from literature could be misleading due to human biases and incompleteness, for example, lack of failed results [72, 73].

Given the drawbacks of rule or template libraries-based retrosynthetic planning approaches, experts begin to develop template-free strategies. Cases in this field generally employ text, for example, simplified molecular-input line-entry system (SMILES) [74], to represent molecules [75, 76]. For example, Liu et al. reported a template-free data-driven neural sequence-to-sequence (seq2seq) model for retrosynthetic prediction [77]. This end-to-end trained model, which used SMILES as both input and output, had an encoder-decoder architecture consisting of two recurrent neural networks (RNN), which was previously successfully demonstrated for forward reaction prediction tasks [78–80]. This approach has shown a comparable predicted accuracy with a rule-based expert system baseline model.

Using a graph convolutional neural network (GCNN), Jin et al. reported an efficient approach to explore the space of product molecules by first pinpointing the reaction center instead of generating candidate products by reaction templates [81]. Both the reaction center identifier and the candidate ranking model are built from Weisfeiler-Lehman Network (WLN) and its variants, such as Weisfeiler-Lehman Difference Network (WLDN) [82].

Although SMILES has been widely used to represent molecules, the grammar of SMILES is so fragile and complex that even a single character change is enough to invalidate the entire SMILES string. Recently, Lee's group developed a new template-free retrosynthesis prediction model, RetroTRAE, by the usage of the Transformer

architecture [83, 84] associated with the atom environments [85] representation [86]. The approach showed comparable performance compared to other SMILES-based Transformer models [87–94].

Besides the synthesis of conventional organic molecules, chemists also use planning algorithm for peptide synthesis. Since that peptide designs are traditionally guided by human expertise and intuition, which are difficultly scalable and are susceptible to human bias, very recently, Fry, Sankaranarayanan, and their colleagues documented a machine learning workflow that combined MCTS and random forest (RF) with molecular dynamics simulations to develop a fully autonomous computational search engine to discover peptide sequences with high potential for self-assembly [95]. The workflow, called AI-expert, could efficiently search large spaces of tripeptides and pentapeptides, with better ability in peptide prediction than human experts. A matter of non-intuitive sequences with high self-assembly propensity were obtained through the AI-expert, suggesting its potential to overcome human bias and accelerate peptide discovery. However, AI-expert did not receive any feedback from actual experiments, and only a modest correlation was found between the computational and experimental scores. In addition, the AI-expert seemed to have its own bias: the preference for proline and the limited proclivity for valine.

8.3 Reaction Prediction

Besides computer-assisted retrosynthetic planning, using AI to predict reaction outcomes, such as reaction products of given reactants, yield, and selectivity, is also appealing. There are three essential parts for achieving the goal of reaction prediction (Fig. 8.1): availability of sufficient high-quality data set for model training, appropriate descriptors to describe reaction variables, and fitting algorithm to build the prediction model.

Training data can be obtained from the Internet, such as SciFinder^n, Reaxys®, patents, published literature, proprietary databases, or platforms that have extracted and tabulated data from publications [96–98]. However, the quality of such data can be unreliable, and metadata such as temperature or critical variables such as stereochemistry may be omitted. Data imbalance is another problem because negative results are always dropped away when chemists publish their research achievements

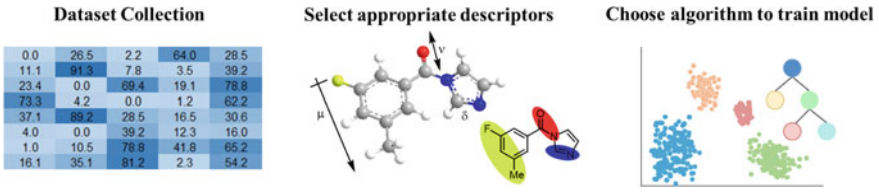


Fig. 8.1 Procedure for establishment of reaction outcomes prediction model

[99–101]. Another path to access required data is in-house experiments. In order to generate a large data set (at least hundreds of data points), HTE is required. HTE is a technique for screening chemical experiments on a miniaturized scale. Commonly, the term "HTE" refers to a miniaturized well-plate batch format, which includes 24-well, 96-well, as well as 384-well arrays, while ultraHTE, which requires specialized equipment [7, 102, 103], refers to 1536 arrays or more. When performing HTE reactions, solvents or liquid reagents in small volumes can be handled through single-channel or multichannel micropipettes manually, or commercial liquid-handling automatic systems, such as Chemspeed SWING [104, 105], Tecan Freedom EVO [5], Labcyte Echo [106, 107], Thermo Matrix [108], SPT Labtech mosquito [102, 108–111], and Unchained Labs Junior [5]. On the other hand, owing to the diversity of properties, solid dispensing is problematic in HTE [112]. One of the solutions is coating solid chemicals on the surface of tiny glass beads evenly [113, 114]. However, our in-house practice suggests that some solid chemicals are inapplicable for the "ChemBeads" strategy because the solid cannot evenly coat on the glass beads. Numerous examples of using HTE for condition screening or data collection have been documented [102, 108, 110, 115–119]. For those who are interested in HTE in chemistry can refer to corresponding comprehensive reviews [2–14].

Molecular descriptors are used to characterize the molecules. Such descriptors are generated by algorithms as mathematical representations of molecules' properties [120] according to their "dimensionality" [121]. One-dimensional (1D) descriptors contain bulk physicochemical properties parameters like molecular weight, log P, polar surface area, etc. Two-dimensional (2D) descriptors comprise structural fragments or connectivity indices derived from the 2D representation of the molecule. Three-dimensional (3D) descriptors, such as molecular shape, are derived from 3D molecular structures. Diverse types of molecular fingerprints, a vector of digits or floats that correspond to the molecule's properties, were applied as descriptors in algorithms for reaction prediction. Morgan molecular fingerprints (MMF) [122] and extended circular fingerprints (ECFP) [123] have been widely used to predict molecular properties, including HOMO–LUMO gaps [124], protein–ligand binding affinity [125], drug toxicity levels [126], as well as to predict synthetic accessibility [127]. Plentiful developments in fingerprinting methods have already been reported [128–135]. Practically, one can use software libraries like RDKit or Mordred [136] to convert a given molecule into a 1D or 2D descriptor. The usage of neural networks can provide a molecular graph representation for reaction prediction models from 2D molecular structures [75, 129, 134, 135]. Density functional theory (DFT) calculation and molecular mechanics (MM) calculation are used to generate physically meaningful descriptors that express electronic and steric properties in 3D space [137], including HOMO–LUMO energies, torsion angles, Sterimol parameters [138, 139], and buried volume [140, 141]. Several attempts were reported to predict the reaction's likelihood through the reaction energy barriers [142–147]. Although quantum calculations can guide the exploration to a broad range of reactions than the heuristic-based methods, new calculations for each additional reaction family will cost a lot of time and money. Baldi and co-workers documented predicting reactions through

reaction mechanism perspectives [148–150]. They used the ReactionPredictor algorithm to identify potential electron sources and sinks of molecules, enumerate and rank possible interactions [64, 151]. However, manually encoding mechanistic rules would be problematic and limit scalability. Jiang, Luo, and co-workers recently designed a chemistry-informed molecular graph (CIMG) to describe chemical reactions [152]. Chemical information, including NMR chemical shifts as vertex features, bond dissociation energies (BDEs) as edge features, and catalysts/solvents information as global features, was used to represent molecules and reactions. Associated with graph neural network (GNN) models and MCTS, full multi-step retrosynthetic planning routes with recommended catalysts/solvents are predicted efficiently.

With a high-quality data set and appropriate descriptors in hand, determining a fitting algorithm is the final step towards constructing a reaction prediction model. Mainly, the given data set will be divided into two subsets: the training data set that is used to fit the reaction algorithm and the test data set that is used to validate the trained model. In some cases, the data set will be partitioned into the training set (to initially fit the model), the validation set (to provide an unbiased evaluation of the model while tuning hyperparameters of the model), and the test set (an unseen data set that provides an unbiased evaluation of the model). Generally, multiple algorithms will be tested and tuned for the given data set to find the best performer. To test the robustness of a trained model, external validation, meaning that testing the model in a new out-of-sample data set, is usually performed. There are several metrics to evaluate the performance of a prediction model. Simply put, *accuracy* is used to evaluate a classification model (e.g., to predict whether a reaction will occur), while R^2 *value* (also known as the coefficient of determination) and *root mean squared error* (*RMSE*) are generally used to appraise a regression model (e.g., to predict a reaction yield). Accuracy and R^2 are always the larger, while a low RMSE is preferred.

It is a general myth that the larger data set used to train the algorithm, the better performance the model will demonstrate. In fact, it is the quality of the data set that influences the performance of the reaction prediction models, not the amount of the data set. Several examples have proven that a good performance prediction model can be obtained using a small high-quality data set if taking advantage of an appropriate algorithm [153–156].

Generally, there are two primary purposes for reaction prediction models. One is to predict a reaction yield or regio-/stereo-/enantio-selectivity of the product under given conditions that have been parameterized. The other is using the algorithms to propose the product, or conditions for successfully transforming a given set of reactants.

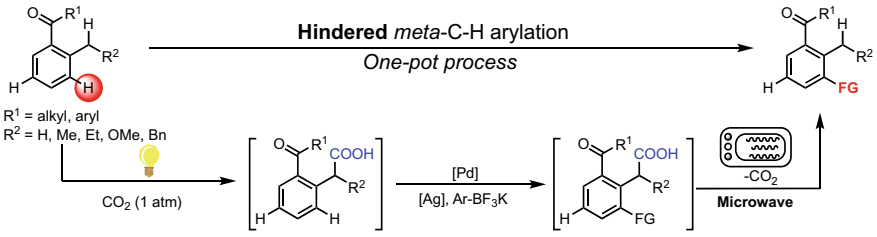
Reaction Yield Prediction

A recent example that takes the usage of ML to predict reaction yield was reported in 2018, Doyle and co-workers described how to utilize machine learning to predict the reaction yield of Pd-catalyzed Buchwald-Hartwig reaction of aryl halides with 4-methylaniline in multidimensional chemical space using data obtained via THE [109]. They created atomic, molecular, and vibrational descriptors for the reaction components in the presence of various potentially inhibitory additives. Applying

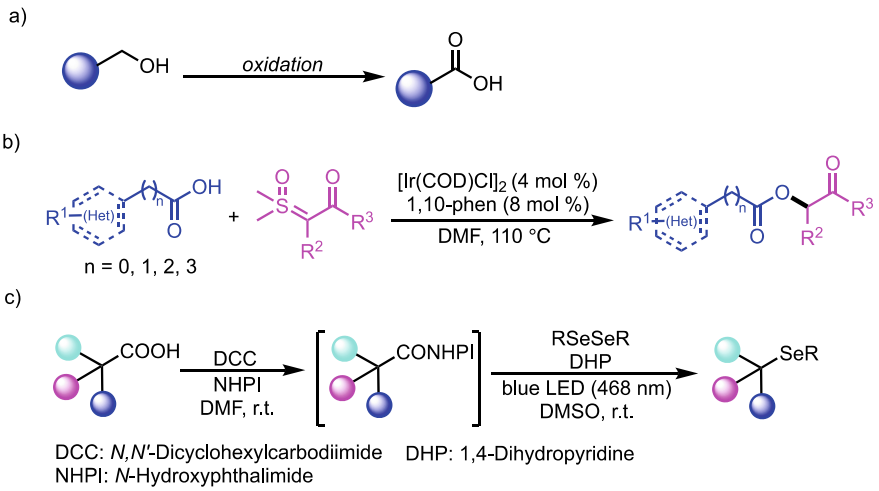
these descriptors as inputs and reaction yield as output, it was shown that the RF algorithm significantly improved the predictive performance over linear regression analysis. The RF model was also successfully applied to sparse training sets and out-of-sample prediction, suggesting its potential to facilitate the adoption of synthetic methodology. The corresponding protocol was then further applied to discover optimal conditions for diverse classes of alcohols with various sulfonyl fluoride scaffolds in deoxyfluorination [155]. However, these algorithms could not be applied for predictions beyond the range of data in the training set.

Deep learning (DL) [157, 158] techniques have been applied to many reaction prediction tasks. For example, Liao and co-workers constructed a deep learning-based model for reaction yield prediction of a novel selective functionalization of hindered *meta*-C–H bond of *o*-alkylaryl ketones using CO_2 as a traceless directing group (Scheme 8.2) [159]. This one-pot three-step transformation includes a photo-induced Csp^3–H carboxylation, an aryl-acetic acid-directed Pd-catalyzed hindered *meta*-C–H bond functionalization, and a microwave-assisted decarboxylation. HTE was used in this work for condition screening, substrate scope exploration, and experimental data conditions for machine learning. Based on the communicative message passing neural network (CMPNN) architecture, they developed a model, namely communicative message passing neural network for reaction yield prediction (CMPRY), to predict the reaction yield of this tandem reaction. The CMPRY model could successfully predict reaction yield over a range of unseen substrates, with a reasonable R^2 value of 0.706 and a mean absolute error (MAE) in yield of 8.4%.

Very recently, Liao's group employed AutoGluon [160], an open-source auto machine learning framework, to build a yield prediction model for TEMPO-mediated oxidation of primary alcohols (Scheme 8.3a) [161a]. With 3444 experimental data points accessed via HTE, they used AutoGluon to automatically train, tune, and ensembled several different models, including neural networks, LightGBM boosted trees, CatBoost boosted trees, Random Forests, Extremely Randomized Trees, and k-Nearest Neighbors. In this multi-label strategy, AGMP first classifies the reaction as reactive (Y) or non-reaction (N), then outputs predicted yields as regression results with the generated label "Y/N" as an additional descriptor. By combining HTE and



Scheme 8.2 Selective functionalization of hindered meta-C–H bond of *o*-alkylaryl ketones



Scheme 8.3 Reaction scheme of prediction models from Liao's group

ML, they also reported reaction yield prediction models for iridium(I)-catalyzed highly selective O–H bond insertion reaction of carboxylic acids and sulfoxonium ylides (Scheme 8.3b) [161b], and 1,4-dihydropyridine (DHP) derivative-catalyzed decarboxylative selenation reaction (Scheme 8.3c) [161c].

Reaction Selectivity Prediction

In 2019, Sigman and coworkers reported a multivariate linear regression model to predict enantioselectivities of chiral phosphoric acid-catalyzed nucleophilic additions to imines [162]. The difference in free energy of activation $\Delta\Delta G‡$ value between the transition states of each isomer (*E*-imine or *Z*-imine) was assigned as output in a comprehensive multivariate linear regression model with a data set of 367 literature reactions. The data were subsequently partitioned by transition state geometry to establish two focused multivariate linear regression models. The out-of-sample prediction revealed the ability of this enantioselectivity model to predict outcomes for varied reactions and novel catalyst structures.

Denmark et al. developed a chemoinformatics-guided workflow for chiral catalyst selection through machine learning [137]. They first constructed an in silico library of a collection of BINOL-derived families of chiral phosphoric acids and calculated relevant chemical descriptors for each chiral phosphoric acid scaffold. A new descriptor called average steric occupancy (ASO) was invented to describe the steric environment around the scaffold structure. A representative subset of the catalysts was then selected as a universal training set (UTS), which was agnostic to reaction or mechanism and could be used to optimize any reaction catalyzed by such scaffold. Following the training data collection, machine learning methods, including SVM and deep feed-forward neural networks, were applied to generate prediction models

for enantioselectivity prediction. An external test set of catalysts were used to evaluate these models. A range of chiral phosphoric acid-catalyzed thiol addition to *N*-acylimines demonstrated the accurate prediction ability of these models, suggesting their usage in enantioselectivity prediction as well as chiral catalyst selection.

Prediction models for regioselectivity are also developed. Traditionally, regioselectivity was basically predicted by chemists' intuition [163]. Harmmett [164] and Taft parameters [165], as well as calculated NMR shifts [166], were then used to approximate the nucleophilicity of aromatic cores. DFT [167] and semi-empirical quantum mechanics (SQM) [168, 169] were utilized to achieve better accuracy for the regioselectivity prediction of electrophilic aromatic substitution. In 2020, Jensen et al. presented a multitask approach to site selectivity of electrophilic aromatic substitution reactions [170]. Based on the WLN architecture, a site selectivity prediction model for electrophilic aromatic substitution reactions over 127 reaction types was developed, with overall Top-1 accuracy of 84%. Employing quantum mechanical descriptors, they recently developed a prediction platform for regioselectivity prediction of aromatic C–H functionalization [171].

Hong and coworkers developed a rapid and reliable model for regioselectivity prediction of radical C–H functionalization to heterocycles [172]. DFT was employed to generate physical organic features of molecules, while RF was applied to provide a satisfying regression model. A 94.2% accuracy in out-of-sample validation was achieved.

Very recently, Hartwig's group reported a hybrid computational model to predict the site selectivity of Ir-catalyzed Csp^2–H bond borylation in complex molecules [173]. DFT, SQM, cheminformatics, linear regression, and ML are involved in establishing the prediction model (SoBo). Compared to another prediction model (i.e., WLN) and human intuition, SoBo demonstrated a better performance in hitting the exact site of the out-of-sample validation set with full accuracy (Fig. 8.2).

Reaction Intermediate, Reaction Type, Reaction Product, and Reaction Condition Prediction

ML can also be used to propose the reaction intermediate or product of given reactants. In 2015, Kabeshov and Ley et al. reported a novel integrated web-based system that could compute, visualize and store systematized key reaction parameters from open-source components (Fig. 8.3) [174]. Users could draw a 2D structure that would be processed to generate the 3D structures of the most stable conformers through a web interface, followed by creating all possible intermediates for each conformer through DFT calculation. The most plausible intermediate was then confirmed by quick and full geometry optimization. Numerous error checks on the calculation, with restarts using different input options and initial guesses, were added to make the computations entirely automatic and robust. The rationalization and prediction of the regioselectivity of lithiation reactions and the relative reactivity of substrates were performed by this web-based platform as proof of concept. Superb agreement between the in silico algorithm evaluation and experimental data was obtained.

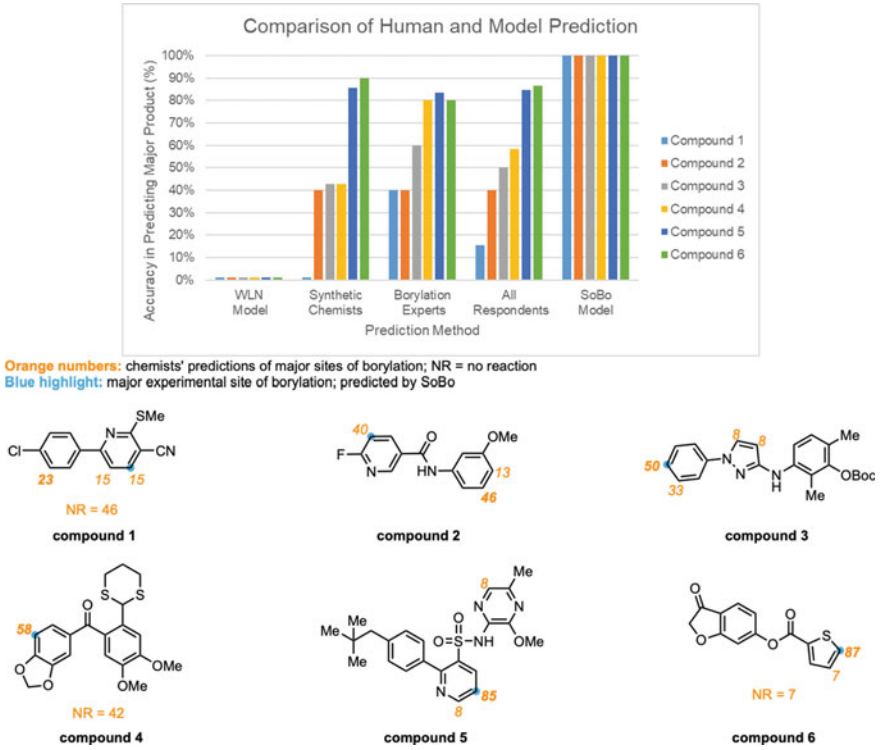


Fig. 8.2 Comparison of human and model site-selectivity prediction. Normalized out of 100

In 2016, Aspuru-Guzik's group applied fingerprinting methods, including neural molecular fingerprints, to predict organic chemistry reaction type, even the product molecules, for a given set of reactants and reagents [175]. An accuracy of 85% of the test reactions was achieved, while 80% of selected textbook questions were correctly identified by this fingerprint-based neural network algorithm. Jensen's group also reported a GCNN model for reactivities of given reactants [176].

Jensen's group described a model to predict the major products of chemical reactions given a set of reactant molecules by combining rigid reaction templates and machine learning (Scheme 8.4) [177]. Template-based forward enumeration and ML-based candidate ranking are combined in the model framework. All reaction templates are applied to enumerate candidate reactions of given reactants, followed by the soring of candidate probabilities to find out the most possible product. Although this two-step approach can predict the recorded product with 72% accuracy in a fivefold cross-validation, predictions cannot be made outside the template library scope.

Jensen's group also developed a neural network model to predict the reaction conditions, including catalyst(s), ligand(s), solvent(s), reagent(s), as well as temperature [58]. The model was capable of predicting the exact conditions or conditions

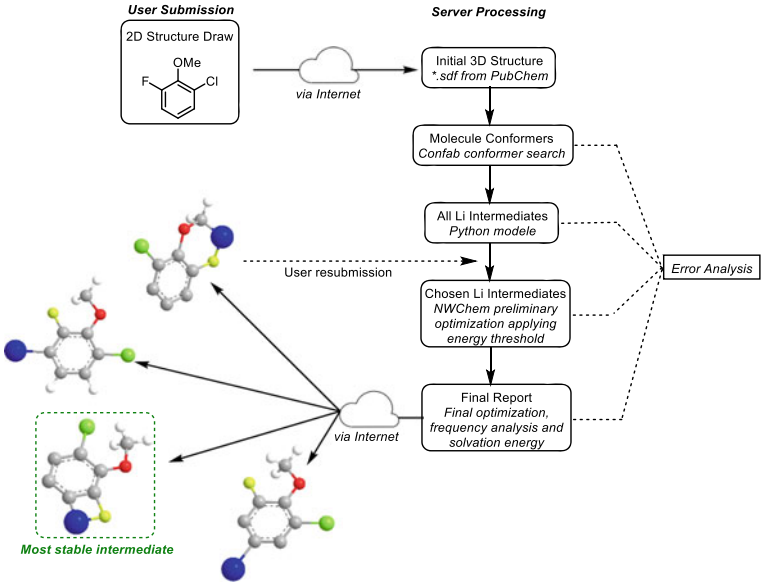
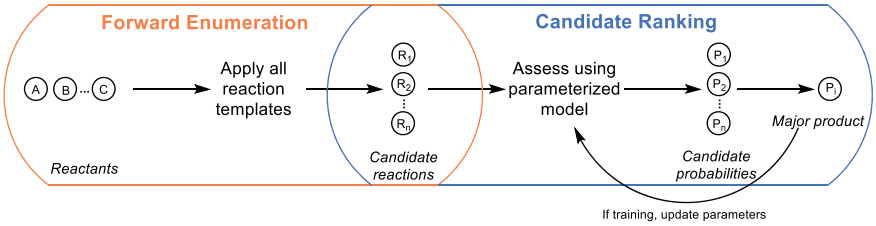


Fig. 8.3 Lithiation reaction outcome prediction workflow



Scheme 8.4 Model framework combining forward enumeration and candidate ranking for major product prediction

that have the same functionality as the reported conditions. ML models for conditions prediction of Michael reaction [178] and protective group deprotection [179] were also reported by Varnek and coworkers.

Miscellaneous

Besides reaction outcomes, ML can be used for physiochemical properties prediction. For example, Jiang et al. used ML for the prediction of the catalyst surface-molecular adsorbate interactions [180], infrared spectra (IR) of protein [181], and the recognition of protein secondary structures [182]. Grzybowski and coworkers used GCNN to predict the pKa values of C–H bonds [183]. Another example was recently described

by Mo's group, who used an automated platform to conduct Thin-layer chromatography (TLC) analysis, facilitating the high-throughput collection of 4944 standardized R_f values from 387 compounds under 17 solvent conditions [184]. ML methods were employed to construct surrogate models correlating organic compound structures and their polarity reflected by retardation factor (R_f). The trained ML models could provide general guidelines for selecting purification conditions according to the prediction of the R_f value curve of organic compounds in different solvent combinations. Moreover, the model could predict the probability of the separation of multiple compounds as reflected by the predicted difference in the R_f values.

8.4 Reaction Discovery and Optimization

Reaction Discovery

It is inherently unpredictable, time-consuming, and laborious for new reaction discovery. For example, while exploring a new reaction, the initial conditions are usually suboptimal, which can be easily overlooked if the desired molecule appears as a trace product, especially in a micro- or nanoscale. In such cases, deconvolution algorithms can be used to highlight these trace peaks as potentially novel reaction products [185, 186]. On the other hand, the application of ML toward the navigation of new chemical space remains underexplored [187]. The reason is that it is much more challenging to assess the reactivity of an unknown reaction with unpredictable products than to use metrics meant for the optimization of a known target compound, such as yield or selectivity [188].

Cronin's group showed that a synthetic robot controlled by SVM could significantly accelerate the discovery of organic reactions [189]. Reactions were carried out by the liquid-handling robot by choosing reactants from the pool of starting materials, with online analytics for real-time monitoring of reaction outcomes. ML was employed to build a model of the chemical space and recommend the next experiments and control the robot. The system could not only perform and analyze reactions faster than manually but predict the reactivity of about 1000 reaction combinations with an accuracy greater than 80% through a machine learning-based decision-making process. Additionally, chemists successfully discovered four reactions by manually following up on the predictions. This study represents a crucial step towards developing intelligent automated approaches for reaction space exploration through AI-driven chemical robots trained by humankind experts.

Recently, Zahrt and colleagues applied ML to guide the discovery of electrochemical reactions [190]. A molecular representation that enabled the production of general models with limited training data was first developed, followed by the data collection of electrochemical reactions through automated experimentation. Such a

significant number of reactions were then categorized as competency or incompetency and used to train a classification model. This reactivity prediction model was evaluated by screening 38,865 potential reactions in silico and identifying a number of new reactions as competency, with 80% of which were found to be reactive.

ML-Assisted Reaction Optimization

Traditionally, reaction optimization studies are mainly based on the knowledge of chemists and eventually use the principles of design of experiment (DoE) methods. Although DoE can be automated to accelerate the screening of predetermined experimental conditions and allows response surface modeling, it requires a high number of experiments to shoot the optimal condition [191, 192]. Despite the application of HTE and synthetic automation, accessing a large number of data points is not a simple mission. As a result, the employment of ML in organic chemistry is an attractive strategy to assist [45].

In 2018, Aspuru-Guzik's group reported an algorithm for global optimization in the context of chemistry and experimentation based on Bayesian optimization [193]. The algorithm, called *Phoenics*, can propose new conditions to locate the optimal conditions based on all previous observations, thus avoiding redundant evaluations. For example, Phoenics showed its applicability in the Oregonator, a complex case study describing a nonlinear chemical reaction network. Although the search space is large, Phoenics could quickly identify the conditions that yield the desired target dynamic behavior.

Sunoj and coworkers developed a practical ML model to discover catalysts for asymmetric hydrogenation of alkenes and imines in axially chiral BINAP-derived catalyst families [194]. Utilizing the molecular parameters of a set of 368 substrate–catalyst combinations, RF was capable of making accurate predictions, with an RMSE in the predicted enantiomeric excess (*e.e.*) of 8.4 $\pm$ 1.8 compared to the experimental data. This RF model could also successfully predict a range of out-of-sample data, indicating its potential in the discovery of asymmetric catalysts as well as choice-making of substrates for a particular catalyst.

In 2021, Adams and Doyle et al. reported a Bayesian reaction optimization modular framework to integrate optimization algorithms into daily laboratory practices [195]. They gathered a large benchmark dataset for a palladium-catalyzed direct arylation reaction, performed a systematic investigation of algorithm optimization compared to human decision-making in reaction optimization, and applied the Bayesian optimization to Mitsunobu and deoxyfluorination reactions. This work might facilitate the more efficient synthesis of value-added compounds by enabling better-informed, data-driven decisions about which experiments to conduct.

In 2021, Wang and coworkers used machine learning for the acceleration of the discovery and optimization of Cu catalysts for CO_2 reduction [196]. This ML-assisted approach could first find out the notable features for the reduction of CO_2 and then refine the critical features, followed by the catalyst design and validation.

Besides small molecules and peptides synthesis, ML is also employed in the optimization of materials synthesis. Hereby are two selected examples. In 2016, Norquist and co-workers reported an ML-assisted materials discovery approach from

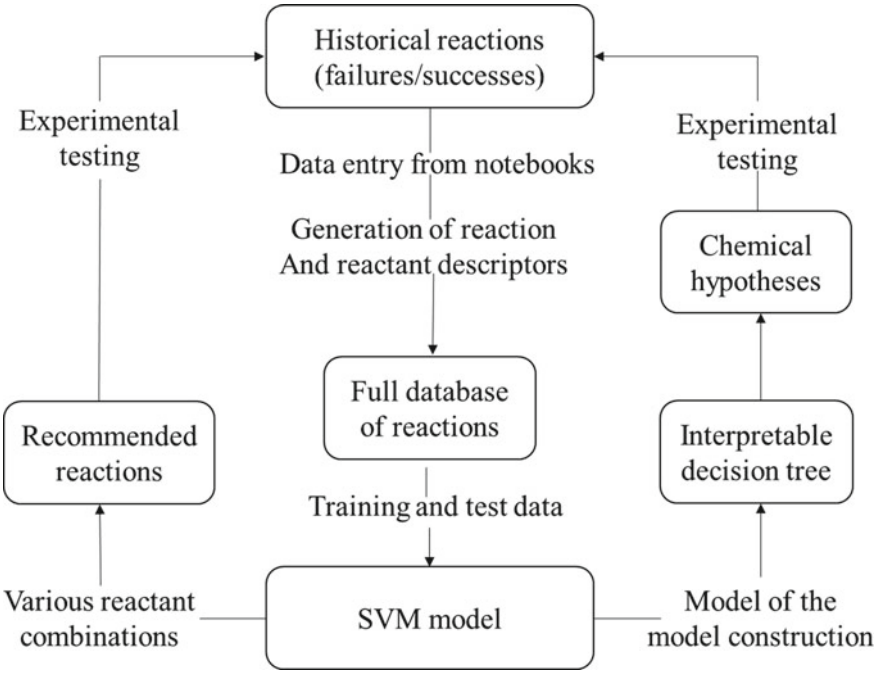


Fig. 8.4 Schematic representation of ML-assisted materials discovery

failed experiments with the help of an SVM (Fig. 8.4) [197]. ML models generated from historical reaction data are used to recommend new reactions to perform and to generate human-interpretable hypotheses about crystal formation.

Cooper's group integrated robotic experimentation with high-throughput computation to explore high-activity linear polymers as hydrogen evolution photocatalysts [198]. Four variables were used in an ML model to navigate the structure–property relationship between different polymers, including the predicted electron affinity, the predicted ionization potential, the optical gap, and the dispersibility of the polymer particles in solution.

Reaction Self-optimization with ML and PAT Integration

Process analytical technologies (PAT) [199], including ultraviolet–visible spectroscopy (UV–vis) [200], infrared spectroscopy (IR) [201], Raman [202], nuclear magnetic resonance spectroscopy (NMR) [203], mass spectrometry (MS) [204], gas chromatography (GC) [205], and high-performance liquid chromatography (HPLC) [204c, 206], are playing an increasingly critical role in continuous flow processing [19, 199b, 207], which enables real-time reaction monitoring for a variety of applications, such as dynamic [201e, 208] and kinetic [209] analysis, online chiral monitoring [210], process control [201b, 211], and most importantly, self-optimization [21, 212].

In 2018, Abolhasani's group designed an autonomous microscale flow chemistry platform with an online HPLC unit coupled to an evaporative light scattering detector (ELSD) and MS module for fundamental and exploratory investigations of homogeneous Rh-catalyzed hydroformylation of alkenes [204c]. The single-droplet tube-in-tube microreactor offered superior heat and mass transfer rates and removed the start-up/shut-down time delay of batch reactors. This approach allowed 43 hydroformylation experiments to be performed continuously with less than 1 mL of total reagent volume. The catalytic activity, chemo-selectivity, and regioselectivity of a library of ligands were systematically studied by means of this microfluidic platform.

Kappe's group demonstrated the integration of three distinct real-time analytical instruments, i.e., inline IR and NMR, as well as online UPLC, within a modular microreactor system, for the optimization and in-process monitoring of a multistep organometallic reaction [213]. This modular flow reactor has enabled real-time reaction monitoring with numerous sensors, alongside an integrated control system.

While attempting to optimize a palladium-catalyzed Suzuki–Miyaura reaction, Christensen and Jason et al. developed an automated kinetic profiling system integrated with an online HPLC and a Chemspeed liquid handling robot to facilitate automated reaction optimization and monitoring [214]. By the utilization of this automated system, the authors successfully dug the mechanistic insight and mapped a much more efficient precatalyst for the Suzuki–Miyaura coupling reaction.

A much more attractive strategy for reaction optimization is integrating inline/online analysis feedback with intelligent algorithms, which enables optimization systems to analyze previous reaction results in order to adapt and modify the following experiments in a short timeframe [207c, 212a, 215–217]. Such a "closed-loop" system, in which exploration of chemical space in a trial-and-error fashion will be performed automatically by imitating human experimenters, requires three essential parts: a robot to perform and analyze the reactions, a program to interpret analytical data, and an algorithm to correlate the outcome of the reaction with the input and process parameters, which will recommend the predicted optimal parameters for the next reactions and closes the loop (Fig. 8.5) [103, 188, 206b, 218–220].

In 2007, deMello's group pioneered a line of this concept by applying online fluorescence spectroscopy as a monitoring tool for the controlled synthesis of fluorescent nanoparticles [200]. Several research groups soon began to employ this "closed-loop" strategy for reaction self-optimization. Poliakoff's group developed a self-optimizing automated reactor in supercritical CO_2 ($scCO_2$). The technique that was capable of continuously adjusting key reaction parameters and monitoring the effect on the reaction outcome by online gas–liquid chromatography was applied to a range of reactions, such as etherification [221], hydrogenation [222, 223], aldol condensation [224], and methylation [225–228]. Poliakoff and colleagues also constructed an automated feedback loop system with a stable noisy optimization by branch and fit (SNOBFIT) algorithm [229] based on the super modified simplex (SMSIM) algorithm to optimize reactions in $scCO_2$ [205]. This approach was possible to optimize for multiple products from the same reaction mixture and access a broader range of reaction conditions than previously reported literature.

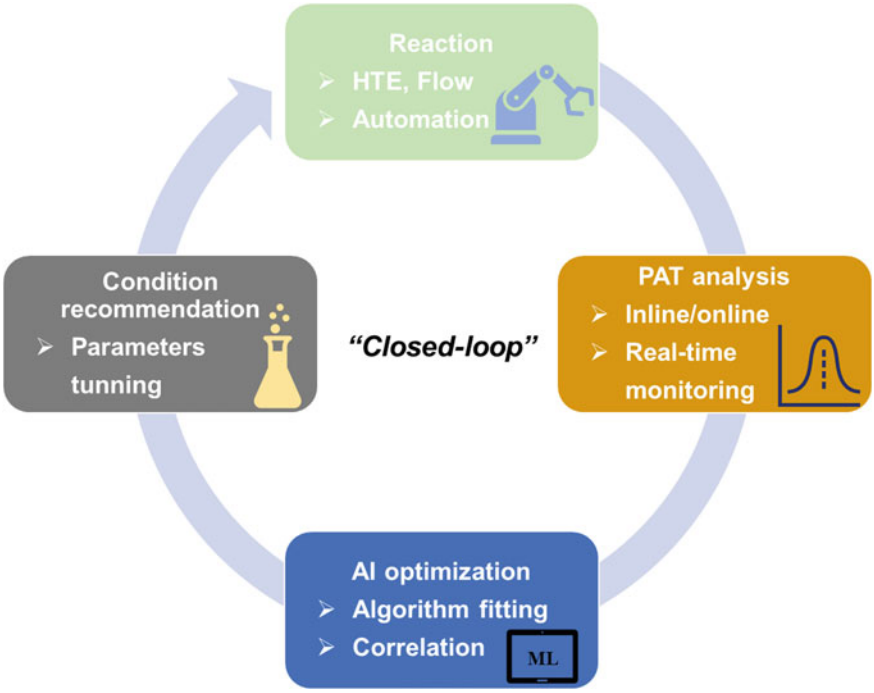


Fig. 8.5 "Closed-loop" reaction optimization

In 2010, Buchwald and Jensen et al. described a self-optimizing automated flow-based microreactor system that employed inline HPLC monitoring feedback control and optimization algorithm directed by the "black box" algorithms. The system was demonstrated to maximize the yield of Knoevenagel condensation [206], oxidation of benzyl alcohol and benzaldehyde [206], and Heck reaction [230]. The same group used an analogous strategy integrated with online IR monitoring for the optimization of the Paal-Knorr reaction [231]. In 2016, based on sequential adaptive response surface methodology (ARSM) coupled with optimal design of experiments, they developed an automated feedback droplet-flow microfluidic system involving discrete variables, i.e., precatalyst and ligand, as well as continuous variables, temperature, time, and loading, to explore and optimize Pd-catalyzed Suzuki–Miyaura cross-coupling reactions [232].

Cronin reported an integrated self-optimizing synthetic flow platform incorporating a real-time inline benchtop NMR for monitoring and controlling the organic reaction, with a modified Nelder-Mead algorithm [233] employed for self-optimization [234]. Felpin's group used the modified Nelder-Mead optimization algorithm as well to perform Heck-Matsuda reaction optimizations in a four-dimensional space with a variety of objective functions [235], as well as the self-optimizing synthesis of the natural product Carpanone [236]. Recently, Felpin's

group described an automatic continuous flow setup using a home-built photochemical reactor for the self-optimization and functionalization of cinchona alkaloids with a variety of organic thiols through the photo-thiol-ene reaction [237]. The reaction was optimized in a 3-dimensional space, i.e., the residence time, the equivalents of this, and the base loading, utilizing a modified version of the Nelder-Mead method they developed previously.

In 2016, Bourne's group documented an automated continuous reactor integrated the online MS for reaction monitoring and product quantification (Fig. 8.6) [238]. Due to its rapid quantification (< 1 min analysis duration), the online MS, which was calibrated by HPLC using linear relative response values, could provide real-time qualitative identification of compounds and intermediates or analysis of relative composition. They soon described a similar automated reactor and adaptive feedback control for optimizing the last step in synthesizing Osimertinib, an irreversible epidermal growth factor receptor kinase inhibitor [239]. The self-optimizing flow reactors combined inline HPLC analysis with the feedback SNOBBFIT algorithm to rapidly achieve optimum conditions. In their study, conditions generated by SNOBFIT in each optimization cycle were ranked in ascending temperature to minimize the time taken for heating up and cooling down. Once the reactor reached the desired temperature, all the pumps were set to 0.02 mL per minute, and the correct flow rates were set. A real-time yield was calculated using the ratio of HPLC peaks between the product and the internal standard. The system would loop the cycle until the optimum condition was found (Fig. 8.7).

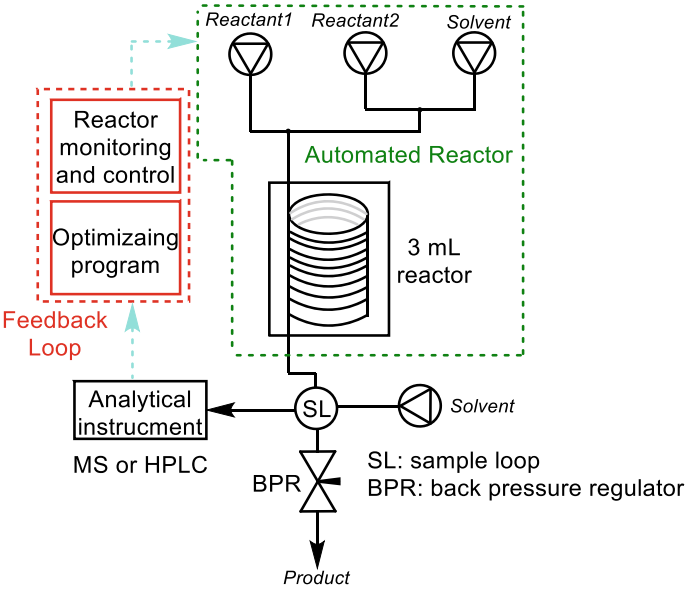


Fig. 8.6 Automated flow reactor with feedback control and optimization algorithm

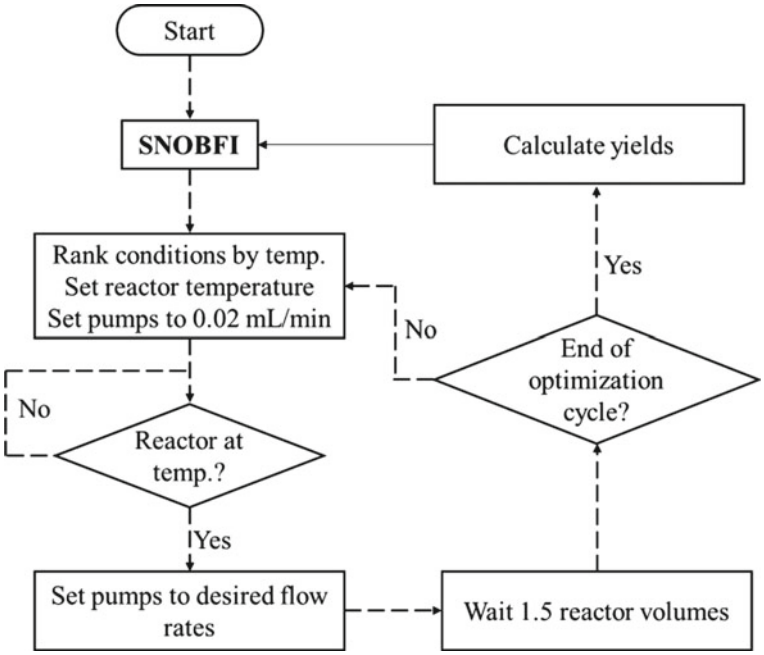


Fig. 8.7 Self-optimizing algorithm workflow

Lapkin and co-workers developed a model-based design of experiment (MBDoE) and self-optimization approach for a Csp^3–H activation in flow (Fig. 8.8) [240]. An inline UV and online GC modules were utilized for real-time analysis.

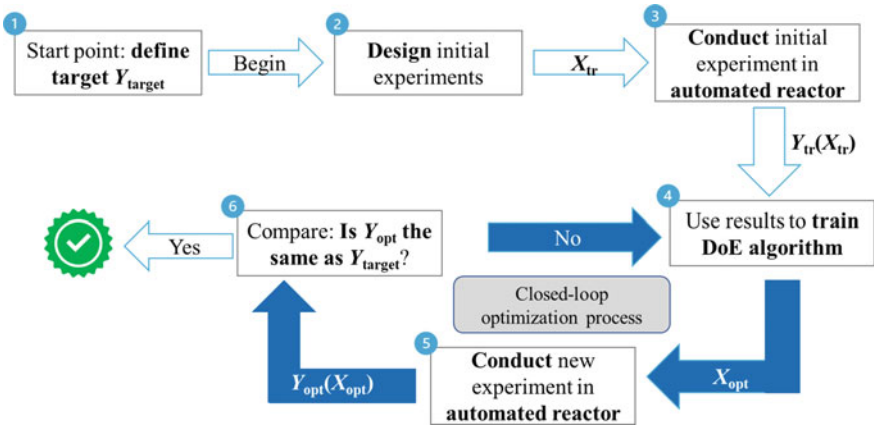


Fig. 8.8 The closed-loop optimization process of $C(sp^3)$–H activation by MBDoE

Zare's group developed a model, Deep Reaction Optimizer (DRO), to guide the interactive decision-making procedure in chemical reaction optimization by employing deep reinforcement learning with the domain knowledge of chemistry [241]. Using the output of MS as the optimization objective, the DRO could iteratively interact with chemical reactions to obtain the current reaction condition and yield, then determines new experimental conditions to attempt.

In 2018, Bédard et al. developed a fully integrated and versatile system for condition optimization and substrate scope evaluation [242]. Plenty of reactions, including Buchwald-Hartwig amination, Horner-Wadsworth-Emmons olefination, reductive amination, Suzuki–Miyaura coupling, nucleophilic aromatic substitution (S_NAr), as well as a visible light photoredox reaction, were successfully demonstrated by this convenient reconfigurable system. The platform was also used to perform a two-step process involving a limited stable ketene intermediate.

Based on their previous studies, in 2018, Jensen et al. implemented and validated a mixed-integer nonlinear program (MINLP) algorithm to optimize catalyst turnover number (TON) and product yield by simultaneously modulating discrete variables, such as catalyst types and continuous variables, including temperature, residence time, and catalyst loading [243]. With the ability to determine optimal conditions in the presence of side reactions, this algorithm was then applied to the real-time optimization of a Suzuki–Miyaura cross-coupling reaction in an automated microfluidic reaction platform comprising a liquid handler, an oscillatory flow reactor, and an online LC/MS.

Pentelute and coworkers applied deep learning (DL) over data collected from 35,427 individual highly reproducible deprotection steps through inline UV–vis analysis to predict peptide synthesis and minimize aggregation for aggregation-prone peptides [244]. This DL approach was also able to predict the synthesis outcome for sequences from unseen data, suggesting its ability to predict and optimize peptide synthesis in automated flow synthesis platforms.

Recently, Collins et al. designed an automated multi-step chemical synthesizer, *AutoSyn*, comprising a fixed configuration flow-based synthesis platform, a reagent delivery system, and process analytical technologies with an integrated software control system for automatic end-to-end process operations and monitoring (Fig. 8.9) [245]. The synthesizer was capable of assembling milligram-to-gram-scale amounts of virtually drug-like small molecules in several hours.

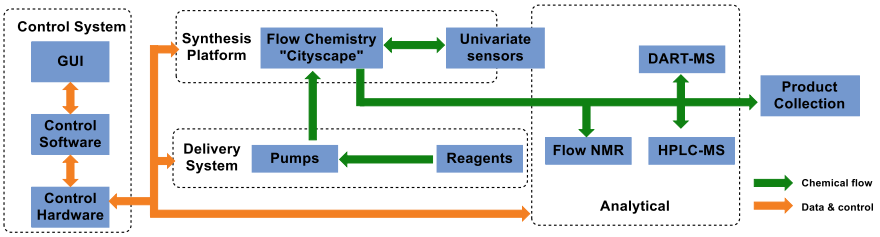


Fig. 8.9 Schematic representation of *AutoSyn*

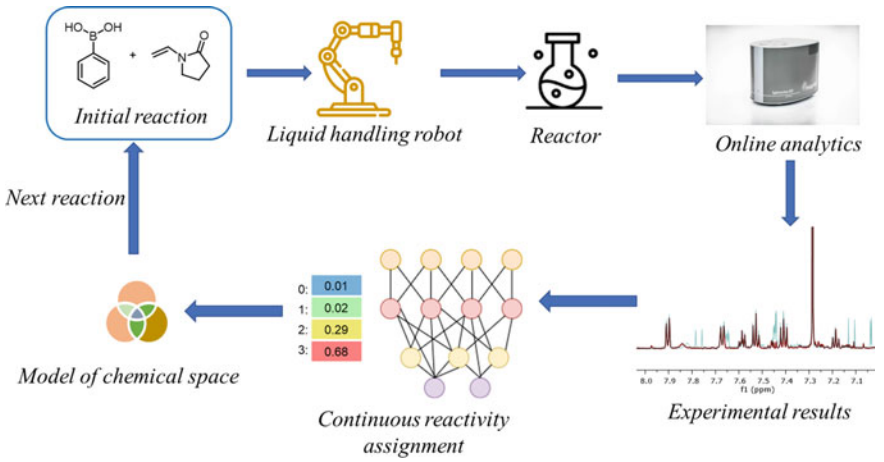


Fig. 8.10 New reaction discovery by neural network-driven autonomous platform

Cronin's group presented another closed-loop reactivity-seeking neural network-driven autonomous robotic platform for discovering new reactions (Fig. 8.10) [188]. The convolutional neural network devised by Cronin and co-workers, called *Reactify*, which had the ability to assign reactivity values to "unseen" reaction mixture through NMR spectra, was coupled with a closed-loop system for exploring the reactivity of experimental space. A range of chemical reactions and products, including several new reactions as well as new reactivity modes, were identified through the system.

Synthetic chemists typically use a small set of similar starting compounds to explore reactions. The optimized conditions that work best for those model substrates are then applied to numerous other, often dissimilar, compounds, even if alternate conditions might improve outcomes. Very recently, Grzybowski and Burke et al. reported an iterative protocol of ML and automated synthesis to explore general reaction conditions [246]. This simple closed-loop workflow leveraged data-guided matrix down-selection to render the large search space tractable while retaining validity to the whole, utilized uncertainty-minimizing machine learning to efficiently drive prediction optimization, and employed robotic experimentation to increase throughput, precision, and reproducibility of datasets recursively generated on demand, which provided a practical solution for multidimensional chemical optimization problems with vast search spaces. The workflow successfully identified general reaction conditions for the (hetero)aryl Suzuki–Miyaura cross-coupling reaction.

Human input was still required to fill in key procedural details and manually optimize approximate reaction conditions generated by CASP. In 2022, Jensen's group reported an algorithm-guided multistep reaction optimization protocol to identify optimal process conditions for approximate synthesis recipes generated by CASP [247]. They used ASKCOS, an open-source CASP software based on a multi-objective Bayesian optimization algorithm for computer-assisted synthesis [58, 176,

248–250], to iteratively generate recommendations for synthesis routes and reaction conditions for the target molecule. A high-ranked synthetic pathway was selected and completed by the automated synthesis platform after a manual assessment of synthetic feasibility. Human intervention is still required owing to the limited data availability.

Current machine-learning implementations count on the utilization of massive datasets coupled with black-box algorithms and retrospective benchmark statistics. Additionally, given the requirement of expert chemistry knowledge to effectively deal with non-routine tasks, most algorithms have experienced questionable applicability to organic chemistry practice. In 2020, Bernardes and Rodrigues et al. developed *LabMate.ML*, a computational framework, to navigate unknown reactivity spaces by leveraging unbiased, random experiment selection for initialization and adaptive machine learning. LabMate.ML selects small data, like 5–10 random reactions (0.03–0.04% of search space), for initialization (Fig. 8.11). The reactions are then performed, and the collected data are treated as an initial knowledge base. Subsequent reactions are carried out with computer-suggested conditions, based on active learning heuristics, and assessed by liquid chromatography-mass spectrometry/ultraviolet–visible (LC–MS/UV–vis). The area under the curve (AUC) value for the required m/z or UV absorption peak, which is used as a proxy for the reaction outcome (e.g., total product amount or yield), is treated as the objective target value in machine learning. The reaction conditions are used as features for LabMate.ML to train in the Optimizer. LabMate.ML uses RF to generate models that rationalize the available data and suggest a new condition set. The software is autonomous and re-trainable after each iteration. It takes only 5–10 min for LabMate.ML to run a full re-training cycle and prediction on a personal computer.

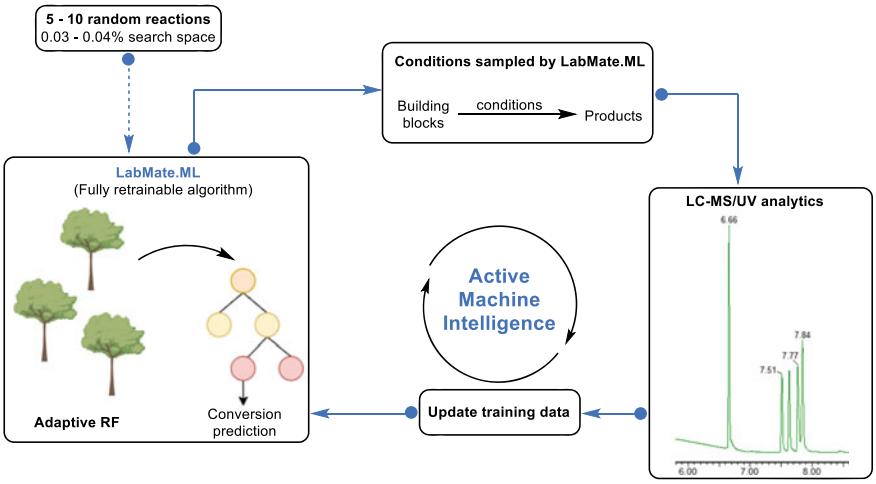
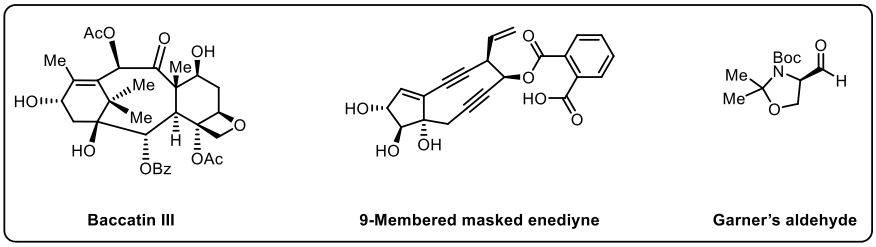


Fig. 8.11 Workflow for reaction optimization by LabMate.ML



Scheme 8.5 Examples of the application of ChemKonzert

8.5 Synthetic Automation, Robo-chemist, and Cloud Laboratory

Automation provides an avenue to transfer organic synthesis from a labor-intensive job to a machine-driven process [251]. The first concept of automated synthesis can trace back to the 1960s when Merrifield and Stewart reported an automated system for solid-phase peptide synthesis [252–255]. In terms of yield, purity, and biological activity, the peptides obtained by the automated system were comparable with those synthesized manually, while the time and material losses were significantly reduced. Taking advantage of a similar strategy, DNA [256, 257], RNA [258] fragments, as well as polysaccharides [259, 260] could be synthesized through an automated procedure.

The synthesis protocol in peptides and oligonucleotides automated synthesizers is fundamentally the same for every molecule. In contrast, general automated organic synthesis instruments need to adjust the synthetic routes to holistic, interdependent, and multistep processes, which are mostly unique for each molecule [245]. Given that automation in chemical research is rare and the commercially available systems were usually designed for specific purposes and only valid for repetitive work, in 1978, Legrand and Foucard developed an automation kit for synthetic chemists [261]. The actuators could execute various operations, such as heating, cooling, and reagent addition. The sensors could measure various parameters, including temperature, pressure, volume, weight, pH, and redox potential.

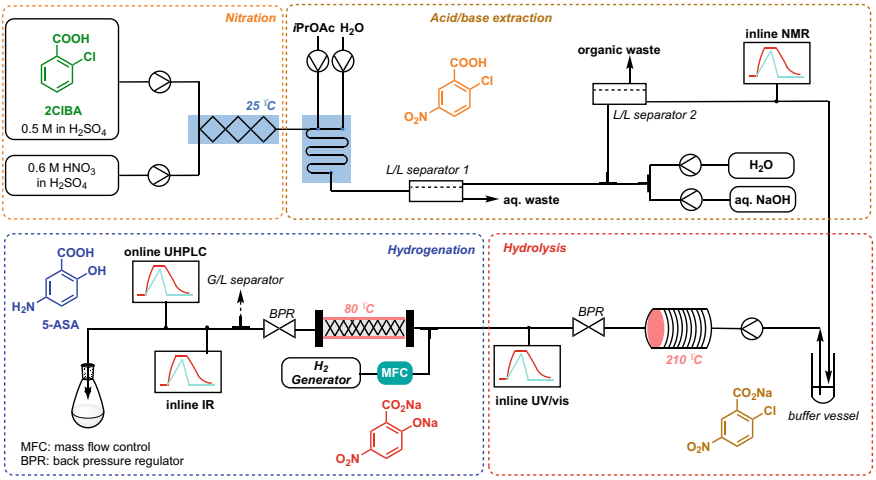
Ley's group devised a convenient and efficient prototype for evaporating, concentrating, and switching solvents in continuous flow processes and batch mode [262]. The system has been successfully applied to a telescoped process. In 2013, researchers from AbbVie *Inc.* developed an efficient compound-synthesis system with integrated components and automated sample-handling modules [263]. The system took only three days for the preparation of a 48-member library of 10-mg compounds, including synthesis, purification, validation, and sample distribution. Tu et al. soon developed a fully automated synthesis-purification station based on the SWAVE platform and in-house developed robotics [264]. Two widely used reactions, i.e., Sonogashira coupling and Buchwald-Hartwig amination, were selected as proof of the effectiveness of the integrated system. Very recently, Ahmed's group developed

a robot-assisted acoustofluidic end effector (RAEE) system consisting of a robotic arm and an acoustofluidic end effector [265]. The RAEE system was demonstrated through fluid pumping, particle and zebra fish embryo trapping, as well as mobile mixing of complex viscous liquids in well plates.

Continuous flow manufacture, which is now widely accepted in the synthesis of active pharmaceutical ingredients (APIs) as well as other fine chemicals [266–269], is an efficient and attractive technology platform no matter in laboratories and pharmaceutical industries [269–273]. Thanks to its inherent benefits, such as precise control of reaction times, temperatures, and composition, continuous flow chemistry is an ideal technology for computer-controlled synthesis, and a large number of examples of automated continuous flow platforms have been reported till now [21, 26, 216, 217, 269, 272, 274, 275]. For example, in 2010, Machida et al. developed a solution-phase automated synthesizer, *ChemKonzert*, to prepare a wide range of organic compounds [276]. A centrifugal separator was integrated into the system for efficient liquid–liquid extraction. Several examples were carried out to demonstrate the performance of this automated synthesizer (Scheme 8.5), including the synthesis of baccatin III [277], the key intermediate of taxol through a 36-step route, a 9-membered masked enediyne [278], and Garner's aldehyde [279], the widely used multistep synthesis intermediate.

Pentelute's lab reported an automated flow-based system for polypeptide synthesis in 2017 [280]. The approach took only 40 s for the total synthesis of an amino acid residue. The authors claimed that tens of thousands of 30-mer peptides could be produced annually at full capacity through this method. Similarly, researchers from Pfizer developed an automated flow-based synthesis platform [103]. The system, which integrated rapid nanomole-scale reaction screening and micromole-scale synthesis into one single modular unit, was validated by exploring a distinct range of reaction variables in a Suzuki–Miyaura coupling on a nanoscale at elevated temperatures and was able to perform over 1500 experiments per 24 h. Very recently, Li and coworkers invented a fully automated fast-flow device called Tiny Tides that can produce on-demand customized antisense phosphorodiamidate morpholino oligomers (PMOs) by reducing coupling times by up to 22-fold compared to previously reported methods [281]. This high-efficiency synthesizer can also synthesize cell-penetrating peptides-conjugated peptide-nucleic acids (PPNAs) in a single shot [282], which takes only 10 s for each amide bond formation between peptide-nucleic acid (PNA) monomers, compared with 10 min per amide bond for microwave peptide synthesizers; or 32 min per coupling cycle as with the DNA synthesizer Expedite 8909. They also used the synthesis data from Tiny Tides to train effective ML models to guide efficient PNA sequence design [283].

In 2020, Dömling's group used a positive-pressure-based low-volume non-contact dispensing technology, *I-DOT*, to synthesize more than 1000 different iminopyrrolidine-2-carboxylic acid derivatives through Ugi-3-component reaction in a fully automated fashion in a nanoscale [284]. The total chemical consumption in precious reagents and solvent was less than 0.6 mg for 1152 reactions, while approximately 400 g of reagents and 12 L of solvents would be employed when the same number of reactions is done on a millimole scale.



Scheme 8.6 PAT workflow towards 5-ASA

Williams and Kappe et al. recently designed a fully integrated multistep reaction and real-time analysis platform [285], which was used to synthesize the API mesalazine (5-aminosalicylic acid, 5-ASA) in a fully controlled and robust manner, over three synthetic steps (nitration, hydrogenation, and hydrolysis) and three phase separations (one gas/liquid separation, two liquid/liquid separation), with a throughput of 1.6 g per hour (Scheme 8.6). The use of four real-time analytical tools, including inline NMR, inline UV/vis, inline IR, and online UHPLC, was augmented by advanced data processing techniques.

General flow approaches are based on a linear strategy (Fig. 8.12a), which always leads to low efficiency. In 2021, Kim's group developed a parallel flow synthesizer capable of multiplex synthesis and optimization of compound libraries (Fig. 8.12b) [286]. Compared to the linear flow approach, this parallel flow synthesizer is much swifter. The synthesizer could perform its function without interruption upon clogging in flow lines. It allows independent control of the configured capillaries by adopting different reaction parameters in feeding rate, concentration, and temperature. Using this platform, Kim et al. screened and obtained the optimal conditions of aromatic substitution reactions (C–C, C–N, and C–S bonds) and azo-coupling reactions in less than one hour from 96 different conditions.

Given that most reported automated multi-step synthetic processes depend on iterative or linear procedures and require compromises regarding versatility and the equipment, Gilmore and co-workers detailed an automatic radial synthesizer based on multiple continuous flow modules arranged around a central core (Fig. 8.12c) [287]. This platform allowed remote control and could provide both linear and convergent syntheses stably and reproductively without requiring manual reconfiguration. In addition, IR and NMR spectroscopy were integrated into this automated instrument for inline monitoring, which enables post-reaction analysis and feedback.

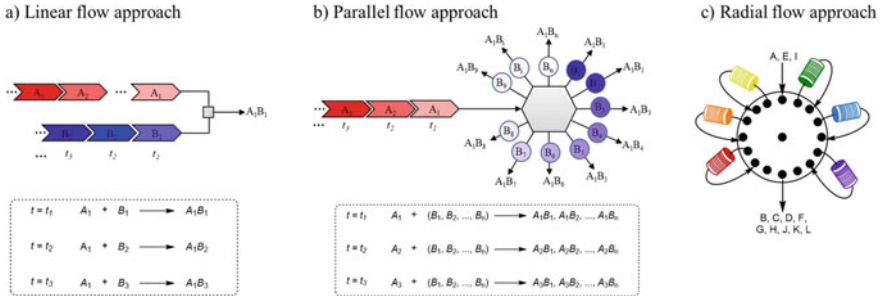
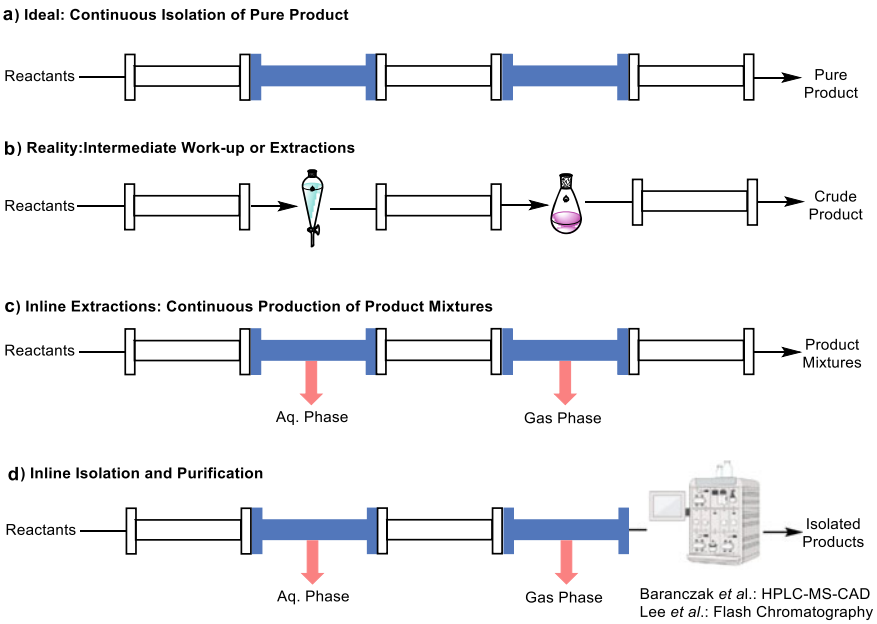


Fig. 8.12 Different kinds of flow. **a** General flow approach. **b** Parallel flow approach. **c** Radial flow approach

Since the solids often cause reactor channel clogging, it remains difficult when handling solids in research-scale flow reactors. Although the usage of acid-scavenging organic bases to generate low- to moderate-melting ionic liquids (ILs) upon protonation has been reported as a solution to prevent clogging [288], these additives may let the reaction mixture complex or cause side reactions. In 2022, Jensen et al. developed a continuous stirred-tank reactor (CSTR) flow platform that can handle slurries and solids during a chemical transformation in flow [289]. They also employed MINLP for single automated optimization and Bayesian optimization algorithms for automated multi-objective optimization routines. This fully automated flow platform for reaction self-optimization with solids/slurries feeding and handling will accelerate the identification of reaction conditions optimization for the manufacturing process development.

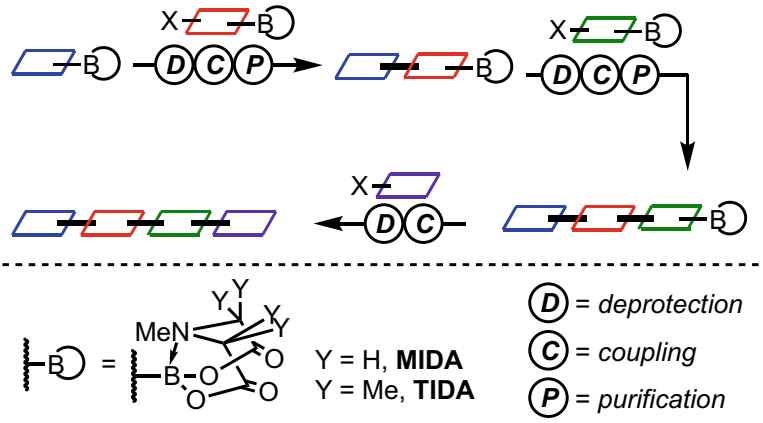
Isolation and purification of the target products are essential steps in organic synthesis. An ideal process for flow chemistry is that reactants are input to a flow system, and pure products are output from the process (Scheme 8.7a). An example of this was demonstrated by George et al., who used a supercritical CO_2 flow system for the continuous synthesis of artemisinin [290]. Nevertheless, the fact is that lots of multi-step reaction flow systems must be interrupted for workups and extractions to remove byproducts prior to proceeding to the next step (Scheme 8.7b). With the development of inline solid phase extraction [291], gas–liquid and liquid–liquid separation [292] technologies, the majority of these workups become to be incorporated into a single continuous process (Scheme 8.7c). However, offline purification, typically column chromatography, is still required to obtain the pure final product after the reaction. Baranczak and co-workers developed a fully automated and integrated platform for expedited synthesis-purification-testing of small molecule libraries (Scheme 8.7d) [293]. The platform enabled batch-supported compound synthesis based on a broad array of chemical transformations with testing in various biochemical assay formats. An HPLC-MS-CAD (charged aerosol detector) integration was utilized to separate and determine the concentration of purified fractions of the desired product. The compounds produced through library synthesis were sufficient for further evaluation in secondary assays, thereby contributing significantly



Scheme 8.7 Diverse kinds for isolation and purification of products in flow chemistry

to the shortening of medicinal chemistry discovery cycles. Recently, Lee and Vilela et al. reported an inline chromatographic purification automated flow synthesis platform to accelerate automated flow synthesis with continuous isolation of the final product (Scheme 8.7d) [294]. Three distinct reactions were successfully applied with inline normal- or reversed-phase flash chromatography in this system, with 97–99% purity of the continuously isolating products.

Burke employed a unified and entirely automated process analogous to iterative peptide coupling to achieve 14 diverse classes of small molecules through Suzuki–Miyaura Csp^2–Csp^2 bond formation (Scheme 8.8) [295]. This Lego-like synthetic approach used *N*-methyliminodiacetic acid (MIDA) as a building block assembled into a boronic ester group of starting material to reduce the reactivity of boronic acids for undesired reaction prevention. Additionally, MIDA boronates will selectively elute from silica gel when tetrahydrofuran (THF) is used as an eluent [296, 297], allowing easy purification through a "catch-and-release" protocol after the reaction. With more than 5000 compatible building blocks already commercially available, numerous small molecules were accessible by this synthesis machine. Nevertheless, this iterative assembly strategy was incompatible with stereospecific Csp^3–Csp^2 or Csp^3–Csp^3 bond-forming reactions owing to the sensitivity of MIDA boronates to Csp^3–C bond-forming conditions, which require either aqueous basic conditions or nucleophilic reagents that will react with MIDA boronates. Very recently, Burke's team developed a new class of tetramethyl *N*-methyliminodiacetic acid (TIDA) boronates that are stable to conditions for Csp^3–Csp^2 and Csp^3–Csp^3 bond formation



Scheme 8.8 Automated iterative synthetic strategy for C–C coupling

[298]. The new approach allowed Csp^3 boronate building blocks to be assembled through automated synthesis, including the synthesis of natural products via automated stereospecific Csp^3–C bond formation. The methodology for the iterative synthesis of small molecules has been reviewed [20, 299].

The microfluidic format is a miniaturized flow type of HTE technics. Jensen and co-workers have developed a range of droplet microfluidic automated platforms [300–305]. For example, in 2017, they reported a droplet-based automated segmented flow chemistry platform capable of efficient reaction screening, small-scale product isolation, and quantifying diverse medicinal chemistries under a 100 μg scale [300]. In 2019, Jensen and Buchwald et al. utilized an automated microfluidic platform to optimize Pd-catalyzed C–N couplings conditions in the presence of soluble organic bases, covering various *N*-nucleophiles and aryl triflates [301]. Based on the optimization algorithm, they exploited a number of ligand and base combinations for this transformation. They also developed an automated microfluidic single-droplet screening and analysis platform that enabled automated HTE for accelerating the discovery of new electroorganic processes [302]. In 2020, Kennedy and Stephenson et al. developed a droplet microfluidic platform that enabled flow-based high-throughput reaction discovery for pharmaceutically relevant compound libraries establishment [306]. The platform allows reactions to be performed on a picomole scale (i.e., 10^{-12} mol), with high-throughput data collection via online electrospray ionization-mass spectrometry (ESI–MS) for rapid analysis at acquisition rates of 0.3 samples per second. Very recently, collaborating with Pidko et al., Jensen's team reported a catalytic asymmetric hydrogenation reaction of a sensitive β-amino-ketone substrate by means of an automated microfluidic platform [305].

In pharmaceutical industries, generally, molecular fragments from various sources are utilized for the assembly of APIs, followed by the formation of the final drug product in a different plant, which led to the increased interest in the continuous manufacturing of APIs and drug products, and the safety issues around the storage

and transport of such materials [307]. In addition, the limitations of batchwise manufacturing, such as variations in quality control and supply chain interruption [308, 309], generally cause drug shortages. To address these problems, in 2016, Adamo and colleagues developed a continuous manufacturing platform combining both synthesis and final drug product formulation into a single, highly compact unit [310]. The refrigerator-sized continuous end-to-end synthetic system enabled efficient heat and mass transfer, as well as process intensification and automation. Hundreds to thousands of oral or topical liquid doses of high-profile APIs that meet U.S. Pharmacopeia standards could be produced by the system daily.

In 2021, Bode's group reported an integrated system for the automated capsule-based synthesis of saturated *N*-heterocycles and reductive amination with minimal human involvement [311]. The reported console, which combined reagent design, hardware engineering, and a simple operating system, could execute complex organic reactions with prepacked capsules and provide drug-like molecules by implementing multi-step sequences in a fully automated manner.

Virtual screening is a computational technique used in drug discovery to automatically search and evaluate exceptionally large libraries of small molecules in order to identify those structures which are most likely to bind to a drug target [312]. However, due to its high false positive rates in virtual screening campaigns, prospective hits must be synthesized for validation, which is time-consuming and laborious when done manually. In 2022, Bode et al. reported an iterative automated synthesis console capable of rapidly assembling molecules from virtual libraries exceeding 7 $\times$ 10^{12} members, which contained novel scaffolds with drug-like physicochemical properties [313]. Compared to manual workflows, this approach provided a tenfold efficiency improvement, suggesting that this automated synthesis console could facilitate the fast and trustworthy preparation of attractive virtual screening hits from large virtual libraries.

Cronin et al. provided another protocol to alleviate the above issues [314]. They reported a methodology that customized a blueprint for the synthesis of a target compound in a series of interconnected plastic modules, which could be fabricated by using three-dimensional (3D) printing technologies. The devices were designed and constructed by using a chemical to computer-automated design (ChemCAD) approach, which was capable of translating traditional bench-scale synthesis procedures into a platform-independent digital code. The successful demonstration of the γ-aminobutyric acid receptor agonist and (±)-baclofen showcased the possibility of this approach for the local manufacture of drugs outside of specialist facilities. This digitizing synthesis approach in 3D printed reactionware was also applied in various other examples [315, 316]. Recently, Cronin et al. reported a portable suitcase-sized platform containing all the required modules for synthesis and purification for universal chemical synthesis [317]. The system uses chemical markup language (χDL) coupled to a digital reactor generator to produce reactors through 3D printing and executable protocols based on text-based literature syntheses. This charming platform provides a blueprint for future automated synthetic platform designs.

Despite the development of synthetic automation and AI, the establishment of an entirely automated intelligent platform or laboratory which is less human-intensive

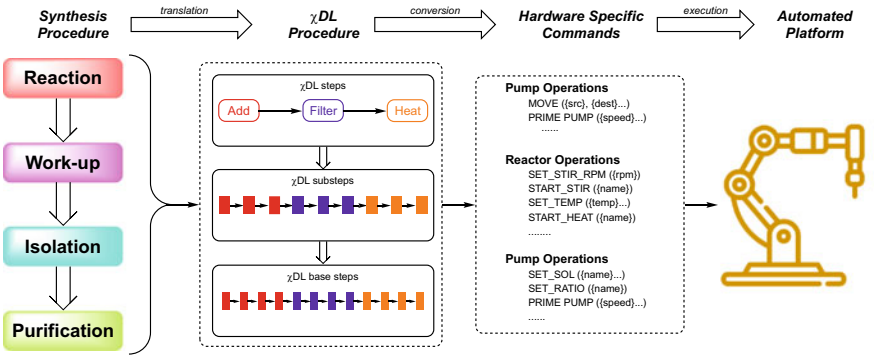
and can be controlled and monitored remotely is still calling for demand. During the early stage of the COVID-19 pandemic, most laboratories around the world were temporarily shut down owing to public health reasons, which significantly influenced the process of scientific research, especially experiment-based subjects like organic chemistry. As a result, constructing a "cloud chemistry laboratory," i.e., allowing chemists to perform chemical experiments remotely, is an extremely attractive idea. In fact, several research groups have already explored the possibility of building a "cloud lab." Poliakoff's group has showcased a proof-of-concept that researchers could operate the self-optimizing system remotely [318]. In 2016, Ley's group built an Internet-based modular software system, *LeyLab*, that enables chemists to monitor and control chemical reactions via the Internet with any device from anywhere around the world [319]. It facilitates the automation of synthetic procedures and is capable of autonomously self-optimizing reaction parameters to find the best conditions meeting customizable multi-component optimization functions. Researchers could enjoy the system's benefits regardless of their background in computer knowledge. The utility of *LeyLab* was demonstrated by the application of reaction automation for maximization of the output from a fixed volume of catalyst, as well as optimization of a three-dimensional heterogeneous catalytic reaction and a five-dimensional Appel reaction against various target functions.

In 2018, Aspuru-Guzik and co-workers described a portable and modular software framework called *ChemOS* for operating closed-loop systems with integrated robotics, sensors, and AI [320]. This single-software framework was applied to a number of different tasks, including learning the color space of dyes and autonomously calibrating high-performance liquid chromatography analysis.

Cronin's group soon developed an architecture and abstraction, the *Chemputer*, that maps reported methodological instructions into individual steps amenable to automation with traditional bench-scale techniques (Scheme 8.9) [321]. In their study, the automated units were implemented in a robotic platform that runs a complied chemical language to control the syntheses of the molecules. Users could code the necessary language without effort, even if they lacked programming knowledge. As validation of this concept, three high-quality pharmaceutical compounds were assembled by *Chemputer* without human intervention, with better yields and purities than those achieved manually. Using a natural language processing (NLP) algorithm, Cronin's group also presented a system to digitize the synthetic procedure from chemical literature into chemical codes (χDL) that was executable on the automated synthesis platform [37, 322–324].

In 2020, Zhu's group reported an intelligent robotics system, *Materials Acceleration Operation System* (*MAOS*), for materials synthesis and scientific discovery [325]. Human chemists can monitor and control the system via the VR-robot interaction through the 4G/5G network. For the post-trained experiments, MAOS can finish the experimental task by fully automatic procedures with AI-assured quality control.

Although people have started to explore the possibilities of chemical laboratory robots since the 1980s [326–328], the first AI-integrated mobile robot was developed by Cooper's group in 2020 [329]. This Bayesian search algorithm-driven robochemist could autonomously operate over eight days to carry out 688 reactions within



Scheme 8.9 Schematic representation of Chemputer

a ten-variable experimental space. Lacking the computational brain, however, the robotic search could neither capture the existing chemical knowledge nor generate or assess scientific hypotheses.

Jiang's group lately documented an all-round AI-Chemist that was able to perform all essential steps required for chemical research [330]. Based on a service platform, the AI-Chemist could automatically read the literature from the cloud database and propose experimental plans accordingly. It could execute the experimental process by controlling a mobile robot and simultaneously analyzing the experimental data by the computational brain through machine learning and Bayesian optimization. With a "scientific mind," this AI-Chemist could also propose a new hypothesis for the next iteration, which showcases its ability to perform high-level chemical research like a well-trained human chemist.

8.6 Outlook

Imagine that you wake up on a Monday morning, sitting at the table and enjoying your coffee. Suddenly you notice a lately online scientific publication saying a new group of bioactive natural products was just isolated and characterized recently, and their total syntheses are still under investigation. Excitedly, you pick up your phone, send a single command line to your lab robots' server, and continue your breakfast. Then you drive to work. When you enter your lab, that group of newly discovered natural products is already in the vials laying on your fume hood with the label of their names, structures, purities, and weights—your lab robots had already synthesized the corresponding compounds autonomously when they received your command in the morning. You rush to your office and check the reports on your table. It is about the details of syntheses and characteristics of those natural compounds, as well as two novel reactions found during the total synthesis, with all optimization details, substrate scope expanding, and mechanism studies. After double-checking the data,

you begin to prepare and submit your manuscripts on the first total syntheses of those natural products and discoveries of two new reactions. After a two-week peer review, your manuscripts are accepted and published online in high-impact journals.

With a single finger click and all synthetic things will be done. What an exciting dream for synthetic chemists. But let's go back to reality. Although there are a great many impressive examples demonstrating the power and infinite possibilities of ML and synthetic automatics in synthetic chemistry in the current day and the future, several issues and concerns remain to be cleared out, as discussed below.

Data Availability and Quality. As the essential part of ML, data can be accessed through online databases such as SciFinder^n or Reaxys. Nevertheless, the existing datasets suffer several limitations: (i) Missing information. Reaction details are always omitted. For example, many publications did not report the condition details like temperature, pressure, solvent consumption, and so on, which may be critical variables for the success of transformations. In addition, chemists usually focus much on reporting yields, which means that the stereo- or regioselectivities are often discarded in the publications, even if these data can be helpful in training a selectivity prediction model. (ii) Data bias [72, 73]. Researchers always prefer exhibiting their successful results, while negative outcomes are put away from the publics, even though these failures are essential to design reliable synthetic routes to unknown compounds [17]. This leads to extensive data cleaning and curation before using such datasets [331]. (iii) Data quality. Although data in such online databases are extracted from peer-reviewed publications, the reliability and reproducibility [332] of the data cannot be guaranteed. Unlike biologists, chemists seldom reperform the experiments published by other research groups to validate the experimental results. As a result, we would rather believe the results published in high-impact journals. However, retractions still happen in such high-impact journals. Additionally, tiny mistakes like typos appear on research papers. For instance, "CuCl" and "$CuCl_2$", "0.2 mmol/L" and "0.2 mol/mL", etc. Sometimes such fault can be neglected, but sometimes this tiny error will point to the opposite side. (iv) Most online reaction databases are commercial, which has set a barrier for users to access.

An alternative to obtaining datasets for ML is in-house experiments, carried out by automatic machines, not graduate students. Thanks to HTE, flow chemistry, and corresponding automatic technics, researchers can rapidly obtain an extensive data set with little effort. Numerous cases have already demonstrated the productivity of these synthetic automations. Since that synthetic requirements are diverse, the platforms and the procedures may be distinct in every laboratory. The question is, how can the data obtained from different labs communicate with each? Or let it be more specific: how do we deal with our experiment records and make them detailed, traceable, as well as sharable? One solution is using electronic laboratory notebooks (ELNs), which are widely used in industry but still rare in academia [17, 333]. But several questions come after: (i) Each group has their own preference in ELN systems; some even use home-built ELN. How can data in diffident ELN systems transfer to another? (ii) ELN systems are mostly commercial. For those labs that cannot afford it, how can they share their research data with others? (iii) how to balance shared

contents and research privacy when research folks are requested to share as much as raw data?

Recently, Kearnes et al. presented a paradigm for structuring and sharing organic reaction data [334]. Such open-access databases, including a centralized data repository, namely Open Reaction Database (ORD), are publicly available on GitHub. Supporting Information (SI) in publications will be standardized and readable by anyone under the ORD schema. Moreover, the omission of procedural details required for reproducibility will be more obvious, while comparisons between reactions will be easier. Conversion tools between the ORD format and other emerging data standards [98, 335], as well as ELN formats, are being released. The ORD is a splendid start and trial for reaction data repository in the chemistry community. Given that data sharing beyond required or peer-reviewed publication is not a custom in chemical research, data storage and sharing platforms like ORD still face social and cultural barriers. On the other hand, chemists should overcome the unwillingness to share unsatisfied results with low yields or selectivities, regardless of natural reactivity, procedural errors, or human faults.

Representation of molecules and reactions. Descriptors are critical to ML. Whether excessive or insufficient descriptors can limit a prediction model's predictive ability [30]. Proper descriptors can lead to an excellent performance of the prediction model [336], e.g., major regio-, site-, and diastereoselective outcomes prediction for Diels–Alder reactions [337]. As a result, it should be careful to select an appropriate way to represent molecules. As mentioned above, there are many methods to represent molecules and reactions for the establishment of ML models. Although DFT and MM calculation can provide multitudinous descriptors, the quantum calculation for each molecule will cost a tremendous budget of time and computer resources. Besides, the creation of a comprehensive descriptor set remains a challenge because physical parameters of each molecular scaffold are varied [30, 162]. On the other hand, present descriptors endure obstacles in depicting some molecular scaffolds, especially for organometallic complexes, and molecules' stereo-electronic properties and 3D conformations [338].

Translating molecules or reactions into the natural language (like a sentence) is another option to represent molecules and reactions. In fact, algorithms such as Bert RXN and AutoGluon Text Prediction that use natural language processing (NLP) have already been employed in reaction prediction [339–341]. However, the relationship between molecular structures and reactivities is hardly reflected via NLP, which limits the application of such approaches in reaction predictions. Therefore, the development of new descriptors to characterize molecules and reactions accurately and concisely is an important part in future research.

Prediction applicability and reliability. Although current reaction prediction models (including retrosynthesis planning) have shown their incredible prediction abilities as claimed by publications, we would like to ask what if such prediction models were applied to a much larger chemical space (applicability) and how likely will a chemist believe the prediction outcome made by AI even if the evaluation metrics of the model are impressive as stated by the publication (reliability)? For the former one, according

to the "no free lunch theorem" [342], there is no single optimal algorithm for solving all problems. Thus, the statement on the prediction applicability is essential. For instance, a team who reported a reaction outcomes prediction model had better figure out the applicable space of their model through external validation (by means of synthetic automatics and HTE, of course), which can make their models much more practical instead of theoretical, instead of simply claiming they have developed a prediction model but without any proof. Chemists should realize that publications are not the purpose of science; chasing the truth is.

The last question is a little complex. Models that predict reaction outcomes such as yields, selectivities, and possible products are effortless to be validated, as a single reaction is enough to tell. The problem is much more complicated for retrosynthesis planning because the predicted synthetic route will only be proven to be feasible or impractical if the final product is successfully synthesized or if an intermediate step fails, which is a resource-consuming process (time, manpower, money, etc.). Additionally, if the retrosynthesis planning model proposes a novel strategy beyond the chemist's awareness instead of an already known transformation, how likely will the chemist process this synthetic route? Providing a reliable and widely accepted prediction model should be a goal for organic-intelligent communities. The more successful examples of AI-driven total synthesis are reported, the greater chances that chemists will choose to believe the prediction model.

Chemists' background. The tendency towards automatic and intelligent in organic synthesis is indubitable. Someone may smell the anxiety and fear surrounding the organic community while the publications on organic chemistry crossed ML are emerging these years, because most organic chemistry experts are also greenhorns in programming and ML. Two questions may be concerned a lot of the mentioned majority: (i) will traditional chemists be replaced by the rising AI and robots? (ii) how can a chemist who lacks corresponding background begin to dive into the field of ML? For the first question, the answer is NO. Indeed, human action in the traditional synthetic routines will be taken place by automated machines and AI gradually since more productive and reliable results compared to manual work can be achieved by synthetic automatics and AI. However, the popularity of automated machines and AI will definitely liberate chemists from routine manual tasks, allowing investigators to focus more on the creative aspect of their studies and improve their work-life balance. Besides, with the usage of synthetic automatics (or robots), chemists can perform reactions in a safer situation, especially when toxic or explosive chemicals are involved. Else, the rise of automation and AI technics allows researchers to continue their investigation without interruption by situations such as laboratory closures during COVID-19. Human chemists are irreplaceable. Instead, we will play another role in the future synthetic chemistry field. We should change our minds from "how" to "what" and "why": from thinking about "how to synthesize this molecule" to "what should we synthesize and why should we synthesize it."

Chemists without corresponding backgrounds may feel stressed when they first get started in ML. Actually, there are user-friendly tools for chemists without programming backgrounds to employ ML in their studies. For instance, AutoGluon [160]

enables ML beginners and experts to quickly prototype highly accurate ML models with a few lines of Python code. With automatic hyperparameters tuning, model selection/ensembling, architecture search, and data processing, this open-source framework allows chemists easily employ ML in their research. An example that used AutoGluon to build a reaction prediction model has already been documented [161a] (vide supra). For retrosynthetic planners, both open-source (e.g., AiZynthFinder, ASKCOS, IBM RXN for Chemistry) and commercial (e.g., Chemical.AI, ICSYNTH, Reaxys, SciFinder^n, Spaya, SYNTHIA) platforms are accessible. In addition, a free and open-source data analytics platform, KNIME, can be helpful for chemists without expert backgrounds to analyze data as well as execute codes for ML. We believe that with the development and ubiquity of these beginner-friendly tools, programming background will no longer be a barrier between chemists and ML.

Convenience of synthetic automatics and robots. Generally, an automated synthetic platform contains following modules: the reagent storage, reaction preparation module, reactor, analytics and purification system, compound management system, monitoring unit, and console. In order to reduce human intervention when performing automated syntheses, these modules should integrate seamlessly. Meanwhile, given that most research groups do not have enough room for sizable machines, these highly integrated automatics should have a reasonable size, ideally as small as a fume hood, or even a suitcase size [317]. In addition, since demands for lab automation are diverse in every laboratory, and not all chemists have knowledge of automation, the automatic system should be highly and easily customized. A good automation platform should not only satisfy its users' needs but also be adaptable during usage. Moreover, regardless of hardware or software, an ideal automation platform should be convenient and straightforward to use as well as remotely controllable. Considering that most synthetic chemists do not have a background in automatics or programming, a universal chemical programming language based on the user habit of chemists is desired. Last but not least, the design, production, and maintenance of automated equipment, including hardware and software, require enormous amounts of time and money, which are unaffordable for most laboratories. Machines can run all day and night without a single complaint. But if it spends significantly more on using laboratory automation than hiring human operators, most labs will rather pay for human labor instead of robots. As a result, decreasing the costs for configuring automated platforms will popularize the automated platforms, especially at the grassroots level, and better promote chemical automation development. A paradigm of customizable and user-friendly laboratory robots for liquid handling has been commercial by Opentrons. As they claimed, their commercial lab robots and workstations (OT-1 and OT-2) are open-source and easily tailored, with a price as less as $10,000.

Cost in using synthetic automatics and ML. Training a greenhand into an ML/automation expert will obviously take time. Graduate students spend years becoming skilled chemists, and no one complains about the time it takes. Why are they so concerned about the time it takes when it turns to ML and synthetic automation? With the pervasiveness of user-friendly automatic synthetic platforms and feasible ML tools (such as AutoGluon, vide supra), we DO NOT think training a beginner

to use synthetic automatics and ML is time-consuming. Instead, the employment of synthetic automatics and ML will certainly accelerate the investigation process, with appreciably more reliable and reproducible results acquired.

Expenditure is another avoidable issue because almost every laboratory has a limited budget. As described above, deploying an automated synthesis platform or laboratory robot is expensive. In the past, chemists spent most of their budget on purchasing chemicals, glassware, reactors, and other analysis and experimental equipment such as GC–MS, rotary evaporators, and so on. Since synthetic platforms are primarily integrated with several modules and reaction scales are smaller on synthetic platforms than on traditional synthesis, chemists may adjust their cost on chemicals and equipment to set up synthetic platforms. Meanwhile, we hope that more economical automated synthetic modules like OT-2 can enter the market, which can reduce the economic burden of chemists who would like to drive on the automated synthesis avenue.

On the other hand, the main cost of ML should be the computation resource. Although sometimes a PC is enough to train a prediction model, better computation power is always required for a precise model, especially when quantum calculation is involved. A solution is to use cloud servers. The cloud server is computers from somebody else (hypercloud providers). Hypercloud providers can offer a smaller total cost of ownership while delivering superior features from scalability to security. Cloud servers can help chemists to get access to the latest graphics processing units (GPUs) or even tensor processing units (TPUs), which can hardly be afforded or maintained on chemists' own. Amazon Web Services (AWS), Google Cloud Platform (GCP), IBM Cloud, Microsoft Azure, Alibaba Cloud, and Tencent Cloud are several well-known cloud platforms for ML. The utilization of these cloud servers will release the pressure on those chemists who have limited budgets but desire to chase the pavement of ML.

Organic synthesis is unquestionable in a "cyber-chemistry era." We believe that with the endeavor of chemists and computer scientists, this brand-new synthetic mode will make our "finger-click-synthesis-done" dream come true.

Abbreviation

Abbreviation	Definition
1D	One-dimensional, one dimension
2D	Two-dimensional, two-dimension
3D	Three-dimensional, three-dimension
5-ASA	5-aminosalicylic acid, mesalazine
AI	Artificial intelligent
API	Active pharmaceutical ingredient
ARSM	Adaptive response surface methodology
ASO	Average steric occupancy
AUC	Area under the curve

AWS	Amazon web services
BDE	Bond dissociation energy
BINOL	1,1′-bi-2-naphthol
CAD	Charged aerosol detector
CAMEO	Computer-assisted mechanistic evaluation of organic reactions
CASP	Computer-aided synthesis planning
ChemCAD	Chemical to computer-automated design
CIMG	Chemistry-informed molecular graph
CMPNN	Communicative message passing neural network
CMPRY	Communicative message passing neural network for reaction yield prediction
CSTR	Continuous stirred-tank reactor
DFT	Density functional theory
DHP	1,4-dihydropyridine
DL	Deep learning
DNA	Deoxyribonucleic acid
DoE	Design of experiment
DRO	Deep reaction optimizer
e.e.	Enantiomeric excess
ECFP	Extended circular fingerprints
ELN	Electronic laboratory notebook
ELSD	Evaporative light scattering detector
ESI-MS	Electrospray ionization-mass spectrometry
GC	Gas chromatography
GCNN	Graph convolutional neural network
GCP	Google cloud platform
GNN	Graph neural network
GPU	Graphics processing unit
HOMO	Highest occupied molecular orbital
HPLC	High-performance liquid chromatography
HTE	High-throughput experimentation
IL	Ionic liquid
IR	Infrared spectroscopy
LC	Liquid chromatography
LHASA	Logic and heuristics applied to synthetic analysis
LUMO	Lowest unoccupied molecular orbital
MAE	Mean absolute error
MAOS	Materials acceleration operation system
MBDoE	Model-based design of experiment
MCTS	Monte Carlo tree search
MIDA	*N*-methyliminodiacetic acid
MINLP	Mixed-integer nonlinear programming
ML	Machine learning
MM	Molecular mechanics
MMF	Morgan molecular fingerprints

MS	Mass spectrometry
NLP	Natural language processing
NMR	Nuclear magnetic resonance spectroscopy
NN	Neural networks
ORD	Open reaction database
PAT	Process analytical technology
PMO	Phosphorodiamidate morpholino oligomer
PNA	Peptide-nucleic acid
PPNA	Peptides-conjugated peptide-nucleic acid
RAEE	Robot-assisted acoustofluidic end effector
RF	Random forest
R_f	Retardation factor
RMSE	Root mean squared error
RNA	Ribonucleic acid
RNN	Recurrent neural networks
$scCO_2$	Supercritical CO_2
seq2seq	Sequence-to-sequence
SI	Supporting information
SMILES	Simplified molecular-input line-entry system
SMSIM	Super modified simplex
S_NAr	Nucleophilic aromatic substitution
SNOBFIT	Stable noisy optimization by branch and fit
SQM	Semi-empirical quantum mechanics
SVM	Support-vector machines
TEMPO	(2,2,6,6-tetramethylpiperidin-1-yl)oxyl
THF	Tetrahydrofuran
TIDA	Tetramethyl *N*-methyliminodiacetic acid
TLC	Thin-layer chromatography
TON	Turnover number
TPU	Tensor processing unit
UPLC, UHPLC	Ultra-performance liquid chromatography
UTS	Universal training set
UV-vis	Ultraviolet-visible spectroscopy
WLDN	Weisfeiler-Lehman difference network
WLN	Weisfeiler-Lehman Network

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Chapter 9
Machine Learning for Protein Engineering



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Abstract Directed evolution of proteins has been the most effective method for protein engineering. However, a new paradigm is emerging, fusing the library generation and screening approaches of traditional directed evolution with computation through the training of machine learning models on protein sequence fitness data. This chapter highlights successful applications of machine learning to protein engineering and directed evolution, organized by the improvements that have been made with respect to each step of the directed evolution cycle. Additionally, we provide an outlook for the future based on the current direction of the field, namely in the development of calibrated models and in incorporating other modalities, such as protein structure.

Keywords Protein engineering · Directed evolution · Enzyme engineering · Sequence models · Machine learning · Deep learning

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Proteins are the molecular drivers for many of the processes necessary for life, performing functions such as binding small molecules, stabilizing other proteins, or catalyzing vital reactions. They are synthesized as linear sequences of individual building blocks called amino acids, of which there are twenty canonical types; these sequences fold into complex and dynamic three-dimensional structures to accomplish their functions. Existing proteins perform diverse functions, but nature only selects for a small fraction of them, leaving many functions unexplored, including many in which humans are interested.

Thus, protein engineers pursue these functions, most often with a powerful functional optimization strategy known as directed evolution. Directed evolution adapts nature's evolutionary process to enable the rapid improvement of proteins for human-desired functions [1, 2]. Specifically, directed evolution iterates between modifying proteins at the sequence level and identifying protein sequences with improved level of function. This process has yielded many impactful engineered proteins, ranging from antibodies for combating disease to enzymes capable of catalyzing non-natural reactions, and its success has been recognized by the 2018 Nobel Prize in Chemistry.

Fundamentally, directed evolution is an optimization process for protein sequences, and historically, it has been relatively straightforward but effective. More recently, advances in machine learning have been successful in improving directed evolution's ability to identify better proteins. In this chapter, we review the core concepts that have enabled successful integration of machine learning into directed evolution and protein engineering at-large by interpreting the process through the directed evolution cycle.

We begin with background on directed evolution, outlining the steps involved and casting them in the context of navigating protein fitness landscapes. We then highlight recent developments in machine learning's contribution to each of these steps as well as the experimental constraints, and finally we end with an outlook for the field. We focus on methods employing sequence-based machine learning for the directed evolution of proteins, but there are many other aspects of protein modeling that can be explored in biology [3] and chemistry that we leave out [4]. For example, proteins do not function in a vacuum, and other molecules, such as the nucleotides that encode our genomes [5, 6] and small molecules [7], can have profound effects on proteins' biological functions. Another view of understanding proteins is through their three-dimensional structures [8]. Additionally, protein dynamics [9] and structure prediction [10] are two rapidly growing fields, but the connection to protein function remains opaque [11]. Nevertheless, the goal of protein structure-based design [12] is to replace much of the work required for protein engineering, instead generating optimized proteins in a single step. While these approaches hold much promise, particularly in light of the paradigm-shifting success of AlphaFold [13], a protein structure prediction network, much of the established work in protein engineering is rooted in protein sequence.

9.1 Background

Directed evolution is an adaption of natural (and artificial) selection to the modern laboratory. First, consider a simplified view of how a new protein or function may arise naturally. In nature, evolution occurs within populations and is largely driven by competition to survive, grow, and reproduce. Therefore, an attribute which confers an advantage for any of these processes is selected for, enabling an organism to outcompete the organisms around it. The advantage could come from a mutation in a protein sequence that imparts a low level of activity for a new, beneficial function, making the organism and its offspring more successful. Thus, this protein sequence becomes more common as well. Through many iterations of this process, that protein sequence and its function could be further optimized. This process is similar to many engineering strategies, but contains notable distinctions for proteins, discussed in the following Sects. 9.1.1 and 9.1.1.1.

9.1.1 Protein Sequence Space

The number of possible protein sequences is immense, presenting a significant challenge to any protein engineering strategy. A typical protein is several hundred amino acids long, and at each position there are twenty canonical amino acids possible. This results in 20^{300} (10^{390}) possible sequences, which is around 10^{300} times more than the total number of atoms in the universe. There will never be enough resources to synthesize all of these sequences let alone screen them.

Interestingly, this does not mean that finding functional proteins or improving protein fitness is a hopeless endeavor. Maynard Smith made the critical observation that for natural selection to be possible, functional protein sequences must neighbor other functional sequences in protein space, implying that functional proteins exist clustered together within a vast sea of non-functional ones [14]. Therefore, rather than throwing metaphorical darts at this astronomical space and hoping for a hit, protein engineers can begin with a small level of activity for their function of interest and leverage methods for local exploration to improve it.

9.1.1.1 Protein Fitness Landscapes

Due to the iterative nature of the optimization cycle that is directed evolution, it is commonly conceptualized as a greedy, uphill walk on a **protein fitness landscape** towards a fitness peak [2]. Each round of mutagenesis and screening searches through the local landscape, typically sampling only a few mutations away from the current position in sequence space. When a hit is identified, a step toward the fitness peak is taken and the local search is repeated, with the entire process continuing until

the fitness is satisfactory or a peak is reached. Importantly, no downward steps into valleys of the fitness landscape are typically allowed in directed evolution.

Fitness landscapes are often visualized as smooth, easy-to-navigate surfaces, but in reality they are discrete, high-dimensional spaces, with many of the dimensions being quite rugged. This ruggedness is due to a phenomenon known in biology as epistasis, where mutational effects are dependent on higher order interactions rather than their individual contributions [15]. Epistasis arises most commonly from direct structural contacts, but interactions between residues can also be modulated by ligands, substrates, allostery, cofactors, or conformational dynamics [16]. As a result, it is often reasonable to assume that distant mutations are mostly independent, but there are important cases where this assumption breaks down and epistasis must be considered. Otwinowski et al. explored the prevalence of global epistasis, where the mapping of genotype to phenotype showed inherent non-linearity not just due to pairwise interactions between residues [17]. This effect varied in magnitude depending on the protein being studied.

Intertwined with the idea of global epistasis, it is important to consider the prerequisite protein properties that must be satisfied to take fitness measurements, such as expression, stability, and substrate binding. This means that fitness landscapes are a result of some combination of these factors, and changes in any of them can modulate fitness or cause epistasis. For example, Romero and Arnold outline how negative epistasis can arise from a protein stability threshold, where beneficial, but destabilizing mutations combine to completely ablate activity [2]. In fact, any mutation is more likely to be destabilizing than neutral or stabilizing, and therefore, most activity-improving mutations that improve activity are also likely to be destabilizing [17]. Therefore, more stable proteins can be easier to evolve [18]. The protein fitness landscape is a useful analogy for both intrinsic proteins as well as the methods in this chapter, and we come back to this concept often.

9.1.2 Protein Engineering, Directed Evolution, and Machine Learning

In this chapter, we view the protein engineering process through the directed evolution cycle, which adapts natural and artificial selection to the modern laboratory. In this process, the desired properties of engineered proteins are directly queried (assayed) by man-made techniques, instead of interpreted through survival and reproduction (as in natural selection or digital evolution [19]). This allows the engineer to rapidly evolve proteins for desired properties (such as thermal stability or expression) and functions (such as binding or enzymatic activity).

9.1.2.1 The Directed Evolution Cycle

The directed evolution cycle (Fig. 9.1) can be segmented into three basic steps, with two additional steps in cases where protein engineering is driven by machine learning. In Sects. 9.2.1 and 9.2.2, we highlight examples where machine learning has enabled advances in directed evolution, and in Sect. 9.2 we elaborate on the experimental constraints and ML considerations for these steps. The steps are as follows:

1. First, a protein with a measurable amount of the property of interest must be identified. Directed evolution relies on accurate physical interrogation of this scalar value, termed fitness, and a measurable starting point is required to begin this process. Many protein engineering projects fail at this stage, since biochemical ingenuity is required to identify a protein able to accomplish a new goal such as breaking down polyethylene terephthalate (in man-made plastics [20]) or selectively modifying DNA for targeted gene therapy [21] (more on identifying a starting point with ML in Sect. 9.2.1).
2. Second, the protein sequence is randomized to generate a pool of variants, often called a **library**. In nature, this process typically occurs through random mutagenesis, where random mutations in DNA correspond to random changes in the protein sequence, or recombination of existing protein fragments. In the laboratory setting, current experimental techniques give protein engineers more freedom in selecting which sequences to test (more on library generation in Sect. 9.2.2.1).
3. Third, the library is tested for the desired property. This step captures a wide range of assays that have been developed to probe biomolecules. Some examples of approaches capable of generating the largest amounts of data (high-throughput assays), are based on protein fluorescence or binding, and reach hundreds of thousands of labels per month. However, other properties such as enzymatic activity for generating small molecule products are measured in much lower throughput, typically on the scale of hundreds per month. This step often limits the *amount* of data labels the modeler can expect to obtain.

9.1.2.2 How Does Machine Learning Fit into the Directed Evolution Cycle?

In traditional directed evolution, the top k variants (often $k = 1$) are fed back into the second step for further improvement. For methods where directed evolution is further enabled by machine learning, two other steps may follow. The advantages that machine learning confers currently depends on the protein being evolved and its fitness assay. Some protein engineering projects are already able to test millions of variants in a single round, minimizing the value of a machine learning approach. Other projects require a day or more to acquire a single data point and would not even be possible without further guidance from sources such as machine learning.

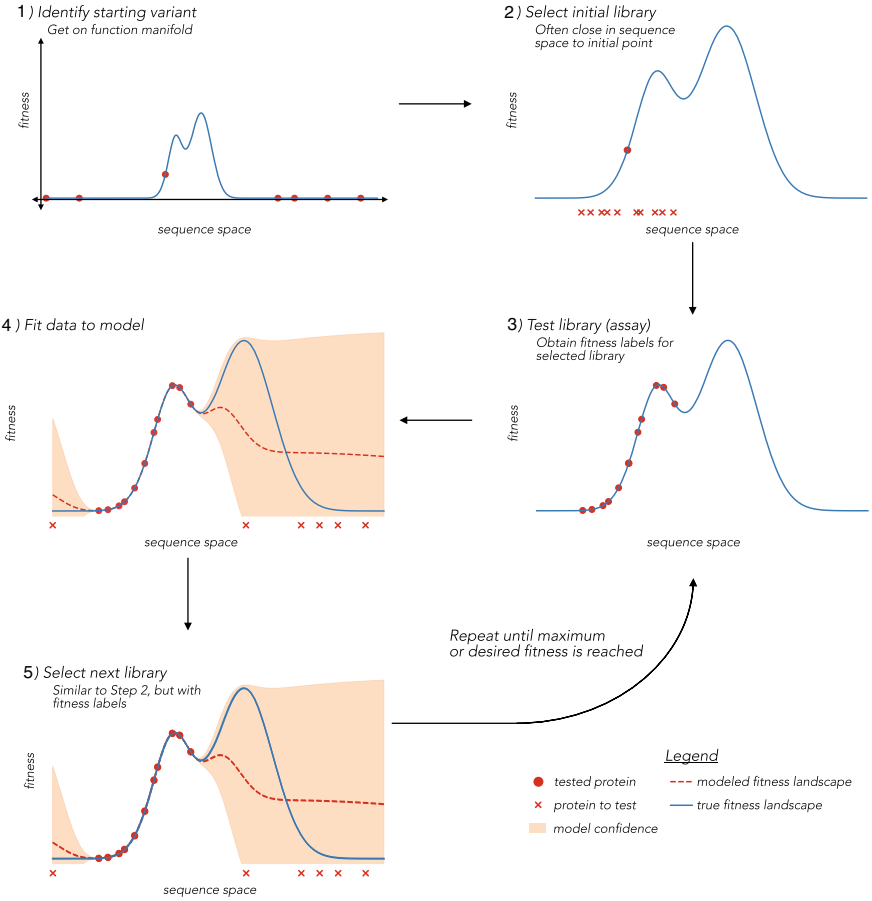


Fig. 9.1 Protein engineering steps viewed as steps in exploring a 1D fitness landscape. In each plot, the protein fitness (scalar measurement of a function or property), is on the y-axis, and the protein sequence is on the x-axis. In (1), a protein with measurable function is identified. In (2), an initial library of protein variants is built. In (3), the library is tested (assayed) for desired property. The next steps are additional for protein engineering methods guided by machine learning, including (4) fitting the models and (5) selecting the next library based on these models. Finally, the process is repeated until a maximum is reached, or the desired fitness satisfies a desired threshold

4. In these machine-learning-assisted approaches, the next step is to fit models to the relationship between proteins and their fitness labels. A wide variety of approaches are available to the machine learning practitioner here, and there are multiple sources of prior knowledge that can be leveraged for proteins. One example is the rich historical record of protein sequences, which can be obtained from sequence databases such as UniRef [22, 23]. From such a database, sets of evolutionarily related sequences (**homologs**) can be obtained and aligned in Multiple Sequence Alignments (**MSA**s), which can be used as priors on viable

sequences. However, while this history represents sequences retained in nature, it does not necessarily represent the distribution of allowed sequences for a specific protein on an engineered task, especially for non-natural activities. For example, mutating the axial ligand of cytochrome $P450_{BM3}$ from a cysteine to a serine unlocked multiple non-natural activities, but an MSA would show high conservation of the cysteine residue and disfavor mutation at this position [24]. However, the exact implications of this for machine learning have not yet been thoroughly examined and more research is needed to elucidate the potential biases of these methods.

5. Finally, these models are used to select optimal proteins for experimental validation. Again, a variety of approaches have been employed in this step such as gradient-based, reinforcement-learning-inspired, and active learning methods (Sect. 9.2.6.3). The sampling strategy often depends on the modeling approach used in the previous step. Additionally, this step is similar to the second in that both may be constrained by current molecular biology techniques. These constraints are often enforced manually, but can be encoded in the design process as well. From here, directed evolution re-enters the third step, and the cycle is repeated until the optimum is reached or design criteria are satisfied.

The process of directed evolution with machine learning is depicted in Fig. 9.1, which interprets each step in the analogy of climbing protein fitness landscapes.

9.2 Applying Machine Learning in Directed Evolution

The directed evolution cycle powered by machine learning methods can be segmented into five steps, as described previously. In this section, we summarize important contributions machine learning has made, and promises to make, to improve over the directed evolution baseline over each of its steps.

Additionally, we discuss experimental constraints to proposed methods, which are imposed by a variety of factors, including available technology, cost, and time. Current laboratory methods for directed evolution have been reviewed extensively [25, 26], so we focus here on facets that can impact the usefulness of machine learning methods. There is often a trade-off between lab-work complexity or cost and efficiency of ML-guided engineering, and we discuss these trade-offs below. Notably, both the input protein sequences and the output experimental labels both depend on physical constraints.

9.2.1 Identifying Starting Variants

The first consideration in a directed evolution experiment is selecting the protein variant to evolve, which can be a nontrivial task. In the most extreme setting, there may be no known proteins that perform the desired function; a more common but related setting is when the desired output of the directed evolution experiment needs to be substantially different from all known proteins due to considerations of scientific novelty or intellectual property.

9.2.1.1 Experimental Considerations for Finding Starting Variants

There are a number of traditional methods for finding starting points for directed evolution, and many of them draw inspiration from Maynard-Smith's observation that functional protein sequences must neighbor other functional sequences in protein space. Thus, when screening for new activities, protein engineers typically begin by searching annotated databases of existing sequences and structures, looking for proteins which perform a similar function to the desired one. Such proteins are hoped to have at least a small level of "promiscuous" activity for the new function. For enzymes, this could mean looking for ones that have the desired mechanism but act on a different substrate. As another approach, plates of variants from previous evolution campaigns might be screened for a new but similar activity. These methods ultimately rely on the latent promiscuity of extant sequences, as the probability that a functional sequence is found through a random search is approximately zero.

9.2.1.2 Applying Machine Learning to Finding Starting Variants

Alternatively, machine learning can be a useful tool for identifying a novel starting variant, typically with weak or suboptimal fitness, that can subsequently be given to a traditional or machine-learning-guided directed evolution pipeline. Existing approaches fall somewhere between using machine learning for finding such variants (1) to screen a large and diverse collection of proteins for any that have a nonzero fitness value [27, 28] or (2) to de-novo design the initial, functional protein [29, 30], with some additional variations summarized in Fig. 9.2.

As an example of the first approach, Shin et al. use an autoregressive language model, trained on approximately 1.2 million natural llama nanobody sequences, to generate a nanobody library that is screened to potentially identify novel binders to a target protein [27]. The generative model enables improved sequence diversity over previous synthetic libraries, enabling the authors to identify new proteins with high binding affinity using an efficient set of approximately 10^5 generated sequences, which is 1000-fold smaller than other libraries.

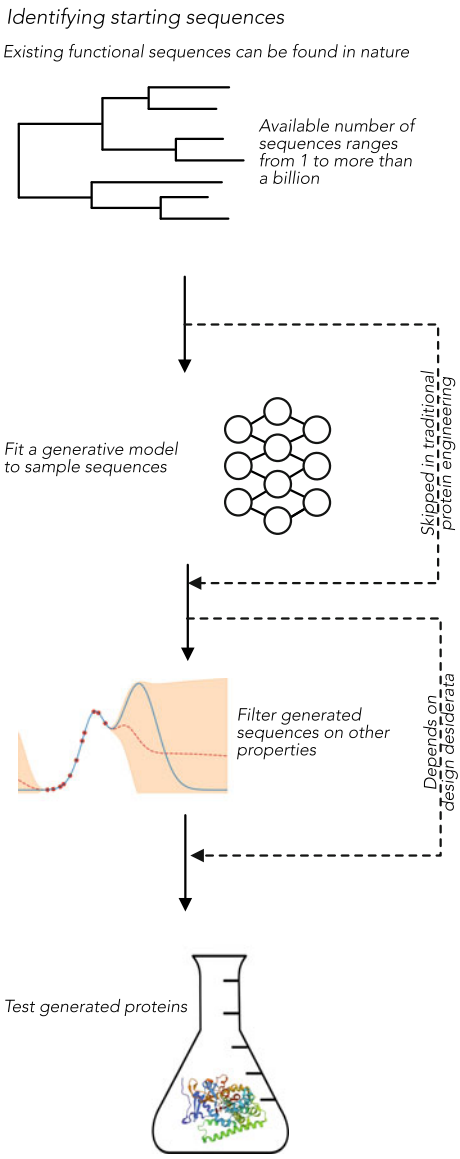


Fig. 9.2 Nature contains a wide variety of natural proteins from which a starting variant can be selected. Machine learning has also been used in various ways to improve the starting variants selected for further protein engineering

The second approach, which promises robust starting points with desired properties without the need to physically screen generated proteins, is an ideal yet unreached. Nonetheless, ProGen [31] is a noteworthy example that has generated functional protein sequences with high (> 50%) success rate on three proteins families [32]. However, there is room for improvement in extending to more difficult functions and protein properties.

These approaches are cases in which the starting variant is unknown or must be substantially different from existing proteins, but this is not a constraint in many other settings where a good, functional variant exists and the goal is simply to improve fitness. Directed evolution could be used directly to evolve the best-known variant, but it may not be the most effective strategy. A known, high-fitness variant may be in a **local** optimum of the fitness landscape such that the evolutionary path to the unknown **global** optimum (or the highest-fitness variant) requires crossing a **fitness valley** (where some intermediate mutations along the path decrease fitness).

Therefore, it may be desirable to begin evolution at a lower-fitness variant from which multiple fitness peaks (both locally- and globally-optimal) are accessible via a greedy uphill walk strategy typical in protein engineering. Engineering a more **evolvable** starting variant, even at some fitness cost, is currently an open question [33] for which machine learning could play an important role. Proteins with high intrinsic stability are thought to be more evolvable [18], and many machine learning models have been developed that either directly (via supervision) or indirectly (as an emergent property of an unsupervised model) predict stability [34–37]. Moreover, we note that directed evolution that begins at a local optimum could still cross fitness valleys via more complex, model-guided strategies beyond greedy exploration, which we review in the following sections.

9.2.2 Building Initial Libraries

The data used to train a machine learning model determines what it learns and, by extension, in what situations it can be used to make effective predictions. For protein engineering, this means that the design of the library that will provide training data is critical to the eventual effectiveness of the trained model in finding improved sequences.

9.2.2.1 Experimental Methods for Library Building

After a starting point for evolution is identified, a variety of molecular biology methods are available for generating local sequence diversity. While these steps are physically separate from the downstream assay, which couples sequence to a fitness label, the assay throughput is a key factor in selecting the method of diversity generation. We introduce several methods and their biases for the ML practitioner below, but notably, these methods are often combined in protein engineering campaigns.

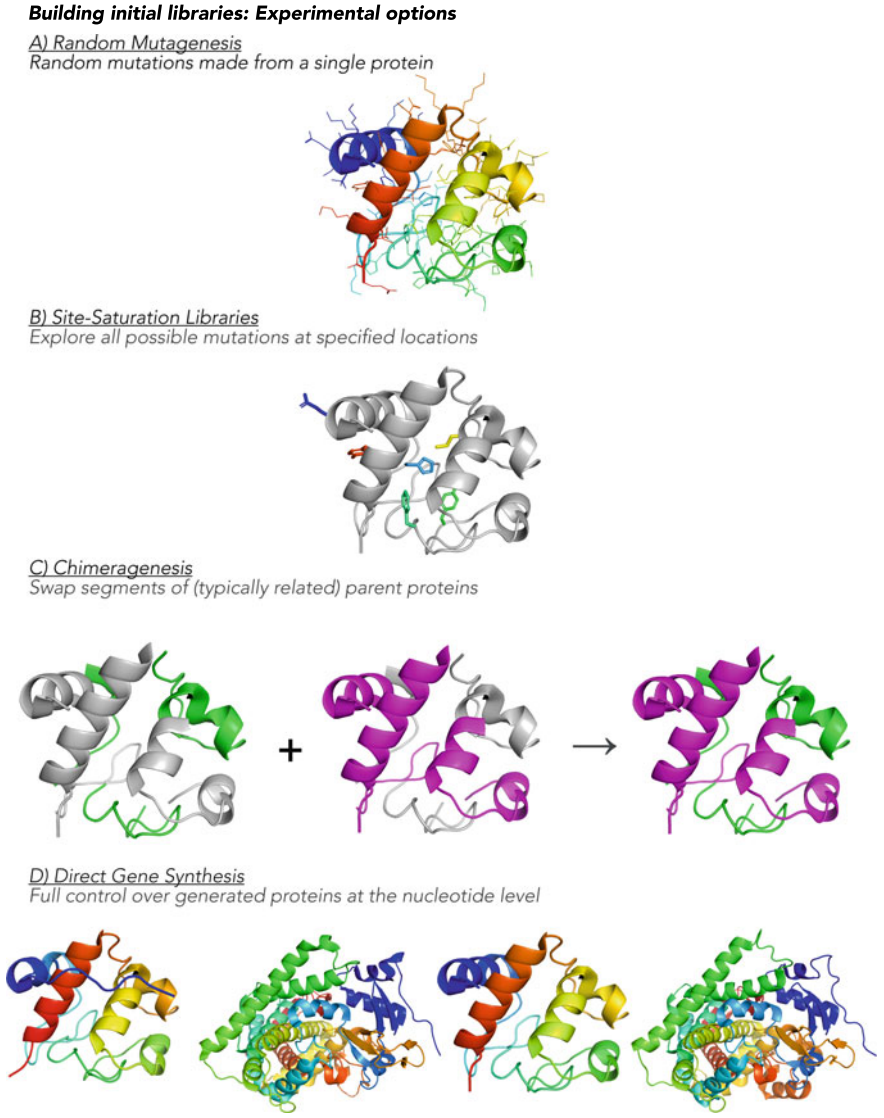


Fig. 9.3 Protein sequence diversity can be generated through a combination of various techniques. **a** Random mutagenesis introduced random mutations throughout the protein. **b** Targeted mutagenesis focuses either the available amino acids or the residue positions at which mutations are made. **c** Chimeragenesis combines fragments from various parent proteins. **d** Finally, direct gene synthesis is a relatively new approach that is becoming affordable for protein engineering campaigns

Random mutagenesis (Fig. 9.3a) is one of the most straightforward methods for creating initial sequence diversity. Errors are introduced throughout an initial DNA sequence by either randomly damaging DNA or by introducing errors during replication such as via epPCR, a process that introduces mutations randomly by increasing the error rate of the copying enzyme, the polymerase. However, it is important to note that errors introduced via "random" mutagenesis are not perfectly random in two major ways. First, mutations from one nucleotide to another do not occur at identical frequencies, so the original base can dictate what mutations are most likely at a given position [38]. Second, the genetic code is redundant, with the twenty canonical amino acids encoded by 61, three-nucleotide codons. Although it is possible for multiple nucleotides within a single codon to mutate simultaneously, this is rare, and generally only one nucleotide mutation occurs per codon during random mutagenesis, limiting the mutations available at the amino acid level. In comparison to other mutagenesis methods, the protein engineer has much less control over the generated library—only the rate of mutations can be changed.

Targeted mutagenesis, of which the focused mutagenesis discussed in the previous section is a subset, is an alternative to random mutagenesis that affords more control over the final library. Unlike random mutagenesis, targeted mutagenesis typically assumes that either (1) specific sites in the protein sequence are important to mutate or (2) it is important to be able to access all amino acids at a given position. Site selection usually requires structural knowledge or other biochemical insights into the protein system and, unlike random mutagenesis, any amino acid can be accessed with equal probability. The pool of degenerate oligos can also be modified to achieve a relatively even distribution across all twenty canonical amino acids [39] or to achieve a different distribution of interest [40]. Importantly, as the desired amino acid distribution becomes more complex and more sites are mutated simultaneously, both difficulty of laboratory implementation and the cost of oligos can become untenable, approaching direct gene synthesis costs. Therefore, ML methods relying on targeted mutagenesis for either training set design or evaluation of predicted designs must keep in mind these constraints.

Another strategy for library generation is recombination, which pieces together, or "recombines", initial diversity into different arrangements to create new diversity. The choice of recombination strategy typically relies on the type of initial diversity on hand. Such diversity could be comprised of a set of functional, **homologous** proteins, the top variants from a random mutagenesis library, or the top variants from a targeted mutagenesis library.

Another recombinatorial approach of note is the use of SCHEMA libraries [41], where fragments of multiple parent proteins (selected by conserving contacts) are swapped, and which has been successfully engineered with machine learning methods [42, 43]. Importantly, some recombination strategies are quite experimentally straightforward and a single round of recombination on top variants can yield much higher improvements in fitness than a single round of random or targeted mutagenesis. Overall, recombination is a very broad category of diversity generation, and due to the array of recombination strategies available, we have only briefly mentioned a few. Further strategies are well reviewed by Packer and Liu [25].

Library design has typically been constrained by methods of library generation, but there is promise to disrupt these traditional paradigms with the advent of cheap gene synthesis technologies [44]. Rather than starting with an initial sequence or pool of sequences and building diversity with mutagenesis or recombination, a set of desired sequences can be synthesized directly for under $100 per protein—a cost which is currently dropping, particularly for proteins shorter than 100 residues. Ordering pools of sequences with targeted or random mutations is also possible, and emerging synthesis technologies can impart more control over the final distribution. As these DNA synthesis technologies improve, ML methods for protein engineering can begin to leverage and propose precisely-defined libraries that were previously cost-prohibitive.

9.2.2.2 Applying Machine Learning to Library Building

Machine learning models are more effective at making predictions when applied to the same data distribution that was used to train them. In general, for a given **design space** of allowed proteins, this means aiming to fully cover the design space by maximizing the sequence diversity of the training data. Because random collection of sequences from a fixed design space will generally yield diverse sequences, it can be effective for initial training data collection. Random collection of training data is also an attractive approach based on available lab methods (see Sect. 9.2.2.1), and this strategy has been combined with machine learning to engineer halohydrin dehalogenase [45], fluorescent proteins [46, 47], and an adenovirus capsid protein [48].

Unfortunately, experimental data collection is expensive and limited for most applications, making it generally impossible to build a perfect map of a fitness landscape through random collection. This is particularly true for fitness landscapes that cover large swaths of sequence space; a small, randomly selected training dataset is unlikely to contain enough information to build a complete map of the design space. Fortunately, the goal of ML-assisted protein engineering is not to perfectly map the design space, but rather to identify protein variants improved in fitness, as discussed further in Sect. 9.2.5.2. As a result, rather than collecting random training datasets, it can instead be preferable to focus initial data acquisition on regions of the design space believed *a priori* to contain higher-fitness variants. Assuming such focused libraries can be built, a model trained on them will have greater predictive ability in the design regions of interest (those containing high-fitness variants) than others. This approach better aligns with the overall goals of ML-assisted protein engineering, as it allows greater error in prediction of the lowest-fitness variants (where high resolution is not needed—simply knowing that those variants are not fit is good enough) and encourages lower error in prediction of the highest-fitness variants (where there are typically less examples and ranking is more important).

Focused libraries can be particularly helpful when navigating hole-filled protein fitness landscapes [49]. When compared to design spaces built by mutating one position in a protein at a time, **combinatorial** fitness landscapes built by simultaneously mutating multiple positions tend to be enriched in non-functional variants, which

are commonly referred to as "**holes**" in the fitness landscape. These holes are distinct from the fitness **valleys** discussed previously, as holes provide no information about the effect magnitude that mutations have on protein fitness, which is valuable information for training the regression models typically employed for ML-assisted protein engineering.

Wittmann et al. demonstrated that by using so-called zero-shot predictors—models or strategies that can predict protein fitness prior to collection of new experimental data—focused training sets can be constructed that minimize inclusion of holes in training data [49]. Through simulation on a complex, hole-filled, combinatorial fitness landscape, they showed that models trained with these focused training sets tend to be far more effective at identifying the highest-fitness variants than models trained with data drawn randomly from the landscape.

The prior information needed to construct focused libraries can come from many sources. For instance, prediction of protein thermal stability [49], use of meta-predictors of protein fitness [50], or strategies based on evolutionary conservation can all be used to make zero-shot predictions of protein fitness [36, 51–54]. The exact strategy that will be most effective, however, will vary depending on the fitness and protein being optimized. Take, for instance, zero-shot strategies that rely on sequence conservation. Such strategies assume that evolutionary fitness aligns with whatever fitness is being predicted; that is, they assume that mutant proteins more closely resembling known protein sequences (found in databases of protein sequence such as UniProt [55]) are more likely to be functional than others. Should this assumption not hold (for instance, the fitness of a protein being engineered for a new-to-nature activity may not correlate well with evolutionary fitness), or if there are simply not enough homologous protein sequences available to build an effective sequence-based zero-shot prediction model, then the zero-shot predictions are likely to be inaccurate. Inaccurate zero-shot predictions are unhelpful for focused library design; indeed, they may even be detrimental to effective learning by focusing training data collection on regions of the fitness landscape dominated by holes.

Ultimately, the decision between random library design and focused library design will depend on a number of factors. If the fitness landscape to be explored is expected to be minimally complex with few holes and large amounts of training data can be easily collected for it, then random library design is a reasonable approach, as the library itself will be simple to construct, and training data gathered from it will be sufficient to build a comprehensive map of the fitness landscape. If the fitness landscape to be explored is complex, full of holes, or it is challenging to gather training data for it, then focused libraries may be more viable, particularly if high-confidence zero-shot predictions can be made for the fitness landscape. Such libraries may be more challenging to construct in the laboratory, but they will likely result in more efficient machine learning-guided engineering. These are the applications upon which the dropping costs of gene synthesis will have the largest impact.

9.2.3 *Assaying Protein Fitness Creates Labels for Machine Learning*

With proteins performing such a wide variety of different functions, the protein engineering community has had to devise countless different assays to measure them all. As such, assays for protein function vary widely in both accuracy and throughput, with ranges from tens to millions of protein variants (Fig. 9.4). This amount is typically dependent on the project definition of fitness. For instance, if the measurement of fitness can be directly coupled to a sequencing assay (as in deep mutational scanning), then large datasets (10^5–10^6) can be rapidly created. Many assays for fitness, however, are limited to comparatively low-throughput chromatographic methods (e.g., HPLC, LCMS, GCMS, etc.), which rely on physical separation of a mixture through a column, producing smaller datasets (10^1–10^4). The amount of data available for training a model will dictate how much of sequence space can be explored and how accurate the predictions of fitness for new sequences will be. At the same time, however, the goal of ML in protein engineering is to reduce the burden of experimental screening and expedite the process, so a balance must be found between the amount of data collected and the accuracy of predictions of models trained on that data.

Additionally, one should consider where an ML method for protein engineering may have the biggest impact on efficiency, cost, and time. Protein functions that can be assayed in high-throughput provide more data for training downstream ML models; however, applying an ML-based engineering method to such a function may not impart as much benefit as it would being applied to engineering a function with a low-throughput assay. Therefore, many recent efforts for building ML methods for protein engineering have focused on the **low-N** regime, where few samples are used for training [43, 49, 56–59].

Final considerations, for both the protein engineer choosing an assay and the ML scientist choosing a dataset, are the assay bias and noise. As interest has grown in applying ML to protein engineering, method developers have sought out sequence-fitness datasets with which to benchmark their approaches. This has resulted in the pursuit of large datasets built by multiplexed assays of variant effect like deep mutational scanning (DMS) [60] and using some fitness-related screen or selection such as fluorescence-activated cell sorting (FACS) for a fluorescent protein [61]. Inherent biases in such protein fitness datasets are left under-discussed, making it difficult for the ML community to discern what datasets are appropriate for a given application. As a couple of examples of such biases, DMS fitness measurements can be heavily impacted by the input library and can result in highly non-uniform error across a dataset [62] and FACS-based assays have inherent bias due to fluorescence-based binning. Post-binning sequence processing attempts to smooth the categories into a continuous function, but systemic bias persists, leading Trippe et al. to propose a random gating strategy to reduce it [63]. Noise can be more difficult to address as there tends to be a balance between throughput and measurement accuracy for biological systems. However, examining and quantifying the noise within a dataset

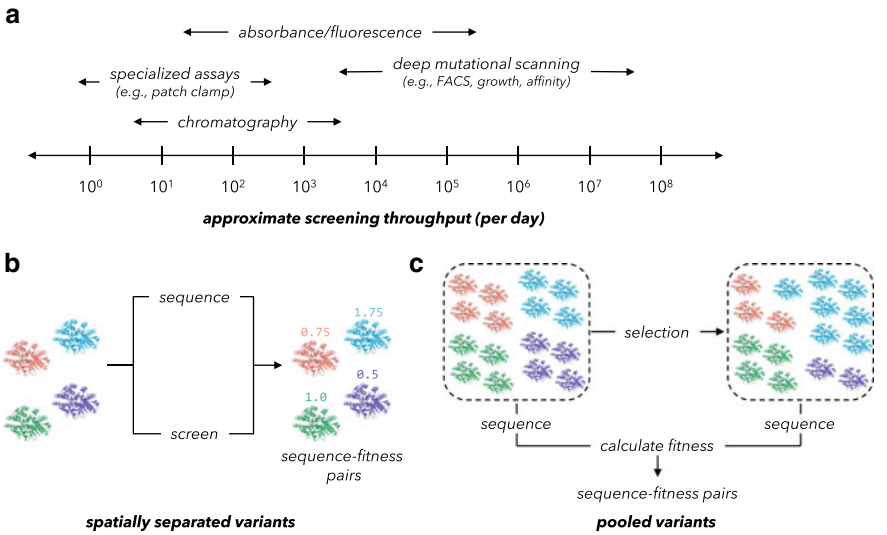


Fig. 9.4 There are many types of screens for protein fitness. **a** Screening throughput can range from less than one variant per day for some specialized experiments to over 10^7 per day for some DMS-based screens. **b** Many protein screens require spatial separation of variants, including many chromatographic methods. This then requires separate sequencing and screening processes that can be later be combined into complete sequence-fitness datasets. **c** With deep-mutational scanning assays sequencing and fitness are linked. Sequencing is performed pre- and post-selection and changes in read frequency are used to calculate a fitness value for every variant

can be important for setting expectations for the success of downstream ML models. This could be done by comparing a subset of high-throughput measurements to more accurate, low-throughput measurements or using published statistical packages for data processing and analysis [62, 64]. Note that all fitness assays have their own noise and bias which should be thoroughly considered when developing an ML method on a particular dataset.

9.2.4 Pairing Sequences to Fitness Data Through Sequencing Creates Features for Machine Learning

Uncommon in the broader ML discipline, datasets that result from typical protein engineering campaigns are label-rich and feature-poor, with many assayed fitnesses and relatively few variant sequences. Because only function is being optimized in directed evolution, this process does not technically require sequencing. Indeed, only the top few variants from each round are sequenced for validation in practice, and sequencing all of the variants is considered an unnecessary, unjustifiable expense. Thus, the remainder of the unimproved variants are discarded without sequencing,

resulting in fitness labels that are rendered useless. Notably, this is not true for libraries where fitness is directly coupled to sequencing, which is a large part of the reason many ML approaches are currently developed on these types of datasets [31, 49, 58, 65, 66].

Variant sequencing, especially for low-throughput assays, has traditionally been done via Sanger sequencing, but this method scales linearly with the number of variants, costing a few dollars per sequence. Depending on the assay used for evaluating protein fitness, sequencing could easily become the most expensive part of a protein engineering campaign. Fortunately, next generation sequencing (NGS) technologies [67] have begun disrupting this paradigm. New sequencing methods that incorporate NGS show promise to continue shifting the sequencing paradigm by spreading reads over many, multiplexed sequences, enabling large-scale sequencing for independently screened fitnesses. For such methods, there is a trade-off between read length and cost per sequence. Both long- and short-read methods can return full-length sequences, but they typically require filling an entire flow cell, which can cost upwards of $1000 per run and a few dollars per variant [68, 69]. Alternatively, short amplicons from within an entire gene, which are restricted to only a few hundred amino acids, can be sequenced for cents per variant and are a cost-effective compromise when variation is confined to a smaller region of a gene [70]. Nonetheless, both sequencing technologies and sequencing methods for the protein engineering community have been growing rapidly in the past few years, showing promise to address these shortcomings in the near future.

9.2.5 Training a Protein Sequence-Fitness Model

Fitness datasets obtained as described above and can be used to fit models of the sequence-fitness relationship. Here, we first describe the methods for representing a protein sequence before diving into the model classes used to predict fitness from sequence.

9.2.5.1 Representing Protein Sequences

Proteins are variable-length sequences of twenty canonical amino acids. Due to both the variability in sequence length and the categorical nature of amino acids, an important and long-standing problem has been how to best represent a protein sequence to facilitate machine learning. Roughly speaking, there are two general approaches for doing this: (1) hand-crafting features based on, for example, statistical summaries or biophysical properties of a sequence or its constituent amino acids, and (2) using evolutionarily related sequences to learn likelihoods or embeddings as representations.

Statistical summaries of a sequence: higher-order terms and k-mers. One of the simplest ways to represent a protein sequence is to concatenate one-hot encodings

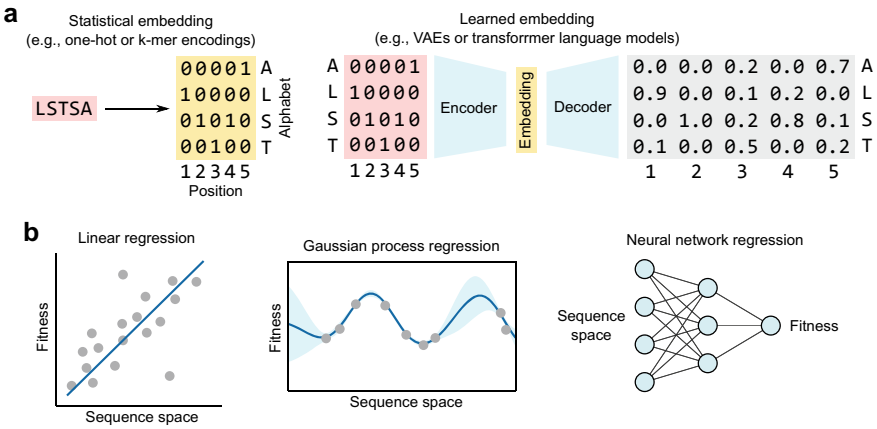


Fig. 9.5 **a** Methods for converting a protein sequence into a representation that can be used as input to a machine learning model include statistical summaries of a sequence (shown here is a cartoon depicting a one-hot encoding, in which sequences are converted to binary values at each sequence position) or learned sequence embeddings, often by training a machine learning model on large collections of sequences (shown here is a cartoon depicting a VAE-based embedding, in which the latent variable learned by a VAE is used as the sequence representation). **b** Once sequence representations have been computed, they can be used as input to supervised models that predict fitness from these representations. These models include linear regression from sequence space to fitness, Gaussian process regression, or a regressor implemented by a neural network

of the amino acid at each position in the sequence (Fig. 9.5a). Despite its simplicity, even a linear model with this representation can be a surprisingly effective baseline in some settings—for example, when we focus on predictive performance on sequences near a wild type [56, 65]. Amino acid similarities may also be directly encoded through matrices based on substitution probabilities of one amino acid for another [71]. Beyond encoding just the amino acid identities at each sequence position, one can also encode the identities of the amino acids at pairs of structurally contacting positions [42, 72].

Generalizing this idea further, one-hot encodings of groups of amino acids, which we call *higher-order terms*—for example, the identities of the amino acids at three positions for a third-order term—have been shown to be both theoretically principled and empirically useful representations for learning protein fitness functions [66, 73–79]. In particular, it has been shown that any fitness function defined over sequences of a fixed length can be expressed exactly as a linear function of all possible higher-order terms of all orders between sequence positions [73]. The number of such terms grows exponentially with sequence length. However, there is increasing evidence that many realistic protein fitness functions can be well-approximated by a linear model of just a small number of these terms [78, 80, 81]; an effective baseline representation may therefore be to one-hot encode all terms up to, for example, second- or third-order [56, 76, 79].

Beyond higher-order terms, the frequencies of different *k-mers*—contiguous subsequences of k amino acids—are another kind statistical summary that has been used to represent sequences and learn fitness functions [82, 83].

Physicochemical features. Beyond one-hot encodings of a sequence's residues, or the higher-order terms between them, features based on physicochemical properties of a sequence have been used to learn fitness functions. For example, each amino acid can be featurized as a vector of various physicochemical properties [84, 85] or by a low-dimensional representation thereof [86–88]. These properties can also be combined with structural information [89–91], or with the aforementioned statistical summary representations [82].

However, it not generally clear which physicochemical properties will be relevant for any particular fitness function. Consequently, in recent years considerable attention has turned to *learning* relevant features by fitting low-dimensional representations of evolutionarily related sequences.

Representations learned from evolutionarily related sequences. In directed evolution, one generally starts by identifying a **wild-type** protein (or multiple such wild types) that exists naturally in the genome of some organism and exhibits the function of interest to an appreciable degree. In many cases, that function of interest is needed by more than one organism, and natural selection has yielded multiple homologs residing in a variety of organisms. Though the precise fitness value differs across homologs, due to evolutionary constraints homologs are generally functional as opposed to non-functional. Based on the premise that homologs are mostly functional, a great body of work in recent years has focused on learning the distribution of homologous sequences, as a proxy for the distribution of functional sequences. The density, or approximations thereof, of the resulting learned sequence distribution has been shown to correlate with fitness in many cases, whether modeling the homologs with a hidden Markov model [56, 92, 93], Potts model [53, 94], variational autoencoder [54, 95, 96], or autoregressive model [27], leading to its use as an informative correlate of fitness that does not require any experimental measurements.

Currently, variational autoencoders such as [54] appear to provide the highest correlations with fitness for a variety of fitness datasets, presumably due to their ability to learn evolutionary constraints on higher-order interactions between sequence positions, though autoregressive models [27] and transformers [36] also achieve comparable performance. Due to this correlation, approximations of the density such as the evidence lower bound (ELBO) have shown to be an effective feature when used with supervised learning approaches [56]; the latent space learned by variational autoencoders has also been used to represent proteins for learning fitness functions [95] (Fig. 9.5a). Finally, generative models fit to homologs have also been used to sample novel functional proteins [32, 97, 98].

Beyond homologs, which, by virtue of natural selection, contain sequence patterns necessary for the function of interest, the set of all proteins sequenced so far may similarly reveal patterns necessary for basic properties of all functional proteins, such as the ability to fold into a stable structure. Based on this hypothesis, a growing body of work has trained various deep learning models on massive databases of known proteins, such as UniProt [55], UniRef [23], and UniParc [55], which contain up

to hundreds of millions of sequence as of this writing. The model architectures are almost all borrowed from natural language processing (NLP), ranging from relatively simple skip-gram neural networks [99, 100] and its extensions to embedding text documents [101, 102] to recurrent [57, 103, 104] and convolutional neural networks [105] to, most recently, transformers [31, 36, 106–109]. These models are generally trained using self-supervision tasks, where the goal is, for example, to predict the amino acid at a certain position given the amino acids at other positions (though other approaches such as multi-task learning using function labels [105] and incorporating protein structural information [104] have also been used). The outputs of intermediate layers in these models, which we will call deep embeddings, can then be used to represent protein sequences for learning fitness functions.

Estimates of the density from transformers have also been shown to correlate with a variety of fitness functions [36, 108], presumably both because the model learns sequence patterns required for basic properties of all functional proteins, such as stability, and because the training data contains all homologs corresponding to all known functions. Whether using them for deep embeddings or density estimates, if homologs of a functional wild type are also available for a function of interest, these models can also be further trained or "fine-tuned" on them [32, 57, 108] to focus on a particularly relevant region of sequence space.

Despite promising results on capturing phylogenetic relationships and various clustering tasks, these deep embeddings have shown limited success in learning fitness functions. For example, when used with ridge regression or Gaussian process regression, they do not appear to systematically outperform simpler baselines [56, 65, 102] such as concatenating the aforementioned density of a homolog sequence distribution to a one-hot encoding [56], particularly when there is limited training data.

9.2.5.2 Modeling the Sequence-Fitness Relationship

Given a representation of protein sequences, a variety of model classes have been used to learn the sequence-fitness relationship (Fig. 9.5b). In general, however, note that models that excel under more traditional evaluation metrics—for example, prediction error on held-out sequences from the same distribution as the training data—may or may not be appropriate for use with directed evolution, since in the latter we focus on sequences the model predicts to have high fitness. That is, a model that facilitates directed evolution needs to be reliable even in regions of sequence space far from the training data, particularly in regions on which it predicts high fitness.

With that in mind, a few domain-specific models have been designed with the particular goal of jointly learning (1) a density model over homologs and (2) the fitness function from experimentally labeled sequences [110, 111], under the premise that the density of homologs is positively correlated with the fitness and that therefore both models should stand to gain from sharing information via joint training.

In the more common setting of using only experimentally labeled sequences, two models have been shown to be very strong baselines across a variety of fitness

datasets, particularly in the regime of limited training data (i.e., at most a few hundred labeled sequences): linear regression, for example on one-hot encodings [45, 65, 112], optionally concatenated with the density of a homolog sequence distribution [56], and Gaussian process regression with a variety of kernel functions, which measure the similarity between proteins. These methods include both domain-specific kernels (e.g., structurally informed [42, 72, 113] or mismatch string [102, 114] kernels, which count shared subsequences between proteins) or more general-purpose kernels [43, 46, 83, 91, 102, 115].

Other methods using kernels have been used as well, such as support vector machines for modeling enzyme enantioselectivity [116], membrane protein expression [117], and protein thermostability [90, 118–124]. The latter task, which has been of widespread interest due to its necessity in industrial applications, has also been approached using regression trees and their extensions [118–120].

When labeled sequences are abundant (that is, at least tens to hundreds of thousands), deep learning models can also be effective for learning the fitness function. Models range from simple feedforward networks, which have been used to predict fluorescence [125, 126], histidine synthesis efficacy [127], and viral capsid packaging ability [77], to convolutional neural networks [48, 65, 128, 129], recurrent neural networks [48, 103, 130], and transformers [65, 130], which have been assessed on a variety of protein tasks including fluorescence, thermostability, and binding affinity prediction.

Limitations of current model comparisons. Our understanding of what representations and models are most effective is limited by the constraints of the molecular biology techniques available to generate fitness datasets. For example, the vast majority of currently available fitness datasets only assess the fitnesses of mutants one or two mutations away from a wild type, excluding insertions and deletions [27, 50, 53, 54, 56]. This choice is usually justified by the assumption that most mutations are deleterious [131]; therefore, to generate data with informative, non-zero labels, it is beneficial to stay near sequences already known to be functional. Pragmatically, the resource and throughput constraints on mutagenesis and sequencing protocols described in Sect. 9.2.3 further discourage protein engineers from making more simultaneous mutations. If one intends on generating training data near a functional wild type, comparing representations and models using these datasets may be sufficiently informative. However, beyond this limited setting, fitness datasets with measurements for proteins more broadly dispersed in sequence space are needed to comprehensively understand what machine learning strategies are most effective for learning fitness functions. Dallago et al. [65] have collected several such datasets in recent work; there are also a few datasets with fitness measurements for all possible protein sequences varying at a small number (typically less than twenty) of positions [76, 127, 132], which comprehensively expose the effects of all possible combinations of (albeit a limited number of) mutations.

Current evidence suggests that, when focusing on relatively small regions of sequence space—for example, clustered around a wild type—simple models such as ridge regression on one-hot encodings can be much more effective than more complex models [56, 65], but when considering predictive performance on a broader

region, given sufficient training data there may be benefits from using transformers pretrained on massive protein databases [65]. Further work is needed to systematically investigate the machine learning strategies appropriate for the different types of training and test data that arise in protein engineering.

9.2.6 Protein Selection and Optimization

After obtaining a trained sequence-fitness model, the goal is to use it as part of an **acquisition function** (in the Bayesian optimization sense) to decide which sequences from the theoretical library to characterize next.

9.2.6.1 Specifying the Available Design Space

While calling a sequence-fitness model is much faster than experimental validation, it is still impossible to exhaustively evaluate all possible protein sequences, and the design space must be restricted or implicitly defined. One method of restriction is to limit the **design space** of proteins either by biomolecular or pragmatic constraints. Common examples of design spaces inspired by molecular biology constraints include all single- or double-mutants, all the possible mutations at a small number of sites [49, 58, 132], recombining sequence or structure segments from multiple parent proteins [41, 42, 133, 134], or stochastic libraries that are easy to make using current DNA synthesis techniques [40].

The design space can also be implicitly defined by a *generative model*, allowing examples to be directly generated from the learned distribution. The most straightforward encodings of proteins consider individual or pairs of positions [53, 97]. More recently, researchers have used deep generative models, such as variational autoencoders (VAEs) [54, 135], generative adversarial networks (GANs) [136–138], and autoregressive language models [27, 31, 32, 104, 139], to learn more complex distributions over sequences from which to generate sequences [140].

9.2.6.2 Defining the Acquisition Function

An acquisition function combines sequence information and the sequence-function model to select sequences from the design space for experimental measurements. A simple example of an acquisition function is greedy selection of the top prediction(s) from the sequence-function model. Greedy acquisition is common and can work well in protein engineering [45, 48, 49, 57, 58, 141], but it can become trapped in local optima in more complex fitness landscapes.

Additionally, many standard acquisition functions select only a single example in each round before updating the sequence-fitness function. However, biology techniques are fairly parallelized, and it is often faster and cheaper to obtain data in

batches. Simply acquiring several of the top-ranked sequences according to the acquisition function may result in evaluating many similar sequences, leaving the design space underexplored. Batched acquisition functions are therefore designed to encourage intra-batch diversity [42, 142–146], but this remains an open area of research.

Given a theoretical library and an acquisition function, the simplest method for choosing sequences is to predict the fitness for as many members of the library as is computationally feasible, and then to use those scores to select sequences. However, when the library is defined by a generative model, it can be shifted towards more optimal sequences with *adaptive sampling*, which uses predicted fitness values for previously-generated sequences (from a sequence-fitness model) to re-estimate the parameters of the generative model until convergence [147]. For example, adaptive sampling with a VAE was used to optimize protein expression at the nucleotide level [147] and with a GAN to design antimicrobial peptides [148].

Naive adaptive sampling assumes that the sequence-fitness model is trustworthy; however, these models are inaccurate in many regions of the design space. These regions can be avoided by constraining the generative model to sample sequences close to the training distribution for the sequence-fitness model [126] or by retraining the sequence-fitness model to account for covariate shift [149]. Additionally, sampling from an ensemble of generative models, instead of a single model, also improves generation quality [150, 151].

Similarly, genetic algorithms can be used to produce new sequences [152] by adaptively querying a sequence-fitness model to preserve sequence designs [146]. Alternatively, the sequence-fitness model can be inverted to identify the most desirable of the design space. An inverse of the sequence-fitness model can be trained via an iterative procedure similar to adaptive sampling [153], or, given a *differentiable* sequence-fitness model, can be obtained with gradient-based methods [28, 154, 155].

While it is computationally convenient to exactly specify the sequences to measure next, specifying a new library and sampling from that library may allow many more sequences to be measured for the same cost as directly synthesizing a small number of sequences. This library may be optimized for expected improvement [145], to balance diversity and expected function [77], or to match a desired distribution over sequences [40].

9.2.6.3 Updating the Sequence-Fitness Model with Experimental Knowledge

So far, we have discussed methods for using an acquisition function based on a sequence-fitness model to choose sequences from a theoretical library that are optimized for a desired function. However, in some cases it may be possible to sequentially update the sequence-fitness model with new measurements. The simplest method for sequential optimization is greedy selection of the best-predicted sequence at each step, which is used widely in practice [45]. More sophisticated and

principled methods balance **exploiting** the knowledge in the sequence-fitness model and **exploring** in order to improve the sequence-fitness model [156–159].

Bayesian optimization is a popular framework for sequential optimization [158] that relies on an acquisition function that systematically combines the sequence-fitness model's prediction with its associated uncertainty to navigate this exploration-exploitation tradeoff. For example, the upper confidence bound (UCB) acquisition function uses a weighted sum of the predicted value and uncertainty, with a larger uncertainty weight encouraging more exploration [157, 160]. UCB is used widely in protein engineering [42, 72, 161, 162] and enjoys strong theoretical guarantees [158]. A popular alternative category of acquisition functions selects the example that is predicted to have the largest improvement in expectation over the current best example or compared to a randomly-drawn example from the previously observed set [163]. Uncertainty can also inform exploration in batch or parallel acquisition, in which multiple protein sequences are acquired and evaluated before updating the sequence-function model [42, 142–145].

Gaussian process models [164] are a popular family of sequence-fitness functions for Bayesian optimization because of their simplicity, flexibility, and strong performance. However, Bayesian neural networks, in which the parameters of the network are themselves random variables with associated prior distributions, can also combine accurate predictions with uncertainty estimates, although efficient and accurate inference in these models can be challenging [165]. One particularly interesting approach is to combine a fitness-space prior (e.g. stability predictions) with the standard neural network weight-space prior [166], and more examples are described in Sect. 9.2.5.2.

Probabilistic sequence-fitness models can also be implemented by model ensembles [167], which train multiple sequence-to-fitness models on the same data and estimate uncertainty from the variance in model predictions [28]. Ensembles are not Bayesian by default, so incorporating prior information into these models can be challenging [168, 169].

Finally, conformal prediction (CP) is an approach for rigorous uncertainty quantification that can be used with any predictive model in a black-box manner, including the aforementioned Gaussian process models or any deep learning model [170–172]. The framework leverages the assumption of exchangeable (e.g., independently and identically distributed) training and test data to construct confidence sets around predictions that provably satisfy *coverage*, a frequentist guarantee on how often the confidence sets contain the true label of test inputs. Beyond exchangeable data, CP has been generalized to handle various forms of distribution shift, including label shift [173], covariate shift [174, 175], distribution shifts in an online setting [176], and test distributions that are close to the training distribution [177]. Notably, in recent work Fannjiang et al. generalized CP for the setting of protein design, in which any of the aforementioned optimization procedures induces a distribution shift wherein the distribution of test (i.e., designed) sequences is dependent on the training data [79]. This method enables practitioners to assess the predictive uncertainty of any

optimization procedure, using any machine learning model, and as such can be a principled approach for selecting an optimization procedure (or any component thereof, including any hyperparameters or the machine learning model class).

9.3 Outlook and Concluding Remarks

In this chapter, we have discussed current efforts to improve protein engineering with machine learning. The field has grown rapidly over the past five years, as the development of deep learning methods and methods to obtain and organize biomolecular data have reached inflection points. We anticipate continued growth in this area of research from both directions and expand on two areas of particular interest here.

9.3.1 Model Calibration

The space of possible amino acid combinations is vast, but the space of functional proteins is much smaller, with estimates ranging from one in 10^{11} [178] to as low as one in 10^{77} [179]. The field is regularly identifying new proteins through metagenomic analysis [180], but even if we had a record of all naturally existing proteins, nature may not have completely explored the manifold of functional proteins. Therefore, it is unclear whether characterizing identified natural proteins is sufficient to identify the manifold of functional proteins.

As the goal of protein engineering is to achieve properties beyond what has been thus far observed in proteins, it is particularly critical that machine learning methods navigate sequence space in a way that strikes a balance between exploring new sequences and staying near sequences that a predictive model was trained on, where its predictions tend to be more trustworthy. Methods have characterized and leveraged these trust regions in a variety of ways [57, 79, 126, 150, 181]. In order to characterize more useful trust regions, which will allow for more effective exploration of sequence space, methods for model calibration—accurately quantifying the uncertainty of a model—will become increasingly important. Additionally, there are few combinatorially-comprehensive fitness datasets with which to evaluate existing approaches. A nearly comprehensive GB1 study [132] has been remarkably useful, as have synthetic datasets, but the field would benefit from a wide and diverse set of epistatic fitness landscapes with which to evaluate machine learning methods.

Another reason model calibration is critical is the measurement noise present in fitness data. Common sample preparation methods often introduce noise, may have limited linear range of measurement, and can be difficult to reproduce, particularly for newly developed assays designed to probe more complex biological phenomena. While training methods that account for noisy or subjective labels have gained some traction in other deep learning applications [182, 183], in protein engineering noisy labels are typically only addressed during data pre-processing (e.g., by labeling the

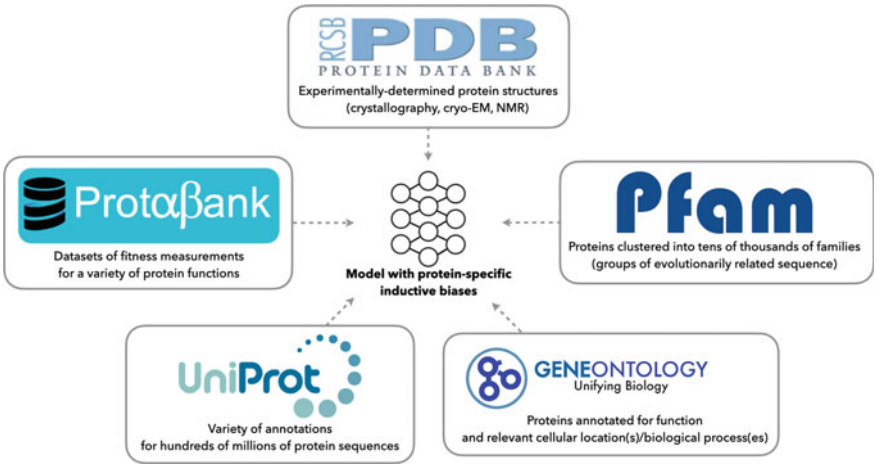


Fig. 9.6 Future machine learning methods for protein engineering will integrate the diverse types of information available from different protein databases, in ways that will improve both predictive performance and calibration in unexplored regions of sequence space

data with Gaussian Mixture Models [27, 48]), although a recent method demonstrates that introducing biophysical priors on the labels can improve predictions [166].

9.3.2 Protein-Specific Data Modalities and Inductive Biases

Much of this chapter is focused on the guided acquisition of data for a particular protein function, which is being collected and curated by ProtaBank [184]. However, the amount of information available for proteins in other data modalities is also rapidly increasing (Fig. 9.6). Many databases exist for various properties of proteins, ranging from protein stability to binding. Of particular note are UniProt [55] and the Protein Data Bank (PDB) [185, 186].

UniProt contains a large amount of information about proteins beyond their sequence, including cross-references to functional labels, disease-association and Protein family (Pfam) classifications at the per-residue level [187], Gene Ontologies on the per-protein example [188, 189], and links to several other databases. UniProt also releases reference protein clusters (UniRef [55, 190]), which are currently almost ubiquitous as the training set for large protein language models [107]. This dataset will continue to grow as more metagenomes are sequenced [180] and more unique proteins are identified. Some approaches to modeling proteins are conditioned on protein functional labels [30] or other data available in UniProt [31], but there is no clear optimal approach to incorporating annotations about all proteins for protein engineering campaigns, which are often focused on specific protein families.

The PDB is the primary source of experimental protein structure data, and it largely consists of static protein structures obtained through protein crystallography, although the number of structures obtained through cryo-EM and NMR are also increasing. Protein structures are invaluable to biologists in providing much-needed context for molecules that are otherwise difficult to probe. However, mutations may have effects that are not captured by static structures. For example, they may bias the protein's Boltzmann distribution toward different conformational states in the ensemble without perturbing the ground state crystal structure, or the crystal structure may simply have too low resolution to capture small changes. Nonetheless, structure can be a useful prior in directed evolution [49, 104] and can also be directly applied to identify beneficial mutations [191].

While protein data continues to grow, methods with inductive biases tailored for biology are comparatively underdeveloped, particular for protein engineering applications. A few notable exceptions have been developed in application to protein structure modeling. For example, the MSA Transformer [108] develops a variant of axial attention specific for MSAs, which contain an aligned homolog sequence in each row. The protein sequences in an MSA likely share similar structure, and the MSA transformer is able to leverage this structure by tying row attention maps.

AlphaFold [13] has the potential to enable protein sequence-based protein engineering methods to incorporate structural information. AlphaFold incorporated several biophysical inductive biases in the CASP14 protein structure prediction contest. These include a variant of attention to account for the triangle inequality on distances, a lowered emphasis on the linear input sequence of a protein (which folds into a three-dimensional structure), and a variant of axial attention for the MSA. These methods have been successful for improving protein structure prediction, and it is likely that machine learning methods adapted from them (or even the predictions themselves) will improve each step of the protein engineering cycle.

9.3.3 Summary

Protein engineering is an optimization strategy, and many recent advances in machine learning have been successfully applied or developed to the directed evolution of proteins. In this chapter, we have highlighted studies where machine learning contributes to each step of the optimization process. As we continue to gather more protein information and develop methods to bootstrap and guide engineering from these data, machine learning will play an increasingly important role in protein engineering.

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